



US012412642B2

(12) **United States Patent**
Cowens et al.

(10) **Patent No.: US 12,412,642 B2**
(45) **Date of Patent: *Sep. 9, 2025**

(54) **METHOD TO USE GENE EXPRESSION TO DETERMINE LIKELIHOOD OF CLINICAL OUTCOME OF RENAL CANCER**

(71) Applicant: **Genomic Health, Inc.**, Redwood City, CA (US)

(72) Inventors: **Wayne Cowens**; **Steven Shak**, Hillsborough, CA (US); **Audrey Goddard**, San Francisco, CA (US); **Dejan Knezevic**, Palo Alto, CA (US); **Joffre Baker**, Montara, CA (US); **Michael C. Kiefer**, Walnut Creek, CA (US); **Tara Maddala**, Sunnyvale, CA (US); **Frederick L. Baehner**, San Francisco, CA (US)

(73) Assignee: **Genomic Health, Inc.**, Redwood City, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/454,312**

(22) Filed: **Aug. 23, 2023**

(65) **Prior Publication Data**

US 2024/0096451 A1 Mar. 21, 2024

Related U.S. Application Data

(63) Continuation of application No. 16/952,566, filed on Nov. 19, 2020, now Pat. No. 11,776,664, which is a continuation of application No. 15/368,979, filed on Dec. 5, 2016, now Pat. No. 10,892,038, which is a continuation of application No. 12/987,023, filed on Jan. 7, 2011, now Pat. No. 9,551,034.

(60) Provisional application No. 61/346,230, filed on May 19, 2010, provisional application No. 61/294,038, filed on Jan. 11, 2010.

(51) **Int. Cl.**
G01N 33/574 (2006.01)
G06N 7/01 (2023.01)
G16B 25/10 (2019.01)
G16B 99/00 (2019.01)

(52) **U.S. Cl.**
CPC **G16B 99/00** (2019.02); **G01N 33/57438** (2013.01); **G06N 7/01** (2023.01); **G16B 25/10** (2019.02); **C12Q 2600/112** (2013.01); **C12Q 2600/118** (2013.01); **C12Q 2600/158** (2013.01)

(58) **Field of Classification Search**
CPC G16B 25/10; C12Q 2600/112; C12Q 2600/158

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,135,314	B1	11/2006	Leung et al.
7,611,839	B2	11/2009	Twine et al.
2002/0173461	A1	11/2002	Pennica et al.
2003/0109434	A1	6/2003	Algate et al.
2003/0180770	A1	9/2003	Damokosh et al.
2003/0224374	A1	12/2003	Dai et al.
2004/0088746	A1	5/2004	Grimm et al.
2004/0110197	A1	6/2004	Skinner et al.
2004/0110221	A1	6/2004	Twine et al.
2005/0002904	A1	1/2005	Wary et al.
2005/0048542	A1	3/2005	Baker et al.
2006/0088823	A1	4/2006	Haab et al.
2006/0183120	A1	8/2006	Teh et al.
2006/0281122	A1	12/2006	Bryant et al.
2007/0037186	A1	2/2007	Jiang et al.
2007/0099209	A1	5/2007	Clarke et al.
2007/0105133	A1	5/2007	Clarke et al.
2007/0224596	A1	9/2007	Nacht et al.
2008/0032299	A1	2/2008	Burczynski et al.
2008/0064055	A1	3/2008	Bryant et al.
2008/0119367	A1	5/2008	Vasmatzis et al.
2008/0182255	A1	7/2008	Baker et al.
2008/0242606	A1	10/2008	Jiang
2008/0286273	A1	11/2008	Starmans et al.
2009/0023149	A1	1/2009	Knudsen
2009/0035312	A1	2/2009	Griffioen et al.
2009/0186924	A1	7/2009	Billen et al.
2009/0258002	A1	10/2009	Barrett et al.
2009/0280490	A1	11/2009	Baker et al.
2010/0093768	A1	4/2010	Nelson et al.
2010/0152055	A1	6/2010	Kozono et al.
2011/0123990	A1	5/2011	Baker et al.
2011/0129833	A1	6/2011	Baker et al.
2011/0171633	A1	7/2011	Cowens et al.
2012/0142553	A1	6/2012	Smit et al.

FOREIGN PATENT DOCUMENTS

CA	2531091	1/2005
CA	2530738	5/2005
CA	2530738 A1	5/2005
CA	2621932	3/2007

(Continued)

OTHER PUBLICATIONS

Kondo et al. Cancer Sci. 2006. 97(8):780-786. (Year: 2006).*

(Continued)

Primary Examiner — Joseph G. Dauner

(74) *Attorney, Agent, or Firm* — McNeill PLLC

(57) **ABSTRACT**

The present disclosure provides gene and gene sets, the expression of which is important in the classification and/or prognosis of cancer, in particular of renal cell carcinoma.

20 Claims, 9 Drawing Sheets

Specification includes a Sequence Listing.

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CA	2631236	6/2007
CA	2633593	10/2007
JP	2005211023 A	8/2005
JP	2008529554 A	8/2008
JP	2009515148 A	4/2009
WO	WO-2002079411 A2	10/2002
WO	WO-2004048933 A2	6/2004
WO	WO-2004097052 A2	11/2004
WO	2005024603 A2	3/2005
WO	WO-2005117943 A2	12/2005
WO	WO-2006089185 A2	8/2006
WO	WO-2006124022 A1	11/2006
WO	WO-2006124836 A1	11/2006
WO	WO-2007026896 A1	3/2007
WO	2007045996 A1	4/2007
WO	WO-2007072225 A2	6/2007
WO	WO-2008021115 A2	2/2008
WO	WO-2008138579 A1	11/2008
WO	WO-2009105640 A1	8/2009
WO	WO-2010/056374	5/2010
WO	2011085263 A2	7/2011
WO	2012174282 A2	12/2012
WO	2013028807 A2	2/2013

OTHER PUBLICATIONS

Shioi et al. Clin Cancer Res. 2006. 12(24):7339-7346. (Year: 2006).*

Anders M., et al., "Microarray Meta-analysis Defines Global Angiogenesis-related Gene Expression Signatures in Human Carcinomas," Molecular Carcinogenesis, 2011.

Annual Meeting of the Japanese Cancer Association, 2006, vol. 65, pp. 367, P-888.

Chan E., et al., "Integrating Transcriptomics and Proteomics," G & P Magazine, 2006, vol. 6 (3), 4 pgs.

De Kok J.B., et al., "Normalization of Gene Expression Measurements in Tumor Tissues: Comparison of 13 Endogenous Control Genes," Laboratory Investigation, 2005, vol. 85 (1), pp. 154-159.

Extended European Search Report dated Jul. 28, 2016, for European Application No. 15203193.6, 12 pages.

GenBank: AB385541.1 (Oct. 3, 2008).

GenBank: AF043329.1 (Jan. 5, 1999).

Hata M., et al., "Expression of Th2-skewed Pathology Mediators in Monocyte-Derived Type 2 of Dendritic Cells (DC2)," Immunol Lett, vol. 126(1-2), 2009, pp. 29-36.

Hoshikawa Y., et al., "Hypoxia Induces Different Genes in the Lungs of Rats Compared with Mice," Physiological Genomics, 2003, vol. 12 (3), pp. 209-219.

International Search Report and Written Opinion for Application No. PCT/US2011/020596, mailed on Nov. 25, 2011, 10 pages.

NCBI Reference Sequence: NM_003713.3 (Oct. 22, 2008).

Partial European Search Report for Application No. EP15203193, mailed on Apr. 11, 2016, 7 pages.

Search Report and Written Opinion for Singapore Patent Application No. 201204514-2 mailed Jul. 29, 2013.

Sengupta S., et al., "Histologic Coagulative Tumor Necrosis as a Prognostic Indicator of Renal Cell Carcinoma Aggressiveness," Cancer, 2005, vol. 104 (3), pp. 511-520.

Takahashi M., et al., "Gene Expression Profiling of Clear Cell Renal Cell Carcinoma: Gene Identification and Prognostic Classification," Proceedings of the National Academy of Sciences, 2001, vol. 98 (17), pp. 9754-9759.

Tan X., et al., "Global Analysis of Metastasis-associated Gene Expression in Primary Cultures from Clinical Specimens of Clear-cell Renal-cell Carcinoma," International Journal of Cancer, 2008, vol. 123 (5), pp. 1080-1088.

Unwin R.D., et al., "Proteomic Changes in Renal Cancer and Co-ordinate Demonstration of both the Glycolytic and Mitochondrial Aspects of the Warburg Effect", Proteomics, 2003, vol. 3 (8), pp. 1620-1632.

Whitehead A., et al., "Variation in Tissue-specific Gene Expression Among Natural Populations," Genome Biology, 2005, vol. 6 (2), R13.

Yang H., et al., "Caffeine Suppresses Metastasis in a Transgenic Mouse Model: A Prototype Molecule for Prophylaxis of Metastasis," Clinical and Experimental Metastasis, 2004, vol. 21 (8), pp. 719-735.

Yao M., et al., "A Three-gene Expression Signature Model to Predict Clinical Outcome of Clear Cell Renal Carcinoma," International Journal of Cancer, 2008, vol. 123 (5), pp. 1126-1132.

Zhao H., et al., "Gene Expression Profiling Predicts Survival in Conventional Renal Cell Carcinoma," PLOS Medicine, 2006, vol. No. 3 (1), 11pp.

International Search Report and Written Opinion mailed Oct. 10, 2014 for International Patent Application No. PCT/US2014/040003, 9 pages.

Tomsig, et al., "Lipid Phosphate Phosphohydrolase Type 1 (LPP1) Degrades Extracellular Lysophosphatidic Acid In Vivo", Biochem J., vol. 419, 2009, pp. 611-618.

Affymetrix, Information retrieved on Jan. 8, 2019 from the internet: <https://www.affymetrix.com/analysis/netaffx/showresults.affx#> (2019).

Baldewijns et al. British Journal of Cancer 95:1888-1895 (2007).

Dekel et al. Cancer Res 66(12):6040-6049 (2006).

Dell'Agnola et al., "Clinical utilization of chemokines to combat cancer: the double-edged sword", Expert Review of Vaccines, 6(2):267-283 (2007).

Examination Report issued in Canadian Application No. 3,123, 103, mailed Aug. 24, 2022, 6 pages.

Extended European Search Report dated Aug. 10, 2017, European Patent Application No. 17153152.8.

Extended European Search Report dated Dec. 23, 2016, for European Patent Application No. 14804772.3.

Extended European Search Report issued in European Application No. 19173150.4, dated Aug. 2, 2019, 9 pages.

Extended European Search Report issued in European Application No. 20210111.9, dated Sep. 16, 2021, 17 pages.

Gao et al., "Immune cell recruitment and cell-based system for cancer therapy", Pharmaceutical Research, 25(4):752-768 (2007).

Haller et al., "Equivalence Test in Quantitative Reverse Transcription Polymerase Chain Reaction: Confirmation of Reference Genes Suitable for Normalization", Analytical Biochemistry, 2004, vol. 335, No. 1, pp. 1-9.

Hoshikawa et al. Physical Genomics 12:209-219 (2003).

Jansson et al., "Nitric Oxide Synthase Activity in Human Renal Cell Carcinoma," J. Urol. 160(2):556-560, 1998.

Jones et al., Clin Cancer Res 11(16):5730-5739 (2005).

Jung et al. BMC Molecular Biology 8:47 (2007).

Kondo et al., "Favorable prognosis of renal cell carcinoma with increased expression of chemokines associated with a Th1-type immune response," Cancer Sci 97(9):780-786 (2006).

Kondo et al., High Expression of Chemokine Gene as a Favorable Prognostic Factor in Renal Cell Carcinoma, J. Urology 171(6):2171-2175 (2004).

Kosari et al. Clin Cancer Res 11(14):5128 (2005).

Kruizinga R.C. et al., "Role of Chemokines and their receptors in cancer", Current Pharmaceutical Design, 15(29):3396-3416 (2009).

Lane et al., "Differential Expression in Clear Cell Renal Cell Carcinoma Identified by Gene Expression Profiling", J. Urology 181(2):849-860 (2009).

Lenburg et al. BMC Cancer 3:31 (2003).

Li et al., Clin Cancer Res 9:6441-6446 (2003).

Maeurer et al., Cancer Immunol Immunother. 41:111-121 (1995).

Nogueira et al., "Molecular markers for predicting prognosis of renal cell carcinoma", Urologic Oncology 26(2):113-124 (2007).

Nukiwa et al., "Dendritic cells modified to express fractalkine/CX3CL1 in the treatment of preexisting tumors", European Journal of Immunology, 36(4):1019-1027 (2006).

Office Action issued in Canadian Application No. 2783858, dated Dec. 6, 2018, 6 pages.

Osunkoya et al., Human Pathology 40:1671-1678 (2009).

Partial European Search Report dated May 8, 2017, European Patent Application No. 17153152.8.

(56)

References Cited

OTHER PUBLICATIONS

- Rini et al., "4501: Identification of prognostic genomic markers in patients with localized clear cell renal cell carcinoma (ccRCC)," *J. Clin. Oncol.* 28(15):4501, Suppl., 2010.
- Rini et al., *Lancet Oncol* 16:676-685 (2015).
- Sanjmyatav et al., "A Specific Gene Expression Signature Characterizes Metastatic Potential in Clear Cell Renal Cell Carcinoma", *Journal of Urology*, vol. 186, No. 1, 2011, pp. 289-294.
- Strieter et al., "CXC chemokines in angiogenesis of cancer," *Seminars in Cancer Biology* 14(3):195-200 (2004).
- Supplementary Data for Yang et al. *Cancer Res.* 65(13):5628-5637 (2005).
- Tang et al., "MYC pathway is activated in clear cell renal cell carcinoma and essential for proliferation of clear cell renal cell carcinoma cells," *Cancer Letters*, 273(1):35-43, 2009.
- The Human Protein Atlas. Retrieved on Jun. 27, 2018 from the internet: <https://www.proteinatlas.org/ENSG000001133638-LMNBL1/tissue> (2018).
- Vasselli et al., *PNAS* 100(12):6958-6963 (2003).
- Xin et al., "Antitumor immune response by CX3CL1 fractalkine gene transfer depends on both NK and T cells," *Eur. J. Immunol.* 35(5):1371-1380 (2005).
- Yang, et al., "A Molecular Classification of Papillary Renal Cell Carcinoma", *Cancer Research*, 2005, vol. 65, pp. 5628-5637.

* cited by examiner

Consistency between Stage I and III for Exemplary Genes Associated with RFI

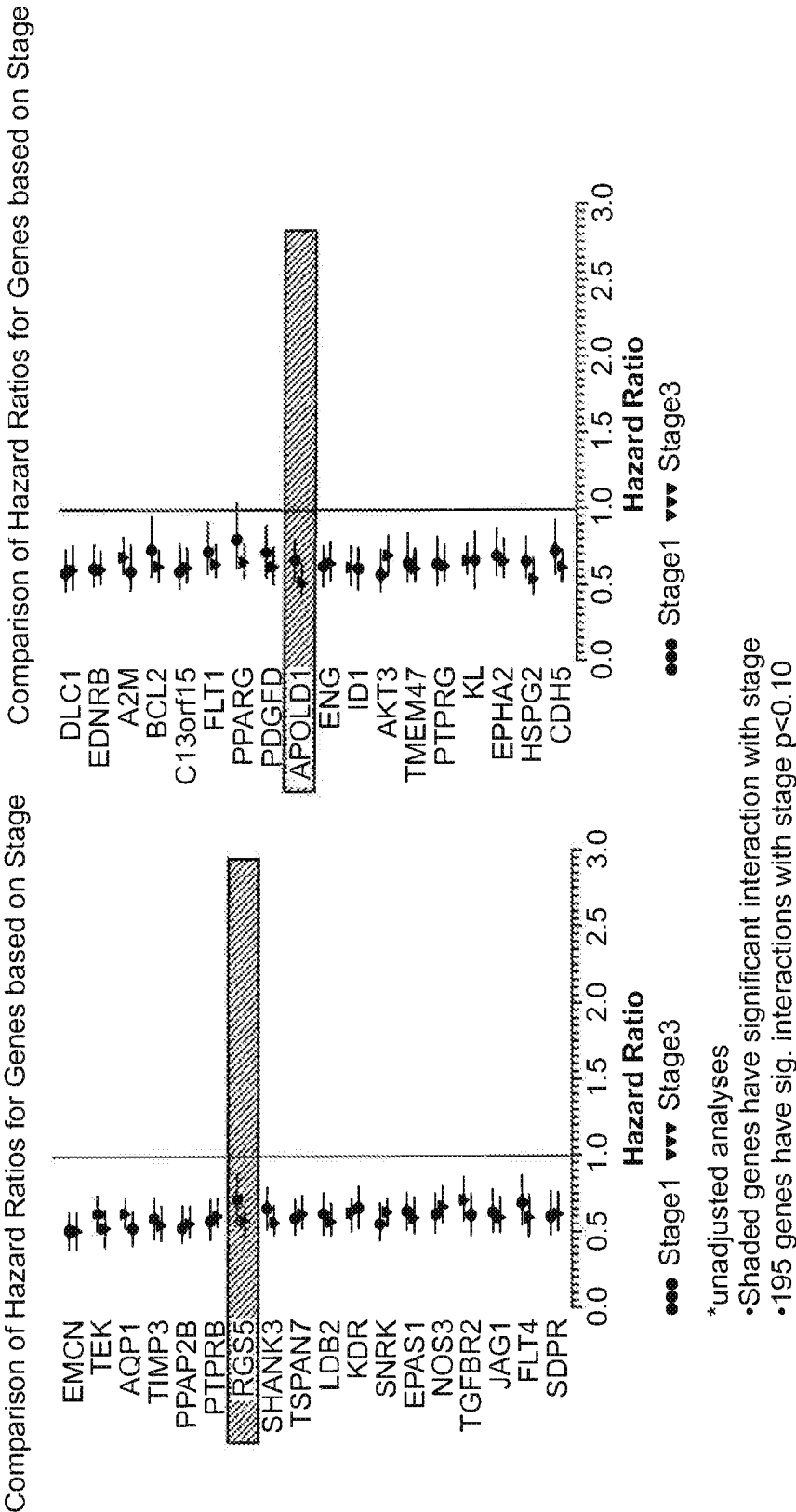


FIG. 1a

Consistency between Stage I and III for Exemplary Genes Associated with RFI

Comparison of Hazard Ratios for Genes based on Stage

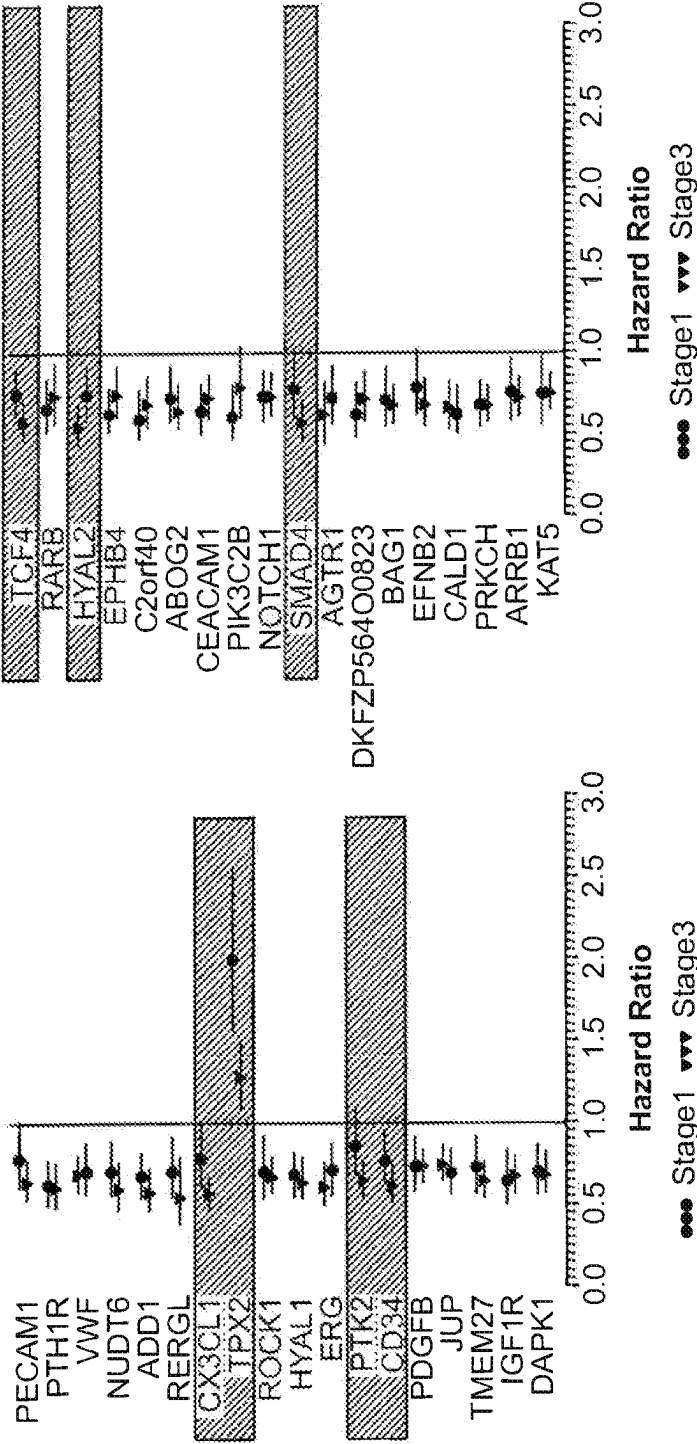


FIG. 1b

Consistency between Stage I and III for Exemplary Genes Associated with RFI

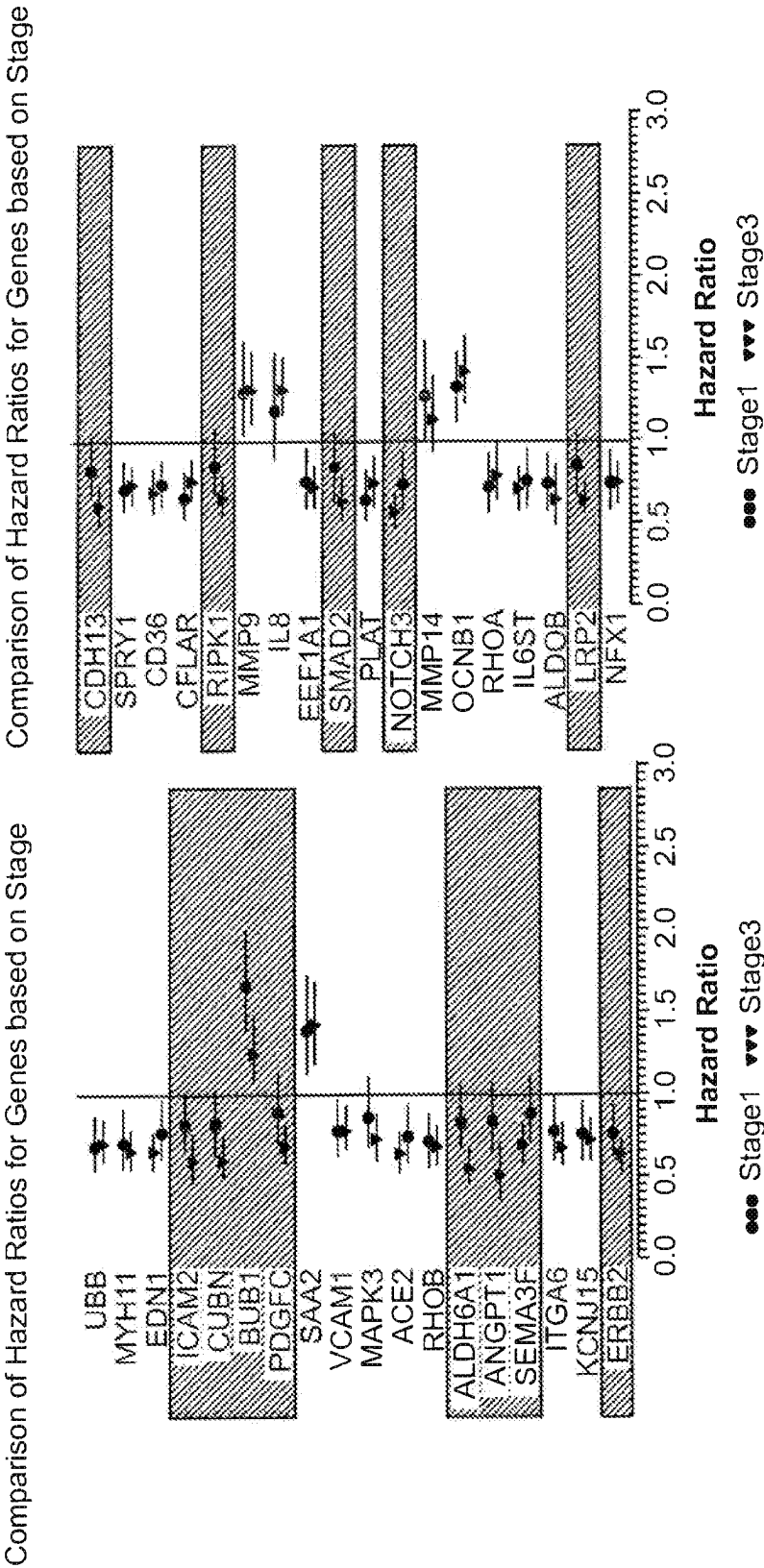


FIG. 1c

Consistent Result across Endpoints (RFI and OS) for Exemplary Genes

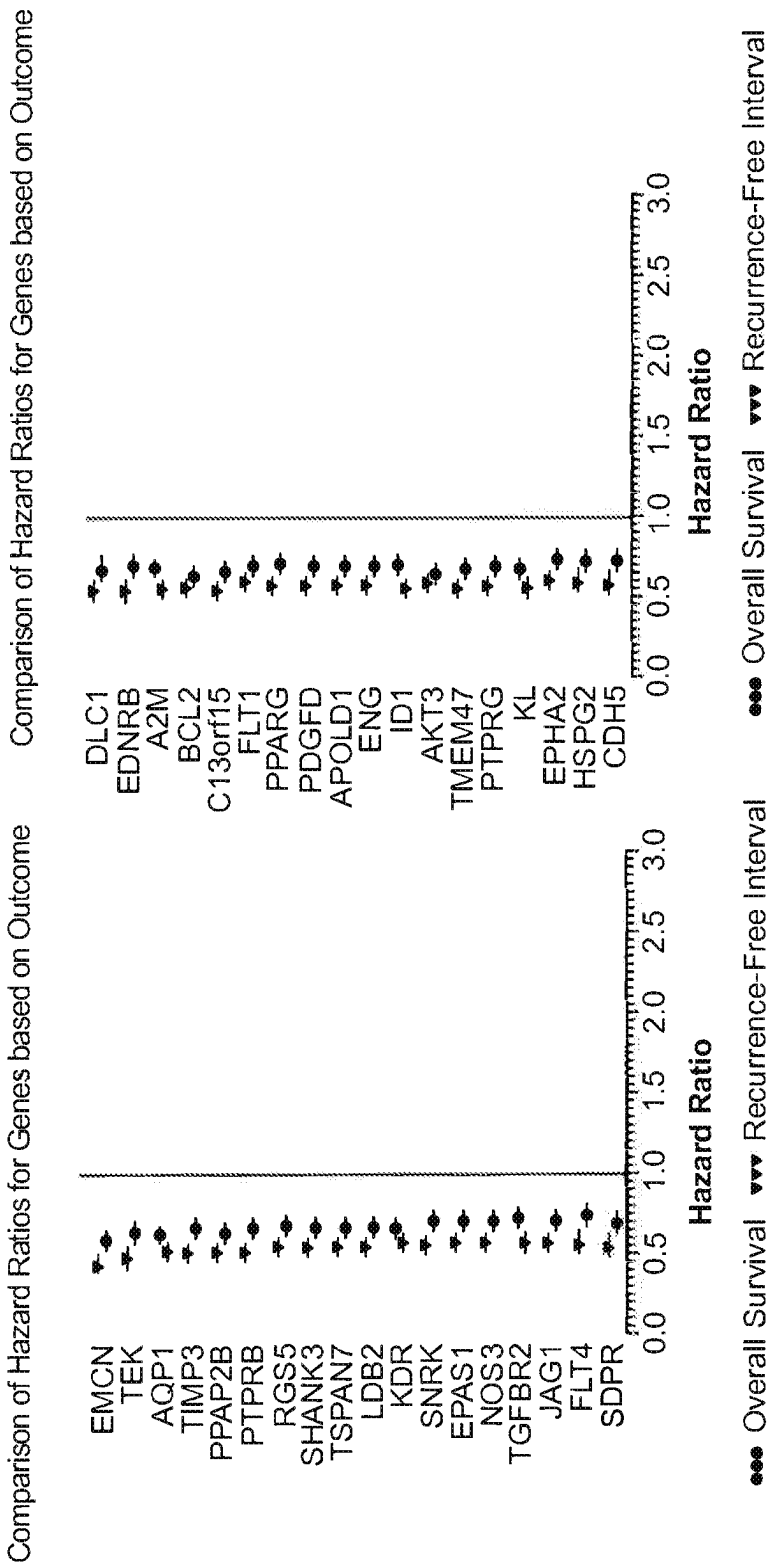


FIG. 2

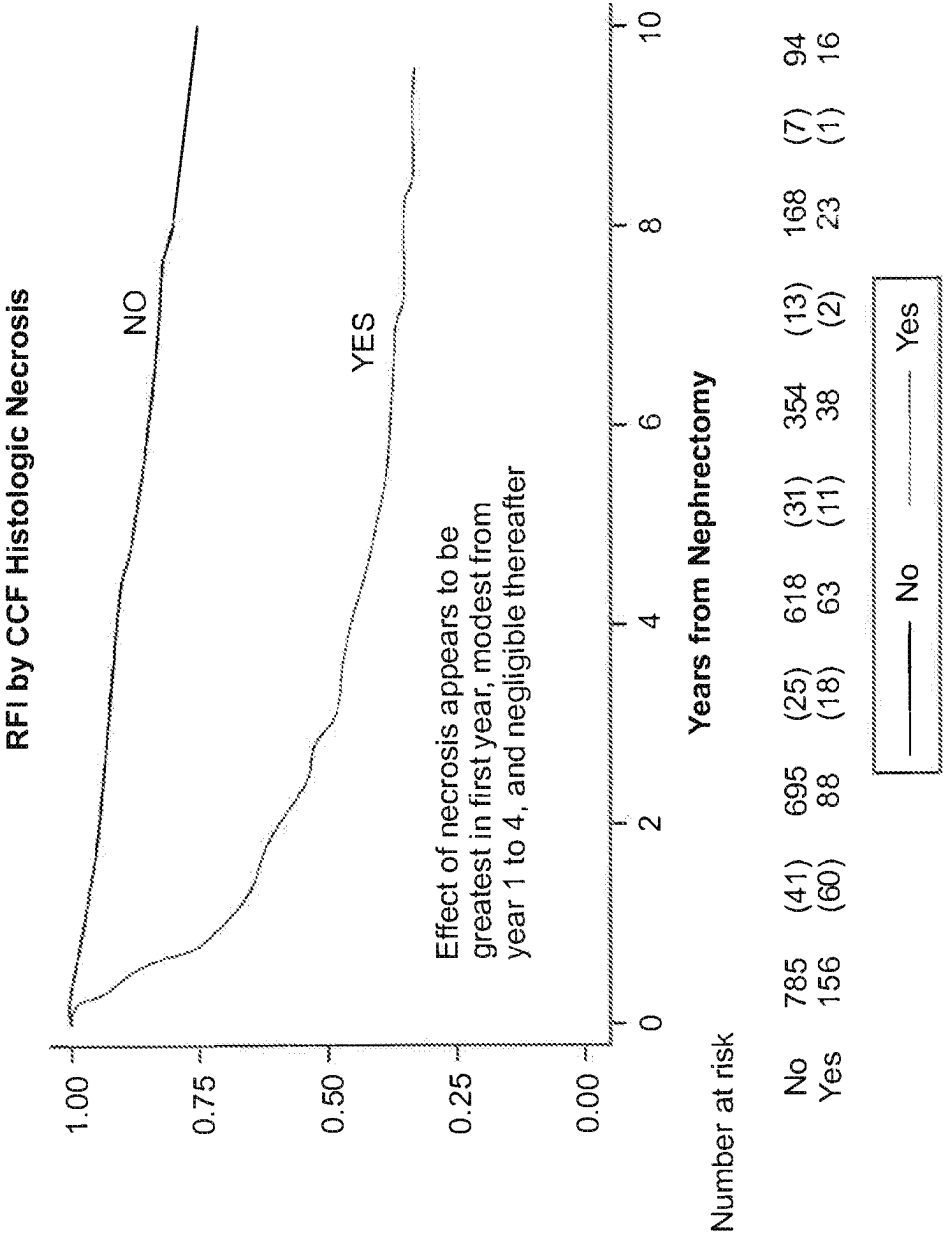
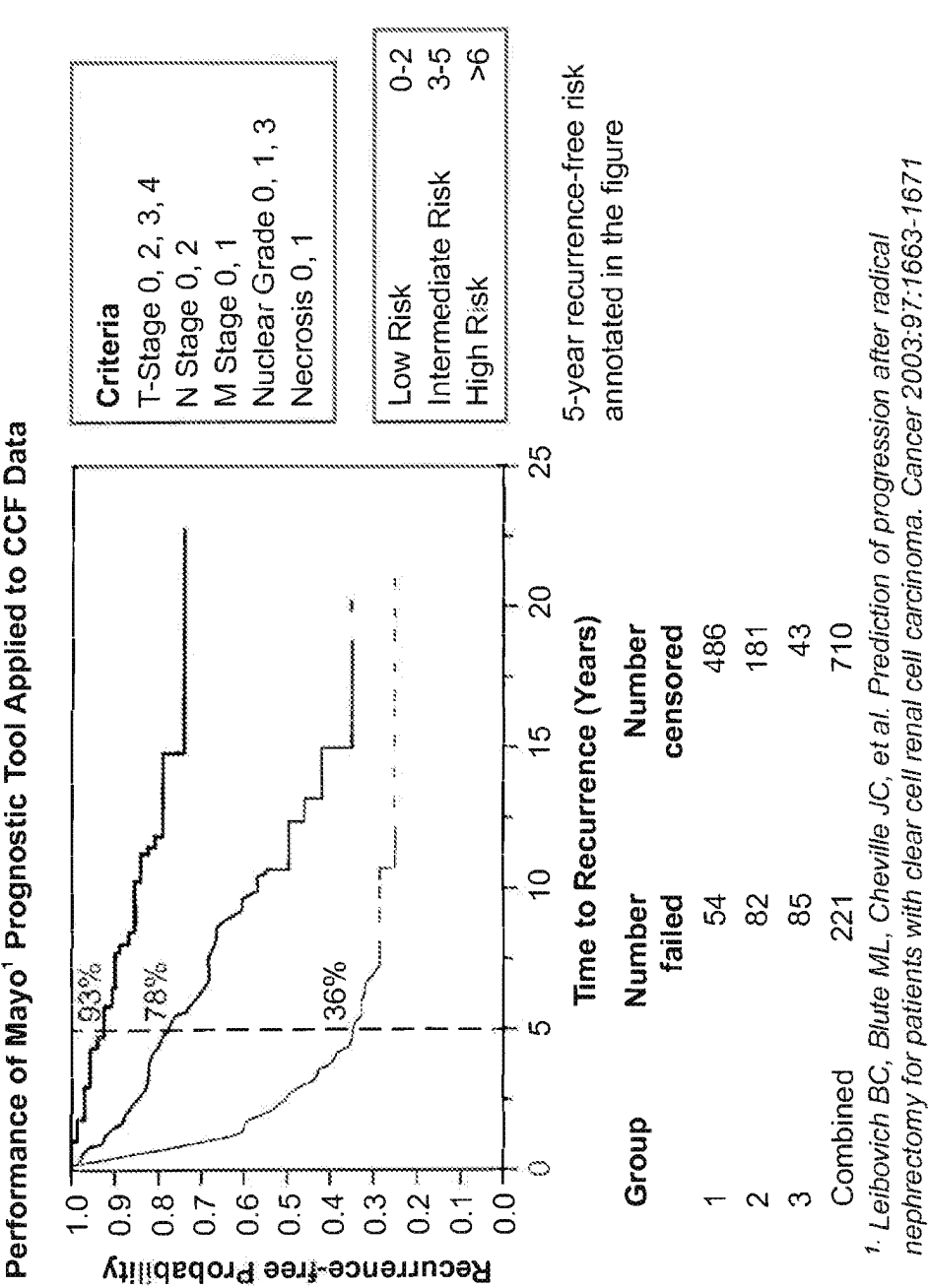


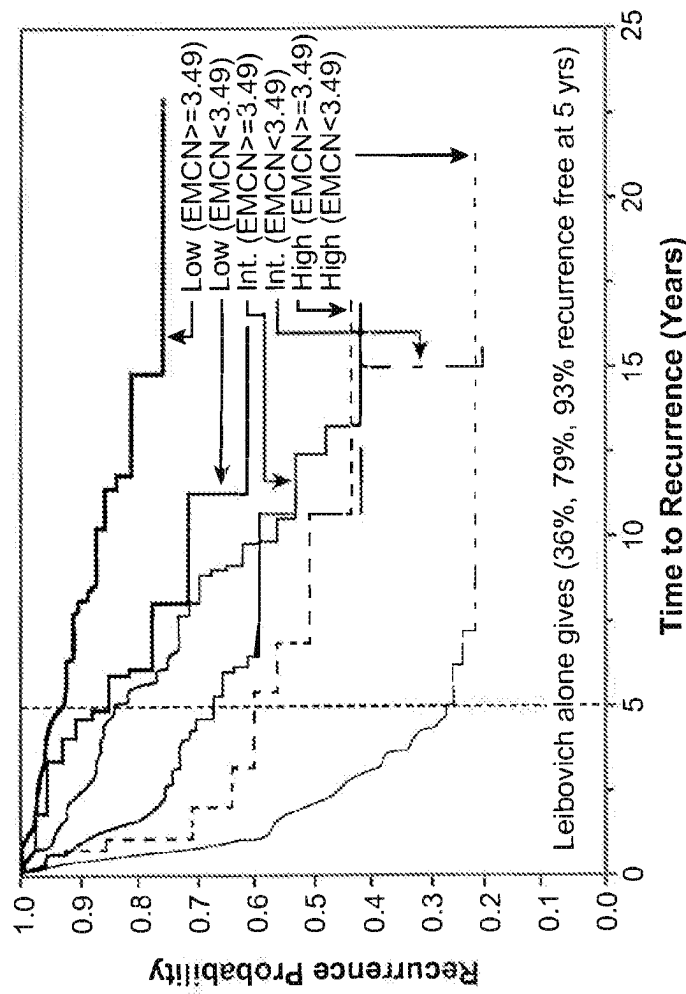
FIG. 3



“5-year recurrence-free risk” means proportion without recurrence at 5 years

FIG. 4

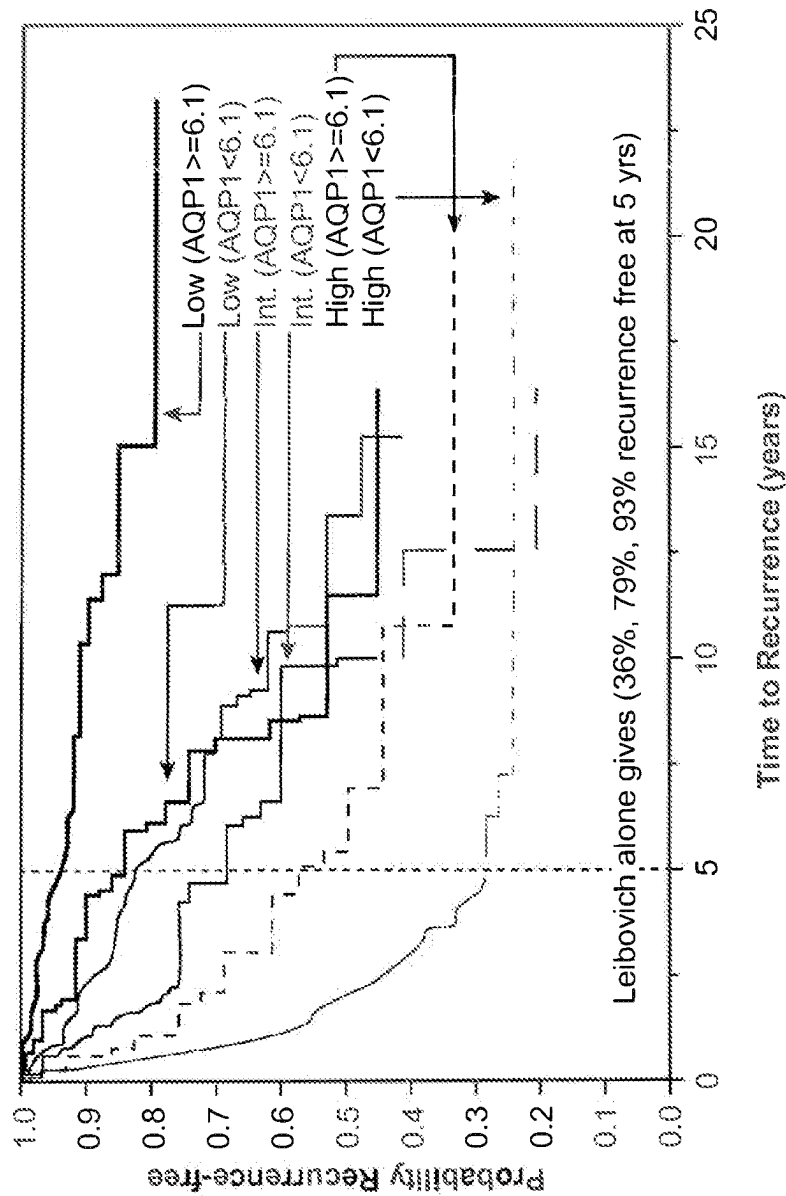
Example of Using 1 Gene to Improve Prediction: EMCN in addition to Mayo Criteria
(Leibovich et. al.)



- Cutpoint for EMCN derived by looking at quartiles of EMCN expression and their KM curves for RFI
- EMCN gives good separation beyond clinical/pathology covariates
- Similar results with other key genes named earlier

FIG. 5

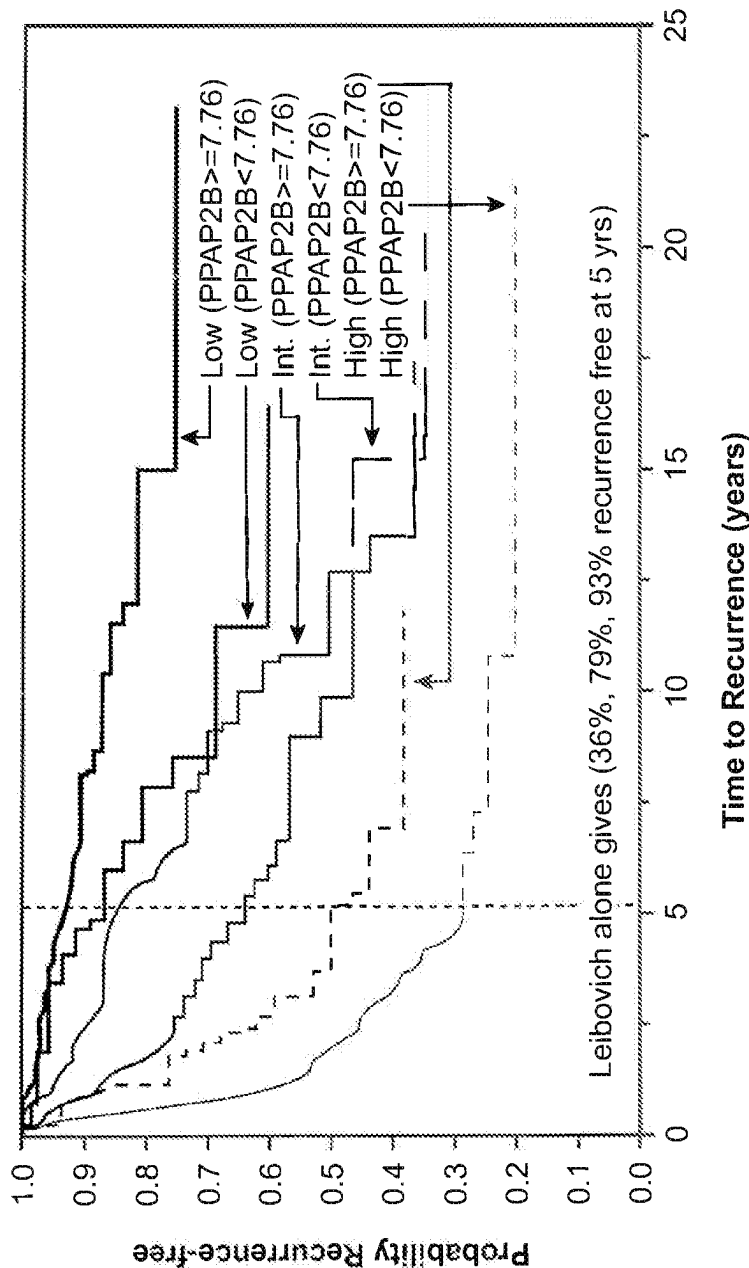
Example of Using 1 Gene to Improve Prediction: AQP1 in addition to Mayo Criteria
(Leibovich et. al.)



Cutpoint for AQP1 derived by looking at quartiles of AQP1 expression and their KM curves for RFI

FIG. 6

Example of Using 1 Gene to Improve Prediction: PPAP2B in addition to Mayo Criteria
(Leibovich et. al.)



Cutpoint for PPAP2B derived by looking at quartiles of PPAP2B expression and their KM curves for RFI

FIG. 7

1

METHOD TO USE GENE EXPRESSION TO DETERMINE LIKELIHOOD OF CLINICAL OUTCOME OF RENAL CANCER

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. application Ser. No. 16/952,566, filed Nov. 19, 2020, which is a continuation of U.S. application Ser. No. 15/368,979, filed Dec. 5, 2016, which is a continuation of U.S. application Ser. No. 12/987,023, filed Jan. 7, 2011, now U.S. Pat. No. 9,551,034, which claims priority benefit of U.S. provisional application Ser. No. 61/294,038, filed Jan. 11, 2010 and priority benefit of U.S. provisional application Ser. No. 61/346,230, filed May 19, 2010. All of the above applications are incorporated herein in their entireties.

SEQUENCE LISTING

The present application contains a Sequence Listing which has been submitted electronically in XML format. Said XML copy, created on Aug. 10, 2023, is named "2023-08-10_ST26_GH1043USCON3.xml" and is 3,747, 780 bytes in size. The information in the electronic format of the sequence listing is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to molecular diagnostic assays that provide information concerning prognosis in renal cancer patients.

INTRODUCTION

Each year in the United States there are approximately 51,000 cases of renal cell carcinoma (kidney cancer) and upper urinary tract cancer, resulting in more than 12,900 deaths. These tumors account for approximately 3% of adult malignancies. Renal cell carcinoma (RCC) represents about 3 percent of all cancers in the United States. Predictions for the United States for the year 2007 were that 40,000 new patients would be diagnosed with RCC and that 13,000 would die from this disease.

The clinical outcome for a renal cell carcinoma patient depends largely on the aggressiveness of their particular cancer. Surgical resection is the most common treatment for this disease as systemic therapy has demonstrated only limited effectiveness. However, approximately 30% of patients with localized tumors will experience a relapse following surgery, and only 40% of all patients with renal cell carcinoma survive for 5 years.

In the US, the number of adjuvant treatment decisions that will be made by patients with early stage renal cell carcinoma in 2005 exceeded 25,000. The rates in the European Union are expected to be similar. Physicians require prognostic information to help them make informed treatment decisions for patients with renal cell carcinoma and recruit appropriate high-risk patients for clinical trials. Surgeons must decide how much kidney and surrounding tissue to remove based, in part, on predicting the aggressiveness of a particular tumor. Today, cancer tumors are generally classified based on clinical and pathological features, such as stage, grade, and the presence of necrosis. These designations are made by applying standardized criteria, the sub-

2

jectivity of which has been demonstrated by a lack of concordance amongst pathology laboratories.

SUMMARY

The present disclosure provides biomarkers, the expression of which has prognostic value in renal cancer.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1a-1c: Consistency between Stage I and III for exemplary genes associated with RFI

FIG. 2: Consistent results across endpoints (OS and RFI) for exemplary genes

FIG. 3: Kaplan-Meier curve: Recurrence Free Internal (RFI) by Cleveland Clinic Foundation (CCF) histologic necrosis

FIG. 4: Performance of Mayo prognostic tool applied to CCF data

FIG. 5: Example of using one gene to improve estimate: EMCN in addition to Mayo Criteria

FIG. 6: Example of using one gene to improve estimate: AQP1 in addition to Mayo Criteria

FIG. 7: Example of using one gene to improve estimate: PPA2B in addition to Mayo Criteria

DETAILED DESCRIPTION

Definitions

Unless defined otherwise, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Singleton et al., Dictionary of Microbiology and Molecular Biology 2nd ed., J. Wiley & Sons (New York, NY 1994), and March, Advanced Organic Chemistry Reactions, Mechanisms and Structure 4th ed., John Wiley & Sons (New York, NY 1992), provide one skilled in the art with a general guide to many of the terms used in the present application.

One skilled in the art will recognize many methods and materials similar or equivalent to those described herein, which could be used in the practice of the present invention. Indeed, the present invention is in no way limited to the methods and materials described. For purposes of the present invention, the following terms are defined below.

The term "tumor" is used herein to refer to all neoplastic cell growth and proliferation, and all pre-cancerous and cancerous cells and tissues. The term "primary tumor" is used herein to refer to a tumor that is at the original site where it first arose. For example, a primary renal cell carcinoma tumor is one that arose in the kidney. The term "metastatic tumor" is used herein to refer to a tumor that develops away from the site of origin. For example, renal cell carcinoma metastasis most commonly affects the spine, ribs, pelvis, and proximal long bones.

The terms "cancer" and "carcinoma" refer to or describe the physiological condition in mammals that is typically characterized by unregulated cell growth. The pathology of cancer includes, for example, abnormal or uncontrollable cell growth, metastasis, interference with the normal functioning of neighboring cells, release of cytokines or other secretory products at abnormal levels, suppression or aggravation of inflammatory or immunological response, neoplasia, premalignancy, malignancy, invasion of surrounding or distant tissues or organs, such as lymph nodes, blood vessels, etc.

As used herein, the terms “renal cancer” or “renal cell carcinoma” refer to cancer that has arisen from the kidney.

The terms “renal cell cancer” or “renal cell carcinoma” (RCC), as used herein, refer to cancer which originates in the lining of the proximal convoluted tubule. More specifically, RCC encompasses several relatively common histologic subtypes: clear cell renal cell carcinoma, papillary (chromophil), chromophobe, collecting duct carcinoma, and medullary carcinoma. Further information about renal cell carcinoma may be found in Y. Thyaviahally, et al., *Int Semin Surg Oncol* 2:18 (2005), the contents of which are incorporated by reference herein. Clear cell renal cell carcinoma (ccRCC) is the most common subtype of RCC. Incidence of ccRCC is increasing, comprising 80% of localized disease and more than 90% of metastatic disease.

The staging system for renal cell carcinoma is based on the degree of tumor spread beyond the kidney. According to the tumor, node, metastasis (TNM) staging system of the American Joint Committee on Cancer (AJCC) (Greene, et al., *AJCC Cancer Staging Manual*, pp. 323-325 (6th Ed. 2002), the various stages of renal cell carcinoma are provided below. “Increased stage” as used herein refers to classification of a tumor at a stage that is more advanced, e.g., Stage 4 is an increased stage relative to Stages 1, 2, and 3.

Description of RCC Stages

Stages for Renal Cell Carcinoma

Stage 1: T1, N0, M0

Stage 2: T2, N0, M0

Stage 3: T1-T2, N1, M0; T3, N0-1, M0; T3a, N0-1, M0; T3b, N0-1, M0; and T3c, N0-1, M0

Stage 4: T4, N0-1, M0; Any T, N2, M0; and Any T, any N, M1

Primary Tumor (T)

TX: Primary tumor cannot be assessed

T0: No evidence of primary tumor

T1: Tumor 7 cm or less in greatest dimension and limited to the kidney

T1a: Tumor 4 cm or less in greatest dimension and limited to the kidney

T1b: Tumor larger than 4 cm but 7 cm or less in greatest dimension and limited to the kidney

T2: Tumor larger than 7 cm in greatest dimension and limited to the kidney

T3: Tumor extends into major veins or invades adrenal gland or perinephric tissues but not beyond Gerota fascia

T3a: Tumor directly invades adrenal gland or perirenal and/or renal sinus fat but not beyond Gerota fascia

T3b: Tumor grossly extends into the renal vein or its segmental (i.e., muscle-containing) branches, or it extends into the vena cava below the diaphragm

T3c: Tumor grossly extends into the vena cava above the diaphragm or invades the wall of the vena cava

T4: Tumor invades beyond Gerota fascia

Regional Lymph Nodes (N)

NX: Regional lymph nodes cannot be assessed

N0: No regional lymph node metastasis

N1: Metastasis in a single regional lymph node

N2: Metastasis in more than one regional lymph node

Distant Metastasis (M)

MX: Distant metastasis cannot be assessed

M0: No distant metastasis

M1: Distant metastasis

The term “early stage renal cancer”, as used herein, refers to Stages 1-3, as defined in the American Joint Committee on Cancer (AJCC) *Cancer Staging Manual*, pp. 323-325 (6th Ed. 2002).

Reference to tumor “grade” for renal cell carcinoma as used herein refers to a grading system based on microscopic appearance of tumor cells. According to the TNM staging system of the AJCC, the various grades of renal cell carcinoma are:

GX (grade of differentiation cannot be assessed);

G1 (well differentiated);

G2 (moderately differentiated); and

G3-G4 (poorly differentiated/undifferentiated).

“Increased grade” as used herein refers to classification of a tumor at a grade that is more advanced, e.g., Grade 4 (G4) is an increased grade relative to Grades 1, 2, and 3. Tumor grading is an important prognostic factor in renal cell carcinoma. H. Rauschmeier, et al., *World J Urol* 2:103-108 (1984).

The terms “necrosis” or “histologic necrosis” as used herein refer to the death of living cells or tissues. The presence of necrosis may be a prognostic factor in cancer. For example, necrosis is commonly seen in renal cell carcinoma (RCC) and has been shown to be an adverse prognostic factor in certain RCC subtypes. V. Foria, et al., *J Clin Pathol* 58(1):39-43 (2005).

The terms “nodal invasion” or “node-positive (N+)” as used herein refer to the presence of cancer cells in one or more lymph nodes associated with the organ (e.g., drain the organ) containing a primary tumor. Nodal invasion is part of tumor staging for most cancers, including renal cell carcinoma.

The term “prognostic gene,” when used in the single or plural, refers to a gene, the expression level of which is correlated with a good or bad prognosis for a cancer patient. A gene may be both a prognostic and predictive gene, depending on the association of the gene expression level with the corresponding endpoint.

The terms “correlated” and “associated” are used interchangeably herein to refer to the strength of association between two measurements (or measured entities). The disclosure provides genes and gene subsets, the expression levels of which are associated with a particular outcome measure, such as between the expression level of a gene and the likelihood of a recurrence event or relapse. For example, the increased expression level of a gene may be positively correlated (positively associated) with an increased likelihood of good clinical outcome for the patient, such as a decreased likelihood of recurrence of cancer. Such a positive correlation may be demonstrated statistically in various ways, e.g. by a hazard ratio less than 1.0. In another example, the increased expression level of a gene may be negatively correlated (negatively associated) with an increased likelihood of good clinical outcome for the patient. In that case, for example, a patient with a high expression level of a gene may have an increased likelihood of recurrence of the cancer. Such a negative correlation could indicate that the patient with a high expression level of a gene likely has a poor prognosis, or might respond poorly to a chemotherapy, and this may be demonstrated statistically in various ways, e.g., a hazard ratio greater than 1.0.

“Co-expression” is used herein to refer to strength of association between the expression levels of two different genes that are biologically similar, such that expression level of a first gene may be substituted with an expression level of a second gene in a given analysis in view of their correlation

of expression. Such co-expressed genes (or correlated expression) indicates that these two genes are substitutable in an expression algorithm, for example, if a first gene is highly correlated, positively or negatively, with increased likelihood of a good clinical outcome for renal cell carcinoma, then the second co-expressed gene also correlates, in the same direction as the first gene, with the same outcome. Pairwise co-expression may be calculated by various methods known in the art, e.g., by calculating Pearson correlation coefficients or Spearman correlation coefficients or by clustering methods. The methods described herein may incorporate one or more genes that co-express, with a Pearson correlation co-efficient of at least 0.5. Co-expressed gene cliques may also be identified using graph theory. An analysis of co-expression may be calculated using normalized or standardized and normalized expression data.

The terms “prognosis” and “clinical outcome” are used interchangeably herein to refer to an estimate of the likelihood of cancer-attributable death or progression, including recurrence, and metastatic spread of a neoplastic disease, such as renal cell carcinoma. The terms “good prognosis” or “positive clinical outcome” mean a desired clinical outcome. For example, in the context of renal cell carcinoma, a good prognosis may be an expectation of no local recurrences or metastasis within two, three, four, five or more years of the initial diagnosis of renal cell carcinoma. The terms “poor prognosis” or “negative clinical outcome” are used herein interchangeably to mean an undesired clinical outcome. For example, in the context of renal cell carcinoma, a poor prognosis may be an expectation of a local recurrence or metastasis within two, three, four, five or more years of the initial diagnosis of renal cell carcinoma.

The term “predictive gene” is used herein to refer to a gene, the expression of which is correlated, positively or negatively, with likelihood of beneficial response to treatment.

A “clinical outcome” can be assessed using any endpoint, including, without limitation, (1) aggressiveness of tumor growth (e.g., movement to higher stage); (2) metastasis; (3) local recurrence; (4) increase in the length of survival following treatment; and/or (5) decreased mortality at a given point of time following treatment. Clinical response may also be expressed in terms of various measures of clinical outcome. Clinical outcome can also be considered in the context of an individual’s outcome relative to an outcome of a population of patients having a comparable clinical diagnosis, and can be assessed using various endpoints such as an increase in the duration of Recurrence-Free interval (RFI), an increase in the duration of Overall Survival (OS) in a population, an increase in the duration of Disease-Free Survival (DFS), an increase in the duration of Distant Recurrence-Free Interval (DRFI), and the like.

The term “treatment”, as used herein, refers to therapeutic compounds administered to patients to cease or reduce proliferation of cancer cells, shrink the tumor, avoid progression and metastasis, or cause primary tumor or metastases regression. Examples of treatment include, for example, cytokine therapy, progestational agents, anti-angiogenic therapy, hormonal therapy, and chemotherapy (including small molecules and biologics).

The terms “surgery” or “surgical resection” are used herein to refer to surgical removal of some or all of a tumor, and usually some of the surrounding tissue. Examples of surgical techniques include laproscopic procedures, biopsy, or tumor ablation, such as cryotherapy, radio frequency ablation, and high intensity ultrasound. In cancer patients, the extent of tissue removed during surgery depends on the

state of the tumor as observed by a surgeon. For example, a partial nephrectomy indicates that part of one kidney is removed; a simple nephrectomy entails removal of all of one kidney; a radical nephrectomy, all of one kidney and neighboring tissue (e.g., adrenal gland, lymph nodes) removed; and bilateral nephrectomy, both kidneys removed.

The terms “recurrence” and “relapse” are used herein, in the context of potential clinical outcomes of cancer, to refer to a local or distant metastases. Identification of a recurrence could be done by, for example, CT imaging, ultrasound, arteriogram, or X-ray, biopsy, urine or blood test, physical exam, or research center tumor registry.

The term “recurrence-free interval” as used herein refers to the time from surgery to first recurrence or death due to recurrence of renal cancer. Losses to follow-up, second primary cancers, other primary cancers, and deaths prior to recurrence are considered censoring events.

The term “overall survival” is defined as the time from surgery to death from any cause. Losses to follow-up are considered censoring events. Recurrences are ignored for the purposes of calculating overall survival (OS).

The term “disease-free survival” is defined as the time from surgery to first recurrence or death from any cause, whichever occurs first. Losses to follow-up are considered censoring events.

The term “Hazard Ratio (HR)” as used herein refers to the effect of an explanatory variable on the hazard or risk of an event (i.e. recurrence or death). In proportional hazards regression models, the HR is the ratio of the predicted hazard for two groups (e.g. patients with or without necrosis) or for a unit change in a continuous variable (e.g. one standard deviation change in gene expression).

The term “Odds Ratio (OR)” as used herein refers to the effect of an explanatory variable on the odds of an event (e.g. presence of necrosis). In logistic regression models, the OR is the ratio of the predicted odds for a unit change in a continuous variable (e.g. one standard deviation change in gene expression).

The term “prognostic clinical and/or pathologic covariate” as used herein refers to clinical and/or prognostic covariates that are significantly associated ($p \leq 0.05$) with clinical outcome. For example, prognostic clinical and pathologic covariates in renal cell carcinoma include tumor stage (e.g. size, nodal invasion, etc.), and grade (e.g., Fuhrman grade), histologic necrosis, and gender.

The term “proxy gene” refers to a gene, the expression of which is correlated (positively or negatively) with one or more prognostic clinical and/or pathologic covariates. The expression level(s) of one or more proxy genes may be used instead of, or in addition to, classification of a tumor by physical or mechanical examination in a pathology laboratory.

The term “microarray” refers to an ordered arrangement of hybridizable array elements, preferably polynucleotide probes, on a substrate.

The term “polynucleotide,” when used in singular or plural, generally refers to any polyribonucleotide or polydeoxyribonucleotide, which may be unmodified RNA or DNA or modified RNA or DNA. Thus, for instance, polynucleotides as defined herein include, without limitation, single- and double-stranded DNA, DNA including single- and double-stranded regions, single- and double-stranded RNA, and RNA including single- and double-stranded regions, hybrid molecules comprising DNA and RNA that may be single-stranded or, more typically, double-stranded or include single- and double-stranded regions. In addition, the term “polynucleotide” as used herein refers to triple-

stranded regions comprising RNA or DNA or both RNA and DNA. The strands in such regions may be from the same molecule or from different molecules. The regions may include all of one or more of the molecules, but more typically involve only a region of some of the molecules. One of the molecules of a triple-helical region often is an oligonucleotide. The term "polynucleotide" specifically includes DNAs (e.g., cDNAs) and RNAs that contain one or more modified bases. Thus, DNAs or RNAs with backbones modified for stability or for other reasons are "polynucleotides" as that term is intended herein. Moreover, DNAs or RNAs comprising unusual bases, such as inosine, or modified bases, such as tritiated bases, are included within the term "polynucleotides" as defined herein. In general, the term "polynucleotide" embraces all chemically, enzymatically and/or metabolically modified forms of unmodified polynucleotides, as well as the chemical forms of DNA and RNA characteristic of viruses and cells, including simple and complex cells.

The term "oligonucleotide" refers to a relatively short polynucleotide, including, without limitation, single-stranded deoxyribonucleotides, single- or double-stranded ribonucleotides, RNA:DNA hybrids and double-stranded DNAs. Oligonucleotides, such as single-stranded DNA probe oligonucleotides, are often synthesized by chemical methods, for example using automated oligonucleotide synthesizers that are commercially available. However, oligonucleotides can be made by a variety of other methods, including in vitro recombinant DNA-mediated techniques and by expression of DNAs in cells and organisms.

The term "expression level" as applied to a gene refers to the normalized level of a gene product.

The terms "gene product" or "expression product" are used herein interchangeably to refer to the RNA transcription products (RNA transcript) of a gene, including mRNA, and the polypeptide translation product of such RNA transcripts. A gene product can be, for example, an unspliced RNA, an mRNA, a splice variant mRNA, a microRNA, a fragmented RNA, a polypeptide, a post-translationally modified polypeptide, a splice variant polypeptide, etc.

"Stringency" of hybridization reactions is readily determinable by one of ordinary skill in the art, and generally is an empirical calculation dependent upon probe length, washing temperature, and salt concentration. In general, longer probes require higher temperatures for proper annealing, while shorter probes need lower temperatures. Hybridization generally depends on the ability of denatured DNA to re-anneal when complementary strands are present in an environment below their melting temperature. The higher the degree of desired homology between the probe and hybridizable sequence, the higher the relative temperature that can be used. As a result, it follows that higher relative temperatures would tend to make the reaction conditions more stringent, while lower temperatures less so. For additional details and explanation of stringency of hybridization reactions, see Ausubel et al., *Current Protocols in Molecular Biology*, (Wiley Interscience, 1995).

"Stringent conditions" or "high stringency conditions", as defined herein, typically: (1) employ low ionic strength solutions and high temperature for washing, for example 0.015 M sodium chloride/0.0015 M sodium citrate/0.1% sodium dodecyl sulfate at 50° C.; (2) employ during hybridization a denaturing agent, such as formamide, for example, 50% (v/v) formamide with 0.1% bovine serum albumin/0.1% Ficoll/0.1% polyvinylpyrrolidone/50 mM sodium phosphate buffer at pH 6.5 with 750 mM sodium chloride, 75 mM sodium citrate at 42° C.; or (3) employ 50%

formamide, 5×SSC (0.75 M NaCl, 0.075 M sodium citrate), 50 mM sodium phosphate (pH 6.8), 0.1% sodium pyrophosphate, 5×Denhardt's solution, sonicated salmon sperm DNA (50 µg/ml), 0.1% SDS, and 10% dextran sulfate at 42° C., with washes at 42° C. in 0.2×SSC (sodium chloride/sodium citrate) and 50% formamide at 55° C., followed by a high-stringency wash consisting of 0.1×SSC containing EDTA at 55° C.

"Moderately stringent conditions" may be identified as described by Sambrook et al., *Molecular Cloning: A Laboratory Manual* (Cold Spring Harbor Press, 1989), and include the use of washing solution and hybridization conditions (e.g., temperature, ionic strength and % SDS) less stringent than those described above. An example of moderately stringent condition is overnight incubation at 37° C. in a solution comprising: 20% formamide, 5×SSC (150 mM NaCl, 15 mM trisodium citrate), 50 mM sodium phosphate (pH 7.6), 5×Denhardt's solution, 10% dextran sulfate, and 20 mg/ml denatured sheared salmon sperm DNA, followed by washing the filters in 1×SSC at about 37-50° C. The skilled artisan will recognize how to adjust the temperature, ionic strength, etc. as necessary to accommodate factors such as probe length and the like.

In the context of the present invention, reference to "at least one," "at least two," "at least five," etc. of the genes listed in any particular gene set means any one or any and all combinations of the genes listed.

The terms "splicing" and "RNA splicing" are used interchangeably and refer to RNA processing that removes introns and joins exons to produce mature mRNA with continuous coding sequence that moves into the cytoplasm of an eukaryotic cell.

In theory, the term "exon" refers to any segment of an interrupted gene that is represented in a mature RNA product (B. Lewin, *Genes IV* (Cell Press, 1990)). In theory the term "intron" refers to any segment of DNA that is transcribed but removed from within the transcript by splicing together the exons on either side of it. Operationally, exon sequences occur in the mRNA sequence of a gene as defined by Ref. SEQ ID numbers. Operationally, intron sequences are the intervening sequences within the genomic DNA of a gene, bracketed by exon sequences and usually having GT and AG splice consensus sequences at their 5' and 3' boundaries.

A "computer-based system" refers to a system of hardware, software, and data storage medium used to analyze information. The minimum hardware of a patient computer-based system comprises a central processing unit (CPU), and hardware for data input, data output (e.g., display), and data storage. An ordinarily skilled artisan can readily appreciate that any currently available computer-based systems and/or components thereof are suitable for use in connection with the methods of the present disclosure. The data storage medium may comprise any manufacture comprising a recording of the present information as described above, or a memory access device that can access such a manufacture.

To "record" data, programming or other information on a computer readable medium refers to a process for storing information, using any such methods as known in the art. Any convenient data storage structure may be chosen, based on the means used to access the stored information. A variety of data processor programs and formats can be used for storage, e.g. word processing text file, database format, etc.

A "processor" or "computing means" references any hardware and/or software combination that will perform the functions required of it. For example, a suitable processor may be a programmable digital microprocessor such as available in the form of an electronic controller, mainframe,

server or personal computer (desktop or portable). Where the processor is programmable, suitable programming can be communicated from a remote location to the processor, or previously saved in a computer program product (such as a portable or fixed computer readable storage medium, whether magnetic, optical or solid state device based). For example, a magnetic medium or optical disk may carry the programming, and can be read by a suitable reader communicating with each processor at its corresponding station.

The present disclosure provides methods for assessing a patient's risk of recurrence of cancer, which methods comprise assaying an expression level of at least one gene, or its gene product, in a biological sample obtained from the patient. In some embodiments, the biological sample can be a tumor sample obtained from the kidney, or surrounding tissues, of the patient. In other embodiments, the biological sample is obtained from a bodily fluid, such as blood or urine.

The present disclosure provides genes useful in the methods disclosed herein. The genes are listed in Tables 3a and 3b, wherein increased expression of genes listed in Table 3a is significantly associated with a lower risk of cancer recurrence, and increased expression of genes listed in Table 3b is significantly associated with a higher risk of cancer recurrence. In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a gene listed in Tables 3a or 3b with which it co-expresses.

The present disclosure further provides genes significantly associated, positively or negatively, with renal cancer recurrence after adjusting for clinical/pathologic covariates (stage, tumor grade, tumor size, nodal status, and presence of necrosis). For example, Table 8a lists genes wherein increased expression is significantly associated with lower risk of renal cancer recurrence after adjusting for clinical/pathologic covariates, and Table 8b lists genes wherein increased expression is significantly associated with a higher risk of renal cancer recurrence after adjusting for clinical/pathologic covariates. Of those genes listed in Tables 8a and 8b, 16 genes with significant association, positively or negatively, with good prognosis after adjusting for clinical/pathologic covariates and controlling the false discovery rate at 10% are listed in Table 9. One or more of these genes may be used separately, or in addition to, at least one of the genes listed in Tables 3a and 3b, to provide information concerning a patient's prognosis.

The present disclosure also provides proxy genes that are useful for assessing the status of clinical and/or pathologic covariates for a cancer patient. Proxy genes for tumor stage are listed in Tables 4a and 4b, wherein increased expression of genes listed in Table 4a is significantly associated with higher tumor stage, and increased expression of genes listed in Table 4b is significantly associated with lower tumor stage. Proxy genes for tumor grade are listed in Tables 5a and 5b wherein increased expression of genes listed in Table 5a is significantly associated with higher tumor grade, and increased expression of genes listed in Table 5b is significantly associated with lower tumor grade. Proxy genes for the presence of necrosis are listed in Tables 6a and 6b, wherein expression of genes listed in Table 6a is significantly associated with the presence of necrosis, and increased expression of genes listed in Table 6b is significantly associated with the absence of necrosis. Proxy genes for nodal involvement are listed in Tables 7a and 7b wherein higher expression of genes listed in Table 7a are significantly associated with the presence of nodal invasion, and increased expression of genes listed in Table 7b are significantly associated with the absence of nodal invasion. One or

more proxy genes may be used separately, or in addition to, at least one of the genes listed in Tables 3a and 3b, to provide information concerning a patient's prognosis. In some embodiments, at least two of the following proxy genes are used to provide information concerning the patient's prognosis: TSPAN7, TEK, LDB2, TIMP3, SHANK3, RGS5, KDR, SDPR, EPAS1, ID1, TGFBR2, FLT4, SDPR, ENDRB, JAG1, DLC1, and KL. In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a proxy gene with which it co-expresses.

The present disclosure also provides sets of genes in biological pathways that are useful for assessing the likelihood that a cancer patient is likely to have a positive clinical outcome, which sets of genes are referred to herein as "gene subsets". The gene subsets include angiogenesis, immune response, transport, cell adhesion/extracellular matrix, cell cycle, and apoptosis. In some embodiments, the angiogenesis gene subset includes ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; the immune response gene subset includes CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1, and; the transport gene subset includes AQP1 and SGK1; the cell adhesion/extracellular matrix gene subset includes ITGB1, A2M, ITGB5, LAMB1, LOX, MMP14, TGFBR2, TIMP3, and TSPAN7; the cell cycle gene subset includes BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; the apoptosis gene subset includes CASP10; the early response gene subset includes EGR1 and CYR61; the metabolic signaling gene subset includes CA12, ENO2, UGCG, and SDPR; and the signaling gene subset includes ID1 and MAP2K3.

The present disclosure also provides genes in biological pathways targeted by chemotherapy that are correlated, positively or negatively, to a risk of cancer recurrence. These genes include KIT, PDGFA, PDGFB, PDGFC, PDGFD, PDGFRb, KRAS, RAF1, MTOR, HIF1AN, VEGFA, VEGFB, and FLT4. In some embodiments, the chemotherapy is cytokine and/or anti-angiogenic therapy. In other embodiments, the chemotherapy is sunitinib, sorafenib, temsirolimus, bevacizumab, everolimus, and/or pazopanib.

In some embodiments, a co-expressed gene may be used in conjunction with, or substituted for, a gene with which it co-expresses.

In some embodiments, the cancer is renal cell carcinoma. In other embodiments, the cancer is clear cell renal cell carcinoma (ccRCC), papillary, chromophobe, collecting duct carcinoma, and/or medullary carcinoma.

Various technological approaches for determination of expression levels of the disclosed genes are set forth in this specification, including, without limitation, reverse-transcription polymerase chain reaction (RT-PCR), microarrays, high-throughput sequencing, serial analysis of gene expression (SAGE), and Digital Gene Expression (DGE), which will be discussed in detail below. In particular aspects, the expression level of each gene may be determined in relation to various features of the expression products of the gene, including exons, introns, protein epitopes, and protein activity.

The expression levels of genes identified herein may be measured in tumor tissue. For example, the tumor tissue may be obtained upon surgical resection of the tumor, or by tumor biopsy. The expression level of the identified genes may also be measured in tumor cells recovered from sites distant from the tumor, including circulating tumor cells or body fluid (e.g., urine, blood, blood fraction, etc.).

The expression product that is assayed can be, for example, RNA or a polypeptide. The expression product may be fragmented. For example, the assay may use primers that are complementary to target sequences of an expression product and could thus measure full transcripts as well as those fragmented expression products containing the target sequence. Further information is provided in Tables A and B, which provide examples of sequences of forward primers, reverse primers, probes and amplicons generated by use of the primers.

The RNA expression product may be assayed directly or by detection of a cDNA product resulting from a PCR-based amplification method, e.g., quantitative reverse transcription polymerase chain reaction (qRT-PCR). (See e.g., U.S. Pub. No. US2006-0008809A1.) Polypeptide expression product may be assayed using immunohistochemistry (IHC). Further, both RNA and polypeptide expression products may also be assayed using microarrays.

Clinical Utility

Currently, of the expected clinical outcome for RCC patients is based on subjective determinations of a tumor's clinical and pathologic features. For example, physicians make decisions about the appropriate surgical procedures and adjuvant therapy based on a renal tumor's stage, grade, and the presence of necrosis. Although there are standardized measures to guide pathologists in making these decisions, the level of concordance between pathology laboratories is low. (See Al-Ayanti M et al. (2003) Arch Pathol Lab Med 127, 593-596) It would be useful to have a reproducible molecular assay for determining and/or confirming these tumor characteristics.

In addition, standard clinical criteria, by themselves, have limited ability to accurately estimate a patient's prognosis. It would be useful to have a reproducible molecular assay to assess a patient's prognosis based on the biology of his or her tumor. Such information could be used for the purposes of patient counseling, selecting patients for clinical trials (e.g., adjuvant trials), and understanding the biology of renal cell carcinoma. In addition, such a test would assist physicians in making surgical and treatment recommendations based on the biology of each patient's tumor. For example, a genomic test could stratify RCC patients based on risk of recurrence and/or likelihood of long-term survival without recurrence (relapse, metastasis, etc.). There are several ongoing and planned clinical trials for RCC therapies, including adjuvant radiation and chemotherapies. It would be useful to have a genomic test able to identify high-risk patients more accurately than standard clinical criteria, thereby further enriching an adjuvant RCC population for study. This would reduce the number of patients needed for an adjuvant trial and the time needed for definitive testing of these new agents in the adjuvant setting.

Finally, it would be useful to have a molecular assay that could predict a patient's likelihood to respond to treatment, such as chemotherapy. Again, this would facilitate individual treatment decisions and recruiting patients for clinical trials, and increase physician and patient confidence in making healthcare decisions after being diagnosed with cancer.

Reporting Results

The methods of the present disclosure are suited for the preparation of reports summarizing the expected clinical outcome resulting from the methods of the present disclosure. A "report," as described herein, is an electronic or tangible document that includes report elements that provide information of interest relating to a likelihood assessment and its results. A subject report includes at least a likelihood

assessment, e.g., an indication as to the risk of recurrence for a subject with renal cell carcinoma. A subject report can be completely or partially electronically generated, e.g., presented on an electronic display (e.g., computer monitor). A report can further include one or more of: 1) information regarding the testing facility; 2) service provider information; 3) patient data; 4) sample data; 5) an interpretive report, which can include various information including: a) indication; b) test data, where test data can include a normalized level of one or more genes of interest, and 6) other features.

The present disclosure thus provides for methods of creating reports and the reports resulting therefrom. The report may include a summary of the expression levels of the RNA transcripts, or the expression products of such RNA transcripts, for certain genes in the cells obtained from the patient's tumor. The report can include information relating to prognostic covariates of the patient. The report may include an estimate that the patient has an increased risk of recurrence. That estimate may be in the form of a score or patient stratifier scheme (e.g., low, intermediate, or high risk of recurrence). The report may include information relevant to assist with decisions about the appropriate surgery (e.g., partial or total nephrectomy) or treatment for the patient.

Thus, in some embodiments, the methods of the present disclosure further include generating a report that includes information regarding the patient's likely clinical outcome, e.g. risk of recurrence. For example, the methods disclosed herein can further include a step of generating or outputting a report providing the results of a subject risk assessment, which report can be provided in the form of an electronic medium (e.g., an electronic display on a computer monitor), or in the form of a tangible medium (e.g., a report printed on paper or other tangible medium).

A report that includes information regarding the patient's likely prognosis (e.g., the likelihood that a patient having renal cell carcinoma will have a good prognosis or positive clinical outcome in response to surgery and/or treatment) is provided to a user. An assessment as to the likelihood is referred to below as a "risk report" or, simply, "risk score." A person or entity that prepares a report ("report generator") may also perform the likelihood assessment. The report generator may also perform one or more of sample gathering, sample processing, and data generation, e.g., the report generator may also perform one or more of: a) sample gathering; b) sample processing; c) measuring a level of a risk gene; d) measuring a level of a reference gene; and e) determining a normalized level of a risk gene. Alternatively, an entity other than the report generator can perform one or more sample gathering, sample processing, and data generation.

For clarity, it should be noted that the term "user," which is used interchangeably with "client," is meant to refer to a person or entity to whom a report is transmitted, and may be the same person or entity who does one or more of the following: a) collects a sample; b) processes a sample; c) provides a sample or a processed sample; and d) generates data (e.g., level of a risk gene; level of a reference gene product(s); normalized level of a risk gene for use in the likelihood assessment. In some cases, the person(s) or entity(ies) who provides sample collection and/or sample processing and/or data generation, and the person who receives the results and/or report may be different persons, but are both referred to as "users" or "clients" herein to avoid confusion. In certain embodiments, e.g., where the methods are completely executed on a single computer, the user or client provides for data input and review of data output. A "user" can be a health professional (e.g., a clini-

cian, a laboratory technician, a physician (e.g., an oncologist, surgeon, pathologist), etc.).

In embodiments where the user only executes a portion of the method, the individual who, after computerized data processing according to the methods of the present disclosure, reviews data output (e.g., results prior to release to provide a complete report, a complete, or reviews an “incomplete” report and provides for manual intervention and completion of an interpretive report) is referred to herein as a “reviewer.” The reviewer may be located at a location remote to the user (e.g., at a service provided separate from a healthcare facility where a user may be located).

Where government regulations or other restrictions apply (e.g., requirements by health, malpractice, or liability insurance), all results, whether generated wholly or partially electronically, are subjected to a quality control routine prior to release to the user.

Methods of Assaying Expression Levels of a Gene Product

Numerous assay methods for measuring an expression level of a gene product are known in the art, including assay methods for measuring an expression level of a nucleic acid gene product (e.g., an mRNA), and assay methods for measuring an expression level of a polypeptide gene product.

Measuring a Level of a Nucleic Acid Gene Product

In general, methods of measuring a level of a nucleic acid gene product (e.g., an mRNA) include methods involving hybridization analysis of polynucleotides, and methods involving amplification of polynucleotides. Commonly used methods known in the art for the quantification of mRNA expression in a sample include northern blotting and in situ hybridization (See for example, Parker & Barnes, *Methods in Molecular Biology* 106:247-283 (1999)); RNase protection assays (Hod, *Biotechniques* 13:852-854 (1992)); and reverse transcription polymerase chain reaction (RT-PCR) (Weis et al., *Trends in Genetics* 8:263-264 (1992)). Alternatively, antibodies may be employed that can recognize specific duplexes, including DNA duplexes, RNA duplexes, and DNA-RNA hybrid duplexes or DNA-protein duplexes. Representative methods for sequencing-based gene expression analysis include Serial Analysis of Gene Expression (SAGE), and gene expression analysis by massively parallel signature sequencing (MPSS).

Expression Methods Based on Hybridization

The level of a target nucleic acid can be measured using a probe that hybridizes to the target nucleic acid. The target nucleic acid could be, for example, a RNA expression product of a response indicator gene associated with response to a VEGF/VEGFR Inhibitor, or a RNA expression product of a reference gene. In some embodiments, the target nucleic acid is first amplified, for example using a polymerase chain reaction (PCR) method.

A number of methods are available for analyzing nucleic acid mixtures for the presence and/or level of a specific nucleic acid. mRNA may be assayed directly or reverse transcribed into cDNA for analysis.

In some embodiments, the method involves contacting a sample (e.g., a sample derived from a cancer cell) under stringent hybridization conditions with a nucleic acid probe and detecting binding, if any, of the probe to a nucleic acid in the sample. A variety of nucleic acid hybridization methods are well known to those skilled in the art, and any known method can be used. In some embodiments, the nucleic acid probe will be detectably labeled.

Expression Methods Based on Target Amplification

Methods of amplifying (e.g., by PCR) nucleic acid, methods of performing primers extension, and methods of assess-

ing nucleic acids are generally well known in the art. (See e.g., Ausubel, et al, Short Protocols in Molecular Biology, 3rd ed., Wiley & Sons, 1995 and Sambrook, et al, Molecular Cloning: A Laboratory Manual, Third Edition, (2001) Cold Spring Harbor, N.Y.)

A target mRNA can be amplified by reverse transcribing the mRNA into cDNA, and then performing PCR (reverse transcription-PCR or RT-PCR). Alternatively, a single enzyme may be used for both steps as described in U.S. Pat. No. 5,322,770.

The fluorogenic 5' nuclease assay, known as the TaqMan® assay (Roche Molecular Systems, Inc.), is a powerful and versatile PCR-based detection system for nucleic acid targets. For a detailed description of the TaqMan assay, reagents and conditions for use therein, see, e.g., Holland et al., Proc. Natl. Acad. Sci., U.S.A. (1991) 88:7276-7280; U.S. Pat. Nos. 5,538,848, 5,723,591, and 5,876,930, all incorporated herein by reference in their entireties. Hence, primers and probes derived from regions of a target nucleic acid as described herein can be used in TaqMan analyses to detect a level of target mRNA in a biological sample. Analysis is performed in conjunction with thermal cycling by monitoring the generation of fluorescence signals. (TaqMan is a registered trademark of Roche Molecular Systems.)

The fluorogenic 5' nuclease assay is conveniently performed using, for example, AmpliTaq Gold® DNA polymerase, which has endogenous 5' nuclease activity, to digest an internal oligonucleotide probe labeled with both a fluorescent reporter dye and a quencher (see, Holland et al., Proc Nat Acad Sci USA (1991) 88:7276-7280; and Lee et al., Nucl. Acids Res. (1993) 21:3761-3766). Assay results are detected by measuring changes in fluorescence that occur during the amplification cycle as the fluorescent probe is digested, uncoupling the dye and quencher labels and causing an increase in the fluorescent signal that is proportional to the amplification of target nucleic acid. (AmpliTaq Gold is a registered trademark of Roche Molecular Systems.)

The amplification products can be detected in solution or using solid supports. In this method, the TaqMan probe is designed to hybridize to a target sequence within the desired PCR product. The 5' end of the TaqMan probe contains a fluorescent reporter dye. The 3' end of the probe is blocked to prevent probe extension and contains a dye that will quench the fluorescence of the 5' fluorophore. During subsequent amplification, the 5' fluorescent label is cleaved off if a polymerase with 5' exonuclease activity is present in the reaction. Excision of the 5' fluorophore results in an increase in fluorescence that can be detected.

The first step for gene expression analysis is the isolation of mRNA from a target sample. The starting material is typically total RNA isolated from human tumors or tumor cell lines, and corresponding normal tissues or cell lines, respectively. Thus RNA can be isolated from a variety of primary tumors, including breast, lung, colon, prostate, brain, liver, kidney, pancreas, spleen, thymus, testis, ovary, uterus, head and neck, etc., tumor, or tumor cell lines, with pooled DNA from healthy donors. If the source of mRNA is a primary tumor, mRNA can be extracted, for example, from frozen or archived paraffin-embedded and fixed (e.g., formalin-fixed) tissue samples.

General methods for mRNA extraction are well known in the art and are disclosed in standard textbooks of molecular biology, including Ausubel et al., *Current Protocols of Molecular Biology* (Wiley and Sons, 1997). Methods for RNA extraction from paraffin embedded tissues are disclosed, for example, M. Cronin, Am J. Pathol 164(1):35-42

(2004), the contents of which are incorporated herein. In particular, RNA isolation can be performed using kits and reagents from commercial manufacturers according to the manufacturer's instructions. For example, total RNA from cells in culture can be isolated using RNeasy® mini-columns (Qiagen GmbH Corp.). Other commercially available RNA isolation kits include MasterPure™ Complete DNA and RNA Purification Kit (EPICENTRE® Biotechnologies, Madison, WI), mirVana (Applied Biosystems, Inc.), and Paraffin Block RNA Isolation Kit (Ambion, Inc.). Total RNA from tissue samples can be isolated using RNA STAT-60 TH (IsoTex Diagnostics, Inc., Friendswood TX). RNA prepared from tumor can be isolated, for example, by cesium chloride density gradient centrifugation. (RNeasy is a registered trademark of Qiagen GmbH Corp.; MasterPure is a trademark of EPICENTRE Biotechnologies; RNA STAT-60 is a trademark of Tel-Test Inc.)

As RNA cannot serve as a template for PCR, the first step in gene expression profiling by RT-PCR is the reverse transcription of the RNA template into cDNA, followed by its exponential amplification in a PCR reaction. The two most commonly used reverse transcriptase enzymes are avian myeloblastosis virus reverse transcriptase (AMV-RT) and Moloney murine leukemia virus reverse transcriptase (MMLV-RT). The reverse transcription step is typically primed using specific primers, random hexamers, or oligo-dT primers, depending on the circumstances and the goal of expression profiling. For example, extracted RNA can be reverse-transcribed using a GeneAmp® RNA PCR kit (Applied Biosystems Inc., Foster City, CA) according to the manufacturer's instructions. The derived cDNA can then be used as a template in a subsequent PCR reaction. (GeneAmp is a registered trademark of Applied Biosystems Inc.)

Although the PCR step can use a variety of thermostable DNA-dependent DNA polymerases, it typically employs the Taq DNA polymerase, which has a 5'-3' nuclease activity but lacks a 3'-5' proofreading endonuclease activity. Thus, TaqMan PCR typically utilizes the 5'-nuclease activity of Taq or Tth polymerase to hydrolyze a hybridization probe bound to its target amplicon, but any enzyme with equivalent 5' nuclease activity can be used. Two oligonucleotide primers are used to generate an amplicon. A third oligonucleotide, or probe, is designed to detect nucleotide sequence located between the two PCR primers. The probe is non-extendible by Taq DNA polymerase enzyme, and is labeled with a reporter fluorescent dye and a quencher fluorescent dye. Any laser-induced emission from the reporter dye is quenched by the quenching dye when the two dyes are located close together as they are on the probe. During the amplification reaction, the Taq DNA polymerase enzyme cleaves the probe in a template-dependent manner. The resultant probe fragments disassociate in solution, and signal from the released reporter dye is free from the quenching effect of the second fluorophore. One molecule of reporter dye is liberated for each new molecule synthesized, and detection of the unquenched reporter dye provides the basis for quantitative interpretation of the data. (TaqMan is a registered mark of Applied Biosystems.)

TaqMan RT-PCR can be performed using commercially available equipment, such as, for example, the ABI PRISM 7700® Sequence Detection System (Applied Biosystems, Foster City, CA, USA), or the Lightcycler® (Roche Molecular Biochemicals, Mannheim, Germany). In a preferred embodiment, the 5' nuclease procedure is run on a real-time quantitative PCR device such as the ABI PRISM 7900™ Sequence Detection System™. The system consists of a thermocycler, laser, charge-coupled device (CCD), camera

and computer. The system amplifies samples in a 96-well format on a thermocycler. During amplification, laser-induced fluorescent signal is collected in real-time through fiber optics cables for all 96 wells, and detected at the CCD. The system includes software for running the instrument and for analyzing the data. (PRISM 7700 is a registered trademark of Applied Biosystems; Lightcycler is a registered trademark of Roche Diagnostics GmbH LLC.)

5' Nuclease assay data are initially expressed as Cr, or the threshold cycle. As discussed above, fluorescence values are recorded during every cycle and represent the amount of product amplified to that point in the amplification reaction. The point when the fluorescent signal is first recorded as statistically significant is the threshold cycle (Ct).

To minimize the effect of sample-to-sample variation, quantitative RT-PCR is usually performed using an internal standard, or one or more reference genes. The ideal internal standard is expressed at a constant level among different tissues, and is unaffected by the experimental treatment. RNAs that can be used to normalize patterns of gene expression include, e.g., mRNAs for the reference genes glyceraldehyde-3-phosphate-dehydrogenase (GAPDH) and β -actin.

A more recent variation of the RT-PCR technique is the real time quantitative PCR, which measures PCR product accumulation through a dual-labeled fluorogenic probe (i.e., TaqMan® probe). Real time PCR is compatible both with quantitative competitive PCR, where internal competitor for each target sequence is used for normalization, and with quantitative comparative PCR using a normalization gene contained within the sample, or a reference gene for RT-PCR. For further details see, e.g., Held et al., *Genome Research* 6:986-994 (1996).

Factors considered in PCR primer design include primer length, melting temperature (T_m), and G/C content, specificity, complementary primer sequences, and 3'-end sequence. In general, optimal PCR primers are generally 17-30 bases in length, and contain about 20-80%, such as, for example, about 50-60% G+C bases. T_m 's between 50 and 80° C., e.g., about 50 to 70° C. can be used.

For further guidelines for PCR primer and probe design see, e.g., Dieffenbach, C. W. et al., "General Concepts for PCR Primer Design" in: *PCR Primer, A Laboratory Manual*, Cold Spring Harbor Laboratory Press, New York, 1995, pp. 133-155; Innis and Gelfand, "Optimization of PCRs" in: *PCR Protocols, A Guide to Methods and Applications*, CRC Press, London, 1994, pp. 5-11; and Plasterer, T. N. PrimerSelect: Primer and probe design. *Methods Mol. Biol.* 70:520-527 (1997), the entire disclosures of which are hereby expressly incorporated by reference.

Other suitable methods for assaying a level of a nucleic acid gene product include, e.g., microarrays; serial analysis of gene expression (SAGE); MassARRAY® analysis; digital gene expression (DGE) (J. Marioni, *Genome Research* 18(9):1509-1517 (2008), gene expression by massively parallel signature sequencing (see, e.g., Ding and Cantor, *Proc. Nat'l Acad Sci* 100:3059-3064 (2003); differential display (Liang and Pardee, *Science* 257:967-971 (1992)); amplified fragment length polymorphism (AFLP) (Kawamoto et al., *Genome Res.* 12:1305-1312 (1999)); BeadArray™ technology (Illumina, San Diego, CA; Oliphant et al., *Discovery of Markers for Disease* (Supplement to *Biotechniques*), June 2002; Ferguson et al., *Analytical Chemistry* 72:5618 (2000)); BeadsArray for Detection of Gene Expression (BADGE), using the commercially available Luminex100 LabMAP system and multiple color-coded microspheres (Luminex Corp., Austin, TX) in a rapid assay for gene

expression (Yang et al., *Genome Res.* 11:1888-1898 (2001)); and high coverage expression profiling (HiCEP) analysis (Fukumura et al., *Nucl. Acids. Res.* 31(16) e94 (2003)).

Introns

Assays to measure the amount of an RNA gene expression product can be targeted to intron sequences or exon sequences of the primary transcript. The amount of a spliced intron that is measured in human tissue samples is generally indicative of the amount of a corresponding exon (i.e. an exon from the same gene) present in the samples. Polynucleotides that consist of or are complementary to intron sequences can be used, e.g., in hybridization methods or amplification methods to assay the expression level of response indicator genes.

Measuring Levels of a Polypeptide Gene Product

Methods of measuring a level of a polypeptide gene product are known in the art and include antibody-based methods such as enzyme-linked immunoabsorbent assay (ELISA), radioimmunoassay (RIA), protein blot analysis, immunohistochemical analysis and the like. The measure of a polypeptide gene product may also be measured in vivo in the subject using an antibody that specifically binds a target polypeptide, coupled to a paramagnetic label or other label used for in vivo imaging, and visualizing the distribution of the labeled antibody within the subject using an appropriate in vivo imaging method, such as magnetic resonance imaging. Such methods also include proteomics methods such as mass spectrometric methods, which are known in the art.

Methods of Isolating RNA from Body Fluids

Methods of isolating RNA for expression analysis from blood, plasma and serum (See for example, Tsui N B et al. (2002) 48, 1647-53 and references cited therein) and from urine (See for example, Boom R et al. (1990) *J Clin Microbiol.* 28, 495-503 and reference cited therein) have been described.

Methods of Isolating RNA from Paraffin-Embedded Tissue

The steps of a representative protocol for profiling gene expression using fixed, paraffin-embedded tissues as the RNA source, including mRNA isolation, purification primer extension and amplification are provided in various published journal articles. (See, e.g., T. E. Godfrey et al., *J. Molec. Diagnostics* 2: 84-91 (2000); K. Specht et al., *Am. J. Pathol.* 158: 419-29 (2001), M. Cronin, et al., *Am J Pathol* 164:35-42 (2004)).

Manual and Computer-Assisted Methods and Products

The methods and systems described herein can be implemented in numerous ways. In one embodiment of particular interest, the methods involve use of a communications infrastructure, for example the Internet. Several embodiments are discussed below. It is also to be understood that the present disclosure may be implemented in various forms of hardware, software, firmware, processors, or a combination thereof. The methods and systems described herein can be implemented as a combination of hardware and software. The software can be implemented as an application program tangibly embodied on a program storage device, or different portions of the software implemented in the user's computing environment (e.g., as an applet) and on the reviewer's computing environment, where the reviewer may be located at a remote site associated (e.g., at a service provider's facility).

For example, during or after data input by the user, portions of the data processing can be performed in the user-side computing environment. For example, the user-side computing environment can be programmed to provide for defined test codes to denote a likelihood "risk score,"

where the score is transmitted as processed or partially processed responses to the reviewer's computing environment in the form of test code for subsequent execution of one or more algorithms to provide a results and/or generate a report in the reviewer's computing environment. The risk score can be a numerical score (representative of a numerical value) or a non-numerical score representative of a numerical value or range of numerical values (e.g., low, intermediate, or high).

The application program for executing the algorithms described herein may be uploaded to, and executed by, a machine comprising any suitable architecture. In general, the machine involves a computer platform having hardware such as one or more central processing units (CPU), a random access memory (RAM), and input/output (I/O) interface(s). The computer platform also includes an operating system and microinstruction code. The various processes and functions described herein may either be part of the microinstruction code or part of the application program (or a combination thereof) that is executed via the operating system. In addition, various other peripheral devices may be connected to the computer platform such as an additional data storage device and a printing device.

As a computer system, the system generally includes a processor unit. The processor unit operates to receive information, which can include test data (e.g., level of a risk gene, level of a reference gene product(s); normalized level of a gene; and may also include other data such as patient data. This information received can be stored at least temporarily in a database, and data analyzed to generate a report as described above.

Part or all of the input and output data can also be sent electronically; certain output data (e.g., reports) can be sent electronically or telephonically (e.g., by facsimile, e.g., using devices such as fax back). Exemplary output receiving devices can include a display element, a printer, a facsimile device and the like. Electronic forms of transmission and/or display can include email, interactive television, and the like. In an embodiment of particular interest, all or a portion of the input data and/or all or a portion of the output data (e.g., usually at least the final report) are maintained on a web server for access, preferably confidential access, with typical browsers. The data may be accessed or sent to health professionals as desired. The input and output data, including all or a portion of the final report, can be used to populate a patient's medical record which may exist in a confidential database at the healthcare facility.

A system for use in the methods described herein generally includes at least one computer processor (e.g., where the method is carried out in its entirety at a single site) or at least two networked computer processors (e.g., where data is to be input by a user (also referred to herein as a "client") and transmitted to a remote site to a second computer processor for analysis, where the first and second computer processors are connected by a network, e.g., via an intranet or internet). The system can also include a user component(s) for input; and a reviewer component(s) for review of data, generated reports, and manual intervention. Additional components of the system can include a server component(s); and a database(s) for storing data (e.g., as in a database of report elements, e.g., interpretive report elements, or a relational database (RDB) which can include data input by the user and data output. The computer processors can be processors that are typically found in personal desktop computers (e.g., IBM, Dell, Macintosh), portable computers, mainframes, minicomputers, or other computing devices.

The networked client/server architecture can be selected as desired, and can be, for example, a classic two or three tier client server model. A relational database management system (RDMS), either as part of an application server component or as a separate component (RDB machine) provides the interface to the database.

In one example, the architecture is provided as a database-centric client/server architecture, in which the client application generally requests services from the application server which makes requests to the database (or the database server) to populate the report with the various report elements as required, particularly the interpretive report elements, especially the interpretation text and alerts. The server(s) (e.g., either as part of the application server machine or a separate RDB/relational database machine) responds to the client's requests.

The input client components can be complete, stand-alone personal computers offering a full range of power and features to run applications. The client component usually operates under any desired operating system and includes a communication element (e.g., a modem or other hardware for connecting to a network), one or more input devices (e.g., a keyboard, mouse, keypad, or other device used to transfer information or commands), a storage element (e.g., a hard drive or other computer-readable, computer-writable storage medium), and a display element (e.g., a monitor, television, LCD, LED, or other display device that conveys information to the user). The user enters input commands into the computer processor through an input device. Generally, the user interface is a graphical user interface (GUI) written for web browser applications.

The server component(s) can be a personal computer, a minicomputer, or a mainframe and offers data management, information sharing between clients, network administration and security. The application and any databases used can be on the same or different servers.

Other computing arrangements for the client and server(s), including processing on a single machine such as a mainframe, a collection of machines, or other suitable configuration are contemplated. In general, the client and server machines work together to accomplish the processing of the present disclosure.

Where used, the database(s) is usually connected to the database server component and can be any device that will hold data. For example, the database can be a any magnetic or optical storing device for a computer (e.g., CDROM, internal hard drive, tape drive). The database can be located remote to the server component (with access via a network, modem, etc.) or locally to the server component.

Where used in the system and methods, the database can be a relational database that is organized and accessed according to relationships between data items. The relational database is generally composed of a plurality of tables (entities). The rows of a table represent records (collections of information about separate items) and the columns represent fields (particular attributes of a record). In its simplest conception, the relational database is a collection of data entries that "relate" to each other through at least one common field.

Additional workstations equipped with computers and printers may be used at point of service to enter data and, in some embodiments, generate appropriate reports, if desired. The computer(s) can have a shortcut (e.g., on the desktop) to launch the application to facilitate initiation of data entry, transmission, analysis, report receipt, etc. as desired.

Computer-Readable Storage Media

The present disclosure also contemplates a computer-readable storage medium (e.g. CD-ROM, memory key, flash memory card, diskette, etc.) having stored thereon a program which, when executed in a computing environment, provides for implementation of algorithms to carry out all or a portion of the results of a response likelihood assessment as described herein. Where the computer-readable medium contains a complete program for carrying out the methods described herein, the program includes program instructions for collecting, analyzing and generating output, and generally includes computer readable code devices for interacting with a user as described herein, processing that data in conjunction with analytical information, and generating unique printed or electronic media for that user.

Where the storage medium provides a program that provides for implementation of a portion of the methods described herein (e.g., the user-side aspect of the methods (e.g., data input, report receipt capabilities, etc.)), the program provides for transmission of data input by the user (e.g., via the internet, via an intranet, etc.) to a computing environment at a remote site. Processing or completion of processing of the data is carried out at the remote site to generate a report. After review of the report, and completion of any needed manual intervention, to provide a complete report, the complete report is then transmitted back to the user as an electronic document or printed document (e.g., fax or mailed paper report). The storage medium containing a program according to the present disclosure can be packaged with instructions (e.g., for program installation, use, etc.) recorded on a suitable substrate or a web address where such instructions may be obtained. The computer-readable storage medium can also be provided in combination with one or more reagents for carrying out response likelihood assessment (e.g., primers, probes, arrays, or other such kit components).

Methods of Data Analysis

Reference Normalization

In order to minimize expression measurement variations due to non-biological variations in samples, e.g., the amount and quality of expression product to be measured, raw expression level data measured for a gene product (e.g., cycle threshold (CO measurements obtained by qRT-PCR) may be normalized relative to the mean expression level data obtained for one or more reference genes. In one approach to normalization, a small number of genes are used as reference genes; the genes chosen for reference genes typically show a minimal amount of variation in expression from sample to sample and the expression level of other genes is compared to the relatively stable expression of the reference genes. In the global normalization approach, the expression level of each gene in a sample is compared to an average expression level in the sample of all genes in order to compare the expression of a particular gene to the total amount of material.

Unprocessed data from qRT-PCR is expressed as cycle threshold (CO, the number of amplification cycles required for the detectable signal to exceed a defined threshold. High C t is indicative of low expression since more cycles are required to detect the amplification product. Normalization may be carried out such that a one-unit increase in normalized expression level of a gene product generally reflects a 2-fold increase in quantity of expression product present in the sample. For further information on normalization techniques applicable to qRT-PCR data from tumor tissue, see, e.g., Silva S et al. (2006) BMC Cancer 6, 200; de Kok J et al. (2005) Laboratory Investigation 85, 154-159. Gene

expression may then be standardized by dividing the normalized gene expression by the standard deviation of expression across all patients for that particular gene. By standardizing normalized gene expression the hazard ratios across genes are comparable and reflect the relative risk for each standard deviation of gene expression.

Statistical Analysis

One skilled in the art will recognize that variety of statistical methods are available that are suitable for comparing the expression level of a gene (or other variable) in two groups and determining the statistical significance of expression level differences that are found. (See e.g., H. Motulsky, *Intuitive Biostatistics* (Oxford University Press, 1995); D. Freedman, *Statistics* (W. W. Norton & Co, 4th Ed., 2007). For example, a Cox proportional hazards regression model may be fit to a particular clinical time-to-event endpoint (e.g., RFI, OS). One assumption of the Cox proportional hazards regression model is the proportional hazards assumption, i.e. the assumption that model effects multiply the underlying hazard. Assessments of model adequacy may be performed including, but not limited to, examination of the cumulative sum of martingale residuals. One skilled in the art would recognize that there are numerous statistical methods that may be used (e.g., Royston and Parmar (2002), smoothing spline, etc.) to fit a flexible parametric model using the hazard scale and the Weibull distribution with natural spline smoothing of the log cumulative hazards function, with effects allowed to be time-dependent. (See, P. Royston, M. Parmar, *Statistics in Medicine* 21(15:2175-2197 (2002).) The relationship between recurrence risk and (1) recurrence risk groups; and (2) clinical/pathologic covariates (e.g., tumor stage, tumor grade, presence of necrosis, lymphatic or vascular invasion, etc.) may also be tested for significance. Additional examples of models include logistic or ordinal logistic regression models in which the association between gene expression and dichotomous (for logistic) or ordinal (for ordinal logistic) clinical endpoints (i.e. stage, necrosis, grade) may be evaluated. (See e.g., D. Hosmer and S. Lemeshow, *Applied Logistic Regression* (John Wiley and Sons, 1989).

In an exemplary embodiment, results were adjusted for multiple hypothesis tests, and allowed for a 10% false discovery rate (FDR), using Storey's procedure, and using TDRAS with separate classes (M. Crager, Gene identification using true discovery rate degree of association sets and estimates corrected for regression to the mean, *Statistics in Medicine* (published online December 2009). In another embodiment, genes with significant association with RFI were identified through cross-validation techniques in which forward stepwise Cox PH regression was employed using a subset of factors identified through Principal Component Analysis (PCA).

Methods for calculating correlation coefficients, particularly the Pearson product-moment correlation coefficient are known in the art. (See e.g., J. Rodgers and W. Nicewander, *The American Statistician*, 42, 59-66 (1988); H. Motulsky, H., *Intuitive Biostatistics* (Oxford University Press, 1995). To perform particular biological processes, genes often work together in a concerted way, i.e. they are co-expressed. Co-expressed gene groups identified for a disease process like cancer can serve as biomarkers for disease progression and response to treatment. Such co-expressed genes can be assayed in lieu of, or in addition to, assaying of the prognostic and/or predictive gene with which they are co-expressed.

One skilled in the art will recognize that many co-expression analysis methods now known or later developed will fall within the scope and spirit of the present invention. These methods may incorporate, for example, correlation coefficients, co-expression network analysis, clique analysis, etc., and may be based on expression data from RT-PCR, microarrays, sequencing, and other similar technologies. For example, gene expression clusters can be identified using pair-wise analysis of correlation based on Pearson or Spearman correlation coefficients. (See, e.g., Pearson K. and Lee A., *Biometrika* 2, 357 (1902); C. Spearman, *Amer. J. Psychol* 15:72-101 (1904); J. Myers, A. Well, *Research Design and Statistical Analysis*, p. 508 (2nd Ed., 2003).) In general, a correlation coefficient of equal to or greater than 0.3 is considered to be statistically significant in a sample size of at least 20. (See, e.g., G. Norman, D. Streiner, *Biostatistics: The Bare Essentials*, 137-138 (3rd Ed. 2007).)

All aspects of the present disclosure may also be practiced such that a limited number of additional genes that are co-expressed with the disclosed genes, for example as evidenced by high Pearson correlation coefficients, are included in a prognostic or predictive test in addition to and/or in place of disclosed genes.

Having described the invention, the same will be more readily understood through reference to the following Examples, which are provided by way of illustration, and are not intended to limit the invention in any way. All citations throughout the disclosure are hereby expressly incorporated by reference.

EXAMPLES

Two studies were performed to demonstrate the feasibility of gene expression profiling from renal tumors obtained from renal cell carcinoma patients. (See Abstract by M. Zhou, et al., Optimized RNA extraction and RT-PCR assays provide successful molecular analysis on a wide variety of archival fixed tissues, AACR Annual Meeting (2007)).

Study Design

Renal tumor tissue was obtained from approximately 1200 patients from the Cleveland Clinic Foundation (CCF) database. This database consists of patients who were diagnosed with renal carcinoma, clear cell type, stage I, II and III between the years of 1985 and 2003, who had available paraffin-embedded tumor (PET) blocks and adequate clinical follow-up, and who were not treated with adjuvant/neoadjuvant systemic therapy. Patients with inherited VHL disease or bilateral tumors were also excluded. Tumors were graded using (1) Fuhrman grading system as noted in the World Health Organization Classification of Tumours: Pathology and Genetics: Tumours of the Urinary System and Male Genital Organs; and (2) the modified Fuhrman grading system (Table 1). In general, if no nodal involvement is expected or observed for patients, inspection of nodal involvement is not conducted and Nx is noted. In this study, Nx was treated as N0 for purposes of stage classification. The expression of 732 genes was quantitatively assessed for each patient tissue sample.

TABLE 1

Fuhrman Grading Systems		
	Fuhrman Grade (Modified)	Fuhrman Grade (WHO)
1	Nuclei small as lymphocyte with condensed chromatin	Small, round, uniform nuclei (~10 um); nucleoli absent or inconspicuous (at 400x)

TABLE 1-continued

Fuhrman Grading Systems		
	Fuhrman Grade (Modified)	Fuhrman Grade (WHO)
2	Nuclei both small as lymphocytes with condensed chromatin and other nuclei demonstrating enlarged, open chromatin	Larger nuclei (~15 μ m) with irregular outline; small nucleoli present (at 400x)
3	All nuclei enlarged with open chromatin	Larger nuclei (approaching 20 μ m) with more irregular outline; prominent nucleoli present (at 100x)
4	Large bizarre nuclei	Grade 3 features with pleomorphic or multilobed nuclei, with or without spindle cells

Inclusion Criteria

- (1) Patients who underwent nephrectomy at CCF and who have a minimum of 6 months clinical follow-up or have recurrent RCC, documented in the clinical chart, database or registry.
- (2) Diagnosed with RCC, clear cell type, stage I, II, or III.
- (3) Renal blocks fixed in Formalin, Hollandes fluid or Zenkers fixative.

Exclusion Criteria

- (1) No tumor block available from initial diagnosis in the Cleveland Clinic archive.
- (2) No tumor or very little tumor (<5% of invasive cancer cell area) in block as assessed by examination of the Hollandes and/or hematoxylin and eosin stained (H&E) slide.
- (3) High cycle threshold (CO values of reference genes. All samples regardless of their RNA amount will be tested by RT-qPCR, but only plates where the average Ct of reference genes is less than 35 will be analyzed.
- (4) Patients with inherited VHL disease and/or bilateral tumors
- (5) Patients who received neo-adjuvant or adjuvant systemic therapy

Concordance for Clinical and Pathologic Factors

Two separate pathology laboratories conducted analyses of several clinical and pathologic factors using the same standardized measures. The level of concordance, by covariate, is provided in Table 2, below. For purposes of the statistical analysis, the determination of only one of the central laboratories was used.

TABLE 2

Concordance between two central laboratories for clinical/pathologic covariates		
Covariate	Method of analysis	Concordance
Presence of necrosis	Microscopic technique per Leibovich B C et al. (2003) Cancer 97(3), 1663-1671.	47%
Tumor grade	Fuhrman	65%

Expression Profile Gene Panel

The RNA from paraffin embedded tissue (PET) samples obtained from 942 patients who met all inclusion/exclusion criteria was extracted using protocols optimized for fixed renal tissue and perform molecular assays of quantitative gene expression using TaqMan® RT-PCR. RT-PCR was performed with RNA input at 1.0 ng per 5 μ L-reaction volume using two 384 well plates.

RT-PCR analysis of PET samples was conducted using 732 genes. These genes were evaluated for association with the primary and secondary endpoints, recurrence-free interval (RFI), disease-free survival (DFS) and overall survival (OS).

All primary and secondary analyses were conducted on reference normalized gene expression levels using the mean of the reference genes for normalization. Three normalization schemes were tested using AAMP, ARF1, ATP5E, EEF1A1, GPX1, RPS23, SDHA, UBB, and RPLP1. Some or all of the other genes in the test panel were used in analyses of alternative normalization schemes.

Of the 732 genes, 647 were deemed evaluable for further consideration. The outcome analyses of the 647 evaluable genes were adjusted for multiple hypothesis tests, by allowing for a 10% false discovery rate (FDR), using Storey's procedure, and using True Discovery Rate Degree of Association Sets (TDRDAS) with separate classes (M. Crager, Gene identification using true discovery rate degree of association sets and estimates corrected for regression to the mean, *Statistics in Medicine* 29:33-45 (2009)). Unadjusted for baseline covariates and without controlling the FDR, a subset of 448 (69%) of genes were identified as significantly associated with RFI ($p \leq 0.05$). Additional analysis was conducted using a supervised principal component analysis (PCA) on a subset of 188 genes that have maximum lower bound (MLB) > 1.2 using TDRDAS analysis, controlling the FDR but not taking into account the separate classes. The top 10 factors were modified by keeping the genes with high factor loadings (loading/max loading > 0.7) were kept. The modified top 10 factors were put into a 5-fold cross validation to assess performance of the gene groups and identify factors that were appearing most frequently.

Genes that had a significant association ($p \leq 0.05$) with risk of recurrence are listed in Tables 3a and 3b, wherein genes that are positively associated with a good prognosis (i.e., increased expression indicates a lower risk of recurrence) are listed in Table 3a. Genes that are negatively associated with a good prognosis (i.e., increased expression indicates a higher risk of recurrence) are listed in Table 3b. Genes that were associated, positively or negatively, with a good prognosis but not associated with clinical/pathologic covariates are BBC3, CCR7, CCR4, and VCAN. In addition, genes associated positively with a good prognosis after adjusting for clinical and pathologic covariates (stage, tumor grade, tumor size, nodal status, and presence of necrosis) are listed in Table 8a. Genes associated negatively with a good prognosis after adjusting for clinical/pathologic covariates are listed in Table 8b. Of those genes listed in Tables 8a and 8b, 16 genes with significant association, positively or negatively, with a good prognosis after adjusting for clinical and pathologic covariates and controlling the false discovery rate at 10% are listed in Table 9.

For the majority of these genes significantly associated with RFI ($p \leq 0.05$) (82%), increased expression is associated with a good prognosis. Most of the genes significantly associated with RFI showed consistency between (1) between stages (I-III); (2) primary and secondary endpoints (RFI and OS). See, e.g., FIGS. 1 and 2, respectively.

From this analysis, certain gene subsets emerged as significantly associated with recurrence and overall survival. For example, increased expression of angiogenesis genes (e.g., EMCN, PPAP2B, NOS3, NUDT6, PTPRB, SNRK, APOLD1, PRKCH, and CEACAM1), cell adhesion/extracellular matrix genes (e.g., ITGB5, ITGB1, A2M, TIMP3), immune response genes (e.g., CCL5, CCXL9, CCR7, IL8, IL6, and CXCL1), cell cycle (e.g., BUB1, TPX2), apop-

tosis (e.g., CASP10), and transport genes (AQP1) were strongly associated, positively or negatively, with RFI.

Also, certain genes that are associated with pathway targets for renal cancer drugs (sunitinib, sorafenib, temsirolimus, bevacizumab, everolimus, pazopanib) were identified as having a significant association with outcome, including: KIT, PDGFA, PDGFB, PDGFC, PDGFD, PDGFRb, KRAS, RAF1, MTOR, HIF1AN, VEGFA, VEGFB, and FLT4.

It was determined that the presence of necrosis in these tumors was associated with a higher risk of recurrence, at least in the first 4 years after surgery. See FIG. 3. However, the prognostic effect of necrosis after year 4 was negligible.

It was also determined that expression of certain genes was correlated, positively or negatively, with pathologic and/or clinical factors ("proxy genes"). For example, increased expression of the proxy genes listed in Tables 4a-7b correlate, positively or negatively, with tumor stage, tumor grade, presence of necrosis, and nodal invasion, respectively. In Tables 4a-7b, gene expression was normalized and then standardized such that the odds ratio (OR) reflects a one standard deviation change in gene expression.

From these, key genes were identified as good proxies for baseline covariates (stage, grade, necrosis), including TSPAN7, TEK, LDB2, TIMP3, SHANK3, RGS5, KDR, SDPR, EPAS1, ID1, TGFB2, FLT4, SDPR, ENDRB, JAG1, DLC1, and KL. Several of these genes are in the hypoxia-induced pathway: SHANK3, RGS5, EPAS1, KDR, JAG1, TGFB2, FLT4, SDPR, DLC1, EDNRB.

FIGS. 4-7 provide a comparison of patient stratification (Low, Intermediate, or High Risk) obtained by applying the Mayo prognostic tool (described in Leibovich et al. "prediction of progression after radical nephrectomy for patients with clear cell renal cell carcinoma" (2003) Cancer 97:1663-1671) CCF expression data. As shown in FIG. 4, the Mayo prognostic tool alone provides for stratification into three populations Low Risk (93% recurrence free at 5 years), Intermediate Risk (79% recurrence free at 5 years), and High Risk (36% recurrence free at 5 years). In contrast, use of expression data from even one gene (as exemplified by EMCN (FIG. 5), AQP1 (FIG. 6), or PPAP2B (FIG. 7)) allowed for more detailed stratification of patients according to risk.

TABLE 3a

Genes for which increased expression is associated with lower risk of cancer recurrence (p-value ≤ 0.05)							
Univariate Cox Analyses (No Covariate Adjustment) with RFI				Univariate Cox Analyses (No Covariate Adjustment) with RFI			
Gene	Official Symbol	p-value for HR	HR	Gene	Official Symbol	p-value for HR	HR
YB-1.2	YBX1	<0.0001	0.75	IGF1R.3	IGF1R	<0.0001	0.58
XIAP.1	XIAP	0.0009	0.81	IFI27.1	IFI27	0.0205	0.85
WWOX.5	WWOX	<0.0001	0.71	ID3.1	ID3	<0.0001	0.69
VWF.1	VWF	<0.0001	0.61	ID2.4	ID2	<0.0001	0.75
VEGF.1	VEGFA	<0.0001	0.75	ID1.1	ID1	<0.0001	0.55
VCAM1.1	VCAM1	<0.0001	0.66	ICAM2.1	ICAM2	<0.0001	0.64
USP34.1	USP34	0.0003	0.79	HYAL2.1	HYAL2	<0.0001	0.60
UMOD.1	UMOD	<0.0001	0.68	HYAL1.1	HYAL1	<0.0001	0.58
UGCG.1	UGCG	<0.0001	0.71	HSPG2.1	HSPG2	<0.0001	0.60
UBB.1	UBB	<0.0001	0.62	HSD11B2.1	HSD11B2	<0.0001	0.63
UBE1C.1	UBA3	0.0003	0.79	Hepsin.1	HPN	0.0001	0.76
TS.1	TYMS	0.0056	0.83	HPCAL1.1	HPCAL1	0.0031	0.82
tusc4.2	TUSC4	<0.0001	0.76	HMGB1.1	HMGB1	<0.0001	0.67
TSPAN7.2	TSPAN7	<0.0001	0.52	HLA-DPB1.1	HLA-DPB1	<0.0001	0.70
TSC2.1	TSC2	0.0043	0.82	HIF1AN.1	HIF1AN	<0.0001	0.77
TSC1.1	TSC1	<0.0001	0.71	HDAC1.1	HDAC1	0.0003	0.78
P53.2	TP53	0.0008	0.81	HAVCR1.1	HAVCR1	0.0003	0.79
TOP2B.2	TOP2B	0.0001	0.80	HADH.1	HADH	<0.0001	0.68
TNFSF12.1	TNFSF12	<0.0001	0.68	GZMA.1	GZMA	0.0125	0.84
TRAIL.1	TNFSF10	0.0065	0.83	GSTp.3	GSTP1	<0.0001	0.76
TNFRSF11B.1	TNFRSF11B	0.0002	0.78	GSTM3.2	GSTM3	<0.0001	0.70
TNFRSF10D.1	TNFRSF10D	<0.0001	0.76	GSTM1.1	GSTM1	<0.0001	0.72
DR5.2	TNFRSF10B	<0.0001	0.75	GRB7.2	GRB7	<0.0001	0.74
TNFAIP6.1	TNFAIP6	0.0005	0.79	GPX3.1	GPX3	<0.0001	0.75
TMEM47.1	TMEM47	<0.0001	0.57	GJA1.1	GJA1	0.0049	0.84
TMEM27.1	TMEM27	<0.0001	0.58	GFRA1.1	GFRA1	0.0003	0.77
TLR3.1	TLR3	<0.0001	0.72	GCLC.3	GCLC	<0.0001	0.68
TIMP3.3	TIMP3	<0.0001	0.50	GBP2.2	GBP2	0.0156	0.85
TIMP2.1	TIMP2	<0.0001	0.73	GATM.1	GATM	<0.0001	0.66
THBS1.1	THBS1	0.0002	0.79	GATA3.3	GATA3	0.0001	0.75
TGFB2.3	TGFB2	<0.0001	0.54	FOS.1	FOS	<0.0001	0.74
TGFB1.1	TGFB1	<0.0001	0.71	FOLR1.1	FOLR1	0.0003	0.79
TGFB2.2	TGFB2	<0.0001	0.65	FLT4.1	FLT4	<0.0001	0.54
TGFA.2	TGFA	<0.0001	0.70	FLT3LG.1	FLT3LG	0.0001	0.73
TEK.1	TEK	<0.0001	0.47	FLT1.1	FLT1	<0.0001	0.59
TCF4.1	TCF4	<0.0001	0.62	FILIP1.1	FILIP1	<0.0001	0.72
TAP1.1	TAP1	0.0024	0.82	FIGF.1	FIGF	0.0014	0.76
TAGLN.1	TAGLN	<0.0001	0.70	FHL1.1	FHL1	<0.0001	0.68
TACSTD2.1	TACSTD2	<0.0001	0.72	FHIT.1	FHIT	<0.0001	0.73
SUCLG1.1	SUCLG1	<0.0001	0.72	FH.1	FH	<0.0001	0.73
STK11.1	STK11	<0.0001	0.76	FGFR2 isoform 1.1	FGFR2	<0.0001	0.70
STAT5B.2	STAT5B	<0.0001	0.71	FGFR1.3	FGFR1	0.0001	0.77
STAT5A.1	STAT5A	<0.0001	0.72	FGF2.2	FGF2	<0.0001	0.73
STAT3.1	STAT3	0.0016	0.80	FGF1.1	FGF1	0.0015	0.78

TABLE 3a-continued

Genes for which increased expression is associated with lower risk of cancer recurrence (p-value ≤ 0.05)							
Univariate Cox Analyses (No Covariate Adjustment) with RFI				Univariate Cox Analyses (No Covariate Adjustment) with RFI			
Gene	Official Symbol	p-value for HR	HR	Gene	Official Symbol	p-value for HR	HR
SPRY1.1	SPRY1	<0.0001	0.68	FDPS.1	FDPS	<0.0001	0.69
SPARCL1.1	SPARCL1	<0.0001	0.71	FBXW7.1	FBXW7	0.0001	0.75
SPARC.1	SPARC	0.0009	0.80	fas.1	FAS	<0.0001	0.73
SOD1.1	SOD1	0.0114	0.84	FABP1.1	FABP1	0.0455	0.86
SNRK.1	SNRK	<0.0001	0.54	ESRRG.3	ESRRG	0.0008	0.79
SNAI1.1	SNAI1	0.0055	0.82	ERG.1	ERG	<0.0001	0.60
MADH4.1	SMAD4	<0.0001	0.60	ERCC1.2	ERCC1	<0.0001	0.78
MADH2.1	SMAD2	<0.0001	0.64	ErbB3.1	ERBB3	0.0001	0.77
SLC34A1.1	SLC34A1	0.0004	0.76	HER2.3	ERBB2	<0.0001	0.65
SLC22A6.1	SLC22A6	0.0003	0.77	EPHB4.1	EPHB4	<0.0001	0.64
SKIL.1	SKIL	0.0004	0.79	EPHA2.1	EPHA2	<0.0001	0.58
SHANK3.1	SHANK3	<0.0001	0.53	EPAS1.1	EPAS1	<0.0001	0.55
SGK.1	SGK1	<0.0001	0.67	ENPP2.1	ENPP2	0.0090	0.84
FRP1.3	SFRP1	0.0355	0.86	ENPEP.1	ENPEP	<0.0001	0.74
SEMA3F.3	SEMA3F	<0.0001	0.67	CD105.1	ENG	<0.0001	0.60
SELENBP1.1	SELENBP1	0.0016	0.81	EMP1.1	EMP1	<0.0001	0.71
SDPR.1	SDPR	<0.0001	0.51	EMCN.1	EMCN	<0.0001	0.43
SDHA.1	SDHA	0.0003	0.77	ELTD1.1	ELTD1	<0.0001	0.76
SCNN1A.2	SCNN1A	0.0134	0.83	EIF2C1.1	EIF2C1	<0.0001	0.67
SCN4B.1	SCN4B	<0.0001	0.68	EGR1.1	EGR1	<0.0001	0.72
KIAA1303	RPTOR	<0.0001	0.77	EGLN3.1	EGLN3	0.0002	0.80
raptor.1							
RPS6KB1.3	RPS6KB1	<0.0001	0.73	EGFR.2	EGFR	0.0005	0.80
RPS6KA1.1	RPS6KA1	0.0291	0.86	EFNB2.1	EFNB2	<0.0001	0.64
RPS23.1	RPS23	<0.0001	0.73	EFNB1.2	EFNB1	<0.0001	0.68
ROCK2.1	ROCK2	<0.0001	0.66	EEF1A1.1	EEF1A1	<0.0001	0.64
ROCK1.1	ROCK1	<0.0001	0.58	EDNRB.1	EDNRB	<0.0001	0.56
RIPK1.1	RIPK1	<0.0001	0.64	EDN2.1	EDN2	0.0005	0.77
rhoC.1	RHOC	0.0213	0.86	EDN1	EDN1	<0.0001	0.62
			endothelin.1				
RhoB.1	RHOB	<0.0001	0.63	EBAG9.1	EBAG9	0.0041	0.82
ARHA.1	RHOA	<0.0001	0.64	DUSP1.1	DUSP1	0.0029	0.82
RG5.1	RG5	<0.0001	0.52	DPYS.1	DPYS	0.0112	0.84
FLJ22655.1	REGL	<0.0001	0.51	DPEP1.1	DPEP1	<0.0001	0.64
NFKBp65.3	RELA	0.0027	0.82	DLL4.1	DLL4	<0.0001	0.76
RB1.1	RB1	<0.0001	0.73	DLC1.1	DLC1	<0.0001	0.55
RASSF1.1	RASSF1	0.0040	0.83	DKFZP564O0823.1	DKFZP564O0823	<0.0001	0.62
RARB.2	RARB	<0.0001	0.57	DICER1.2	DICER1	<0.0001	0.71
RALBP1.1	RALBP1	<0.0001	0.73	DIAPH1.1	DIAPH1	0.0009	0.80
RAF1.3	RAF1	0.0008	0.81	DIABLO.1	DIABLO	0.0134	0.84
PTPRG.1	PTPRG	<0.0001	0.57	DHPS.3	DHPS	<0.0001	0.70
PTPRB.1	PTPRB	<0.0001	0.51	DET1.1	DET1	<0.0001	0.74
PTN.1	PTN	<0.0001	0.71	DEFB1.1	DEFB1	0.0025	0.81
PTK2.1	PTK2	<0.0001	0.61	DDC.1	DDC	<0.0001	0.68
PTH1R.1	PTH1R	<0.0001	0.55	DCXR.1	DCXR	0.0081	0.83
PTEN.2	PTEN	<0.0001	0.74	DAPK1.3	DAPK1	<0.0001	0.60
PSMB9.1	PSMB9	0.0025	0.82	CYR61.1	CYR61	<0.0001	0.70
PSMB8.1	PSMB8	0.0239	0.85	CYP3A4.2	CYP3A4	0.0380	0.86
PRSS8.1	PRSS8	<0.0001	0.71	CXCL9.1	CXCL9	0.0362	0.87
PRPS2.1	PRPS2	0.0156	0.85	CXCL12.1	CXCL12	<0.0001	0.69
PRKCH.1	PRKCH	<0.0001	0.63	CX3CR1.1	CX3CR1	<0.0001	0.72
PPP2CA.1	PPP2CA	<0.0001	0.77	CX3CL1.1	CX3CL1	<0.0001	0.58
PPARG.3	PPARG	<0.0001	0.59	CUL1.1	CUL1	0.0003	0.80
PPAP2B.1	PPAP2B	<0.0001	0.50	CUBN.1	CUBN	<0.0001	0.61
PLG.1	PLG	<0.0001	0.70	CTSS.1	CTSS	0.0016	0.82
PLAT.1	PLAT	<0.0001	0.64	CTSH.2	CTSH	<0.0001	0.78
PLA2G4C.1	PLA2G4C	<0.0001	0.67	B-Catenin.3	CTNNB1	<0.0001	0.74
PIK3CA.1	PIK3CA	<0.0001	0.75	A-Catenin.2	CTNNA1	<0.0001	0.77
PI3K.2	PIK3C2B	<0.0001	0.56	CTGF.1	CTGF	0.0004	0.79
PFKP.1	PFKP	0.0067	0.84	CSF1R.2	CSF1R	0.0015	0.82
CD31.3	PECAM1	<0.0001	0.60	CSF1.1	CSF1	0.0016	0.81
PDZK3.1	PDZK3	0.0005	0.77	CRADD.1	CRADD	0.0021	0.81
PDZK1.1	PDZK1	<0.0001	0.69	COL4A2.1	COL4A2	0.0016	0.80
PDGFRb.3	PDGFRB	<0.0001	0.74	COL18A1.1	COL18A1	0.0007	0.79
PDGFD.2	PDGFD	<0.0001	0.58	CLU.3	CLU	0.0151	0.85
PDGFC.3	PDGFC	<0.0001	0.66	CLDN7.2	CLDN7	0.0023	0.81
PDGFB.3	PDGFB	<0.0001	0.63	CLDN10.1	CLDN10	<0.0001	0.69
PDGFA.3	PDGFA	<0.0001	0.75	CLCNKB.1	CLCNKB	0.0002	0.74
PCK1.1	PCK1	<0.0001	0.65	CFLAR.1	CFLAR	<0.0001	0.65
PCCA.1	PCCA	<0.0001	0.66	CEACAM1.1	CEACAM1	<0.0001	0.59
PARD6A.1	PARD6A	0.0001	0.76	p27.3	CDKN1B	0.0018	0.83

TABLE 3a-continued

Genes for which increased expression is associated with lower risk of cancer recurrence (p-value ≤ 0.05)							
Univariate Cox Analyses (No Covariate Adjustment) with RFI				Univariate Cox Analyses (No Covariate Adjustment) with RFI			
Gene	Official Symbol	p-value for HR	HR	Gene	Official Symbol	p-value for HR	HR
Pak1.2	PAK1	0.0011	0.81	p21.3	CDKN1A	0.0027	0.81
PAH.1	PAH	0.0296	0.86	CDH6.1	CDH6	<0.0001	0.75
OGG1.1	OGG1	0.0051	0.84	CDH5.1	CDH5	<0.0001	0.59
BFGF.3	NUDT6	<0.0001	0.54	CDH16.1	CDH16	<0.0001	0.75
NRG1.3	NRG1	0.0049	0.81	CDH13.1	CDH13	<0.0001	0.66
NPR1.1	NPR1	<0.0001	0.73	CD4.1	CD4	0.0009	0.81
NPM1.2	NPM1	<0.0001	0.73	CD36.1	CD36	<0.0001	0.65
NOTCH3.1	NOTCH3	<0.0001	0.70	CD34.1	CD34	<0.0001	0.62
NOTCH2.1	NOTCH2	0.0040	0.84	CCR7.1	CCR7	0.0271	0.86
NOTCH1.1	NOTCH1	<0.0001	0.64	CCR4.2	CCR4	0.0106	0.83
NOS3.1	NOS3	<0.0001	0.55	CCND1.3	CCND1	<0.0001	0.70
NOS2A.3	NOS2	<0.0001	0.66	CCL4.2	CCL4	0.0012	0.80
NOL3.1	NOL3	0.0132	0.85	MCP1.1	CCL2	<0.0001	0.75
NFX1.1	NFX1	<0.0001	0.67	CAT.1	CAT	<0.0001	0.72
NFKBp50.3	NFKB1	0.0001	0.77	CASP6.1	CASP6	0.0369	0.87
NFATC2.1	NFATC2	0.0003	0.78	CASP10.1	CASP10	<0.0001	0.69
NFAT5.1	NFAT5	0.0010	0.82	CALD1.2	CALD1	<0.0001	0.61
MYRIP.2	MYRIP	0.0002	0.75	CA9.3	CA9	0.0035	0.84
MYH11.1	MYH11	<0.0001	0.61	CA2.1	CA2	0.0006	0.79
eMYC.3	MYC	0.0002	0.79	C7.1	C7	0.0030	0.82
MVP.1	MVP	0.0052	0.83	ECRG4.1	C2orf40	<0.0001	0.57
MUC1.2	MUC1	0.0405	0.87	C13orf15.1	C13orf15	<0.0001	0.57
FRAP1.1	MTOR	<0.0001	0.75	BUB3.1	BUB3	0.0002	0.77
MSH3.2	MSH3	<0.0001	0.75	BTRC.1	BTRC	0.0006	0.81
MSH2.3	MSH2	<0.0001	0.71	CIAP1.2	BIRC2	0.0030	0.82
GBL.1	MLST8	0.0246	0.87	BIN1.3	BIN1	0.0005	0.80
MIF.2	MIF	0.0012	0.80	BGN.1	BGN	<0.0001	0.76
MICA.1	MICA	<0.0001	0.71	BCL2L12.1	BCL2L12	0.0322	0.86
MGMT.1	MGMT	<0.0001	0.68	Belx.2	BCL2L1	<0.0001	0.74
MCM3.3	MCM3	0.0188	0.85	Bcl2.2	BCL2	<0.0001	0.57
MCAM.1	MCAM	<0.0001	0.71	BBC3.2	BBC3	0.0449	0.87
MARCKS.1	MARCKS	0.0301	0.87	BAG1.2	BAG1	<0.0001	0.64
ERK1.3	MAPK3	<0.0001	0.65	BAD.1	BAD	0.0076	0.85
ERK2.3	MAPK1	0.0005	0.79	ATP6V1B1.1	ATP6V1B1	0.0001	0.71
MAP4.1	MAP4	<0.0001	0.65	ASS1.1	ASS1	<0.0001	0.75
MAP2K3.1	MAP2K3	<0.0001	0.69	ARRB1.1	ARRB1	<0.0001	0.62
MAP2K1.1	MAP2K1	0.0046	0.83	ARHGD1B.1	ARHGD1B	<0.0001	0.66
MAL2.1	MAL2	0.0001	0.76	AQP1.1	AQP1	<0.0001	0.50
MAL.1	MAL	<0.0001	0.66	APOLD1.1	APOLD1	<0.0001	0.57
LYZ.1	LYZ	0.0458	0.88	APC.4	APC	<0.0001	0.73
LTF.1	LTF	0.0005	0.76	ANXA4.1	ANXA4	0.0018	0.81
LRP2.1	LRP2	<0.0001	0.67	ANXA1.2	ANXA1	0.0009	0.80
LMO2.1	LMO2	<0.0001	0.74	ANTXR1.1	ANTXR1	0.0043	0.82
LDB2.1	LDB2	<0.0001	0.52	ANGPTL4.1	ANGPTL4	0.0033	0.84
LDB1.2	LDB1	<0.0001	0.71	ANGPTL3.3	ANGPTL3	0.0003	0.75
LAMA4.1	LAMA4	0.0279	0.86	ANGPT1.1	ANGPT1	<0.0001	0.57
KRT7.1	KRT7	<0.0001	0.68	ALDOB.1	ALDOB	<0.0001	0.62
K-ras.10	KRAS	<0.0001	0.72	ALDH6A1.1	ALDH6A1	<0.0001	0.63
KL.1	KL	<0.0001	0.55	ALDH4.2	ALDH4A1	0.0172	0.85
Kitlng.4	KITLG	<0.0001	0.67	AKT3.2	AKT3	<0.0001	0.57
c-kit.2	KIT	<0.0001	0.72	AKT2.3	AKT2	<0.0001	0.75
KDR.6	KDR	<0.0001	0.54	AKT1.3	AKT1	<0.0001	0.71
KCNJ15.1	KCNJ15	<0.0001	0.63	AIF1.1	AIF1	0.0349	0.87
HTATIP.1	KAT5	<0.0001	0.64	AHR.1	AHR	<0.0001	0.74
G-Catenin.1	JUP	<0.0001	0.64	AGTR1.1	AGTR1	<0.0001	0.57
AP-1 (JUN official).2	JUN	0.0001	0.76	ADH1B.1	ADH1B	0.0002	0.77
JAG1.1	JAG1	<0.0001	0.55	ADFP.1	ADFP	0.0332	0.88
ITGB1.1	ITGB1	0.0085	0.83	ADD1.1	ADD1	<0.0001	0.58
ITGA7.1	ITGA7	<0.0001	0.66	ADAMTS5.1	ADAMTS5	0.0010	0.78
ITGA6.2	ITGA6	<0.0001	0.63	ADAMTS1.1	ADAMTS1	0.0056	0.83
ITGA4.2	ITGA4	<0.0001	0.74	ACE2.1	ACE2	<0.0001	0.62
ITGA3.2	ITGA3	0.0002	0.79	ACADSB.1	ACADSB	<0.0001	0.71
IQGAP2.1	IQGAP2	<0.0001	0.69	BCRP.1	ABCG2	<0.0001	0.58

TABLE 3a-continued

Genes for which increased expression is associated with lower risk of cancer recurrence (p-value ≤ 0.05)							
Univariate Cox Analyses (No Covariate Adjustment) with RFI				Univariate Cox Analyses (No Covariate Adjustment) with RFI			
Gene	Official Symbol	p-value for HR	HR	Gene	Official Symbol	p-value for HR	HR
INSR.1	INSR	<0.0001	0.67	MRP4.2	ABCC4	<0.0001	0.74
IMP3.1	IMP3	<0.0001	0.69	MRP3.1	ABCC3	0.0107	0.85
IL6ST.3	IL6ST	<0.0001	0.66	MRP1.1	ABCC1	<0.0001	0.75
IL15.1	IL15	<0.0001	0.70	ABCB1.5	ABCB1	0.0093	0.84
IGFBP6.1	IGFBP6	0.0004	0.79	NPD009 (ABAT official).3	ABAT	0.0001	0.76
IGFBP3.1	IGFBP3	0.0394	0.87	AAMP.1	AAMP	0.0008	0.80
IGFBP2.1	IGFBP2	0.0134	0.84	A2M.1	A2M	<0.0001	0.56

TABLE 3b

Genes for which increased expression is associated with higher risk of cancer recurrence (p-value ≤ 0.05)							
Univariate Cox Analyses (No Covariate Adjustment) with RFI				Univariate Cox Analyses (No Covariate Adjustment) with RFI			
Gene	Official Symbol	p-value for HR	HR	Gene	Official Symbol	p-value for HR	HR
WT1.1	WT1	0.0002	1.25	LGALS1.1	LGALS1	0.0017	1.25
VTN.1	VTN	0.0097	1.17	LAMB3.1	LAMB3	<0.0001	1.34
VDR.2	VDR	0.0031	1.22	LAMB1.1	LAMB1	0.0014	1.25
VCAN.1	VCAN	0.0036	1.22	L1CAM.1	L1CAM	0.0199	1.16
UBE2T.1	UBE2T	<0.0001	1.38	IL-8.1	IL8	<0.0001	1.53
C20 orf1.1	TPX2	<0.0001	1.76	IL6.3	IL6	<0.0001	1.41
TOP2A.4	TOP2A	<0.0001	1.39	ICAM1.1	ICAM1	0.0013	1.23
TK1.2	TK1	0.0018	1.22	HIST1H1D.1	HIST1H1D	0.0066	1.21
TIMP1.1	TIMP1	0.0259	1.16	FN1.1	FN1	0.0105	1.19
TGFBI.1	TGFBI	0.0004	1.26	F3.1	F3	<0.0001	1.31
SQSTM1.1	SQSTM1	0.0089	1.20	F2.1	F2	<0.0001	1.30
OPN, osteopontin.3	SPP1	<0.0001	1.43	ESPL1.3	ESPL1	0.0155	1.17
SPHK1.1	SPHK1	0.0025	1.22	EPHB2.1	EPHB2	0.0456	1.14
SLC7A5.2	SLC7A5	<0.0001	1.38	EPHB1.3	EPHB1	0.0007	1.22
SLC2A1.1	SLC2A1	0.0010	1.26	ENO2.1	ENO2	<0.0001	1.38
SLC16A3.1	SLC16A3	<0.0001	1.38	EIF4EBP1.1	EIF4EBP1	0.0098	1.19
SLC13A3.1	SLC13A3	0.0192	1.16	CXCR4.3	CXCR4	0.0066	1.21
SHC1.1	SHC1	0.0086	1.19	GRO1.2	CXCL1	<0.0001	1.30
SFN.1	SFN	0.0001	1.26	CTSB.1	CTSB	0.0233	1.17
SERPINA5.1	SERPINA5	0.0462	1.13	CRP.1	CRP	0.0314	1.13
SEMA3C.1	SEMA3C	<0.0001	1.45	CP.1	CP	0.0002	1.32
SAA2.2	SAA2	<0.0001	1.59	COL7A1.1	COL7A1	0.0003	1.24
S100A1.1	S100A1	0.0348	1.16	COL1A1.1	COL1A1	0.0029	1.23
RRM2.1	RRM2	0.0002	1.27	Chk1.2	CHEK1	0.0002	1.26
RPLP1.1	RPLP1	0.0049	1.22	CENPF.1	CENPF	<0.0001	1.36
PTTG1.2	PTTG1	<0.0001	1.45	CD82.3	CD82	0.0009	1.25
COX2.1	PTGS2	0.0013	1.22	CD44s.1	CD44_s	0.0065	1.21
PLAUR.3	PLAUR	<0.0001	1.33	CCNE1.1	CCNE1	0.0098	1.17
PF4.1	PF4	0.0034	1.20	CCNB1.2	CCNB1	<0.0001	1.42
PCSK6.1	PCSK6	0.0269	1.17	CCL20.1	CCL20	0.0029	1.22
MYBL2.1	MYBL2	<0.0001	1.33	CA12.1	CA12	<0.0001	1.48
MT1X.1	MT1X	0.0070	1.20	C3.1	C3	0.0176	1.18
MMP9.1	MMP9	<0.0001	1.54	BUB1.1	BUB1	<0.0001	1.59
MMP7.1	MMP7	0.0312	1.15	SURV.2	BIRC5	<0.0001	1.37
MMP14.1	MMP14	<0.0001	1.47	cIAP2.2	BIRC3	0.0484	1.15
Ki-67.2	MKI67	<0.0001	1.33	BCL2A1.1	BCL2A1	0.0483	1.11
mGST1.2	MGST1	<0.0001	1.38	STK15.2	AURKA	0.0002	1.28
MDK.1	MDK	0.0001	1.31	ANXA2.2	ANXA2	0.0315	1.16
LOX.1	LOX	<0.0001	1.42	ALOX5.1	ALOX5	0.0473	1.14
LMNB1.1	LMNB1	<0.0001	1.40	ADAM8.1	ADAM8	0.0002	1.29
LIMK1.1	LIMK1	<0.0001	1.43	MRP2.3	ABCC2	0.0004	1.28

TABLE 4a

Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤ 0.05)							
Stage 3 vs. 1				Stage 3 vs. 1			
Gene	Official Symbol	p-value	OR	Gene	Official Symbol	p-value	OR
WT1.1	WT1	<0.0001	1.41	LAMA3.1	LAMA3	0.0121	1.22
VTN.1	VTN	0.0007	1.29	L1CAM.1	L1CAM	0.0091	1.22
VDR.2	VDR	0.0065	1.25	ISG20.1	ISG20	0.0006	1.34
UBE2T.1	UBE2T	<0.0001	1.61	IL-8.1	IL8	<0.0001	1.89
TSPAN8.1	TSPAN8	0.0072	1.23	IL6.3	IL6	<0.0001	1.68
C20 orf1.1	TPX2	<0.0001	1.89	IGF1.2	IGF1	0.0214	1.20
TOP2A.4	TOP2A	<0.0001	1.55	ICAM1.1	ICAM1	<0.0001	1.42
TK1.2	TK1	0.0001	1.34	HIST1H1D.1	HIST1H1D	0.0005	1.33
TIMP1.1	TIMP1	0.0021	1.29	GPX2.2	GPX2	0.0129	1.22
TGFBI.1	TGFBI	0.0001	1.39	FN1.1	FN1	0.0002	1.36
OPN,	SPP1	0.0001	1.38	FAP.1	FAP	0.0455	1.18
osteopontin.3							
SLC7A5.2	SLC7A5	<0.0001	1.51	F3.1	F3	<0.0001	1.52
SLC2A1.1	SLC2A1	0.0081	1.24	F2.1	F2	<0.0001	1.79
SLC16A3.1	SLC16A3	<0.0001	1.46	ESPL1.3	ESPL1	0.0001	1.35
SFN.1	SFN	0.0001	1.36	EPB41L3.1	EPB41L3	0.0067	1.24
SEMA3C.1	SEMA3C	<0.0001	1.42	ENO2.1	ENO2	0.0016	1.31
SELL.1	SELL	0.0313	1.19	EIF4EBP1.1	EIF4EBP1	0.0036	1.27
SAA2.2	SAA2	<0.0001	2.04	E2F1.3	E2F1	0.0017	1.27
RRM2.1	RRM2	<0.0001	1.47	DCN.1	DCN	0.0152	1.22
RPLP1.1	RPLP1	0.0007	1.33	CXCR6.1	CXCR6	0.0013	1.30
RAD51.1	RAD51	0.0010	1.31	BLR1.1	CXCR5	0.0232	1.19
PTTG1.2	PTTG1	<0.0001	1.61	CXCR4.3	CXCR4	0.0003	1.35
COX2.1	PTGS2	0.0011	1.29	GRO1.2	CXCL1	0.0005	1.31
PTGIS.1	PTGIS	0.0034	1.27	CTSB.1	CTSB	0.0110	1.24
PLAUR.3	PLAUR	<0.0001	1.51	CRP.1	CRP	0.0002	1.31
PF4.1	PF4	0.0027	1.26	CP.1	CP	0.0008	1.34
PDGFRa.2	PDGFR	0.0480	1.17	COL7A1.1	COL7A1	0.0010	1.28
PCSK6.1	PCSK6	0.0041	1.27	COL1A1.1	COL1A1	0.0001	1.40
NNMT.1	NNMT	0.0003	1.34	Chk2.3	CHEK2	0.0050	1.27
NME2.1	NME2	0.0028	1.28	Chk1.2	CHEK1	<0.0001	1.43
MYBL2.1	MYBL2	<0.0001	1.50	CENPF.1	CENPF	<0.0001	1.55
MT1X.1	MT1X	0.0192	1.21	CD82.3	CD82	0.0001	1.38
MMP9.1	MMP9	<0.0001	1.79	CD44s.1	CD44_s	0.0060	1.25
MMP7.1	MMP7	0.0252	1.20	CCNE2.2	CCNE2_2	0.0229	1.19
MMP14.1	MMP14	<0.0001	1.88	CCNB1.2	CCNB1	<0.0001	1.60
Ki-67.2	MKI67	<0.0001	1.48	CCL20.1	CCL20	0.0010	1.30
mGST1.2	MGST1	0.0004	1.37	CA12.1	CA12	<0.0001	1.66
MDK.1	MDK	<0.0001	1.42	C3.1	C3	0.0009	1.32
MDH2.1	MDH2	0.0321	1.19	BUB1.1	BUB1	<0.0001	1.82
LRRC2.1	LRRC2	0.0259	1.19	SURV.2	BIRC5	<0.0001	1.46
LOX.1	LOX	<0.0001	1.78	BCL2A1.1	BCL2A1	<0.0001	1.44
LMNB1.1	LMNB1	<0.0001	1.82	STK15.2	AURKA	0.0002	1.36
LIMK1.1	LIMK1	<0.0001	1.46	APOL1.1	APOL1	0.0028	1.27
LAPTM5.1	LAPTM5	0.0102	1.23	ANXA2.2	ANXA2	0.0174	1.21
LAMB3.1	LAMB3	<0.0001	1.53	ADAM8.1	ADAM8	<0.0001	1.58
LAMB1.1	LAMB1	0.0452	1.18	MRP2.3	ABCC2	<0.0001	1.45

TABLE 4b

Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤ 0.05)							
Stage 3 vs. 1				Stage 3 vs. 1			
Gene	Official Symbol	p-value	OR	Gene	Official Symbol	p-value	OR
YB-1.2	YBX1	<0.0001	0.63	IL-7.1	IL7	0.0444	0.85
XIAP.1	XIAP	<0.0001	0.62	IL6ST.3	IL6ST	<0.0001	0.50
WWOX.5	WWOX	0.0042	0.79	IL15.1	IL15	<0.0001	0.67
WISP1.1	WISP1	0.0096	0.81	IGFBP6.1	IGFBP6	0.0001	0.73
VWF.1	VWF	<0.0001	0.46	IGFBP3.1	IGFBP3	0.0191	0.83
VEGF.1	VEGFA	<0.0001	0.63	IGF1R.3	IGF1R	<0.0001	0.48
VCAM1.1	VCAM1	<0.0001	0.55	ID3.1	ID3	<0.0001	0.54
USP34.1	USP34	0.0001	0.72	ID2.4	ID2	0.0008	0.77
UMOD.1	UMOD	<0.0001	0.47	ID1.1	ID1	<0.0001	0.34
UGCG.1	UGCG	<0.0001	0.68	ICAM2.1	ICAM2	<0.0001	0.58
UBB.1	UBB	<0.0001	0.54	HYAL2.1	HYAL2	<0.0001	0.41
UBE1C.1	UBA3	<0.0001	0.62	HYAL1.1	HYAL1	<0.0001	0.44
TS.1	TYMS	0.0010	0.76	HSPG2.1	HSPG2	<0.0001	0.44

TABLE 4b-continued

Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤ 0.05)						
Stage 3 vs. 1			Stage 3 vs. 1			
Gene	Official Symbol	p-value	OR	Gene	Official Symbol	p-value OR
tusc4.2	TUSC4	<0.0001	0.57	HSD11B2.1	HSD11B2	<0.0001 0.47
TUSC2.1	TUSC2	0.0295	0.83	Hepsin.1	HPN	0.0031 0.79
TSPAN7.2	TSPAN7	<0.0001	0.35	HPCAL1.1	HPCAL1	0.0004 0.75
TSC2.1	TSC2	<0.0001	0.66	HNRNPAB.3	HNRNPAB	0.0039 0.78
TSC1.1	TSC1	<0.0001	0.56	HMGB1.1	HMGB1	<0.0001 0.46
P53.2	TP53	<0.0001	0.67	HLA-DPB1.1	HLA-DPB1	<0.0001 0.55
TOP2B.2	TOP2B	<0.0001	0.69	HIF1AN.1	HIF1AN	<0.0001 0.57
TNIP2.1	TNIP2	0.0359	0.85	HIF1A.3	HIF1A	0.0076 0.80
TNFSF12.1	TNFSF12	<0.0001	0.50	HGF.4	HGF	0.0067 0.80
TRAIL.1	TNFSF10	0.0011	0.77	HDAC1.1	HDAC1	<0.0001 0.61
TNFRSF11B.1	TNFRSF11B	<0.0001	0.63	HAVCR1.1	HAVCR1	0.0001 0.73
TNFRSF10D.1	TNFRSF10D	<0.0001	0.57	HADH.1	HADH	<0.0001 0.62
DR5.2	TNFRSF10B	<0.0001	0.64	GSTT1.3	GSTT1	0.0112 0.82
TNFAIP6.1	TNFAIP6	0.0001	0.74	GSTp.3	GSTP1	<0.0001 0.64
TNF.1	TNF	0.0138	0.80	GSTM3.2	GSTM3	<0.0001 0.57
TMEM47.1	TMEM47	<0.0001	0.41	GSTM1.1	GSTM1	<0.0001 0.55
TMEM27.1	TMEM27	<0.0001	0.53	GRB7.2	GRB7	<0.0001 0.66
TLR3.1	TLR3	<0.0001	0.68	GRB14.1	GRB14	0.0123 0.81
TIMP3.3	TIMP3	<0.0001	0.39	GPX3.1	GPX3	0.0120 0.82
TIMP2.1	TIMP2	<0.0001	0.68	GNAS.1	GNAS	0.0003 0.74
THBS1.1	THBS1	<0.0001	0.65	GJA1.1	GJA1	0.0034 0.79
TGFBR2.3	TGFBR2	<0.0001	0.41	GFR1A.1	GFR1A	0.0164 0.82
TGFBR1.1	TGFBR1	<0.0001	0.59	GCLM.2	GCLM	0.0056 0.80
TGFB2.2	TGFB2	<0.0001	0.50	GCLC.3	GCLC	<0.0001 0.49
TGFB1.1	TGFB1	<0.0001	0.67	GBP2.2	GBP2	0.0388 0.84
TGFA.2	TGFA	<0.0001	0.63	GATM.1	GATM	<0.0001 0.59
TEK.1	TEK	<0.0001	0.34	GATA3.3	GATA3	0.0002 0.72
TCF4.1	TCF4	<0.0001	0.47	FOS.1	FOS	<0.0001 0.65
TAP1.1	TAP1	0.0017	0.77	FOLR1.1	FOLR1	<0.0001 0.65
TAGLN.1	TAGLN	0.0001	0.72	FLT4.1	FLT4	<0.0001 0.37
TACSTD2.1	TACSTD2	<0.0001	0.58	FLT3LG.1	FLT3LG	<0.0001 0.61
SUCLG1.1	SUCLG1	<0.0001	0.56	FLT1.1	FLT1	<0.0001 0.40
STK11.1	STK11	<0.0001	0.58	FILIP1.1	FILIP1	<0.0001 0.47
STAT5B.2	STAT5B	<0.0001	0.50	FIGF.1	FIGF	<0.0001 0.53
STAT5A.1	STAT5A	<0.0001	0.60	FHL1.1	FHL1	<0.0001 0.52
STAT3.1	STAT3	0.0001	0.72	FHIT.1	FHIT	<0.0001 0.54
STAT1.3	STAT1	0.0344	0.84	FH.1	FH	<0.0001 0.67
SPRY1.1	SPRY1	<0.0001	0.57	FGFR2 isoform 1.1	FGFR2	<0.0001 0.62
SPAST.1	SPAST	0.0142	0.82	FGFR1.3	FGFR1	<0.0001 0.63
SPARCL1.1	SPARCL1	<0.0001	0.59	FGF2.2	FGF2	<0.0001 0.58
SPARC.1	SPARC	<0.0001	0.69	FGF1.1	FGF1	<0.0001 0.66
SOD1.1	SOD1	0.0245	0.83	FDPS.1	FDPS	<0.0001 0.47
SNRK.1	SNRK	<0.0001	0.35	FBXW7.1	FBXW7	<0.0001 0.66
SNAI1.1	SNAI1	<0.0001	0.70	fas.1	FAS	<0.0001 0.66
MADH4.1	SMAD4	<0.0001	0.47	ESRRG.3	ESRRG	0.0001 0.72
MADH2.1	SMAD2	<0.0001	0.50	ERG.1	ERG	<0.0001 0.44
SLC9A1.1	SLC9A1	0.0207	0.82	ERCC1.2	ERCC1	<0.0001 0.60
SLC34A1.1	SLC34A1	<0.0001	0.63	ERBB4.3	ERBB4	0.0018 0.74
SLC22A6.1	SLC22A6	0.0010	0.76	ErbB3.1	ERBB3	0.0031 0.79
SKIL.1	SKIL	<0.0001	0.64	HER2.3	ERBB2	<0.0001 0.53
PTPNS1.1	SIRPA	0.0075	0.81	EPHB4.1	EPHB4	<0.0001 0.51
SHANK3.1	SHANK3	<0.0001	0.37	EPHA2.1	EPHA2	<0.0001 0.40
SGK.1	SGK1	<0.0001	0.55	EPAS1.1	EPAS1	<0.0001 0.38
FRP1.3	SFRP1	0.0156	0.82	ENPP2.1	ENPP2	0.0001 0.72
SEMA3F.3	SEMA3F	<0.0001	0.50	ENPEP.1	ENPEP	<0.0001 0.65
SELPLG.1	SELPLG	0.0022	0.78	CD105.1	ENG	<0.0001 0.38
SELENBP1.1	SELENBP1	0.0003	0.75	EMP1.1	EMP1	<0.0001 0.64
SDPR.1	SDPR	<0.0001	0.42	EMCN.1	EMCN	<0.0001 0.27
SDHA.1	SDHA	<0.0001	0.65	ELTD1.1	ELTD1	<0.0001 0.67
SCNN1A.2	SCNN1A	0.0024	0.77	EIF2C1.1	EIF2C1	<0.0001 0.51
SCN4B.1	SCN4B	<0.0001	0.50	EGR1.1	EGR1	<0.0001 0.59
S100A2.1	S100A2	0.0008	0.75	EGLN3.1	EGLN3	0.0002 0.75
KIAA1303	RPTOR	<0.0001	0.60	EGFR.2	EGFR	0.0072 0.81
raptor.1						
RPS6KB1.3	RPS6KB1	<0.0001	0.60	EGF.3	EGF	0.0051 0.77
RPS6KAI.1	RPS6KAI	0.0002	0.71	EFNB2.1	EFNB2	<0.0001 0.45
RPS23.1	RPS23	0.0002	0.73	EFNB1.2	EFNB1	<0.0001 0.55
ROCK2.1	ROCK2	<0.0001	0.52	EEF1A1.1	EEF1A1	<0.0001 0.55
ROCK1.1	ROCK1	<0.0001	0.37	EDNRB.1	EDNRB	<0.0001 0.44
RIPK1.1	RIPK1	<0.0001	0.55	EDN2.1	EDN2	0.0012 0.74
rhoC.1	RHOC	<0.0001	0.66	EDN1	EDN1	<0.0001 0.53
				endothelin.1		
RhoB.1	RHOB	<0.0001	0.57	EBAG9.1	EBAG9	0.0240 0.83

TABLE 4b-continued

Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤ 0.05)						
Stage 3 vs. 1				Stage 3 vs. 1		
Gene	Official Symbol	p-value	OR	Gene	Official Symbol	p-value OR
ARHA.1	RHOA	<0.0001	0.50	DUSP1.1	DUSP1	0.0130 0.82
RHEB.2	RHEB	<0.0001	0.61	DPYS.1	DPYS	0.0355 0.85
RGS5.1	RGS5	<0.0001	0.37	DPEP1.1	DPEP1	<0.0001 0.66
FLJ22655.1	RERGL	<0.0001	0.33	DLL4.1	DLL4	<0.0001 0.66
RB1.1	RB1	<0.0001	0.61	DLC1.1	DLC1	<0.0001 0.42
RASSF1.1	RASSF1	0.0002	0.75	DKFZP564O0823.1	DKFZP564O0823	<0.0001 0.51
RARB.2	RARB	<0.0001	0.37	DICER1.2	DICER1	<0.0001 0.50
RALBP1.1	RALBP1	<0.0001	0.43	DIAPH1.1	DIAPH1	0.0219 0.83
RAF1.3	RAF1	<0.0001	0.59	DIABLO.1	DIABLO	0.0022 0.78
RAC1.3	RAC1	0.0356	0.84	DHPS.3	DHPS	<0.0001 0.55
PTPRG.1	PTPRG	<0.0001	0.35	DET1.1	DET1	0.0005 0.74
PTPRB.1	PTPRB	<0.0001	0.31	DEFB1.1	DEFB1	0.0002 0.73
PTN.1	PTN	<0.0001	0.56	DDC.1	DDC	<0.0001 0.72
PTK2.1	PTK2	<0.0001	0.45	DAPK1.3	DAPK1	<0.0001 0.42
PTH1R.1	PTH1R	<0.0001	0.45	CYR61.1	CYR61	<0.0001 0.59
PTEN.2	PTEN	<0.0001	0.51	CXCL12.1	CXCL12	<0.0001 0.62
PSMB9.1	PSMB9	0.0139	0.82	CX3CR1.1	CX3CR1	<0.0001 0.46
PSMB8.1	PSMB8	0.0243	0.83	CX3CL1.1	CX3CL1	<0.0001 0.47
PSMA7.1	PSMA7	0.0101	0.81	CUL1.1	CUL1	<0.0001 0.64
PRSS8.1	PRSS8	0.0025	0.78	CUBN.1	CUBN	<0.0001 0.55
PRPS2.1	PRPS2	0.0190	0.82	CTSS.1	CTSS	0.0007 0.76
PRKCH.1	PRKCH	<0.0001	0.48	CTSH.2	CTSH	<0.0001 0.64
PRKCD.2	PRKCD	0.0001	0.72	B-Catenin.3	CTNNB1	<0.0001 0.54
PPP2CA.1	PPP2CA	<0.0001	0.58	A-Catenin.2	CTNNA1	<0.0001 0.65
PPARG.3	PPARG	<0.0001	0.40	CTGF.1	CTGF	<0.0001 0.71
PPAP2B.1	PPAP2B	<0.0001	0.31	CSF2RA.2	CSF2RA	0.0037 0.78
PLG.1	PLG	<0.0001	0.53	CSF1R.2	CSF1R	<0.0001 0.67
PLAT.1	PLAT	<0.0001	0.54	CSF1.1	CSF1	<0.0001 0.65
PLA2G4C.1	PLA2G4C	<0.0001	0.65	CRADD.1	CRADD	0.0032 0.79
PIK3CA.1	PIK3CA	<0.0001	0.53	COL4A2.1	COL4A2	<0.0001 0.65
PI3K.2	PIK3C2B	<0.0001	0.40	COL4A1.1	COL4A1	0.0067 0.80
PFKP.1	PFKP	0.0124	0.82	COL18A1.1	COL18A1	0.0001 0.72
CD31.3	PECAM1	<0.0001	0.42	CLU.3	CLU	0.0004 0.75
PDZK3.1	PDZK3	0.0003	0.72	CLDN10.1	CLDN10	<0.0001 0.65
PDZK1.1	PDZK1	<0.0001	0.58	CLCNKB.1	CLCNKB	<0.0001 0.52
PDGFRb.3	PDGFRB	<0.0001	0.62	CFLAR.1	CFLAR	<0.0001 0.60
PDGFD.2	PDGFD	<0.0001	0.43	CEACAM1.1	CEACAM1	<0.0001 0.55
PDGFC.3	PDGFC	<0.0001	0.51	p21.3	CDKN1A	0.0002 0.73
PDGFB.3	PDGFB	<0.0001	0.40	CDH6.1	CDH6	<0.0001 0.72
PDGFA.3	PDGFA	<0.0001	0.57	CDH5.1	CDH5	<0.0001 0.44
PCK1.1	PCK1	<0.0001	0.60	CDH2.1	CDH2	0.0392 0.85
PCCA.1	PCCA	<0.0001	0.58	CDH16.1	CDH16	<0.0001 0.70
PARD6A.1	PARD6A	0.0042	0.79	CDH13.1	CDH13	<0.0001 0.53
Pak1.2	PAK1	<0.0001	0.69	CDC25B.1	CDC25B	0.0037 0.79
PAH.1	PAH	0.0309	0.84	CD4.1	CD4	<0.0001 0.72
OGG1.1	OGG1	0.0024	0.78	CD36.1	CD36	<0.0001 0.51
BFGF.3	NUDT6	<0.0001	0.46	CD34.1	CD34	<0.0001 0.48
NPR1.1	NPR1	<0.0001	0.58	CD24.1	CD24	0.0206 0.83
NPM1.2	NPM1	<0.0001	0.65	CD14.1	CD14	0.0152 0.82
NOTCH3.1	NOTCH3	<0.0001	0.58	CCND1.3	CCND1	<0.0001 0.51
NOTCH2.1	NOTCH2	<0.0001	0.64	CCL4.2	CCL4	0.0017 0.77
NOTCH1.1	NOTCH1	<0.0001	0.44	MCP1.1	CCL2	<0.0001 0.66
NOS3.1	NOS3	<0.0001	0.44	CAT.1	CAT	<0.0001 0.53
NOS2A.3	NOS2	<0.0001	0.49	CASP10.1	CASP10	0.0001 0.72
NOL3.1	NOL3	0.0003	0.76	CALD1.2	CALD1	<0.0001 0.55
NFX1.1	NFX1	<0.0001	0.50	CACNA2D1.1	CACNA2D1	0.0352 0.84
NFKBp50.3	NFKB1	<0.0001	0.59	CA9.3	CA9	0.0299 0.84
NFATC2.1	NFATC2	<0.0001	0.64	CA2.1	CA2	<0.0001 0.58
NFAT5.1	NFAT5	<0.0001	0.65	C3AR1.1	C3AR1	0.0010 0.77
MYRIP.2	MYRIP	0.0004	0.72	ECRG4.1	C2orf40	<0.0001 0.47
MYH11.1	MYH11	<0.0001	0.50	C1QA.1	C1QA	0.0119 0.82
eMYC.3	MYC	<0.0001	0.70	C13orf15.1	C13orf15	<0.0001 0.37
MX1.1	MX1	0.0103	0.81	BUB3.1	BUB3	<0.0001 0.67
MVP.1	MVP	0.0002	0.74	BTRC.1	BTRC	<0.0001 0.65
MUC1.2	MUC1	0.0005	0.75	CIAP1.2	BIRC2	<0.0001 0.64
FRAP1.1	MTOR	<0.0001	0.61	BIN1.3	BIN1	0.0001 0.73
MSH3.2	MSH3	<0.0001	0.60	BGN.1	BGN	<0.0001 0.63
MSH2.3	MSH2	<0.0001	0.53	Bclx.2	BCL2L1	<0.0001 0.58
STMY3.3	MMP11	0.0034	0.79	Bcl2.2	BCL2	<0.0001 0.37
GBL1.1	MLST8	0.0011	0.77	Bax.1	BAX	0.0035 0.78
MIF.2	MIF	0.0008	0.76	Bak.2	BAK1	0.0215 0.83
MICA.1	MICA	<0.0001	0.70	BAG1.2	BAG1	<0.0001 0.40
MGMT.1	MGMT	<0.0001	0.60	ATP6V1B1.1	ATP6V1B1	<0.0001 0.54

TABLE 4b-continued

Proxy genes for which increased expression is associated with higher tumor stage (p-value ≤ 0.05)						
Stage 3 vs. 1				Stage 3 vs. 1		
Gene	Official Symbol	p-value	OR	Gene	Official Symbol	p-value OR
MCM3.3	MCM3	<0.0001	0.68	ATP1A1.1	ATP1A1	0.0037 0.78
MCAM.1	MCAM	<0.0001	0.52	ASS1.1	ASS1	<0.0001 0.60
MARCKS.1	MARCKS	0.0001	0.73	ARRB1.1	ARRB1	<0.0001 0.45
ERK1.3	MAPK3	<0.0001	0.48	ARHGDIB.1	ARHGDIB	<0.0001 0.50
ERK2.3	MAPK1	0.0221	0.83	AQP1.1	AQP1	<0.0001 0.40
MAP4.1	MAP4	<0.0001	0.63	APOLD1.1	APOLD1	<0.0001 0.54
MAP2K3.1	MAP2K3	<0.0001	0.59	APC.4	APC	<0.0001 0.57
MAP2K1.1	MAP2K1	0.0002	0.74	ANXA4.1	ANXA4	0.0003 0.74
MAL2.1	MAL2	<0.0001	0.64	ANXA1.2	ANXA1	0.0001 0.73
MAL.1	MAL	<0.0001	0.49	ANTXR1.1	ANTXR1	0.0051 0.80
LYZ.1	LYZ	0.0318	0.84	ANGPTL4.1	ANGPTL4	0.0041 0.80
LTF.1	LTF	0.0131	0.80	ANGPTL3.3	ANGPTL3	0.0258 0.82
LRP2.1	LRP2	<0.0001	0.63	ANGPTL2.1	ANGPTL2	0.0015 0.77
LMO2.1	LMO2	<0.0001	0.56	ANGPT2.1	ANGPT2	<0.0001 0.72
LDB2.1	LDB2	<0.0001	0.41	ANGPT1.1	ANGPT1	<0.0001 0.45
LDB1.2	LDB1	<0.0001	0.54	ALDOB.1	ALDOB	<0.0001 0.60
LAMA4.1	LAMA4	0.0004	0.75	ALDH6A1.1	ALDH6A1	<0.0001 0.55
KRT7.1	KRT7	<0.0001	0.60	ALDH4.2	ALDH4A1	0.0124 0.82
K-ras.10	KRAS	<0.0001	0.68	AKT3.2	AKT3	<0.0001 0.43
KL.1	KL	<0.0001	0.49	AKT2.3	AKT2	<0.0001 0.64
Kitlmg.4	KITLG	<0.0001	0.43	AKT1.3	AKT1	<0.0001 0.58
c-kit.2	KIT	<0.0001	0.60	AIF1.1	AIF1	0.0002 0.74
KDR.6	KDR	<0.0001	0.36	AHR.1	AHR	<0.0001 0.59
KCNJ15.1	KCNJ15	<0.0001	0.54	AGTR1.1	AGTR1	<0.0001 0.36
HTATIP.1	KAT5	<0.0001	0.40	ADH1B.1	ADH1B	0.0015 0.77
G-Catenin.1	JUP	<0.0001	0.42	ADD1.1	ADD1	<0.0001 0.40
AP-1 (JUN official).2	JUN	0.0001	0.73	ADAMTS5.1	ADAMTS5	0.0006 0.73
JAG1.1	JAG1	<0.0001	0.42	ADAM17.1	ADAM17	<0.0001 0.72
ITGB5.1	ITGB5	0.0115	0.81	ACE2.1	ACE2	<0.0001 0.61
ITGB1.1	ITGB1	<0.0001	0.64	ACADSB.1	ACADSB	<0.0001 0.54
ITGA7.1	ITGA7	<0.0001	0.54	BCRP.1	ABCG2	<0.0001 0.41
ITGA6.2	ITGA6	<0.0001	0.51	MRP4.2	ABCC4	<0.0001 0.65
ITGA5.1	ITGA5	0.0325	0.84	MRP3.1	ABCC3	0.0005 0.76
ITGA4.2	ITGA4	<0.0001	0.54	MRP1.1	ABCC1	0.0017 0.78
ITGA3.2	ITGA3	<0.0001	0.61	ABCB1.5	ABCB1	0.0003 0.75
IQGAP2.1	IQGAP2	<0.0001	0.63	NPD009 (ABAT official).3	ABAT	<0.0001 0.70
INSR.1	INSR	<0.0001	0.59	AAMP.1	AAMP	0.0292 0.84
IMP3.1	IMP3	<0.0001	0.54	A2M.1	A2M	<0.0001 0.36

TABLE 5a

Proxy genes for which increased expression is associated with higher tumor grade (p-value ≤ 0.05)							
Official		CCF Grade		Official		CCF Grade	
Gene	Symbol	p-value	OR	Gene	Symbol	p-value	OR
WT1.1	WT1	<0.0001	1.39	HPD.1	HPD	0.0054	1.20
VTN.1	VTN	0.0359	1.16	HIST1H1D.1	HIST1H1D	0.0248	1.16
VDR.2	VDR	0.0001	1.29	HGD.1	HGD	<0.0001	1.41
UBE2T.1	UBE2T	<0.0001	1.81	GZMA.1	GZMA	0.0006	1.26
TP.3	TYMP	<0.0001	1.35	GPX2.2	GPX2	0.0182	1.18
C20 orf1.1	TPX2	<0.0001	2.14	GPX1.2	GPX1	0.0008	1.25
TOP2A.4	TOP2A	<0.0001	2.07	FCGR3A.1	FCGR3A	0.0003	1.27
TNFSF13B.1	TNFSF13B	0.0062	1.20	fasl.2	FASLG	0.0045	1.21
TK1.2	TK1	<0.0001	1.66	FABP1.1	FABP1	<0.0001	1.32
TGFB1.1	TGFB1	0.0452	1.14	F2.1	F2	<0.0001	1.77
STAT1.3	STAT1	<0.0001	1.31	ESPL1.3	ESPL1	<0.0001	1.60
SQSTM1.1	SQSTM1	0.0003	1.27	E2F1.3	E2F1	<0.0001	1.36
OPN, osteopontin.3	SPP1	0.0002	1.29	CXCR6.1	CXCR6	<0.0001	1.50
SLC7A5.2	SLC7A5	0.0002	1.28	BLR1.1	CXCR5	0.0338	1.15
SLC16A3.1	SLC16A3	0.0052	1.21	CXCL9.1	CXCL9	0.0001	1.29
SLC13A3.1	SLC13A3	0.0003	1.27	CXCL10.1	CXCL10	0.0078	1.19
SFN.1	SFN	0.0066	1.20	GRO1.2	CXCL1	<0.0001	1.39
SEMA3C.1	SEMA3C	<0.0001	1.32	CTSD.2	CTSD	0.0183	1.17
SAA2.2	SAA2	<0.0001	2.13	CTSB.1	CTSB	0.0006	1.26
				CRP.1	CRP	0.0342	1.15

TABLE 5a-continued

Proxy genes for which increased expression is associated with higher tumor grade (p-value $\leq .05$)							
Gene	Official Symbol	CCF Grade		Gene	Official Symbol	CCF Grade	
		p-value	OR			p-value	OR
S100A1.1	S100A1	<0.0001	1.40	CP.1	CP	<0.0001	1.37
RRM2.1	RRM2	<0.0001	1.71	Chk2.3	CHEK2	<0.0001	1.37
RPLP1.1	RPLP1	0.0007	1.25	Chk1.2	CHEK1	<0.0001	1.37
RAD51.1	RAD51	<0.0001	1.53	CENPF.1	CENPF	<0.0001	1.78
PTTG1.2	PTTG1	<0.0001	1.89	CD8A.1	CD8A	<0.0001	1.35
PSMB9.1	PSMB9	0.0010	1.25	CD82.3	CD82	<0.0001	1.50
PSMB8.1	PSMB8	0.0181	1.17	TNFSF7.1	CD70	<0.0001	1.43
PRKCB1.1	PRKCB	0.0218	1.16	CCNE1.1	CCNE1	0.0002	1.29
PDCD1.1	PDCD1	<0.0001	1.43	CCNB1.2	CCNB1	<0.0001	1.75
PCSK6.1	PCSK6	0.0009	1.25	CCL5.2	CCL5	<0.0001	1.63
PCNA.2	PCNA	0.0041	1.21	CCL20.1	CCL20	0.0082	1.19
NME2.1	NME2	0.0106	1.19	CAV2.1	CAV2	0.0210	1.17
MYBL2.1	MYBL2	<0.0001	1.70	CA12.1	CA12	<0.0001	1.41
MMP9.1	MMP9	<0.0001	1.46	C3.1	C3	<0.0001	1.34
MMP14.1	MMP14	0.0003	1.28	C1QB.1	C1QB	0.0201	1.17
Ki-67.2	MKI67	<0.0001	1.70	BUB1.1	BUB1	<0.0001	2.16
mGST1.2	MGST1	<0.0001	2.13	BRCA1.2	BRCA1	0.0004	1.26
cMet.2	MET	<0.0001	1.57	SURV.2	BIRC5	<0.0001	1.93
MDK.1	MDK	<0.0001	1.55	cIAP2.2	BIRC3	<0.0001	1.44
MDH2.1	MDH2	<0.0001	1.35	STK15.2	AURKA	<0.0001	1.38
MCM2.2	MCM2	<0.0001	1.33	ATP5E.1	ATP5E	<0.0001	1.45
LOX.1	LOX	<0.0001	1.35	APOL1.1	APOL1	<0.0001	1.47
LMNB1.1	LMNB1	<0.0001	1.76	APOE.1	APOE	<0.0001	1.40
LIMK1.1	LIMK1	<0.0001	1.43	APOC1.3	APOC1	<0.0001	1.42
LAPTM5.1	LAPTM5	0.0044	1.21	ANXA2.2	ANXA2	0.0020	1.23
LAMB3.1	LAMB3	<0.0001	1.49	ANGPTL3.3	ANGPTL3	<0.0001	1.32
L1CAM.1	L1CAM	0.0338	1.15	AMACR1.1	AMACR	0.0382	1.15
KLRK1.2	KLRK1	0.0412	1.15	ALOX5.1	ALOX5	0.0001	1.29
CD18.2	ITGB2	0.0069	1.21	ALDH4.2	ALDH4A1	0.0002	1.28
IL-8.1	IL8	0.0088	1.19	ADAM8.1	ADAM8	0.0006	1.26
IL6.3	IL6	0.0091	1.19	MRP2.3	ABCC2	<0.0001	1.97
ICAM1.1	ICAM1	0.0014	1.24				
HSPA8.1	HSPA8	<0.0001	1.39				

TABLE 5b

Proxy genes for which increased expression is associated with lower tumor grade (p-value $\leq .05$)							
Gene	Official Symbol	CCF Grade		Gene	Official Symbol	CCF Grade	
		p-value	OR			p-value	OR
ZHX2.1	ZHX2	0.0061	0.83	ITGA4.2	ITGA4	<0.0001	0.75
YB-1.2	YBX1	<0.0001	0.62	ITGA3.2	ITGA3	0.0128	0.85
XPNPEP2.2	XPNPEP2	0.0040	0.82	IQGAP2.1	IQGAP2	<0.0001	0.68
XIAP.1	XIAP	<0.0001	0.64	INSR.1	INSR	<0.0001	0.46
WISP1.1	WISP1	<0.0001	0.58	INHBA.1	INHBA	0.0049	0.83
VWF.1	VWF	<0.0001	0.34	IMP3.1	IMP3	<0.0001	0.67
VHL.1	VHL	0.0088	0.84	IL6ST.3	IL6ST	<0.0001	0.43
VEGF.1	VEGFA	<0.0001	0.48	IL1B.1	IL1B	0.0102	0.84
VCAN.1	VCAN	0.0023	0.82	IL15.1	IL15	0.0001	0.76
VCAM1.1	VCAM1	0.0049	0.83	IL10.3	IL10	0.0239	0.86
USP34.1	USP34	<0.0001	0.62	IGFBP6.1	IGFBP6	<0.0001	0.75
UMOD.1	UMOD	<0.0001	0.69	IGFBP5.1	IGFBP5	<0.0001	0.74
UGCG.1	UGCG	<0.0001	0.58	IGFBP2.1	IGFBP2	<0.0001	0.76
UBB.1	UBB	<0.0001	0.62	IGF2.2	IGF2	<0.0001	0.69
UBE1C.1	UBA3	<0.0001	0.67	IGF1R.3	IGF1R	<0.0001	0.43
tusc4.2	TUSC4	<0.0001	0.61	ID3.1	ID3	<0.0001	0.48
TUSC2.1	TUSC2	0.0481	0.88	ID2.4	ID2	<0.0001	0.64
TSPAN7.2	TSPAN7	<0.0001	0.29	ID1.1	ID1	<0.0001	0.37
TSC2.1	TSC2	<0.0001	0.60	ICAM2.1	ICAM2	<0.0001	0.42
TSC1.1	TSC1	<0.0001	0.52	HYAL2.1	HYAL2	<0.0001	0.32
P53.2	TP53	<0.0001	0.66	HYAL1.1	HYAL1	<0.0001	0.52
TOP2B.2	TOP2B	<0.0001	0.68	HSPG2.1	HSPG2	<0.0001	0.33
TNIP2.1	TNIP2	0.0001	0.76	HSPA1A.1	HSPA1A	0.0022	0.81
TNFSF12.1	TNFSF12	<0.0001	0.54	HSP90AB1.1	HSP90AB1	<0.0001	0.68
TNFRSF11B.1	TNFRSF11B	<0.0001	0.73	HSD11B2.1	HSD11B2	<0.0001	0.43
TNFRSF10D.1	TNFRSF10D	<0.0001	0.51	HPCAL1.1	HPCAL1	<0.0001	0.69
TNFRSF10C.3	TNFRSF10C	0.0003	0.78	HNRNPAB.3	HNRNPAB	0.0432	0.87
DR5.2	TNFRSF10B	<0.0001	0.69	HMGB1.1	HMGB1	<0.0001	0.47

TABLE 5b-continued

Proxy genes for which increased expression is associated with lower tumor grade (p-value ≤ 0.05)							
Gene	Official Symbol	CCF Grade		Gene	Official Symbol	CCF Grade	
		p-value	OR			p-value	OR
TNFAIP6.1	TNFAIP6	0.0338	0.87	HIF1AN.1	HIF1AN	<0.0001	0.64
TNFAIP3.1	TNFAIP3	0.0083	0.84	HIF1A.3	HIF1A	<0.0001	0.55
TNF.1	TNF	0.0392	0.87	HGF.4	HGF	0.0022	0.81
TMEM47.1	TMEM47	<0.0001	0.33	HDAC1.1	HDAC1	<0.0001	0.48
TMEM27.1	TMEM27	<0.0001	0.73	HADH.1	HADH	<0.0001	0.63
TIMP3.3	TIMP3	<0.0001	0.29	GSTp.3	GSTP1	<0.0001	0.66
TIMP2.1	TIMP2	<0.0001	0.54	GSTM3.2	GSTM3	<0.0001	0.53
THBS1.1	THBS1	<0.0001	0.58	GSTM1.1	GSTM1	<0.0001	0.61
THBD.1	THBD	<0.0001	0.66	GRB7.2	GRB7	<0.0001	0.73
TGFBR2.3	TGFBR2	<0.0001	0.32	GRB14.1	GRB14	0.0001	0.76
TGFBR1.1	TGFBR1	<0.0001	0.59	GPC3.1	GPC3	0.0012	0.80
TGFB2.2	TGFB2	<0.0001	0.54	GNAS.1	GNAS	<0.0001	0.70
TGFB1.1	TGFB1	<0.0001	0.66	GMNN.1	GMNN	0.0006	0.80
TGFA.2	TGFA	<0.0001	0.68	GJA1.1	GJA1	<0.0001	0.66
TEK.1	TEK	<0.0001	0.34	GCLM.2	GCLM	0.0319	0.87
cripto (TDGF1 official).1	TDGF1	0.0328	0.86	GCLC.3	GCLC	<0.0001	0.57
TCF4.1	TCF4	<0.0001	0.33	GATM.1	GATM	0.0006	0.79
TAGLN.1	TAGLN	<0.0001	0.58	GATA3.3	GATA3	0.0029	0.82
TACSTD2.1	TACSTD2	<0.0001	0.61	GAS2.1	GAS2	0.0136	0.84
SUCLG1.1	SUCLG1	<0.0001	0.76	GADD45B.1	GADD45B	<0.0001	0.54
STK11.1	STK11	<0.0001	0.53	FST.1	FST	0.0197	0.85
STC2.1	STC2	0.0085	0.84	FOS.1	FOS	<0.0001	0.44
STAT5B.2	STAT5B	<0.0001	0.40	FOLR1.1	FOLR1	<0.0001	0.75
STAT5A.1	STAT5A	<0.0001	0.60	FLT4.1	FLT4	<0.0001	0.34
STAT3.1	STAT3	<0.0001	0.48	FLT3LG.1	FLT3LG	<0.0001	0.73
SPRY1.1	SPRY1	<0.0001	0.40	FLT1.1	FLT1	<0.0001	0.29
SPAST.1	SPAST	0.0002	0.78	FILIP1.1	FILIP1	<0.0001	0.47
SPARCL1.1	SPARCL1	<0.0001	0.41	FIGF.1	FIGF	<0.0001	0.70
SPARC.1	SPARC	<0.0001	0.50	FHL1.1	FHL1	<0.0001	0.54
SNRK.1	SNRK	<0.0001	0.29	FHIT.1	FHIT	<0.0001	0.72
SNAI1.1	SNAI1	<0.0001	0.54	FH.1	FH	0.0203	0.86
MADH4.1	SMAD4	<0.0001	0.39	FGFR2 isoform	FGFR2	<0.0001	0.62
MADH2.1	SMAD2	<0.0001	0.40	FGFR1.3	FGFR1	<0.0001	0.48
SLC9A1.1	SLC9A1	<0.0001	0.75	FGF2.2	FGF2	<0.0001	0.62
SLC34A1.1	SLC34A1	0.0001	0.76	FGF1.1	FGF1	<0.0001	0.61
SKIL.1	SKIL	<0.0001	0.52	FDPS.1	FDPS	<0.0001	0.52
SHC1.1	SHC1	0.0063	0.83	FBXW7.1	FBXW7	<0.0001	0.57
SHANK3.1	SHANK3	<0.0001	0.27	FAP.1	FAP	0.0440	0.87
SGK.1	SGK1	<0.0001	0.60	ESRRG.3	ESRRG	0.0340	0.87
FRP1.3	SFRP1	0.0003	0.78	ERG.1	ERG	<0.0001	0.36
PAI1.3	SERPINE1	0.0115	0.85	ERCC4.1	ERCC4	0.0337	0.87
SEMA3F.3	SEMA3F	<0.0001	0.45	ERCC1.2	ERCC1	<0.0001	0.59
SELENBP1.1	SELENBP1	<0.0001	0.63	ERBB4.3	ERBB4	<0.0001	0.66
SELE.1	SELE	<0.0001	0.74	ERBB2	ERBB2	<0.0001	0.57
SDPR.1	SDPR	<0.0001	0.35	EPHB4.1	EPHB4	<0.0001	0.43
SDHA.1	SDHA	<0.0001	0.69	EPHA2.1	EPHA2	<0.0001	0.44
SCNN1A.2	SCNN1A	0.0011	0.80	EPAS1.1	EPAS1	<0.0001	0.26
SCN4B.1	SCN4B	<0.0001	0.47	ENPP2.1	ENPP2	<0.0001	0.62
S100A2.1	S100A2	<0.0001	0.72	ENPEP.1	ENPEP	<0.0001	0.75
RUNX1.1	RUNX1	0.0001	0.77	ENO2.1	ENO2	0.0449	0.88
RRM1.2	RRM1	0.0438	0.87	CD105.1	ENG	<0.0001	0.30
KIAA1303	RPTOR	<0.0001	0.53	EMP1.1	EMP1	<0.0001	0.42
raptor.1				EMCN.1	EMCN	<0.0001	0.31
RPS6KB1.3	RPS6KB1	<0.0001	0.49	ELTD1.1	ELTD1	<0.0001	0.59
RPS23.1	RPS23	<0.0001	0.59	EIF2C1.1	EIF2C1	<0.0001	0.63
ROCK2.1	ROCK2	<0.0001	0.42	EGR1.1	EGR1	<0.0001	0.50
ROCK1.1	ROCK1	<0.0001	0.31	EGLN3.1	EGLN3	<0.0001	0.69
RIPK1.1	RIPK1	<0.0001	0.50	EGF.3	EGF	0.0200	0.85
rhoC.1	RHOC	<0.0001	0.70	EFNB2.1	EFNB2	<0.0001	0.36
RhoB.1	RHOB	<0.0001	0.36	EFNB1.2	EFNB1	<0.0001	0.46
ARHA.1	RHOA	<0.0001	0.34	EEF1A1.1	EEF1A1	<0.0001	0.39
RHEB.2	RHEB	<0.0001	0.72	EDNRB.1	EDNRB	<0.0001	0.33
RGS5.1	RGS5	<0.0001	0.30	EDN2.1	EDN2	<0.0001	0.67
FLJ22655.1	RERGL	<0.0001	0.42	EDN1	EDN1	<0.0001	0.41
NFKBp65.3	RELA	<0.0001	0.69	endothelin.1			
RB1.1	RB1	<0.0001	0.74	EBAG9.1	EBAG9	0.0057	0.83
RASSF1.1	RASSF1	<0.0001	0.60	DUSP1.1	DUSP1	<0.0001	0.54
RARB.2	RARB	<0.0001	0.41	DPEP1.1	DPEP1	0.0001	0.76
RALBP1.1	RALBP1	<0.0001	0.47	DLL4.1	DLL4	<0.0001	0.49
RAF1.3	RAF1	<0.0001	0.48	DLC1.1	DLC1	<0.0001	0.33
RAC1.3	RAC1	<0.0001	0.75	DKFZP564O0823.1	DKFZP564O0823	<0.0001	0.46
PXDN.1	PXDN	<0.0001	0.74	DICER1.2	DICER1	<0.0001	0.41
PTPRG.1	PTPRG	<0.0001	0.29	DIAPH1.1	DIAPH1	0.0022	0.81

TABLE 5b-continued

Proxy genes for which increased expression is associated with lower tumor grade (p-value ≤ 0.05)							
Gene	Official Symbol	CCF Grade		Gene	Official Symbol	CCF Grade	
		p-value	OR			p-value	OR
PTPRB.1	PTPRB	<0.0001	0.27	DIABLO.1	DIABLO	<0.0001	0.73
PTN.1	PTN	<0.0001	0.58	DHPS.3	DHPS	<0.0001	0.41
PTK2.1	PTK2	<0.0001	0.36	DET1.1	DET1	<0.0001	0.63
PTH1.1	PTH1R	<0.0001	0.50	DEFB1	DEFB1	0.0001	0.77
PTEN.2	PTEN	<0.0001	0.40	DAPK1.3	DAPK1	<0.0001	0.48
PSMA7.1	PSMA7	0.0002	0.78	DAG1.1	DAG1	0.0150	0.85
PRPS2.1	PRPS2	0.0012	0.80	CYR61.1	CYR61	<0.0001	0.49
PROM2.1	PROM2	0.0326	0.87	CXCL12.1	CXCL12	<0.0001	0.64
PRKCH.1	PRKCH	<0.0001	0.45	CX3CR1.1	CX3CR1	0.0008	0.80
PRKCD.2	PRKCD	<0.0001	0.76	CX3CL1.1	CX3CL1	<0.0001	0.55
PPP2CA.1	PPP2CA	<0.0001	0.68	CUL1.1	CUL1	<0.0001	0.62
PPARG.3	PPARG	<0.0001	0.42	CUBN.1	CUBN	0.0081	0.84
PPAP2B.1	PPAP2B	<0.0001	0.32	CTSL.2	CTSL	0.0328	0.87
PMP22.1	PMP22	<0.0001	0.61	B-Catenin.3	CTNNB1	<0.0001	0.36
PLG.1	PLG	<0.0001	0.75	A-Catenin.2	CTNNA1	<0.0001	0.73
PLAT.1	PLAT	<0.0001	0.42	CTGF.1	CTGF	<0.0001	0.61
PLA2G4C.1	PLA2G4C	0.0002	0.78	CSF1.1	CSF1	<0.0001	0.66
PIK3CA.1	PIK3CA	<0.0001	0.48	CRADD.1	CRADD	0.0151	0.85
PI3K.2	PIK3C2B	<0.0001	0.53	COL5A2.2	COL5A2	0.0155	0.85
PGF.1	PGF	<0.0001	0.74	COL4A2.1	COL4A2	<0.0001	0.51
PFKP.1	PFKP	0.0008	0.80	COL4A1.1	COL4A1	<0.0001	0.59
CD31.3	PECAM1	<0.0001	0.32	COL1A2.1	COL1A2	0.0005	0.79
PDZK3.1	PDZK3	<0.0001	0.72	COL18A1.1	COL18A1	<0.0001	0.53
PDZK1.1	PDZK1	<0.0001	0.76	CLDN10.1	CLDN10	0.0351	0.87
PDGFRb.3	PDGFRB	<0.0001	0.42	CLCNKB.1	CLCNKB	<0.0001	0.73
PDGFRa.2	PDGFRA	0.0022	0.82	CFLAR.1	CFLAR	<0.0001	0.45
PDGFD.2	PDGFD	<0.0001	0.39	CEACAM1.1	CEACAM1	<0.0001	0.64
PDGFC.3	PDGFC	<0.0001	0.57	p27.3	CDKN1B	<0.0001	0.65
PDGFB.3	PDGFB	<0.0001	0.33	p21.3	CDKN1A	<0.0001	0.52
PDGFA.3	PDGFA	<0.0001	0.42	CDK4.1	CDK4	0.0200	0.86
PCK1.1	PCK1	<0.0001	0.71	CDH5.1	CDH5	<0.0001	0.31
PCCA.1	PCCA	<0.0001	0.75	CDH16.1	CDH16	0.0005	0.79
PARD6A.1	PARD6A	0.0088	0.84	CDH13.1	CDH13	<0.0001	0.40
Pak1.2	PAK1	0.0014	0.81	CD99.1	CD99	0.0001	0.77
PAH.1	PAH	0.0160	0.85	CD44.1	CD44_1	0.0216	0.86
OGG1.1	OGG1	0.0139	0.85	CD36.1	CD36	<0.0001	0.42
BFGF.3	NUDT6	<0.0001	0.63	CD34.1	CD34	<0.0001	0.39
NPR1.1	NPR1	<0.0001	0.42	CD14.1	CD14	0.0021	0.81
NPM1.2	NPM1	<0.0001	0.61	CCND1.3	CCND1	<0.0001	0.63
NOTCH3.1	NOTCH3	<0.0001	0.40	MCP1.1	CCL2	<0.0001	0.69
NOTCH2.1	NOTCH2	<0.0001	0.58	CAT.1	CAT	0.0014	0.81
NOTCH1.1	NOTCH1	<0.0001	0.38	CASP10.1	CASP10	<0.0001	0.65
NOS3.1	NOS3	<0.0001	0.42	CALD1.2	CALD1	<0.0001	0.44
NOS2A.3	NOS2	<0.0001	0.56	CACNA2D1.1	CACNA2D1	0.0003	0.78
NOL3.1	NOL3	<0.0001	0.61	CA9.3	CA9	0.0269	0.86
NFX1.1	NFX1	<0.0001	0.53	CA2.1	CA2	0.0001	0.77
NFKBp50.3	NFKB1	<0.0001	0.59	C7.1	C7	<0.0001	0.71
NFATC2.1	NFATC2	<0.0001	0.73	C3AR1.1	C3AR1	0.0032	0.82
NFAT5.1	NFAT5	<0.0001	0.58	ECRG4.1	C2orf40	<0.0001	0.50
MYRIP.2	MYRIP	<0.0001	0.63	C13orf15.1	C13orf15	<0.0001	0.37
MYH11.1	MYH11	<0.0001	0.48	BUB3.1	BUB3	<0.0001	0.65
cMYC.3	MYC	<0.0001	0.68	BTRC.1	BTRC	<0.0001	0.61
MX1.1	MX1	0.0087	0.84	BNIP3.1	BNIP3	0.0018	0.81
MVP.1	MVP	0.0291	0.87	CIAP1.2	BIRC2	<0.0001	0.61
MUC1.2	MUC1	0.0043	0.83	BGN.1	BGN	<0.0001	0.43
FRAP1.1	MTOR	<0.0001	0.61	Bclx.2	BCL2L1	<0.0001	0.76
MT1X.1	MT1X	0.0276	0.86	Bcl2.2	BCL2	<0.0001	0.45
MSH3.2	MSH3	0.0001	0.76	BAG1.2	BAG1	<0.0001	0.47
MSH2.3	MSH2	<0.0001	0.60	AXL.1	AXL	<0.0001	0.74
MMP2.2	MMP2	<0.0001	0.74	ATP6V1B1.1	ATP6V1B1	<0.0001	0.65
STMY3.3	MMP11	<0.0001	0.70	ASS1.1	ASS1	<0.0001	0.67
MIF.2	MIF	0.0004	0.79	ARRB1.1	ARRB1	<0.0001	0.49
MICA.1	MICA	<0.0001	0.60	ARHGDIB.1	ARHGDIB	<0.0001	0.50
MGMT.1	MGMT	<0.0001	0.57	ARF1.1	ARF1	<0.0001	0.69
MCM3.3	MCM3	<0.0001	0.73	AREG.2	AREG	0.0007	0.80
MCAM.1	MCAM	<0.0001	0.42	AQP1.1	AQP1	<0.0001	0.49
MARCKS.1	MARCKS	<0.0001	0.63	APOLD1.1	APOLD1	<0.0001	0.40
ERK1.3	MAPK3	<0.0001	0.35	APC.4	APC	<0.0001	0.62
ERK2.3	MAPK1	<0.0001	0.71	APAF1.2	APAF1	<0.0001	0.76
MAP4.1	MAP4	<0.0001	0.58	ANXA1.2	ANXA1	<0.0001	0.65
MAP2K3.1	MAP2K3	<0.0001	0.60	ANTXR1.1	ANTXR1	<0.0001	0.61
MAP2K1.1	MAP2K1	<0.0001	0.62	ANGPTL4.1	ANGPTL4	<0.0001	0.75
MAL.1	MAL	<0.0001	0.63	ANGPTL2.1	ANGPTL2	<0.0001	0.60
LRP2.1	LRP2	0.0275	0.86	ANGPT2.1	ANGPT2	<0.0001	0.61

TABLE 5b-continued

Proxy genes for which increased expression is associated with lower tumor grade (p-value ≤ 0.05)							
Official		CCF Grade		Official		CCF Grade	
Gene	Symbol	p-value	OR	Gene	Symbol	p-value	OR
LMO2.1	LMO2	<0.0001	0.68	ANGPT1.1	ANGPT1	<0.0001	0.33
LDB2.1	LDB2	<0.0001	0.28	ALDOB.1	ALDOB	0.0348	0.87
LDB1.2	LDB1	<0.0001	0.52	ALDH6A1.1	ALDH6A1	<0.0001	0.61
LAMB1.1	LAMB1	<0.0001	0.73	AKT3.2	AKT3	<0.0001	0.30
LAMA4.1	LAMA4	<0.0001	0.55	AKT2.3	AKT2	<0.0001	0.65
KRT7.1	KRT7	<0.0001	0.61	AKT1.3	AKT1	<0.0001	0.50
K-ras.10	KRAS	<0.0001	0.54	AHR.1	AHR	<0.0001	0.55
KL.1	KL	<0.0001	0.62	AGTR1.1	AGTR1	<0.0001	0.44
Kitlng.4	KITLG	<0.0001	0.46	ADH1B.1	ADH1B	<0.0001	0.70
c-kit.2	KIT	<0.0001	0.48	ADD1.1	ADD1	<0.0001	0.40
KDR.6	KDR	<0.0001	0.33	ADAMTS9.1	ADAMTS9	<0.0001	0.64
KCNJ15.1	KCNJ15	<0.0001	0.67	ADAMTS5.1	ADAMTS5	<0.0001	0.56
HTATIP.1	KAT5	<0.0001	0.33	ADAMTS4.1	ADAMTS4	<0.0001	0.66
G-Catenin.1	JUP	<0.0001	0.39	ADAMTS2.1	ADAMTS2	<0.0001	0.69
AP-1 (JUN official).2	JUN	<0.0001	0.54	ADAMTS1.1	ADAMTS1	<0.0001	0.51
JAG1.1	JAG1	<0.0001	0.28	ADAM17.1	ADAM17	<0.0001	0.67
ITGB5.1	ITGB5	<0.0001	0.60	ACADSB.1	ACADSB	<0.0001	0.63
ITGB3.1	ITGB3	<0.0001	0.68	BCRP.1	ABCG2	<0.0001	0.43
ITGB1.1	ITGB1	<0.0001	0.46	MRP4.2	ABCC4	0.0337	0.87
ITGA7.1	ITGA7	<0.0001	0.40	AAMP.1	AAMP	<0.0001	0.71
ITGA6.2	ITGA6	<0.0001	0.51	A2M.1	A2M	<0.0001	0.29
ITGA5.1	ITGA5	<0.0001	0.62				

TABLE 6a

Proxy genes for which increased expression is associated with the presence of necrosis (p-value ≤ 0.05)							
Official		CCF Necrosis		Official		CCF Necrosis	
Gene	Symbol	p-value	OR	Gene	Symbol	p-value	OR
WT1.1	WT1	<0.0001	1.57	KRT19.3	KRT19	0.0001	1.43
VTN.1	VTN	<0.0001	1.38	ITGB4.2	ITGB4	0.0492	1.19
VDR.2	VDR	0.0013	1.34	ISG20.1	ISG20	0.0006	1.38
UBE2T.1	UBE2T	<0.0001	2.08	IL-8.1	IL8	<0.0001	2.40
TP.3	TYMP	0.0008	1.37	IL6.3	IL6	<0.0001	2.02
TSPAN8.1	TSPAN8	0.0016	1.29	ICAM1.1	ICAM1	<0.0001	1.75
C20 orf1.1	TPX2	<0.0001	2.64	HSPA8.1	HSPA8	0.0004	1.38
TOP2A.4	TOP2A	<0.0001	2.02	HIST1H1D.1	HIST1H1D	0.0017	1.33
TNFSF13B.1	TNFSF13B	0.0002	1.38	GPX1.2	GPX1	<0.0001	1.46
TK1.2	TK1	<0.0001	1.71	FZD2.2	FZD2	0.0031	1.27
TIMP1.1	TIMP1	0.0076	1.27	FN1.1	FN1	<0.0001	1.45
TGFB1.1	TGFB1	<0.0001	1.69	FCGR3A.1	FCGR3A	0.0001	1.43
OPN, osteopontin.3	SPP1	<0.0001	1.81	FCER1G.2	FCER1G	<0.0001	1.50
SPHK1.1	SPHK1	<0.0001	1.44	FAP.1	FAP	0.0395	1.20
SLC7A5.2	SLC7A5	<0.0001	2.12	F3.1	F3	<0.0001	1.62
SLC2A1.1	SLC2A1	<0.0001	1.46	F2.1	F2	<0.0001	1.76
SLC16A3.1	SLC16A3	0.0001	1.48	ESPL1.3	ESPL1	0.0008	1.33
SLC13A3.1	SLC13A3	0.0019	1.29	EPHB2.1	EPHB2	0.0037	1.28
SHC1.1	SHC1	0.0156	1.24	EPHB1.3	EPHB1	0.0041	1.25
SFN.1	SFN	<0.0001	1.57	EPB41L3.1	EPB41L3	0.0410	1.19
PAI1.3	SERPINE1	0.0164	1.25	ENO2.1	ENO2	0.0001	1.46
SERPINA5.1	SERPINA5	0.0016	1.27	EIF4EBP1.1	EIF4EBP1	<0.0001	1.60
SEMA3C.1	SEMA3C	<0.0001	2.14	E2F1.3	E2F1	<0.0001	1.65
SELL.1	SELL	0.0096	1.26	CXCR6.1	CXCR6	<0.0001	1.48
SAA2.2	SAA2	<0.0001	2.50	CXCR4.3	CXCR4	0.0118	1.26
RRM2.1	RRM2	<0.0001	1.86	GRO1.2	CXCL1	<0.0001	1.83
RPLP1.1	RPLP1	0.0025	1.32	CTSB.1	CTSB	<0.0001	1.70
RND3.1	RND3	0.0002	1.41	CRP.1	CRP	0.0494	1.16
RAD51.1	RAD51	<0.0001	1.51	CP.1	CP	<0.0001	1.98
PTTG1.2	PTTG1	<0.0001	2.52	COL7A1.1	COL7A1	<0.0001	1.47
COX2.1	PTGS2	0.0002	1.36	COL1A1.1	COL1A1	0.0059	1.28
PRKCB1.1	PRKCB	0.0188	1.23	Chk2.3	CHEK2	0.0010	1.33
PRKCA.1	PRKCA	0.0339	1.21	Chk1.2	CHEK1	<0.0001	1.49
PLAUR.3	PLAUR	<0.0001	1.60	CENPF.1	CENPF	<0.0001	2.08
upa.3	PLAU	0.0002	1.41	CD82.3	CD82	<0.0001	2.09
PF4.1	PF4	0.0003	1.34	CD68.2	CD68	0.0163	1.24
PDCD1.1	PDCD1	<0.0001	1.40	CD44s.1	CD44_s	<0.0001	1.66
				CCNE2.2	CCNE2_2	<0.0001	1.50

TABLE 6a-continued

Proxy genes for which increased expression is associated with the presence of necrosis (p-value ≤ 0.05)							
Gene	Official Symbol	CCF Necrosis		Gene	Official Symbol	CCF Necrosis	
		p-value	OR			p-value	OR
PCSK6.1	PCSK6	0.0297	1.22	CCNE1.1	CCNE1	<0.0001	1.44
PCNA.2	PCNA	0.0025	1.33	CCNB1.2	CCNB1	<0.0001	2.28
NNMT.1	NNMT	0.0206	1.22	CCL5.2	CCL5	<0.0001	1.45
NME2.1	NME2	0.0124	1.25	CCL20.1	CCL20	0.0121	1.24
MYBL2.1	MYBL2	<0.0001	1.90	CAV2.1	CAV2	0.0003	1.35
MT1X.1	MT1X	0.0003	1.39	CA12.1	CA12	<0.0001	2.11
MMP9.1	MMP9	<0.0001	1.96	C3.1	C3	<0.0001	1.58
MMP7.1	MMP7	<0.0001	1.50	C1QB.1	C1QB	0.0032	1.31
MMP14.1	MMP14	<0.0001	1.50	BUB1.1	BUB1	<0.0001	2.25
Ki-67.2	MKI67	<0.0001	1.96	BRCA1.2	BRCA1	0.0006	1.35
mGST1.2	MGST1	<0.0001	1.63	SURV2	BIRC5	<0.0001	2.11
cMet.2	MET	0.0357	1.22	cIAP2.2	BIRC3	<0.0001	1.47
MDK.1	MDK	<0.0001	1.78	BCL2A1.1	BCL2A1	0.0004	1.34
MCM2.2	MCM2	0.0003	1.40	STK15.2	AURKA	<0.0001	1.61
LRRC2.1	LRRC2	0.0114	1.22	PRO2000.3	ATAD2	0.0166	1.24
LOX.1	LOX	<0.0001	1.99	APOL1.1	APOL1	<0.0001	1.54
LMNB1.1	LMNB1	<0.0001	2.04	APOC1.3	APOC1	0.0026	1.30
LIMK1.1	LIMK1	<0.0001	2.51	ANXA2.2	ANXA2	<0.0001	1.71
LGALS9.1	LGALS9	0.0136	1.25	ALOX5.1	ALOX5	0.0004	1.38
LGALS1.1	LGALS1	<0.0001	1.46	ADAM8.1	ADAM8	<0.0001	1.89
LAPTM5.1	LAPTM5	<0.0001	1.47	MRP2.3	ABCC2	0.0002	1.39
LAMB3.1	LAMB3	<0.0001	1.96				
L1CAM.1	L1CAM	<0.0001	1.43				

TABLE 6b

Proxy genes for which increased expression is associated with the absence of necrosis (p-value ≤ 0.05)							
Gene	Official Symbol	CCF Necrosis		Gene	Official Symbol	CCF Necrosis	
		p-value	OR			p-value	OR
YB-1.2	YBX1	0.0010	0.74	IGF2.2	IGF2	0.0117	0.79
XIAP.1	XIAP	<0.0001	0.68	IGF1R.3	IGF1R	<0.0001	0.38
WWOX.5	WWOX	<0.0001	0.63	ID3.1	ID3	<0.0001	0.44
WISP1.1	WISP1	0.0002	0.71	ID2.4	ID2	<0.0001	0.68
VWF.1	VWF	<0.0001	0.35	ID1.1	ID1	<0.0001	0.32
VHL.1	VHL	0.0086	0.78	ICAM2.1	ICAM2	<0.0001	0.47
VEGF.1	VEGFA	<0.0001	0.50	HYAL2.1	HYAL2	<0.0001	0.28
VCAM1.1	VCAM1	<0.0001	0.55	HYAL1.1	HYAL1	<0.0001	0.39
USP34.1	USP34	<0.0001	0.64	HSPG2.1	HSPG2	<0.0001	0.33
UMOD.1	UMOD	<0.0001	0.36	HSP90AB1.1	HSP90AB1	0.0004	0.73
UGCG.1	UGCG	<0.0001	0.54	HSD11B2.1	HSD11B2	<0.0001	0.32
UBB.1	UBB	<0.0001	0.48	Hepsin.1	HPN	<0.0001	0.59
UBE1C.1	UBA3	<0.0001	0.59	HPCAL1.1	HPCAL1	<0.0001	0.68
TS.1	TYMS	<0.0001	0.70	HMGB1.1	HMGB1	<0.0001	0.42
tusc4.2	TUSC4	<0.0001	0.68	HLA-DPB1.1	HLA-DPB1	0.0002	0.72
TUSC2.1	TUSC2	0.0462	0.83	HIF1AN.1	HIF1AN	<0.0001	0.54
TSPAN7.2	TSPAN7	<0.0001	0.25	HDAC1.1	HDAC1	<0.0001	0.55
TSC2.1	TSC2	<0.0001	0.45	HAVCR1.1	HAVCR1	0.0012	0.76
TSC1.1	TSC1	<0.0001	0.45	HADH.1	HADH	<0.0001	0.49
P53.2	TP53	<0.0001	0.61	GSTT1.3	GSTT1	0.0067	0.80
TOP2B.2	TOP2B	<0.0001	0.69	GSTp.3	GSTP1	<0.0001	0.55
TNFSF12.1	TNFSF12	<0.0001	0.51	GSTM3.2	GSTM3	<0.0001	0.48
TRAIL.1	TNFSF10	<0.0001	0.68	GSTM1.1	GSTM1	<0.0001	0.54
TNFRSF11B.1	TNFRSF11B	<0.0001	0.59	GRB7.2	GRB7	<0.0001	0.49
TNFRSF10D.1	TNFRSF10D	<0.0001	0.58	GPX3.1	GPX3	<0.0001	0.59
DR5.2	TNFRSF10B	0.0001	0.71	GPC3.1	GPC3	0.0287	0.81
TNFAIP6.1	TNFAIP6	<0.0001	0.67	GJA1.1	GJA1	0.0004	0.74
TMEM47.1	TMEM47	<0.0001	0.29	GFRA1.1	GFRA1	0.0011	0.74
TMEM27.1	TMEM27	<0.0001	0.40	GCLC.3	GCLC	<0.0001	0.50
TLR3.1	TLR3	<0.0001	0.64	GATM.1	GATM	<0.0001	0.45
TIMP3.3	TIMP3	<0.0001	0.23	GATA3.3	GATA3	0.0159	0.79
TIMP2.1	TIMP2	<0.0001	0.52	GADD45B.1	GADD45B	<0.0001	0.67
THBS1.1	THBS1	<0.0001	0.62	FOS.1	FOS	<0.0001	0.56
TGFBR2.3	TGFBR2	<0.0001	0.34	FOLR1.1	FOLR1	<0.0001	0.59
TGFBR1.1	TGFBR1	<0.0001	0.63	FLT4.1	FLT4	<0.0001	0.27
TGFB2.2	TGFB2	<0.0001	0.55	FLT3LG.1	FLT3LG	<0.0001	0.62
TGFB1.1	TGFB1	0.0036	0.76	FLT1.1	FLT1	<0.0001	0.32

TABLE 6b-continued

Proxy genes for which increased expression is associated with the absence of necrosis (p-value ≤ 0.05)							
Gene	Official Symbol	CCF Necrosis		Gene	Official Symbol	CCF Necrosis	
		p-value	OR			p-value	OR
TGFA.2	TGFA	<0.0001	0.56	FILIP1.1	FILIP1	<0.0001	0.42
TEK.1	TEK	<0.0001	0.23	FIGF.1	FIGF	0.0001	0.62
TCF4.1	TCF4	<0.0001	0.36	FHL1.1	FHL1	<0.0001	0.36
TAGLN.1	TAGLN	<0.0001	0.50	FHIT.1	FHIT	<0.0001	0.63
TACSTD2.1	TACSTD2	<0.0001	0.64	FH.1	FH	<0.0001	0.65
SUCLG1.1	SUCLG1	<0.0001	0.50	FGFR2 isoform	FGFR2	<0.0001	0.51
STK11.1	STK11	<0.0001	0.48	1.1			
STAT5B.2	STAT5B	<0.0001	0.36	FGFR1.3	FGFR1	<0.0001	0.55
STAT5A.1	STAT5A	<0.0001	0.56	FGF2.2	FGF2	<0.0001	0.61
STAT3.1	STAT3	<0.0001	0.63	FGF1.1	FGF1	<0.0001	0.49
SPRY1.1	SPRY1	<0.0001	0.42	FDPS.1	FDPS	<0.0001	0.43
SPAST.1	SPAST	0.0004	0.74	FBXW7.1	FBXW7	<0.0001	0.60
SPARCL1.1	SPARCL1	<0.0001	0.48	fas.1	FAS	0.0054	0.79
SPARC.1	SPARC	<0.0001	0.54	ESRRG.3	ESRRG	<0.0001	0.68
SOD1.1	SOD1	<0.0001	0.67	ERG.1	ERG	<0.0001	0.34
SNRK.1	SNRK	<0.0001	0.25	ERCC4.1	ERCC4	0.0197	0.81
SNAI1.1	SNAI1	0.0004	0.71	ERCC1.2	ERCC1	<0.0001	0.60
MADH4.1	SMAD4	<0.0001	0.33	ERBB4.3	ERBB4	0.0037	0.72
MADH2.1	SMAD2	<0.0001	0.40	ErB3.1	ERBB3	<0.0001	0.59
SLC34A1.1	SLC34A1	<0.0001	0.47	HER2.3	ERBB2	<0.0001	0.40
SLC22A6.1	SLC22A6	<0.0001	0.52	EPHB4.1	EPHB4	<0.0001	0.44
SKIL.1	SKIL	<0.0001	0.59	EPHA2.1	EPHA2	<0.0001	0.36
SHANK3.1	SHANK3	<0.0001	0.25	EPAS1.1	EPAS1	<0.0001	0.29
SGK.1	SGK1	<0.0001	0.54	ENPP2.1	ENPP2	<0.0001	0.50
FRP1.3	SFRP1	0.0053	0.77	ENPEP.1	ENPEP	<0.0001	0.56
SEMA3F.3	SEMA3F	<0.0001	0.43	CD105.1	ENG	<0.0001	0.31
SELENBP1.1	SELENBP1	<0.0001	0.62	EMP1.1	EMP1	<0.0001	0.49
SDPR.1	SDPR	<0.0001	0.28	EMCN.1	EMCN	<0.0001	0.23
SDHA.1	SDHA	<0.0001	0.47	ELTD1.1	ELTD1	<0.0001	0.59
SCNN1A.2	SCNN1A	0.0013	0.73	EIF2C1.1	EIF2C1	<0.0001	0.52
SCN4B.1	SCN4B	<0.0001	0.35	EGR1.1	EGR1	<0.0001	0.54
S100A2.1	S100A2	0.0355	0.82	EGLN3.1	EGLN3	<0.0001	0.69
KIAA1303	RPTOR	<0.0001	0.49	EGFR.2	EGFR	<0.0001	0.70
raptor.1				EFNB2.1	EFNB2	<0.0001	0.34
RPS6KB1.3	RPS6KB1	<0.0001	0.55	EFNB1.2	EFNB1	<0.0001	0.41
RPS6KA1.1	RPS6KA1	0.0002	0.68	EEF1A1.1	EEF1A1	<0.0001	0.32
RPS23.1	RPS23	<0.0001	0.47	EDNRB.1	EDNRB	<0.0001	0.31
ROCK2.1	ROCK2	<0.0001	0.39	EDN2.1	EDN2	<0.0001	0.51
ROCK1.1	ROCK1	<0.0001	0.35	EDN1	EDN1	<0.0001	0.41
RIPK1.1	RIPK1	<0.0001	0.45	endothelin.1			
rhoC.1	RHOC	0.0001	0.70	EBAG9.1	EBAG9	0.0007	0.74
RhoB.1	RHOB	<0.0001	0.41	DUSP1.1	DUSP1	<0.0001	0.65
ARHA.1	RHOA	<0.0001	0.45	DPYS.1	DPYS	<0.0001	0.66
RHEB.2	RHEB	0.0002	0.69	DPEP1.1	DPEP1	<0.0001	0.34
RG5.1	RG5	<0.0001	0.26	DLL4.1	DLL4	<0.0001	0.49
FLJ22655.1	REGL	<0.0001	0.26	DLC1.1	DLC1	<0.0001	0.36
NFKBp65.3	RELA	<0.0001	0.71	DKFZP564O0823.1	DKFZP564O0823	<0.0001	0.35
RB1.1	RB1	<0.0001	0.54	DICER1.2	DICER1	<0.0001	0.46
RASSF1.1	RASSF1	<0.0001	0.63	DIAPH1.1	DIAPH1	<0.0001	0.63
RARB.2	RARB	<0.0001	0.28	DIABLO.1	DIABLO	0.0002	0.72
RALBP1.1	RALBP1	<0.0001	0.52	DHPS.3	DHPS	<0.0001	0.46
RAF1.3	RAF1	<0.0001	0.58	DET1.1	DET1	<0.0001	0.61
RAC1.3	RAC1	0.0118	0.80	DEFB1.1	DEFB1	<0.0001	0.69
PTPRG.1	PTPRG	<0.0001	0.27	DDC.1	DDC	<0.0001	0.51
PTPRB.1	PTPRB	<0.0001	0.22	DCXR.1	DCXR	0.0061	0.78
PTN.1	PTN	<0.0001	0.41	DAPK1.3	DAPK1	<0.0001	0.47
PTK2.1	PTK2	<0.0001	0.33	CYR61.1	CYR61	<0.0001	0.52
PTHR1.1	PTH1R	<0.0001	0.32	CYP3A4.2	CYP3A4	0.0398	0.82
PTEN.2	PTEN	<0.0001	0.44	CYP2C8v2.1	CYP2C8_21	0.0001	0.67
PSMA7.1	PSMA7	0.0173	0.81	CXCL12.1	CXCL12	<0.0001	0.53
PRSS8.1	PRSS8	<0.0001	0.62	CX3CR1.1	CX3CR1	<0.0001	0.60
PRKCH.1	PRKCH	<0.0001	0.43	CX3CL1.1	CX3CL1	<0.0001	0.34
PRKCD.2	PRKCD	<0.0001	0.68	CUL1.1	CUL1	<0.0001	0.62
PPP2CA.1	PPP2CA	<0.0001	0.53	CUBN.1	CUBN	<0.0001	0.39
PPARG.3	PPARG	<0.0001	0.37	CTSH.2	CTSH	0.0018	0.77
PPAP2B.1	PPAP2B	<0.0001	0.27	B-Catenin.3	CTNNB1	<0.0001	0.38
PMP22.1	PMP22	0.0155	0.81	A-Catenin.2	CTNNA1	<0.0001	0.58
PLG.1	PLG	<0.0001	0.38	CTGF.1	CTGF	<0.0001	0.64
PLAT.1	PLAT	<0.0001	0.42	CSF1R.2	CSF1R	0.0308	0.83
PLA2G4C.1	PLA2G4C	<0.0001	0.48	CSF1.1	CSF1	0.0002	0.73
PIK3CA.1	PIK3CA	<0.0001	0.47	CRADD.1	CRADD	<0.0001	0.60
P13K.2	PIK3C2B	<0.0001	0.42	COL4A2.1	COL4A2	<0.0001	0.51
PGF.1	PGF	0.0153	0.80	COL4A1.1	COL4A1	<0.0001	0.66

TABLE 6b-continued

Proxy genes for which increased expression is associated with the absence of necrosis (p-value $\leq .05$)							
Gene	Official Symbol	CCF Necrosis		Gene	Official Symbol	CCF Necrosis	
		p-value	OR			p-value	OR
PFKP.1	PFKP	<0.0001	0.70	COL18A1.1	COL18A1	<0.0001	0.50
CD31.3	PECAM1	<0.0001	0.34	CLU.3	CLU	0.0091	0.80
PDZK3.1	PDZK3	<0.0001	0.49	CLDN7.2	CLDN7	0.0029	0.76
PDZK1.1	PDZK1	<0.0001	0.45	CLDN10.1	CLDN10	<0.0001	0.48
PDGFRb.3	PDGFRB	<0.0001	0.45	CLCNKB.1	CLCNKB	0.0001	0.61
PDGFD.2	PDGFD	<0.0001	0.33	CFLAR.1	CFLAR	<0.0001	0.47
PDGFC.3	PDGFC	<0.0001	0.53	CEACAM1.1	CEACAM1	<0.0001	0.43
PDGFB.3	PDGFB	<0.0001	0.33	p27.3	CDKN1B	0.0002	0.73
PDGFA.3	PDGFA	<0.0001	0.43	p21.3	CDKN1A	<0.0001	0.65
PCK1.1	PCK1	<0.0001	0.44	CDH6.1	CDH6	<0.0001	0.67
PCCA.1	PCCA	<0.0001	0.47	CDH5.1	CDH5	<0.0001	0.33
PARD6A.1	PARD6A	0.0045	0.77	CDH2.1	CDH2	0.0003	0.75
Pak1.2	PAK1	0.0003	0.74	CDH16.1	CDH16	<0.0001	0.51
PAH.1	PAH	<0.0001	0.62	CDH13.1	CDH13	<0.0001	0.39
OGG1.1	OGG1	<0.0001	0.62	CD36.1	CD36	<0.0001	0.41
BFGF.3	NUDT6	<0.0001	0.45	CD34.1	CD34	<0.0001	0.34
NRG1.3	NRG1	0.0004	0.69	CD24.1	CD24	0.0148	0.81
NPR1.1	NPR1	<0.0001	0.36	CCND1.3	CCND1	<0.0001	0.51
NPM1.2	NPM1	<0.0001	0.55	MCP1.1	CCL2	<0.0001	0.68
NOTCH3.1	NOTCH3	<0.0001	0.40	CAT.1	CAT	<0.0001	0.48
NOTCH2.1	NOTCH2	<0.0001	0.68	CASP10.1	CASP10	<0.0001	0.62
NOTCH1.1	NOTCH1	<0.0001	0.38	CALD1.2	CALD1	<0.0001	0.42
NOS3.1	NOS3	<0.0001	0.37	CACNA2D1.1	CACNA2D1	0.0006	0.74
NOS2A.3	NOS2	<0.0001	0.42	CA2.1	CA2	<0.0001	0.60
NOL3.1	NOL3	<0.0001	0.67	C7.1	C7	<0.0001	0.65
NFX1.1	NFX1	<0.0001	0.43	ECRG4.1	C2orf40	<0.0001	0.32
NFKBp50.3	NFKB1	<0.0001	0.56	C13orf15.1	C13orf15	<0.0001	0.31
NFATC2.1	NFATC2	<0.0001	0.67	BUB3.1	BUB3	<0.0001	0.65
NFAT5.1	NFAT5	<0.0001	0.55	BTRC.1	BTRC	<0.0001	0.63
MYRIP.2	MYRIP	<0.0001	0.36	BNIP3.1	BNIP3	0.0021	0.77
MYH11.1	MYH11	<0.0001	0.35	CIAP1.2	BIRC2	<0.0001	0.56
eMYC.3	MYC	<0.0001	0.68	BIN1.3	BIN1	<0.0001	0.67
MVP.1	MVP	<0.0001	0.66	BGN.1	BGN	<0.0001	0.47
FRAP1.1	MTOR	<0.0001	0.56	BCL2L12.1	BCL2L12	0.0374	0.82
MSH3.2	MSH3	<0.0001	0.47	Bclx.2	BCL2L1	<0.0001	0.60
MSH2.3	MSH2	<0.0001	0.51	Bcl2.2	BCL2	<0.0001	0.31
MMP2.2	MMP2	0.0229	0.82	BAG1.2	BAG1	<0.0001	0.42
STMY3.3	MMP11	<0.0001	0.66	BAD.1	BAD	0.0187	0.82
GBL.1	MLST8	0.0193	0.82	AXL.1	AXL	0.0077	0.79
MIF.2	MIF	<0.0001	0.69	ATP6V1B1.1	ATP6V1B1	<0.0001	0.52
MICA.1	MICA	<0.0001	0.52	ASS1.1	ASS1	<0.0001	0.61
MGMT.1	MGMT	<0.0001	0.50	ARRB1.1	ARRB1	<0.0001	0.48
MCM3.3	MCM3	0.0105	0.78	ARHGDIB.1	ARHGDIB	<0.0001	0.48
MCAM.1	MCAM	<0.0001	0.41	ARF1.1	ARF1	0.0021	0.75
MARCKS.1	MARCKS	0.0259	0.82	AQP1.1	AQP1	<0.0001	0.28
ERK1.3	MAPK3	<0.0001	0.35	APOLD1.1	APOLD1	<0.0001	0.35
ERK2.3	MAPK1	<0.0001	0.61	APC.4	APC	<0.0001	0.55
MAP4.1	MAP4	<0.0001	0.54	APAF1.2	APAF1	0.0264	0.82
MAP2K3.1	MAP2K3	<0.0001	0.52	ANXA4.1	ANXA4	0.0012	0.76
MAP2K1.1	MAP2K1	0.0172	0.81	ANXA1.2	ANXA1	0.0201	0.81
MAL2.1	MAL2	0.0267	0.82	ANTXR1.1	ANTXR1	<0.0001	0.58
MAL.1	MAL	<0.0001	0.46	ANGPTL4.1	ANGPTL4	<0.0001	0.71
LTF.1	LTF	0.0038	0.74	ANGPTL3.3	ANGPTL3	0.0104	0.77
LRP2.1	LRP2	<0.0001	0.52	ANGPTL2.1	ANGPTL2	<0.0001	0.63
LMO2.1	LMO2	<0.0001	0.60	ANGPT2.1	ANGPT2	<0.0001	0.65
LDB2.1	LDB2	<0.0001	0.26	ANGPT1.1	ANGPT1	<0.0001	0.30
LDB1.2	LDB1	<0.0001	0.52	AMACR1.1	AMACR	0.0080	0.79
LAMA4.1	LAMA4	<0.0001	0.67	ALDOB.1	ALDOB	<0.0001	0.40
KRT7.1	KRT7	<0.0001	0.56	ALDH6A1.1	ALDH6A1	<0.0001	0.38
K-ras.10	KRAS	<0.0001	0.48	ALDH4.2	ALDH4A1	0.0001	0.71
KL.1	KL	<0.0001	0.34	AKT3.2	AKT3	<0.0001	0.30
Kitlg.4	KITLG	<0.0001	0.46	AKT2.3	AKT2	<0.0001	0.53
c-kit.2	KIT	<0.0001	0.41	AKT1.3	AKT1	<0.0001	0.47
KDR.6	KDR	<0.0001	0.28	AHR.1	AHR	<0.0001	0.60
KCNJ15.1	KCNJ15	<0.0001	0.43	AGTR1.1	AGTR1	<0.0001	0.33
HTATIP.1	KAT5	<0.0001	0.27	AGT.1	AGT	0.0032	0.77
G-Catenin.1	JUP	<0.0001	0.32	ADH6.1	ADH6	0.0011	0.71
AP-1 (JUN official).2	JUN	<0.0001	0.64	ADH1B.1	ADH1B	<0.0001	0.69
JAG1.1	JAG1	<0.0001	0.23	ADFP.1	ADFP	0.0001	0.73
ITGB5.1	ITGB5	<0.0001	0.64	ADD1.1	ADD1	<0.0001	0.33
ITGB3.1	ITGB3	0.0468	0.84	ADAMTS9.1	ADAMTS9	<0.0001	0.69
ITGB1.1	ITGB1	<0.0001	0.59	ADAMTS5.1	ADAMTS5	<0.0001	0.55
				ADAMTS1.1	ADAMTS1	<0.0001	0.56

TABLE 6b-continued

Proxy genes for which increased expression is associated with the absence of necrosis (p-value ≤ 0.05)							
Gene	Official	CCF Necrosis		Gene	Official	CCF Necrosis	
	Symbol	p-value	OR		Symbol	p-value	OR
ITGA7.1	ITGA7	<0.0001	0.38	ADAM17.1	ADAM17	0.0009	0.76
ITGA6.2	ITGA6	<0.0001	0.40	ACE2.1	ACE2	<0.0001	0.45
ITGA5.1	ITGA5	0.0298	0.83	ACADSB.1	ACADSB	<0.0001	0.46
ITGA4.2	ITGA4	<0.0001	0.61	BCRP.1	ABCG2	<0.0001	0.27
ITGA3.2	ITGA3	0.0018	0.76	MRP4.2	ABCC4	<0.0001	0.61
IQGAP2.1	IQGAP2	<0.0001	0.52	MRP3.1	ABCC3	0.0011	0.76
INSR.1	INSR	<0.0001	0.38	MRP1.1	ABCC1	0.0008	0.75
IMP3.1	IMP3	<0.0001	0.53	ABCB1.5	ABCB1	<0.0001	0.59
IL6ST.3	IL6ST	<0.0001	0.36	NPD009	ABAT	0.0001	0.70
IL15.1	IL15	<0.0001	0.67	(ABAT official).3			
IGFBP6.1	IGFBP6	<0.0001	0.66	AAMP.1	AAMP	<0.0001	0.62
			A2M.1	A2M		<0.0001	0.28

TABLE 7a

Proxy genes for which increased expression is associated with the presence of nodal invasion (p-value ≤ 0.05)							
Gene	Official	Nodal Invasion		Gene	Official	Nodal Invasion	
	Symbol	p-value	OR		Symbol	p-value	OR
TUBB.1	TUBB2A	0.0242	2.56	IL-8.1	IL8	0.0019	3.18
C20orf1.1	TPX2	0.0333	2.61	IL6.3	IL6	0.0333	2.31
TK1.2	TK1	0.0361	1.75	HSPA1A.1	HSPA1A	0.0498	2.11
SPHK1.1	SPHK1	0.0038	3.43	GSTp.3	GSTP1	0.0272	3.46
SLC7A5.2	SLC7A5	0.0053	4.85	GRB14.1	GRB14	0.0287	2.32
SILV.1	SILV	0.0470	1.54	GMNN.1	GMNN	0.0282	3.00
SELE.1	SELE	0.0311	1.93	ENO2.1	ENO2	0.0190	3.43
upa.3	PLAU	0.0450	2.78	CCNB1.2	CCNB1	0.0387	1.87
MMP9.1	MMP9	0.0110	2.65	BUB1.1	BUB1	0.0429	2.21
MMP7.1	MMP7	0.0491	2.34	BAG2.1	BAG2	0.0346	2.54
MMP14.1	MMP14	0.0155	3.21	ADAMTS1.1	ADAMTS1	0.0193	3.31
LAMB1.1	LAMB1	0.0247	3.04				

TABLE 7b

Proxy genes for which increased expression is associated with the absence of nodal invasion (p-value ≤ 0.05)							
Gene	Official	Nodal Invasion		Gene	Official	Nodal Invasion	
	Symbol	p-value	OR		Symbol	p-value	OR
VWF.1	VWF	0.0221	0.42	HMGB1.1	HMGB1	0.0385	0.45
VCAM1.1	VCAM1	0.0212	0.42	HLA-DPB1.1	HLA-DPB1	0.0398	0.43
UBE1C.1	UBA3	0.0082	0.32	HADH.1	HADH	0.0093	0.33
tusc4.2	TUSC4	0.0050	0.28	GSTM1.1	GSTM1	0.0018	0.18
TSPAN7.2	TSPAN7	0.0407	0.43	GPX2.2	GPX2	0.0211	0.07
TSC1.1	TSC1	0.0372	0.38	GJA1.1	GJA1	0.0451	0.46
TMSB10.1	TMSB10	0.0202	0.43	GATM.1	GATM	0.0038	0.25
TMEM47.1	TMEM47	0.0077	0.32	GATA3.3	GATA3	0.0188	0.06
TMEM27.1	TMEM27	0.0431	0.41	FOLR1.1	FOLR1	0.0152	0.32
TLR3.1	TLR3	0.0041	0.32	FLT4.1	FLT4	0.0125	0.22
TIMP3.3	TIMP3	0.0309	0.40	FLT1.1	FLT1	0.0046	0.35
TGFBR2.3	TGFBR2	0.0296	0.35	FHL1.1	FHL1	0.0435	0.47
TGFB2.2	TGFB2	0.0371	0.30	FHIT.1	FHIT	0.0061	0.29
TGFA.2	TGFA	0.0025	0.32	fas.1	FAS	0.0163	0.39
TEK.1	TEK	0.0018	0.09	ErbB3.1	ERBB3	0.0145	0.33
TCF4.1	TCF4	0.0088	0.38	EPHA2.1	EPHA2	0.0392	0.37
STAT5A.1	STAT5A	0.0129	0.49	EPAS1.1	EPAS1	0.0020	0.28
SPRY1.1	SPRY1	0.0188	0.43	ENPEP.1	ENPEP	0.0002	0.34
SPARCL1.1	SPARCL1	0.0417	0.50	CD105.1	ENG	0.0112	0.38
SOD1.1	SOD1	0.0014	0.23	EMCN.1	EMCN	0.0022	0.19
SNRK.1	SNRK	0.0226	0.43	EIF2C1.1	EIF2C1	0.0207	0.33
MADH2.1	SMAD2	0.0098	0.37	EGLN3.1	EGLN3	0.0167	0.52

TABLE 7b-continued

Proxy genes for which increased expression is associated with the absence of nodal invasion (p-value ≤ 0.05)							
Gene	Official	Nodal Invasion		Gene	Official	Nodal Invasion	
	Symbol	p-value	OR		Symbol	p-value	OR
SLC22A6.1	SLC22A6	0.0051	0.00	EFNB2.1	EFNB2	0.0192	0.30
PTPNS1.1	SIRPA	0.0206	0.43	EFNB1.2	EFNB1	0.0110	0.37
SHANK3.1	SHANK3	0.0024	0.30	EDNRB.1	EDNRB	0.0106	0.36
SGK.1	SGK1	0.0087	0.35	EDN1	EDN1	0.0440	0.42
SELENBP1.1	SELENBP1	0.0016	0.29	endothelin.1			
SCN4B.1	SCN4B	0.0081	0.08	DPYS.1	DPYS	0.0454	0.43
ROCK1.1	ROCK1	0.0058	0.41	DKFZP564O0823.1	DKFZP564O0823	0.0131	0.31
RhoB.1	RHOB	0.0333	0.43	DHPS.3	DHPS	0.0423	0.48
RGS5.1	RGS5	0.0021	0.31	DAPK1.3	DAPK1	0.0048	0.39
FLJ22655.1	REGL	0.0009	0.01	CYP2C8v2.1	CYP2C8_21	0.0003	0.01
RB1.1	RB1	0.0281	0.48	CYP2C8.2	CYP2C8_2	0.0269	0.07
RASSF1.1	RASSF1	0.0004	0.23	CX3CR1.1	CX3CR1	0.0224	0.13
PTPRB.1	PTPRB	0.0154	0.40	CUBN.1	CUBN	0.0010	0.04
PTK2.1	PTK2	0.0158	0.38	CRADD.1	CRADD	0.0193	0.40
PTHR1.1	PTH1R	<0.0001	0.01	CLDN10.1	CLDN10	0.0005	0.21
PRSS8.1	PRSS8	0.0023	0.10	CFLAR.1	CFLAR	0.0426	0.49
PRKCH.1	PRKCH	0.0475	0.48	CEACAM1.1	CEACAM1	0.0083	0.23
PPAP2B.1	PPAP2B	0.0110	0.40	CDKN2A.2	CDKN2A	0.0026	0.13
PLA2G4C.1	PLA2G4C	0.0002	0.03	p27.3	CDKN1B	0.0393	0.60
PI3K.2	PIK3C2B	0.0138	0.13	CDH5.1	CDH5	0.0038	0.33
PFKP.1	PFKP	0.0040	0.41	CDH13.1	CDH13	0.0203	0.39
CD31.3	PECAM1	0.0077	0.38	CD99.1	CD99	0.0173	0.38
PDGFD.2	PDGFD	0.0169	0.35	CD36.1	CD36	0.0015	0.29
PDGFC.3	PDGFC	0.0053	0.36	CD34.1	CD34	0.0250	0.38
PDGFB.3	PDGFB	0.0359	0.47	CD3z.1	CD247	0.0419	0.33
PCSK6.1	PCSK6	0.0103	0.34	CCND1.3	CCND1	0.0381	0.50
PCK1.1	PCK1	0.0003	0.02	CAT.1	CAT	0.0044	0.42
PCCA.1	PCCA	0.0074	0.12	CASP6.1	CASP6	0.0136	0.36
PAR6A.1	PAR6A	0.0243	0.21	CALD1.2	CALD1	0.0042	0.31
BFGF.3	NUDT6	0.0082	0.15	CA9.3	CA9	0.0077	0.48
NRG1.3	NRG1	0.0393	0.19	C13orf15.1	C13orf15	0.0152	0.39
NOS3.1	NOS3	0.0232	0.34	BUB3.1	BUB3	0.0157	0.26
NOS2A.3	NOS2	0.0086	0.11	BIN1.3	BIN1	0.0224	0.38
NFX1.1	NFX1	0.0065	0.35	Bclx.2	BCL2L1	0.0225	0.52
MYH11.1	MYH11	0.0149	0.36	Bcl2.2	BCL2	0.0442	0.45
cMYC.3	MYC	0.0472	0.46	AXL.1	AXL	0.0451	0.44
MUC1.2	MUC1	0.0202	0.26	ATP6V1B1.1	ATP6V1B1	0.0257	0.03
MIF.2	MIF	0.0145	0.39	ARRB1.1	ARRB1	0.0337	0.43
MICA.1	MICA	0.0015	0.15	ARHGDIB.1	ARHGDIB	0.0051	0.32
MGMT.1	MGMT	0.0414	0.50	AQP1.1	AQP1	0.0022	0.31
MAP2K1.1	MAP2K1	0.0075	0.40	APOLD1.1	APOLD1	0.0222	0.42
LMO2.1	LMO2	0.0127	0.07	APC.4	APC	0.0165	0.47
LDB2.1	LDB2	0.0048	0.33	ANXA5.1	ANXA5	0.0143	0.37
Kitlmg.4	KITLG	0.0146	0.18	ANXA4.1	ANXA4	0.0019	0.30
KDR.6	KDR	0.0106	0.34	ANXA1.2	ANXA1	0.0497	0.39
ITGB1.1	ITGB1	0.0469	0.36	ANGPTL7.1	ANGPTL7	0.0444	0.16
ITGA7.1	ITGA7	0.0290	0.38	ANGPTL4.1	ANGPTL4	0.0197	0.55
ITGA6.2	ITGA6	0.0010	0.17	ANGPT1.1	ANGPT1	0.0055	0.15
ITGA4.2	ITGA4	0.0089	0.44	ALDOB.1	ALDOB	0.0128	0.02
INSR.1	INSR	0.0057	0.32	ALDH4.2	ALDH4A1	0.0304	0.33
IMP3.1	IMP3	0.0086	0.43	AGTR1.1	AGTR1	0.0142	0.11
IL6ST.3	IL6ST	0.0484	0.46	ADH6.1	ADH6	0.0042	0.03
IL15.1	IL15	0.0009	0.08	ADFP.1	ADFP	0.0223	0.47
IFI27.1	IFI27	0.0013	0.22	ADD1.1	ADD1	0.0135	0.42
HYAL2.1	HYAL2	0.0099	0.36	BCRP.1	ABCG2	0.0098	0.19
HYAL1.1	HYAL1	0.0001	0.08	MRP3.1	ABCC3	0.0247	0.44
Hepsin.1	HPN	0.0024	0.14	MRP1.1	ABCC1	0.0022	0.32
HPCAL1.1	HPCAL1	0.0024	0.38	NPD009	ABAT	0.0110	0.19
				(ABAT official).3			
				A2M.1	A2M	0.0012	0.29

59

TABLE 8a

Genes for which increased expression is associated with lower risk of cancer recurrence after clinical/pathologic covariate adjustment (p < 0.05)			
Gene	Official Symbol	p-value	HR
ACE2.1	ACE2	0.0261	0.85
ADD1.1	ADD1	0.0339	0.85
ALDOB.1	ALDOB	0.0328	0.84
ANGPTL3.3	ANGPTL3	0.0035	0.79
APOLD1.1	APOLD1	0.0015	0.78
AQP1.1	AQP1	0.0014	0.79
BFGF.3	NUDT6	0.0010	0.77
CASP10.1	CASP10	0.0024	0.82
CAV2.1	CAV2	0.0191	0.86
CCL4.2	CCL4	0.0045	0.81
CCL5.2	CCL5	0.0003	0.78
CCR2.1	CCR2	0.0390	0.87
CCR4.2	CCR4	0.0109	0.82
CCR7.1	CCR7	0.0020	0.80
CD4.1	CD4	0.0195	0.86
CD8A.1	CD8A	0.0058	0.83
CEACAM1.1	CEACAM1	0.0022	0.81
CFLAR.1	CFLAR	0.0308	0.87
CTSS.1	CTSS	0.0462	0.87
CX3CL1.1	CX3CL1	0.0021	0.81
CXCL10.1	CXCL10	0.0323	0.86
CXCL9.1	CXCL9	0.0006	0.79
CXCR6.1	CXCR6	0.0469	0.88
DAPK1.3	DAPK1	0.0050	0.83
DDC.1	DDC	0.0307	0.86
DLC1.1	DLC1	0.0249	0.83
ECRG4.1	C2orf40	0.0244	0.84
EDNRB.1	EDNRB	0.0400	0.86
EMCN.1	EMCN	<0.0001	0.68
EPAS1.1	EPAS1	0.0411	0.84
fas.1	FAS	0.0242	0.87
FH.1	FH	0.0407	0.88
GATA3.3	GATA3	0.0172	0.83
GZMA.1	GZMA	0.0108	0.84
HLA-DPB1.1	HLA-DPB1	0.0036	0.82
HSPG2.1	HSPG2	0.0236	0.84
ICAM2.1	ICAM2	0.0091	0.83
ICAM3.1	ICAM3	0.0338	0.87
ID1.1	ID1	0.0154	0.83
IGF1R.3	IGF1R	0.0281	0.85
IL15.1	IL15	0.0059	0.83
IQGAP2.1	IQGAP2	0.0497	0.88
KL.1	KL	0.0231	0.86
KLRK1.2	KLRK1	0.0378	0.87
LDB2.1	LDB2	0.0092	0.82
LRP2.1	LRP2	0.0193	0.86
LTF.1	LTF	0.0077	0.82
MAP4.1	MAP4	0.0219	0.84
MRP1.1	ABCC1	0.0291	0.87
NOS3.1	NOS3	0.0008	0.78
PI3K.2	PIK3C2B	0.0329	0.83

60

TABLE 8a-continued

Genes for which increased expression is associated with lower risk of cancer recurrence after clinical/pathologic covariate adjustment (p < 0.05)			
Gene	Official Symbol	p-value	HR
PLA2G4C.1	PLA2G4C	0.0452	0.87
PPAP2B.1	PPAP2B	0.0001	0.74
PRCC.1	PRCC	0.0333	0.88
PRKCB1.1	PRKCB	0.0353	0.87
PRKCH.1	PRKCH	0.0022	0.82
PRSS8.1	PRSS8	0.0332	0.87
PSMB9.1	PSMB9	0.0262	0.87
PTPRB.1	PTPRB	0.0030	0.80
RGS5.1	RGS5	0.0480	0.85
SDPR.1	SDPR	0.0045	0.80
SELE.1	SELE	0.0070	0.81
SGK.1	SGK1	0.0100	0.83
SHANK3.1	SHANK3	0.0311	0.84
SNRK.1	SNRK	0.0026	0.81
TEK.1	TEK	0.0059	0.78
TGFBR2.3	TGFBR2	0.0343	0.85
TIMP3.3	TIMP3	0.0165	0.83
TMEM27.1	TMEM27	0.0249	0.86
TSPAN7.2	TSPAN7	0.0099	0.83
UBB.1	UBB	0.0144	0.85
WWOX.5	WWOX	0.0082	0.83

TABLE 8b

Genes for which increased expression is associated with higher risk of cancer recurrence after clinical/pathologic covariate adjustment (p < 0.05)			
Gene	Official Symbol	p-value	HR
CIAP1.2	BIRC2	0.0425	1.14
BUB1.1	BUB1	0.0335	1.15
CCNB1.2	CCNB1	0.0296	1.14
ENO2.1	ENO2	0.0284	1.17
ITGB1.1	ITGB1	0.0402	1.16
ITGB5.1	ITGB5	0.0016	1.25
LAMB1.1	LAMB1	0.0067	1.20
MMP14.1	MMP14	0.0269	1.16
MMP9.1	MMP9	0.0085	1.19
PSMA7.1	PSMA7	0.0167	1.16
RUNX1.1	RUNX1	0.0491	1.15
SPHK1.1	SPHK1	0.0278	1.16
OPN, osteopontin.3	SPP1	0.0134	1.17
SQSTM1.1	SQSTM1	0.0347	1.13
C20 orf1.1	TPX2	0.0069	1.20
TUBB.1	TUBB2A	0.0046	1.21
VCAN.1	VCAN	0.0152	1.18

TABLE 9

16 Significant Genes After Adjusting for Clinical/Pathologic
Covariates and Allowing for an FDR of 10%

Official Symbol	n	Gene Subset (Pathway)	HR	HR (95% CI)	LR ChiSq	LR p-value	LR q-value
EMCN	928	Angiogenesis	0.68	(0.57, 0.80)	19.87	<0.0001	0.0042
PPAP2B	928	Angiogenesis	0.74	(0.65, 0.85)	16.15	0.0001	0.0148
CCL5	928	Immune Response	0.78	(0.68, 0.89)	12.98	0.0003	0.0529
CXCL9	928	Immune Response	0.79	(0.70, 0.91)	11.74	0.0006	0.0772
NOS3	928	Angiogenesis	0.78	(0.68, 0.90)	11.32	0.0008	0.0774
NUDT6	926	Angiogenesis	0.77	(0.66, 0.90)	10.77	0.0010	0.0850
AQP1	928	Transport	0.79	(0.69, 0.91)	10.15	0.0014	0.0850
APOLD1	927	Angiogenesis	0.78	(0.68, 0.91)	10.07	0.0015	0.0850
ITGB5	928	Cell Adhesion/ Extracellular Matrix	1.25	(1.09, 1.43)	9.92	0.0016	0.0850
CCR7	928	Immune Response	0.80	(0.69, 0.92)	9.58	0.0020	0.0850
CX3CL1	926	Immune Response	0.81	(0.70, 0.92)	9.44	0.0021	0.0850

TABLE 9-continued

16 Significant Genes After Adjusting for Clinical/Pathologic Covariates and Allowing for an FDR of 10%							
Official Symbol	n	Gene Subset (Pathway)	HR	HR (95% CI)	LR ChiSq	LR p-value	LR q-value
CEACAM1	928	Angiogenesis	0.81	(0.70, 0.93)	9.37	0.0022	0.0850
PRKCH	917	Angiogenesis	0.82	(0.72, 0.93)	9.36	0.0022	0.0850
CASP10	927	Apoptosis	0.82	(0.73, 0.93)	9.25	0.0024	0.0850
SNRK	928	Angiogenesis	0.81	(0.71, 0.92)	9.09	0.0026	0.0863
PTPRB	927	Angiogenesis	0.80	(0.69, 0.92)	8.83	0.0030	0.0933

63

TABLE A

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
A-Catenin.2	NM_001903	NM_001903.1	CTNNA1	765	CGTTCCGATCCTCT ATACTGCAT	1	AGGTCCTGTGTG GCCTTATAGG	733	ATGCTCAGCAGCC TGATGTGCA	1465
A2M.1	NM_000014	NM_000014.4	A2M	6456	CTCTCCGCTTCC TAGC	2	CCGTTTGCACAG ATGCAG	734	CGTTGTTCCTTCTC CACTGGGAC	1466
AAMP.1	NM_001087	NM_001087.3	AAMP	5474	GTGTGGCAGGTGG ACACTAA	3	CTCCATCCACTC CAGGTCTC	735	CGTTCAAAGGACCA GACCTCCTC	1467
ABCB1.5	NM_000927	NM_000927.2	ABCB1	3099	AAACACCAGTGGAG CATTGA	4	CAAGCCTGGAAC CTATAGCC	736	CTCGCCAATGATGCT GCTCAAGTT	1468
ACADSB.1	NM_001609	NM_001609.3	ACADSB	6278	TGCGGAGAACTA GCCAT	5	AAGACAGCCGAG TCCTCAAAT	737	CCTCTGAAGCCTG CCATCATTTGT	1469
ACE.1	NM_000789	NM_000789.2	ACE	4257	CCGCTGTACGAGG ATTCA	6	CCGTGTCTGTGA AGCCGT	738	TGCCCCCTCAGCAATGA AGCCTAGAA	1470
ACE2.1	NM_021804	NM_021804.1	ACE2	6108	TACAATGAGAGGCT CTGGGC	7	TAATGCTCTCAG CTGCTTG	739	CGACCTCAGATCTCC AGCTTTCCC	1471
ADAM17.1	NM_003183	NM_003183.3	ADAM17	2617	GAGTGCCAGGAG GCGATTA	8	CGGGCACTCACT GCTATTACC	740	TGCTACTTGCAAAGG CGTGTCTCTACTGC	1472
ADAM8.1	NM_001109	NM_001109.2	ADAM8	3978	GTCACCTGTGTCAG CCCA	9	TGATGACCTGCT TTGGTGC	741	TTCCCAGTTCTCTGTC TACACCCGG	1473
ADAMTS1.1	NM_006988	NM_006988.2	ADAMTS1	2639	GGACAGGTGCAAG CTCATCTG	10	ATCTACAACCTT GGGCTGCAA	742	CAAGCCAAAGGCATT GGCTACTTCTTTCG	1474
ADAMTS2.1	NM_014244	NM_014244.1	ADAMTS2	3979	GAGAAATGCTGCC GCTGG	11	ATCGTGGTATTC ATCGTGGC	743	TACCTCCAGCAGAAG CCAGACACG	1475
ADAMTS4.1	NM_005099	NM_005099.3	ADAMTS4	2642	TTTGACAAAGTGCAAT GGTGTG	12	AATTTCTTGAAG GAGCCTGA	744	CTGTTGTGTCAACC AGAAACCGT	1476
ADAMTS5.1	NM_007038	NM_007038.1	ADAMTS5	2641	CACGTGGCTCAC GAAATCG	13	GGAAACCAAAGGT CTCTTCACAGA	745	ATTTACTTGGCTCTT CCCATGACGATTCC	1477
ADAMTS8.1	NM_007037	NM_007037.2	ADAMTS8	2640	GCGAGTTCAAAGT GTTCCGAG	14	CACAGATGGCCA GTGTTTCT	746	CACACAGGGTGCCA TCAATACCT	1478
ADAMTS9.1	NM_182920	NM_182920.1	ADAMTS9	6109	GCACAGGTTACACA ACCCAA	15	CGACATGGCAG TCATCG	747	CCGGCTCCCCTTATA GGGACATTC	1479
ADD1.1	NM_001119	NM_001119.3	ADD1	3980	GTCATTACCAGCAG CTCCG	16	TCGTTACACAGGA GTCACCAT	748	CATGTTTAAGGCAGC CATCCCTCC	1480

64

65

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ADFP.1	NM_001122	NM_001122.2	PLIN2	4503	AAGACCATCACCTC CGTGG	17	CAATTTGCGGCT CTAGCTTC	749	ATGACCAGTGCTCTG CCCATCATC	1481
ADH1B.1	NM_000668	NM_000668.4	ADH1B	6325	AAGCAACAACCT TCCTTC	18	AAATGCAAGAA GTCACAGGAA	750	TTTCTCAATGGCAA AGGTGACACA	1482
ADH6.1	NM_000672	NM_000672.3	ADH6	6111	TGTTGGGAGTAAA CACTTG	19	AACGATTCAGC CCCTTC	751	TCTTGTATCCACCA TCTTGGGCC	1483
ADM.1	NM_001124	NM_001124.1	ADM	3248	TAAGCCCAAGCAC ACGG	20	TGGGCCCTAAA TCCATA	752	CGAGTGAAGTGT CCCACATTTC	1484
AGR2.1	NM_006408	NM_006408.2	AGR2	3245	AGCCAAACATGTGAC TAATTGGA	21	TCTGATCTCCAT CTGCCTCA	753	CAACACGTACCAAC CTTTGCTCT	1485
AGT.1	NM_000029	NM_000029.2	AGT	6112	GATCCAGCCTCACT ATGCC	22	CCAGTTGAGGGA GTTTTCCT	754	TGAGACCTCCACCT TGTCAGGT	1486
AGTR1.1	NM_000685	NM_000685.3	AGTR1	4258	AGCATTCATCGATA CCTGGC	23	CTAAGCATTTG TGCCTCG	755	ATTGTTACCCAATG AAGTCCCGC	1487
AHR.1	NM_001621	NM_001621.2	AHR	3981	CGGCATAGAGAC CGACTT	24	ACATCTTGTTGGG AAAGCA	756	CAGGTAGCCAAAC GGTCCACTC	1488
AIF1.1	NM_032955	NM_032955.1	AIF1	6452	GACGTTGAGTTACC CTGACTTT	25	TCAGGATCATTTT TAGGATGGC	757	ATCTCTTGCCAGCA TCATCCCTGA	1489
AKT1.3	NM_005163	NM_005163.1	AKT1	18	CGCTTCTATGGCG CTGAGAT	26	TCCGGGTACACC ACGTTCTT	758	CAGCCCTGGACTAC CTGGACTCGG	1490
AKT2.3	NM_001626	NM_001626.2	AKT2	358	TCCTGCCACCCTTC AAACC	27	GGCGTAAATTC ATCATCGAA	759	CAGGTCACGTCCGA GGTCGACACA	1491
AKT3.2	NM_005465	NM_005465.1	AKT3	21	TTGTCTCTGCCTTG GACTATCTACA	28	CGAGCATTAGAT TCTCCAACTTGA	760	TCACGGTACACAATC TTTCCGGA	1492
ALDH4.2	NM_003748	NM_003748.2	ALDH4A1	2092	GGACAGGGAAGA CCGTGAT	29	AACCGAAGAAG TCGATGAG	761	CTGCAGCGTCAATCT CCGCTTG	1493
ALDH6A1.1	NM_005589	NM_005589.2	ALDH6A1	6114	GGCTCTTTCAACAG CAGTCC	30	GCATGCTCCACC AGCTCT	762	CAGCACTTTCTTGGC TTCTCCCAAC	1494
AALDOA.1	NM_000034	NM_000034.2	ALDOA	3810	GCCTGTACGTGCC AGCTC	31	TCATCGAGCTT GATCTCG	763	TGCCAGAGCCTCAA CTGTCTCTGC	1495
ALDOB.1	NM_000035	NM_000035.2	ALDOB	6321	CCCTCTACCAAG GACAGC	32	TAACTTGATTCC CACCACGA	764	TCCCTTTTCTCTGA GGATGTTTCTG	1496

66

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ALOX12.1	NM_000697	NM_000697.1	ALOX12	3861	AGTTCCTCAATGGT GCCAAC	33	AGCACTAGCCTG GAGGGC	765	CATGCTGTGAGAC GCTCGACCTC	1497
ALOX5.1	NM_000698	NM_000698.2	ALOX5	4259	GAGCTGAGGACT TCGTGA	34	GAAGCCTGAGGA CTTGCG	766	CCGATGCCGTACA CGTAGACATC	1498
AMACR1.1	NM_014324	NM_014324.4	AMACR	3930	GGACAGTCAGTTTT AGGGTTGC	35	GACAGCCAGAG ACCCAC	767	CAGTAACCTCGGGC CTGTTTCCC	1499
ANGPT1.1	NM_001146	NM_001146.3	ANGPT1	2654	TCTACTTGGGGTGA CAGTGC	36	CTTTTAAAGC CCGACAGT	768	TCACGTGGCTCGAC TATAGAAAATCCA	1500
ANGPT2.1	NM_001147	NM_001147.1	ANGPT2	2655	CCGTGAAAGCTGC TCTGTAA	37	TTGCACTGGGAA GAACAGTC	769	AAGCTGACACAGCC CTCCCAAGTG	1501
ANGPTL2.1	NM_012098	NM_012098.2	ANGPTL2	3982	GCCATCTGCTCAA CTCC	38	TAGTCTCTGCTT ATGCACTCG	770	TCTCCAGAAAGCACCT CAGGCTCCT	1502
ANGPTL3.3	NM_014495	NM_014495.2	ANGPTL3	6505	GTTGCGATTAATGG CAATGT	39	TGCTTTGTGATC CCAAGTAGA	771	CCAATGCAATCCG GAAAACAAAG	1503
ANGPTL4.1	NM_016109	NM_016109.2		3237	ATGACCTCAGATGG AGGCTG	40	CCGGTTGAAGTC CACTGAG	772	CATCGTGGCGCTC TGAATTACTG	1504
ANGPTL7.1	NM_021146	NM_021146.2	ANGPTL7	6115	CTGCACAGACTCCA ACCTCA	41	GCCATCCAGGTG CTTATTGT	773	TCACCCAGGCGGTA GTACACTCCA	1505
ANTXR1.1	NM_032208	NM_032208.1	ANTXR1	3363	CTCCAGGTGTACCT CCAAAC	42	GAGAAAGCTGG GAGACTCTG	774	AGCCTTCTCCACAG CTGGCTACA	1506
ANXA1.2	NM_000700	NM_000700.1	ANXA1	1907	GCCCCTATCCTACC TTCAATCC	43	CTTTAAACCATTA TGGCCTTATGC	775	TCCTCGGATGTCGCT GCCT	1507
ANXA2.2	NM_004039	NM_004039.1	ANXA2	2269	CAAGACACTAAGG GCGACTACCA	44	CGTGTGCGGCTT CAGTCAT	776	CCACCACACAGTA CAGCAGCGCT	1508
ANXA4.1	NM_001153	NM_001153.2	ANXA4	3984	TGGAGGGATGAA GGAAT	45	CTCATACAGGTC CTGGGA	777	TGTCACGAGAGCA TCGTCCAGA	1509
ANXZ5.1	NM_001154	NM_001154.2	ANXA5	3785	GCTCAAGCTTGA AGATGAC	46	AGAACCAACCAAC ATCCGCT	778	AGTACCCTGAAGTGT CCCCACCA	1510
AP-1 (JUN official).2	NM_002228	NM_002228.2	JUN	2157	GACTGCAAGATG GAAACGA	47	TAGCCATAAGGT CCGCTCTC	779	CTATGACGATGCCCT CAACGCCTC	1511
AP1M2.1	NM_005498	NM_005498.3	AP1M2	5104	ACAACGCCGCAC CATCT	48	CTGAGCGGTAT GACATGAG	780	CTTCATCCCGCCTGA TGGTGAATT	1512

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
APAF1.2	NM_181861	NM_181861.1	APAF1	4086	CACAAGGAAGAAG CTGTGA	49	CATCCTGGTTCA CCTTTCAA	781	TGCAATTCAGCAGAA GCTCTCCAAA	1513
APC.4	NM_000038	NM_000038.1	APC	41	GGACAGCAGGAAT GTGTTTC	50	ACCACTCGATT TGTTTCTG	782	CATTGGCTCCCGT GACCTGTA	1514
APOC1.3	NM_001645	NM_001645.3	APOC1	6608	CCAGCCTGATAAAG GTCCTG	51	CACCTGAATCC TTGCTGGA	783	AGGACAGGACCTCC CAACCAAGG	1515
APOE.1	NM_000041	NM_000041.2	APOE	4340	GCCTCAAGAGCTG GTTCCG	52	CCTGCACCTTCT CCACCA	784	ACTGGCGCTGCATG TCTTCCAC	1516
APOL1.1	NM_003661	NM_003661.2	APOL1	6117	CGGACCAAGAACT GTGACC	53	ATTTTGTCTCTGG CCCTCG	785	AGGCATATCTCTCCT GGTGGCTGC	1517
APOLD1.1	NM_030817	NM_030817.1	APOLD1	6118	GAGCAGCTGGAGT CTCGG	54	AGAGATCTTGAG GTGCTGGC	786	CAGCTTCACACAA GTCCAGTCTG	1518
AQP1.1	NM_198098	NM_198098.1	AQP1	5294	GCTTGCTGTATGAC CCCTG	55	AAGCTGACCTC TCCCTCTC	787	ACAGCCTTCCCTCTG CATTGACCT	1519
AREG.2	NM_001657	NM_001657.1	AREG	87	TGTGAGTGAATGC CTTCTAGTAGTGA	56	TGTGTTTCTGTT ATCATACTCTTCT	788	CCGTCCTCGGAGC CGACTATGA	1520
ARF1.1	NM_001658	NM_001658.2	ARF1	2776	CAGTAGAGATCCCC GCAACT	57	ACAAGCACATGG CTATGGAA	789	CTTGTCTCTGGGTCA CCCTGCA	1521
ARG9.1	NM_31920	NM_31920.2		3873	GCATGGGCTACTG CATCC	58	CCACATCGATTCT AGCCAAG	790	AGCTTGTCTCAGTCC GTGCACAAA	1522
ARGHEF18.1	NM_015318	NM_015318.2	ARGHEF18	3008	ACTCTGCTTCCCAA GGGC	59	GAAGCTAGAGGC CCGCTC	791	CTGTTACACGCTCA GCCTGTCTG	1523
ARHA.1	NM_001664	NM_001664.1	RHOA	2981	GGTCCCTCGTCGG TTCTC	60	GTCGCAACTCG GAGACG	792	CCAGGTCTGTCTTT CAGCTACCC	1524
ARHGD1B.1	NM_001175	NM_001175.4	ARHGD1B	3987	TGTTCCCTAGAACAA AGAGGC	61	TGATGGAGGATC AGAGGGAG	793	TAAACCGGGCTTTC ACCCAACCT	1525
ARRB1.1	NM_004041	NM_004041.2	ARRB1	2656	TGCAGGACGCGCT CATCAA	62	GGTTTGGAGGGA TCTCAAAGG	794	CTGGCGAGCAGCG TTACCCTTTC	1526
ASS1.1	NM_054012	NM_054012.3	ASS1	6328	CCCCAGATAAAG GTCATTG	63	TGCGTACTCCAT CAGGTCAAT	795	TCTAGACCGGTTCA AGGGCCG	1527
ATP1A1.1	NM_000701	NM_000701.6	ATP1A1	6119	AGACGCTATTG GAGCTG	64	GGCAGAAAGAGG TGGCAG	796	ACCTAGGACTCGTTC TCCGAGGCC	1528

71

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ATP5E.1	NM_006886	NM_006886.2	ATP5E	3535	CCGCTTTTCGTACA GCAT	65	TGGAGTATCGG ATGTAGCTG	797	TCCAGCCTGTCTCCA GTAGGCCAC	1529
ATP6V1B1.1	NM_001692	NM_001692.3	ATP6V1B1	6293	AACCATGGGAAC GTCTG	66	GGTGATGATCCG CTCGAT	798	CTTCTGAACTTGGC CAATGACCC	1530
AXL.1	NM_001699	NM_001699.3	AXL	3989	TTGCACGCCCTGTCT TCCTAC	67	CTGCACAGAGAA GGGGAGG	799	TATCCACCTCCATC CCAGACAGG	1531
AZU1.1	NM_001700	NM_001700.3	AZU1	6120	CCGAGGCCCTGAC TTCCT	68	GTCCCGGTTGT TGAGAA	800	CCATCGATCCAGTCT CGAAGAGC	1532
B-Catenin.3	NM_001904	NM_001904.1	CTNMB1	769	GGCTCTTGTGCGT ACTGTCCTT	69	TCAGATGACGAA GAGCACAGATG	801	AGGCTCAGTGTGT CTTCCCTGTCAACG	1533
B2M.4	NM_004048	NM_004048.1	B2M	567	GGGATCGAGACAT GTAAGCA	70	TGGAATTCATCC AATCCAAAT	802	CGGCATCTTCAAAC TCCATGATG	1534
BAD.1	NM_032989	NM_032989.2	BAD	7209	GGGTCAGGGGCGCT CGAGAT	71	CTGCTCACTCGG CTCAAACTC	803	TGGGCCACAGAGCAT GTTCCAGATC	1535
BAG1.2	NM_004323	NM_004323.2	BAG1	478	CGTTGTGAGCAGCTT GGAATACAA	72	GTTCAACCTCTT CCTGTGGACTGT	804	CCCAATTAACATGAC CCGGCAACCAT	1536
BAG2.1	NM_004282	NM_004282.2	BAG2	2808	CTAGGGGCAAAA GCATGA	73	CTAAATGCCCAA GGTGAATG	805	TTCCATGCCAGACAG GAAAAAGCA	1537
Bak.2	NM_001188	NM_001188.1	BAK1	82	CCATGCCACCATTT CTACCT	74	CCCAACATAGAC CCACCAAT	806	ACACCCACAGACGTC CTGGCCT	1538
Bax.1	NM_004324	NM_004324.1	BAX	10	CCGCCGTGGACAC AGACT	75	TTGCCGTGAGAA AACATGTCA	807	TGCCACTCGGAAAA GACCTCTCG	1539
BBC3.2	NM_014417	NM_014417.1	BBC3	574	CCTGGAGGTCCT GTACAAT	76	CTAATTTGGGCTC CATCTCG	808	CATCATGGGACTCCT GCCCTTACC	1540
Bcl2.2	NM_000633	NM_000633.1	BCL2	61	CAGATGGACCTAGT ACCCACTGAGA	77	CCTATGATTTAA GGGCATTTTTC	809	TTCCACGCCGAAGG ACAGCGAT	1541
BCL2A1.1	NM_04049	NM_04049.2	BCL2A1	6322	CCAGCTCCATGTA TCATCA	78	TGAAGCTGTTGA GGCAATGT	810	CAGTCAAGCTCAGTG AGCATTTCTCAGC	1542
BCL2L12.1	NM_138639	NM_138639.1	BCL2L12	3364	AACCCACCCCTGTC TTGG	79	CTCAGCTGACGG GAAAGG	811	TCCGGGTAGTCTC AAACTCGAGG	1543
Bclx.2	NM_001191	NM_001191.1	BCL2L1	83	CTTTTGTGGAACCTC TATGGAACA	80	CAGCGGTGAAG CGTTCTT	812	TTCCGCTCTCGGCT GCTGCA	1544

72

73

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
BCRP.1	NM_004827	NM_004827.1	AGCG2	364	TGTACTGGCGAAG AATATTTGGTAAA	81	GCCAGTGATTC TTCCACAA	813	CAGGCATCGATCT CTCACCTGG	1545
BFGF.3	NM_007083	NM_007083.1	NUDT6	345	CCAGGAAGAATGCT TAAGATGTGA	82	TGGTATGGGAG TTGTATTTTCAG	814	TTGCGCAGGTCAATTG AGATCCATCCA	1546
BGN.1	NM_001711	NM_001711.3	BGN	3391	GAGTCGCCAAGG ATGAC	83	CTTGTGTTTCAC CAGGACGA	815	CAAGGGTCTCCAGC ACCTCTACGC	1547
BHLHB3.1	NM_030762	NM_030762.1	BHLHE41	6121	AGGAAGATCCCTC GCAGC	84	TTGAACCTCCGT CCTTCG	816	AGGAAGCTCCCTGA ATCCTTGCGT	1548
BIK.1	NM_001197	NM_001197.3	BIK	2281	ATTCCTATGGCTCT GCAATTGTC	85	GGCAGAGGTGAA TGGCTCTTC	817	CCGGTTAACTGTGG CCTGTGCC	1549
BIN1.3	NM_004305	NM_004305.1	BIN1	941	CCTGCAAAAGGGA ACAAGAG	86	CGTGGTTGACTC TGAATCTCG	818	CTTCGCTCCAGATG GCTCCC	1550
BUR1.1	NM_001716	NM_001716.3	CXCR5	6280	GACCAAGCAGGAA GCTCAGA	87	AGGCTGTGTTTCG GTCAGA	819	CCAGGGCAGCTAC CTGAACTCAA	1551
BNIP3.1	NM_004052	NM_004052.2	BNIP3	3937	CTGACGGAGTAG CTCCAAG	88	GGTATCTGTGG TGTCTGCG	820	CTCTCACTGTGACAG CCCACCTCG	1552
BRCA1.1	NM_007294	NM_007294.3	BRCA1	7481	TCAGGGGGCTAGA AATCTGT	89	CCATTCCAGTTG ATCTGTGG	821	CTATGGGCCCTTCAC CAACATGC	1553
BTRC.1	NM_033637	NM_033637.2	BTRC	2555	GTTGGGACACAGTT GGTCTG	90	TGAAGCAGTCAG TTGTGCTG	822	CAGTCGGGCCAGGA CGGTCTACT	1554
BUB1.1	NM_004336	NM_004336.1	BUB1	1647	CCGAGGTTAATCCA GCACGTA	91	AAGACATGGCGC TCTCAGTTC	823	TGCTGGGAGCCTAC ACTTGGCCC	1554
BUB3.1	NM_004725	NM_004725.1	BUB3	3016	CTGAAGCAGATGG TTCATCAT	92	GCTGATTCCTCAA GAGTCTAACC	824	CCTCGCTTTGTTTAA CAGCCAGG	1556
c-kit.2	NM_000222	NM_000222.1	KIT	50	GAGCAACTGCTTA TGGCTTAATTA	93	GGCACTCGGCTT GAGCAT	825	TTACAGCAGACAGTCA TGGCCGCAT	1557
C13orf15.1	NM_014059	NM_014059.2	C13orf15	6122	TAGAATCTGTGCC AGAGGG	94	CAAGGCTGATT TTAAGGTGA	826	TGCACTCAACCTTCT ACCAGGCCA	1558
C1QA.1	NM_015991	NM_015991.2	C1QA	6123	CGGTCAATCCCAAC CAGG	95	CGGTACAGTGC AGACGA	827	AGAACCCTACCAGAA CCACTCCGG	1559
C1QB.1	NM_000491	NM_000491.3	C1QB	6124	CCAGTGGCTTCAC AGGAC	96	CCCATGGATCT TCATCATC	828	TCCAGGAGGCGTC TGACACAGTA	1560

74

75

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
C20 orf1.1	NM_012112	NM_012112.2	TPX2	1239	TCAGCTGTGAGCT GCGGATA	97	ACGTCCTTAGGT TTGAGGTTAAGA	829	CAGGTCCCAATTGCC GGGCG	1561
C3.1	NM_000064	NM_000064.2	C3	6125	CGTGAAGGAGTGC AGAAAGA	98	ACTCGGTGTCCCG GGACTT	830	ACATCCCACTGCAG ACCTCAGTG	1562
C3AR1.1	NM_004054	NM_004054.2	C3AR1	6126	AAGCCGATCCCA GACTT	99	TGTTAAGTGCCC TTGCTGG	831	CAACCCCAAGAGATT CCGATTGAG	1563
C7.1	NM_000587	NM_000587.2	C7	6127	ATGTCGTAGTGTGA GGCGG	100	AGGCCTTATGCT GGTGACAG	832	ATGCTCTGCCCTCTG CATCTCAGA	1564
CA12.1	NM_001218	NM_001218.3	CA12	6128	CTCTCTGAAGTGT CCTGGC	101	ACAGGAACGTGAG GGGTGCT	833	AGACACCAAGTGTCTT TCCAGGGCT	1565
CA2.1	NM_000067	NM_000067.1	CA2	5189	CAACGTGGAGTTTG ATGACTCT	102	CTGTAAGTGCCA TCCAGGG	834	CCTCCCTTGAGCACT GCTTTGTCC	1566
CA9.3	NM_001216	NM_001216.1	CA9	482	ATCCTAGCCCTGGT TTTTGG	103	CTGCTTCTCAT CTGCACAA	835	TTTGCTGTCAACAGC GTCCG	1567
CACNA2D1.1	NM_000722	NM_000722.2	CACNA2D1	6129	CAACATTAGTAGTG GCCTGT	104	CAGCCAGTGGGT GCCTTA	836	CCATGGGCATAAACACT AAGGCGCAG	1568
CALD1.2	NM_004342	NM_004342.4	CALD1	1795	CACAAAGTTTGAG ACAGTTCAGAA	105	GCAGATTAGCCC TCTACAACTGA	837	AACCAAGCTCAAGA CGCAGGACGAG	1569
CASP1.1	NM_033292	NM_033292.2	CASP1	6132	AGAAAGCCACATA GAGAAGGA	106	TGTGGGATGTCT CCAAGAAA	838	CGCTTTCTGTCTCTTC CACACCAGA	1570
CASP10.1	NM_001230	NM_001230.4	CASP10	6133	ACCTTTCTCTTTGGC CGGAT	107	GTGGGACTGTCT CACTGC	839	TCTACTGCATCTGCC AGCCCTGAG	1571
CASP6.1	NM_032992	NM_032992.2	CASP6	6134	CCTCACACTGGTGA ACAGGA	108	AATTGCACTTGG GTCTTTGC	840	AAAGTCCACTCGGC GCTGAGAAAC	1572
Capase 3.1	NM_032991	NM_032991.2	CASP3	5963	TGAGCCTGAGCAG AGACATGA	109	CCTTCCTGGGTG GTCCAT	841	TCAGCCTGTTCCTATG AAGGCAGAGC	1573
CAT.1	NM_001752	NM_001752.1	CAT	2745	ATCCATTCGATCTC ACCAAGGT	110	TCCGGTTTAAGA CCAGTTTACCA	842	TGGCCTCAAGGA CTACCCCTCTCATCC	1574
CAV1.1	NM_001753	NM_001753.3	CAV1	2557	GTGCTCAACATTG TGTTCC	112	CAATGCTCTCCA TTTACAG	843	ATTTCACTGATCAG TGGGCCTCC	1575
CAV2.1	NM_198212	NM_198212.1	CAV2	6460	CTTCCTGGGACG ACTTG	112	CTCTCGTGCACC CTTCTGG	844	CCCGTACTGTCTATGC CTCAGAGCT	1576

76

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CCL18.1	NM_002988	NM_002988.2	CCL18	3994	GCTCCTGTGCACAA GTTGG	113	TGGAATCTGCCA GGAGGTA	845	CAACAAGAGCTCTG CTGCTCGT	1577
CCL19.1	NM_006274	NM_006274.2	CCL19	4107	GAACGCATCATCCA GAGACTG	114	CCTTCACACGGT CATAGGTT	846	CGTTTCATCTTGGT GAGGTCTCTC	1578
CCL20.1	NM_004591	NM_004591.1	CCL20	1998	CCATGTCTCTGTACC AAGAGTTTG	115	CGCCGCAGAGGT GGAGTA	847	CAGCACTGACATCAA AGCAGCCAGGA	1579
CCL4.2	NM_002984	NM_002984.1	CCL4	4148	GGGTCCAGGAGTA CGTGTATGAC	116	CCTTCCCCTGAAG ACTTCTCTGCT	848	ACTGAAGTGAAGTGC TCA	1580
CCL5.2	NM_002985	NM_002985.2	CCL5	4088	AGGTTCTGAGCTCT GGCTTT	117	ATGCTGACTTCC TTCTCTGGT	849	ACAGAGCCCTGGCA AAGCCAAG	1581
CCNB1.2	NM_031966	NM_031966.1	CCNB1	619	TTCAGGTTGTTGCA GGAGAC	118	CATCTTCTTTGGG CACACAAT	850	TGTCCTCAATTATTGA TCGGTTTATGCA	1582
CCND1.3	NM_053056	NM001758.1	CCND1	88	GCATGTTCTGTCGC CTCTAAGA	119	CGGTGTAGATGC ACAGCTTCTC	851	AAGAGAGCAATCCC CCTGACGGC	1583
CCNE1.1	NM_001238	NM_001238.1	CCNE1	498	AAAGAAGATGATGA CCGGGTTTAC	120	GAGCCTCTGGAT GGTGCAAT	852	CAAACTCAACGTGCA AGCCTCGGA	1584
CCNE2 variant 1.1	NM_057749	NM_057749var1	CCNE2	1650	GGTCACCAAGAAAC ATCAGTATGAA	121	TTCATGTATAATG CAAGGACTGATC	853	CCCAGATAATACAGG TGGCCAACATTCCT	1585
CCNE2.2	NM_057749	NM_057749.1	CCNE2	502	ATGCTGTGGCTCCT TCCTAACT	122	ACCCAAATTGTG ATATACAAAAAG GTT	854	TACCAAGCAACCTAC ATGTCAGAAAAGCCC	1586
CCR1.1	NM_001295	NM_001295.2	CCR1	6135	TCCAAGACCCAATG GGAA	123	TCGTAGGCTTTC GTGAGGA	855	ACTCACCACACCTGC AGCCTTCAC	1587
CCR2.1	NM_000648	NM_000648.1		4109	CTCGGAATCCTG AAACC	124	GACTCTCACTGC CCTATGCC	856	TCCTTCGTTTCGAC ACCGAAGCA	1588
CCR4.2	NM_005508	NM_005508.4	CCR4	6502	AGACCCTGGTGGG GCTAGAA	125	AGAGTTTCTGTG GCCTGGAT	857	TCCTTCAGGACTGCA CCTTTGAAAGA	1589
CCR5.1	NM_000579	NM_000579.1	CCR5	4119	CAGACTGAATGGG GGTGG	126	CTGGTTTGTCTG GAGAAGGC	858	TGGAATAAGTACCTA AGGCGCCCCC	1590
CCR7.1	NM_001838	NM_001838.2	CCR7	2661	GGATGACATGCACT CAGCTC	127	CCTGACATTTC CTTGTCTCT	859	CTCCCATCCCAGTG GAGCCAA	1591
CD105.1	NM_000118	NM_000118.1	ENG	486	GCAGGTGTCAGCA AGTATGATCAG	128	TTTTTCCGCTGT GGTGATGA	860	CGACAGGATATTGAC CACCGCCTCATT	1592

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CD14.1	NM_000591	NM_000591.1	CD14	4341	GTGTGCTAGCGTA CTCCCG	129	GCATGGTGCCG GTTATCT	861	CAAGGAACGACGC TCGAGGACCT	1593
CD18.2	NM_000211	NM_000211.1	ITGB2	49	CGTCAGAGCCAC CATGCT	130	GTTAAATGGTG ACCATCCTCAAGA	862	CGCGCCGAGACAT GGCTTG	1594
CD1A.1	NM_001763	NM_001763.1	CD1A	4166	GGAGTGAAGGAA CTGGAA	131	TCATGGCGGTAT CTACGAAT	863	CGCACCATTCGGTCA TTTGAGG	1595
CD24.1	NM_013230	NM_013230.1	CD24	2364	TCCAAGTAATGCCA CCACCAA	132	GAGAGAGTGAGA CCACGAAGAGAC T	864	CTGTTGACTGCAGG GCACCAACA	1596
CD274.2	NM_014143	NM_014143.2	CD274	4076	GCTGCATGATCAG CTATGGT	133	TGTTGTATGGG CATTGACT	865	CACAGTAATTCGCTT GTAGTCGCACC	1597
CD31.3	NM_000442	NM_000442.1	PECAM1	485	TGTATTTCAAGACC TCTGTGACATT	134	TTAGCCTGAGGA ATTGCTGTGTT	866	TTTATGAACCTGCC TGCTCCACA	1598
CD34.1	NM_001773	NM_001773.1	CD34	3814	CCACTGCACACACC TCAGA	135	CAGAGTTTACC TGCCCTT	867	CTGTTCTTTGGGCC CTACACCTTG	1599
CD36.1	NM_000072	NM_000072.2	CD36	6138	GTAACCCAGGACG CTGAGG	136	AAGTTCGAAGA TGGCACC	868	CACAGTCTCTTTTCT GCAGCCCAA	1600
CD32.1	NM_000734	NM_000734.1	CD247	64	AGATGAAGTGGAA GGCGCTT	137	TGCTCTGTAAT CGGCAACTG	869	CACCGCGGCCATCC TGCA	1601
CD4.1	NM_000616	NM_000616.2	CD4	4168	GTGCTGGAGTCGG GACTAAC	138	TCCTGCATTCA AGAGGC	870	CAGGTCCCTTGTC CAAGTTCAC	1602
CD44.1	NM_000610	NM_000610.3	CD44	4267	GGCACCACTGCTTA TGAAGG	139	GATGCTCATGGT GAATGAGG	871	ACTGGAACCCAGAA GCACACCTC	1603
CD44s.1	M59040	M59040.1		1090	GACGAAGACAGTC CCTGGAT	140	ACTGGGTGGAA TGTGTCTT	872	CACCGACAGCACAG ACAGAAATCCC	1604
CD44v6.1	AJ251595v6	AJ251595v6		1061	CTCATCCAGCCAT CCAATG	141	TTGGGTGAAGA AATCAGTCCC	873	CACCAAGCCAGAG GACAGTTCTT	1605
CD53.1	NM_000560	NM_000560.3	CD53	6139	CGACAGATCCAC CGTTAC	142	TGCAGAAATGAC TGGATGGA	874	CACGCTGCCTTGGT GCTATTGTCT	1606
CD68.2	NM_001251	NM_001251.1	CD68	84	TGTTTCCAGCCCT GTGT	143	CTCTCCACCT GGTTGT	875	CTCCAAGCCAGATT CAGATTGAGTCA	1607
CD82.3	NM_002231	NM_002231.2	CD82	316	GTGAGGCTCAGG TGAAGTG	144	GACCTCAGGCG ATTCAATGA	876	TCAGCTTCTACAAT GGACAGACACGCT G	1608

81

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CD8A.1	NM_171827	NM_171827.1	CD8A	3804	AGGTGAGGTGCT TGAGTCT	145	GGGCACAGTATC CAGGTA	877	CCAAGCGCAAGGA ACAAGTACTTCT	1609
CD99.1	NM_002414	NM_002414.3	CD99	6323	GTTCTCTCCCGTAG CTTTTCA	146	ACCATCACTGCC TCTTTTC	878	TCCACCTGAAACGCC ATCCG	1610
cdc25A.4	NM_001789	NM_001789.1	CDC25A	90	TCTTGCTGGCTACG CCTCTT	147	CTGCATTGTGGC ACAGTTCTG	879	TGTCCCTGTTAGCG TCCTCCGTCCATA	1611
CDC25B.1	NM_021873	NM_021874.1	CDC25B	389	AAACGAGCAGTTTG CCATCAG	148	GTTGGTGATGTT CCGAAGCA	880	CCTCACGGCATAG ACTGGAAGCG	1612
CDH1.3	NM_004360	NM_004360.2	CDH1	11	TGAGTGTCCCCCG GTATCTTC	149	CAGCCCGCTTCA GATTTTCAT	881	TGCCAATCCCGATGA AATTGGAAATTT	1613
CDH13.1	NM_001257	NM_001257.3	CDH13	6140	GCTACTTCTCCTACT GTCCCG	150	CCTCTCTGTGGA CCTGCC	882	AGTCTGAATGCTGCC ACAACCAGC	1614
CDH16.1	NM_004062	NM_004062.2	CDH16	4529	GACTGTCTGTAATGG CCCAG	151	CCAGGGGACTCA GATGGA	883	CAGAGGCCAAGCTC CCAGCTAGAG	1615
CDH2.1	NM_001792	NM_001792.2	CDH2	3965	TGACCGATAAGGAT CAACCC	152	GATCTCCGCCAC TGATTTCTG	884	ATACACCAGCCTGGA ACGCAGTGT	1616
CDH5.1	NM_001795	NM_001795.2	CDH5	6142	ACAGGAGACGTGT TCGCC	153	CAGCAGTGAGGT GGTACTCTGA	885	TATTCTCCCGGTCCA GCCTCTCAA	1617
CDH6.1	NM_004932	NM_004932.2	CDH6	3998	ACACAGGCGACATA CAGGC	154	CTCGAAGGATGT AAACGGGT	886	TTTCTTCCCTGTCCA GCCTCTTGG	1618
CDK4.1	NM_000075	NM_000075.2	CDK4	4176	CCTTCCCATCAGCA CAGTTC	155	TTGGGATGCTCA AAAGCC	887	CCAGTCGCCTCAGTA AAGCCACCT	1619
CDK6.1	NM_001259	NM_001259.5	CDK6	6143	AGTGCCCTGTCTCA CCCA	156	GCAGGTGGGAAT CCAGGT	888	TCCTTGACACCTTCC AGGTCTTGG	1620
CDKN2A.2	NM_000077	NM_000077.3	CDKN2A	4278	AGCACTCACGCCCT AAGC	157	TCATGAAGTCA CAGCTTCC	889	CGCAAGAAATGCCC ACATGAATGT	1621
CEACAM1.1	NM_001712	NM_001712.2	CEACAM1	2577	ACTTGCCCTGTTTCTAG AGCACTCA	158	TGGCAAAATCCGA ATTAGAGTGA	890	TCCTTCCACCCCCA GTCCTGTC	1622
CEBPA.1	NM_004364	NM_004364.2	CEBPA	2961	TTGGTTTTGTCTCGG ATACTTG	159	GTCTCAGACCCT TCCCCC	891	AAAATGAGACTCTCC GTCGGCAGC	1623
CENPF.1	NM_016343	NM_016343.2	CENPF	3251	CTCCCGTCAACAGC GTTTC	160	GGGTGAGTCTGG CCTTCA	892	ACACTGGACCAGGA GTGCATCCAG	1624

82

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CFLAR.1	NM_003879	NM_003879.3	VFLAR	6144	GGACTTTTGTCAG TGACAGC	161	CGCGCTTCTCT CTTACA	893	CTCCTCCCGTGGTC CTTGTGTGCT	1625
CGA (CHGA official).3	NM_001275	NM_001275.2	CHGA	1132	CTGAAGGAGCTCC AAGACCT	162	CAAAACCGCTGT GTTTCTTC	894	TGCTGATGTCCTC TCCTTG	1626
Chk1.2	NM_001274	NM_001274.1	CHEK1	490	GATAAATTGGTACA AGGATCAGCTT	163	GGTGCCAAAGTA ACTGACTATTCA	895	CCAGCCCAATGTC CTGATCATATGC	1627
Chk2.3	NM_007194	NM_007194.1	CHEK2	494	ATGTGGAACCCCCA CCTACTT	164	CAGTCACAGCA CGGTTATACC	896	AGTCCCAACAGAAC AAGAACTTCAGCG	1628
CTAP1.2	NM_001166	NM_001166.2	BIRC2	326	TGCCTGTGGTGG AAGCT	165	GGAAATGCCTC CGGTGTT	897	TGACATAGCATATC CTTTGGTTCCAGTT	1629
cTAP2.2	NM_001165	NM_001165.2	BIRC3	79	GGATATTTCCGTGG CTCTTATTCA	166	CTTCTCATCAAG GCAGAAAAATCT T	898	TCTCCATCAAAATCCT GTAAACTCCAGAGCA	1630
CLCNKB.1	NM_000085	NM_000085.1	CLCNKB	4308	GTGACCTTGAAGC TGTCCC	167	GGTTCAACAGCT CAAGAGGTT	899	AGAGACTTCCCTGCA TGAGGCACA	1631
CLDN10.1	NM_182848	NM_182848.2	CLDN10	6145	GGTGTGGATGA ACTGG	168	GATAGTAAATG CGGTCGGC	900	TGGAAGAAGCCCAAC GCGTTACCT	1632
CLDN7.2	NM_001307	NM_001307.3	CLDN7	2287	GGTGTGCCCTAGT CATCTG	169	GTACCCAGCCTT GCTCTCAT	901	TGCACTGCTCTCCTG TTCCTGTCC	1633
CLU.3	NM_001831	NM_001831.1	CLU	2047	CCCAGGATACCTA CCACTACCT	170	TGCGGACTTGG GAAAGA	902	CCCTTCAGCCTGCC CCACCG	1634
CMet.2	NM_000245	NM_000245.1	MET	52	GACATTTCCAGTCC TGCAGTCA	171	CTCCGATCGAC ACATTTGT	903	TGCCTCTCTGCCCA CCCTTGT	1635
cMYC.3	NM_002467	NM_002467.1	MYC	45	TCCCTCCACTCGGA AGGACTA	172	CGTTTGTGTGCTG ATCTGTCTCA	904	TCTGACACTGTCCAA CTTGACCCCTCTT	1636
COL18A1.1	NM_030582	NM_030582.3	COL18A1	6146	AGCTGCCATCACG CCTAC	173	GTGGCTACTTGG AGGAGTC	905	CGTGCTCTGCATTGA GAACAGCTTC	1637
COL1A1.1	NM_000088	NM_000088.2	COL1A1	1726	TGGCCATCCAGC TGACC	174	CAGTGTAGGTG ATGTTCTGGGA	906	TCCTGCGCTGATGT CCACCG	1638
COL1A2.1	NM_000089	NM_000089.2	COL1A2	1727	CAGCAAGAACTG GTATAGGAGCT	175	AAACTGGCTGCC AGCATTTG	907	TCTCCTAGCCAGACG TGTTTCTTGTCTCTTG	1639
COL4A1.1	NM_001845	NM_001845.4	COL4A1	6147	ACAAAGGCTCTCCA GGAT	176	GAGTCCCAAGAA GACCTGCT	908	CTCCTTTGACACCAG GGATGCCAT	1640

85

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
COL4A2.1	NM_001846	NM_001846.2	COL4A2	6148	CAACCTGGTGAT GTCTGC	177	CGAGTGGTAGA GAGCCAGT	909	ACTATGCCAGCCGG AACGACAAGT	1641
COL5A2.2	NM_000393	NM_000393.3	COL5A2	6653	GGTCGAGGAACCC AAGGT	178	GCTGGAGGTCC AACTCTG	910	CCAGAAATCCTGTA GCACCAGGC	1642
COL7A1.1	NM_000094	NM_000094.2	COL7A1	4984	GGTCACAAAGGAC CTCGG	179	ACCAGGCTCTCC CTTGCT	911	CTTGTACCCAGGT CCCCATTGTC	1643
COX2.1	NM_000963	NM_000963.1	PTGS2	71	TCTGCAGAGTTGGA AGCACTCTA	180	GCCGAGGCTTTT CTACCAGAA	912	CAGGATACAGTCCA CAGCATCGATGTC	1644
CP.1	NM_000096	NM_000096.1	CP	3244	CGTCAGTACACAGA TGCCCTCC	181	CCAGATGCCAA GATGCT	913	TCCTCAGGCGCTCTC TCCTTTGCA	1645
CPB2.1	NM_001872	NM_001872.3	CPB2	9294	GGCACATACGGATT CTTGCT	182	CAGCGCAAAAG CTTCTCTA	914	CGGAGCGTTACATCA AACCACCT	1646
CRADD.1	NM_003805	NM_003805.3	CRADD	6149	GATGTGCCTCCA GCAAC	183	GAGTAAAGTCA GGATTCAGCC	915	CATGACTCAGGAC ACACTCCCA	1647
cripto (TDGF1 official)	NM_003212	NM_003212.1	TDGF1	1096	GGTCTGTGCCCC ATGAC	184	TGACCTGTCCAG CAATTACA	916	CCTGGTGCCCAAG AAGTGTCCCT	1648
CRP.1	NM_000567	NM_000567.2	CRP	4187	GACGTGAACCGACA GGGTGT	185	CTCCAGATAGGG AGCTGGG	917	CTGTCAAGAGAGCC CATCTCCAT	1649
CSF1.1	NM_000757	NM_000757.3	CSF1	510	TGCAGCGGCTGAT TGACA	186	CAACTGTTCTCTG CTCTACAAATC A	918	TCAGATGGAGACCTC GTGCCAAATTACA	1650
CSF1R.2	NM_005211	NM_005211.1	CSF1R	2289	GAGCACACCAAC CTACGA	187	CCTGCAGAGATG GGTATGAA	919	AGCCACTCCCCACG CTGTGT	1651
CSF2.1	NM_000758	NM_000758.2	CSF2	2779	GAACTGAAGGACT TTCTGCTTGT	188	CTCATCTGGCCG GTCTCACT	920	ATCCCCCTTTGACTGC TGGGAGCCAG	1652
CSF2R.2	NM_006140	NM_006140.3	CSF2R	4477	TACCACACCCAGCA TTCCCTC	189	CTAGAGGCTGGT GCCACTGT	921	CGCAGATCCGATTC TCTGGATC	1653
CSF3.2	NM_000759	NM_000759.1	CSF3	377	CCCAGGCTCTGT GTCCTT	190	GGAGGACAGGA GCTTTTCTCA	922	TGCATTTCTGAGTTT CATTCTCCTGCTG	1654
CTGF.1	NM_001901	NM_001901.1	CTGF	2153	GAGTCAAGTGCC GTGACG	191	AGTTGTAATGCC AGGCACAG	923	AACATCATGTTCTTC TTCATGACCTCGC	1655
CTSB.1	NM_001908	NM_001908.1	CTSB	382	GGCCGAGATCTAC AAAAACG	192	GCAGGAAGTCCG AATACACA	924	CCCCGTGGAGGGAG CTTTCTC	1656

86

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CTSD.2	NM_001909	NM_001909.1	CTSD	385	GTACATGATCCCTGTGAGAAAGGT	193	GGACAGCTTGTAGCCTTTGC	925	ACCCTGCCCGCATCACACTGA	1657
CTSH	NM_004390	NM_004390.1	CTSH	845	GCAAGTTCCAACCTGAAAAG	194	CATCGCTTCCTGTGCATAGA	926	TGGCTACATCCTTGA	1658
CTSL.2	NM_001912	NM_001912.1	CTSL1	446	GGAGGCTTTATCTCACTGAGTGA	195	CCATTGCAGCCTTCATTGC	927	TTGAGGCCAGAGCAGTCTACGAGATTCT	1659
CTSL2.1	NM_001333	NM_001333.2	CTSL2	1667	TGCTCTACTGAGCGAGCAGAA	196	ACCATTCGAGCCCTGATTG	928	CTTGAGGACGCGAA	1660
CTSS.1	NM_004079	NM_004079.3	CTSS	1799	TGACAACGGTTTC	197	TCCATCGCTTTGTAGGATAGG	929	TGATAACAAGGGCATCGACTCAGACGCT	1661
CUBN.1	NM_001081	NM_001081.2	CUBN	4110	GAGCCGTTACTGTGGCA	198	GAATCTCAGCGTCAGGGC	930	TGCCCCATCCTATCA	1662
CUL1.1	NM_003592	NM_003592.2	CUL1	1889	ATGCCCTGGTAATGTCTGCAT	199	GGACACACAAGCCTTATCAAG	931	CAGCCACAAGCCA	1663
CUL4A.1	NM_003589	NM_003589.1	CUL4A	2780	AAGCATCTTCCTGTTCTTGGGA	200	AATCCATATCC	932	TATGTGTGCGAGAAC	1664
CX3CL1.1	NM_002996	NM_002996.3	CS3CL1	6150	GACCTTTGCCGTCTACCTG	201	GGAGTGTTCCTAGCACCTGG	933	TAGAACCAGCCCAT	1665
CX3CR1.1	NM_001337	NM_001337.3	CX3CR1	4507	TTCCCAGTTGTGACATGAGG	202	GCTAAATGCAACCGTCTCAGT	934	ACTGAGGGCCAGCC	1666
CXCL10.1	NM_001565	NM_001565.1	CXCL10	2733	GGAGCAAAATCGATGCAAT	203	TAGGGAAGTGATGGAGAGG	935	TCTGTGTGCTCCATC	1667
CXCL12.1	NM_000609	NM_000609.3	CXCL12	2949	GAGCTACAGATGCCCATGC	204	TTTGAGATGCTTGACGTTGG	936	TTCTTCGAAAGCCAT	1668
CXCL14.1	NM_004887	NM_004887.3	CXCL14	3247	TGCGCCCTTTCCTCTGTA	205	CAATGCGGCATATAGTGGG	937	TACCCCTTAAGAACGC	1669
CXCL9.1	NM_002416	NM_002416.1	CXCL9	5191	ACCAGACCATTTGCTCAGAGC	206	GTTACCAGAGGCTAGCCAACA	938	TGCTGGCTCTTTCTT	1670
CXCR4.3	NM_003467	NM_003467.1	CXCR4	2139	TGACCGCTTCTACCCCAATG	207	AGGATAGGCCAACCATGATGT	939	CTGAACTGGAACAC	1671
CXCR6.1	NM_006564	NM_006564.1	CXCR6	4002	CAGAGCCTGACGGATGTGT	208	GCAGATGAGAACCAACACC	940	CTGGTGAACCTACCC	1672

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
CYP2C8.2	NM_000770	NM_000770.2	CYP2C8	505	CCGTGTTCAAGAG GAAGCTC	209	AGTGGATCACA GGGTGAAG	941	TTTTCTCAACTCCTC CACAAAGCA	1673
CYP2C8v2.1	NM_030878	NM_030878.1		5423	GCTGTAGTGCACG AGATCCA	210	CAGTGTGCTACTG CATGGG	942	ATACAGTAGACCTTGT CCCCACCGG	1674
CYP3A4.2	NM_017460	NM_017460.3	CYP3A4	586	AGAAACAAGGACAAC ATAGATCCTTACAT AT	211	GCAAACTCATG CCAATGC	943	CACACCCCTTTGGAAG TGGACCCAGAA	1675
CYP61.1	NM_001554	NM_001554.3	CYP61	2752	TGCTCATTTCTTGAG GAGCAT	212	GTGGCTGCATTA GTGTCCAT	944	CAGCACCCCTTGGCA GTTTCGAAAT	1676
DAG1.1	NM_004393	NM_004393.2	CAG1	5968	GTGACTGGGCTCA TGCTT	213	ATCCCACTTGTG CTCCTGTC	945	CAAGTCAGAGTTTCC CTGGTGCC	1677
DAPK1.3	NM_004938	NM_004938.1	DAPK1	636	CGCTGACATCATGA ATGTTCT	214	TCTCTTTCAGCA ACGATGTGCTT	946	TCATATCCAAACTCG CCTCCAGCCG	1678
DCBLD2.1	NM_080927	NM_080927.3	DCBLD2	3821	TCACCAGGGCAGG AAGTTTA	215	GTTTGCACTAC AGCCCC	947	CATGCCTATGCTGAA CCACTCCCA	1679
DCC.3	NM_005215	NM_005215.1	DCC	3763	AAATGTCCTCCTCG ACTGCT	216	TGAATGCCATCT TTCTTCCA	948	ATCACTGGAACTCCT CGGTGGGAC	1680
DCN.1	NM_001920	NM_001920.3	DCN	6151	GAAGGCCACTATCA TCCTCCT	217	GCCTCTCTGTTG AAACGGTC	949	CTGCTTGCACAAAGTT TCCTGGGCT	1681
DCXR.1	NM_016286	NM_016286.2	DCXR	6311	CCATAGGCTCTACT GCTCCA	218	AGCTTAGGGCC ATCACCT	950	TCAGCATGTCCAGG GCACCC	1682
DDC.1	NM_000790	NM_000790.3	DDC	5411	CAGAGCCGAGACA CCATGA	219	CCACGTAATCCA CCATCTCC	951	CCTCTCCTTTCGGAAT TCACTTGG	1683
DEPB1.1	NM_005218	NM_005218.3	DEPB1	4124	GATGGCCTCAGGT GGTAACT	220	TGCTGACGCAAT TGTATATGAT	952	CTCACAGGCCTTGG CCACAGATCT	1684
DET.1	NM_017996	NM_017996.2	DET1	2643	CTTGTGAGATCAC CCAATCAG	221	CCGCTGTGATC TCAAACT	953	CTATGCCCGGAGT CGGGCTT	1685
DHPS.3	NM_013407	NM_013407.1	DHPS	1722	GGGAGAACGGGAT CAATAGGAT	222	GCATCAGCCAGT CCTCAAACT	954	CTCATTGGGCACCA GCAGTTTCC	1686
DIABLO.1	NM_019887	NM_019887.1	DIABLO	348	CACAATGCCGGCT CTGAAG	223	ACAAAAACACTG TCTGTACCTGAA GA	955	AAGTTACGCTCGC GACAGCCAA	1687
DIAPH1.1	NM_005219	NM_005219.2	DIAPH1	2705	CAAGCAGTCAAGG AGAACCA	224	AGTTTGTCTCGC CTCATCTT	956	TTCTTCTGTCTCCCG CCGCTTC	1688

91

92

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
DICER1.2	NM_177438	NM_177438.1	DICER1	1898	TCCAATTCAGCAT CACTGT	225	GGCAGTGAAGGC GATAAAGT	957	AGAAAAGCTGTTGT CTCCCCAGCA	1689
DKFZP56400823.1	NM_015393	NM_015393.2	PARM1	3874	CAGCTACACTGTCG CAGTCC	226	ATGAGCTGGAG CTTGAGG	958	TGCTGAGCTCCCA CACTCATCTC	1690
DLC1.1	NM_006094	NM_006094.3	DLC1	3018	GATTCAGACGAGG ATGAGCC	227	CACCTCTTGCTG TCCCTTTG	959	AAAGTCCATTGCCA CTGATGGCA	1691
DLL4.1	NM_019074	NM_019074.2	DLL4	5273	CACGAGGTATAA GGCAGGAG	228	AGAAGGAAGTGC CAGCCG	960	CTACTGGACATCCG TGCTCAGCC	1692
DPEP1.1	NM_004413	NM_004413.2	DPEP1	6295	GGACTCCAGATGC CAGGA	229	TAAGCCCAGGCG TCCCTCT	961	CACATGCAAGGACC AGCATCTCCT	1693
DPYS.1	NM_001385	NM_001385.1	DPYS	6152	AAAGAATGGCACCA TGCAG	230	AGTCGGGTGTTG AGGGGT	962	CACCATGTCATGGGT CCACCTTGG	1694
DR4.2	NM_003844	NM_003844.1	TNFRSF10A	896	TGCACAGAGGGTG TGGGTTAC	231	TCTTCATCTGATT TACAAGCTGTAC ATG	963	CAATGCTTCCAACAA TTTGTGTGCTTGCC	1695
DR5.2	NM_003842	NM_003842.2	TNFRSF10B	902	CTCTGAGACAGTG CTTCGATGACT	232	CCATGAGGCCCA ACTTCTT	964	CAGACTTGGTGCCC TTTGACTCC	1696
DUSP1.1	NM_004417	NM_004417.2	DUSP1	2662	AGACATCAGCTCCT GGTTCA	233	GACAAACACCCT TCCTCCAG	965	CGAGGCCATTGACTT CATAGACTCCA	1697
DUSP9.1	NM_001395	NM_001395.1	DUSP9	6324	CGTCTAATCAACG TGCCTA	234	CCCGCAAGAAA AGTAACAG	966	CGCTCGGAGCCTGC CTCTTC	1698
E2F1.3	NM_005225	NM_005225.1	E2F1	1077	ACTCCCTCTACCTT TGAGCA	235	CAGGCCTCAGTT CCTTCAGT	967	CAGAAGAACAGCTCA GGGACCCCT	1699
EBAG9.1	NM_004215	NM_004215.3	EBAG9	4151	CGCTCCTGTTTTTC TCATCTGT	236	ACCGAACTGGG TGATGG	968	CAGTGGGTTTTGATT CCCACCATG	1700
ECRG4.1	NM_032411	NM_032411.1	C2orf40	3869	GCTCCTGCTCCTGT GCTG	237	TTTTGAAGCATC AGCTTGAGTT	969	ATTTCCACTTATGCC ACCTGGGCC	1701
EDC2.1	NM_001401	NM_001401.3	LPAR1	4673	ACGAGTCCATTGCC TTCCTTT	238	GCTTGCTGACTG TGTTCAT	970	CGAAGTGAAAAGCA TCTTGCCACA	1702
EDN1 endothelin.1	NM_001955	NM_001955.1	EDN1	331	TGCCACCTGGACAT CATTTG	239	TGGACCTAGGGC TTCCAAGTC	971	CACTCCCAGACACG TTGTTCGGT	1703
DEN2.1	NM_001956	NM_001956.2	EDN2	2646	CGACAAGAGGTGC GTCTACTTCT	240	CAGGCCGTAAAG AGCTGTCT	972	CCACTTGGACATCAT CTGGTGAACACTC	1704

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
EDNRA.2	NM_001957	NM_001957.1	EDNRA	3662	TTTCCTCAAAATTG CCTCAAG	241	TTACACATCCAA CAGTGCC	973	CCTTTGCCTCAGGG CATCCTTTT	1705
EDNRB.1	NM_000115	NM_000115.1	EDNRB	3185	ACTGTGAAGTCCT GGTGC	242	ACCACAGCATGG GTGAGAG	974	TGTACCTGCCCTT TGTATGTG	1706
EEF1A1.1	NM_001402	NM_001402.5	EEF1A1	5522	CGAGTGGAGACTG GTGTTCTC	243	CCGTTGTAAGT TGACTGGA	975	CAAAGTGACCACC ATACGGGTT	1707
EFNB1.2	NM_004429	NM_004429.3	DFNB1	3299	GGAGCCCGTATCC TGGAG	244	GGATAGATCACC AAGCCCTTC	976	CCCTCAACCCCAAGT TCCTGAGTG	1708
EFNB2.1	NM_004093	NM_004093.2	EFNB2	2567	TGACATTATCATCC CGCTAAGGA	245	GTAGTCCCCCGCT GACCTTCTC	977	CGGACAGCGTCTTC TGCCCTCACT	1709
EGF.3	NM_001963	NM_001963.2	EGF	158	CTTTGCCTTGCTCT GTCACAGT	246	AAATACCTGACA CCCTTATGACAA ATT	978	AGAGTTTAAACAGCC TGCTCTGGCTGACTT	1710
EGFR.2	NM_005228	NM_005228.1	EGFR	19	TGTCGATGGACTTC CAGAAC	247	ATTGGACAGCT TGGATCA	979	CACCTGGCAGCTG CCAA	1711
EGLN3.1	NM_022073	NM_022073.2	RGLN3	2970	GCTGGTCTCTTACT GCGG	248	CCACCATGCGT TAGACCTC	980	CCGGCTGGGCAAAAT ACTAGTCAA	1712
EGR1.1	NM_001964	NM_001964.2	EGR1	2615	GTCCCGCTGCAG ATCTCT	249	CTCCAGCTTAGG GTAGTTGTCCAT	981	CGGATCCTTTCTCTCA CTCGCCCA	1713
EIF2C1.1	NM_012199	NM_012199.2	EIF2C1	6454	CCCTCAGGACTCT CAGC	250	TGGTGACTTCC ACCTTCA	982	CGTTTCGCTTCACCAA GGAGATCAA	1714
EIF4EBP1.1	NM_004095	NM_004095.2	EIF4EBP1	4275	GGCGGTGAAGAGT CACAGT	251	TTGGTAGTGTCT CACACGAT	983	TGAGATGGACATTTA AAGCACGAGCC	1715
ELTD1.1	NM_022159	NM_022159.3	ELTD1	6154	AGGTCTTTGTGCAAG AGGAGC	252	AACCCCAAAGAT CCAGGTG	984	CTCGCTCTTCTGTTC CTTCTCGGC	1716
EMCN.1	NM_016242	NM_016242.2	EMCN	3875	AGGCACTGAGGGT GGAAA	253	CACCGCAAAAT AATACTGGA	984	AATGCAAGCACTTCA GCAACCAGC	1717
EMP1.1	NM_001423	NM_001423.1	EMP1	986	GCTAGACTTTTGAT GCTCCCTTGAT	254	GAACAGCTGGAG GCCAAGTC	986	CCAGAGAGCCTCCC TGCAGCCA	1718
ENO2.1	NM_001975	NM_001975.2	ENO2	6155	TCCTTGGCTTACCT GACCTC	255	AACCCCAATGAG TAGGGCA	987	CTGTCTCTGTCTCGCC CTCCTTTCT	1719
ENPEP.1	NM_001977	NM_001977.3	ENPEP	6156	CACCTACCGGAG AACGGAC	256	CCTGGCATCTGT TGGTTCA	988	TCAAAGCATAGTGG CCACCGATC	1720

95

96

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ENPP2.1	NM_006209	NM_006209.3	ENPP2	6174	CTCCTGGCACTAA TACCTTC	257	TCCTCGATAAT TGGGTCTG	989	TAACTTCTCTGGCA TGGTTGGCC	1721
EPAS1.1	NM_001430	NM_001430.3	EPAS1	2754	AAGCCTTGGAGG TTTCATTG	258	TGCTGATGTTT CTGACAGAAAAG	990	TGTGCCCATCTGG GTCACCACG	1722
EPB41L3.1	NM_012307	NM_012307.2	EPB41L3	4554	TCACTGCCATACGC TCTCAC	259	CTTGGCTCCAG GTAGCA	991	CTCTCTTCCCTCTG GCTCTGTGC	1723
EPHA2.1	NM_004431	NM_004431.2	EPHA2	2297	CGCTGTTCACAA GATTGAC	260	GTGGCTGTGCTC GAAGTC	992	TGCGCCGATGAGA TCACCG	1724
EPHB1.3	NM_004441	NM_004441.3	EPHB1	6508	CCTTGGAGGGAA GATCC	261	GAACTGAATTG CGGTAGGC	993	ATGGCTCTGGAGC TGTCCATCTC	1725
EPHB2.1	NM_004442	NM_004442.4	EPHB2	2967	CAACAGGCAGCT CCATC	262	GTAATGCTGTCC ACGGTGC	994	CACCTGATGATGAT GGACACTGC	1726
EPHB4.1	NM_004444	NM_004444.3	EPHB4	2620	TGAACGGGTATC CTCCTTA	263	AGTACTCTCTCG GTCAGTGG	995	CGTCCCATTTGAGCC TGTCAAATGT	1727
EPO.1	NM_000799	NM_000799.2	EPO	5992	CAGTGCAGCAAT GACATCT	264	CAAGTTGGCCCT GTGACAT	996	CTCTCTGGACAGTTC CTCTGGCCC	1728
Erbb3.1	NM_001982	NM_001982.1	ERBB3	93	CGTTATGTCAATG CAGATACAC	265	GAACTGAGACCC ACTGAAGAAAAG	997	CCTCAAAGGTACTCC CTCCTCCCG	1729
ERBB4.3	NM_005235	NM_005235.1	ERBB4	407	TGGCTCTTAATCAG TTTCGTTACCT	266	CAAGGCATATCG ATCCTCATAAAG T	998	TGTCCACAGAAATAT GCGTAAATTTCCAG	1730
ERCC1.2	NM_001983	NM_001983.1	ERCC1	869	GTCCAGGTGGATG TGAAAGA	267	CGCCAGGATAC ACATCTTA	999	CAGCAGGCCCTCAA GGAGCTG	1731
ERCC4.1	NM_005236	NM_005236.1	ERCC4	5238	CTGCTGGAGTACG AGCGAC	268	GGCGCACACTA CTAGCC	1000	CTGGTGTGGAAC GCTCGACACT	1732
EREG.1	NM_001432	NM_001432.1	EREG	309	ATAACAAAGTGTAG CTCTGACATGAATG	269	CACACCTGCAGT AGTTTGTACTCA	1001	TTGTTTGCATGGACA GTGCATCTATCTGCT	1733
ERG.1	NM_004449	NM_004449.3	ERG	3884	CCAACACTAGGCTC CCCA	270	CTCCGCCAGGT CTTTAGT	1002	AGCCATATGCCCTTCT CATCTGGGC	1734
ERK1.3	NM_002746	Z11696.1	MAPK3	548	ACGATCACAGTGT GAGGAAG	271	CTCATCCGTGG GTCCATAGT	1003	CGTGGCTCACCCC TACCTG	1735
ERK2.3	NM_002745	NM_002745.1	MAPK1	557	AGTTCTTGAACCCCT GGTCCT	272	AAACGGTCAAA GGAGTCAA	1004	TCTCCAGCCCGTCTT GGCTT	1736

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ESPL1.3	NM_012291	NM_012291.1	ESPL1	2053	ACCCCGACCGG ATCAG	273	TGTAGGCGAGC TTCTCAAACA	1005	CTGGCCCTCATGTC CCCTTCAG	1737
ESRRG.3	NM_001438	NM_001438.1	ESRRG	2225	CCAGCACCAATGTT GAAGAT	274	AGTCTCTGGGC ATCGAGTT	1006	CCCAGACCAAAGTG TGAATACATGCT	1738
F2.1	NM_000506	NM_000506.2	F2	2877	GCTCATGTCTGG AAGGTAATG	275	CTGACCGGGTG ATGTTTAC	1007	CCTCGGTAGTTCGTA CCCAGACCCCTCAG	1739
F3.1	NM_001993	NM_001993.2	F3	2871	GTGAAGGATGTGA AGCAGACGTA	276	AACCGGTGCTCT CCACATTC	1008	TGGCACGGGTCTTC TCCTACC	1740
FABP1.1	NM_001443	NM_001443.1	FABP1	6175	GGTCCAAAGTGA TCCAAA	277	CCCTGTCAATTGT CTCCAGC	1009	ACATTCTCTCCCCAC CGTGAATTC	1741
FABP7.1	NM_001446	NM_001446.3	FABP7	4048	GGAGACAAAGTGG TCATCAGG	278	CTCTTCTCCAG CTGGAACCT	1010	TCTCAGCACATTCAA GAACACGGAGA	1742
FAP.1	NM_004460	NM_004460.2	FAP	3403	CTGACCAAGACCAC GGCT	279	GGAAGTGGGTCA TGTGGG	1011	CGGCCTGTCCACGA ACCACTTATA	1743
fas.1	NM_000043	NM_000043.1	FAS	42	GGATTGCTCAACAA CCATGCT	280	GGCAATTAACACT TTTGACGATAA	1012	TCTGACCCCTCCTAC CTCTGGTTCTTACGT	1744
fas1.2	NM_000639	NM_000639.1	FSLG	94	GCACTTTGGGATTC TTTCCATTAT	281	GCAATGAAGAAG ACCCTCACTGAA	1013	ACAACATTCCTCGTG CCTGTACAAAGA	1745
FBXW7.1	NM_033632	NM_033632.1	FBXW7	2644	CCCAGTTTCAACG AGACTT	282	GTTCAGGAATG AAAGCACA	1014	TCATTGCTCCCTAAA GAGTTGGCACTC	1746
FCER1G.2	NM_004106	NM_004106.1	FCER1G	4073	TGCCATCCTGTTC TGTAATGA	283	TGCCTTTCGCAC TTGGATCT	1015	TTGTCTCACCCCTCC TCTACTGTCTGACTG	1747
FCGR3S.1	NM_000569	NM_000569.4	FCGR3A	3080	GTCTCCAGTGGAA GGGAAA	284	AGGAATGCAGCT ACTCACTGG	1016	CCCATGATCTTCAAG CAGGGAAGC	1748
FDPS.1	NM_002004	NM_002004.1	FDPS	516	GGATGATTACCTTG ACCTCTTTTG	285	TGCATTTGTTGT CCTGGATGTC	1017	CAGTGTGACCGGA AAATTGGCAC	1749
FEN1.1	NM_004111	NM_004111.4	FEN1	3938	GTGAGAGAGGTA CGCCAG	286	CTCATGGCAACC AGTCCC	1018	CGCTGAGAGACTCT GTTCTCCCTGG	1750
FGF1.1	NM_000800	NM_000800.2	FGF1	4561	GACACCACGGGC TTTTTA	287	CAGCCTTTCAG GAACAAAC	1019	ACGGCTCACAGACA CCAAATGAGG	1751
FGF2.2	NM_002006	NM_002006.2	FGF2	681	AGATGCAGGAGAG AGGAAGC	288	GTTTTGCAGCCT TACCCAAAT	1020	CCTGCAGACTGCTTT TTGCCCAAT	1752

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
FGF9.1	NM_0002010	NM_002010.1	FGF9	6177	CACAGCTGCCATAC TTCGAC	289	AAGTAAGACTGC ACCTCGC	1021	AGGCCACCCAGCCAG AATCTGATA	1753
FGFR1.3	NM_023109	NM_023109.1		353	CACGGGACATTAC CACATC	290	GGTGCCATCCA CTTCACA	1022	ATAAAAAGACAACCA ACGCCGACTGC	1754
FGFR2 isoform 1.1	NM_000141	NM_000141.2	FGFR2	2632	GAGGACTGTTGG CATGCA	291	GAGTGAGAAATC GATCCAAGTCTT C	1023	TCCCAGAGACCAAC GTTCAAGCAGTTG	1755
FH.1	NM_000143	NM_000143.2	FH	4938	ATGGTTGCAGCCC AAGTC	292	CAAAATGTCCAT TGCTGCC	1024	ACAGTGACAGCAACA TGGTTCCCC	1756
FHIT.1	NM_002012	NM_002012.1	FHIT	871	CCAGTGGAGCGT TCCAT	293	CTCTCTGGGTGG TCTGAACAA	1025	TCGGCCACTTCATCA GGACGCAG	1757
FHL1.1	NM_001449	NM_001449.3	FHL1	4005	ATCCAGCCTTTGCC GAATA	294	CCTTGTAGCTGG AGGACC	1026	TCCTATCTGCCACAC ATCCAGCGT	1758
FIGF.1	NM_004469	NM_004469.2	FIGF	3160	GGTTCCAGCTTTCT GTAGCTGT	295	GCCGAGGTTCT AGTTGCT	1027	ATTGTTGGCCACAC CACCTCCTTA	1759
FILIP1.1	NM_015687	NM_015687.2	FILIP1	4510	ACACCGGTCAACA GTCAT	296	CTGGATGACCC GTCTTG	1028	CCTGACACTGACTG GGTTCTCTGA	1760
FKBP1A.1	NM_000801	NM_000801.2	FKBP1A	6330	CTGCCCTGACTGAA TGTTGT	297	TACGAGGAGAAA GGGAAGA	1029	TCACTCAGCTTTGCT TCCGACACC	1761
FLJ22655.1	NM_024730	NM_024730.2	RERGL	3870	CTCCTTCACACAGA ACCTTTCA	298	AGCAAACTGGG ATCGCT	1030	CACACTCACCCCTAAC CTACTGGCGG	1762
FLT1.1	NM_002019	NM_002019.3	FLT1	6062	GGCCTGAATCTA TCTTTG	299	TCCACAGCAAT ACTCCGTA	1031	CTACAGCACCAAGA GCGACGTGTG	1763
FLT3LG.1	NM_001459	NM_001459.2	FLT3LG	6178	TGGTCCCAAGATG CAAGG	300	GAAAGGCACATT TGGTGACA	1032	AGTGTATCTCCGTGT TCACGCGCT	1764
FLT4.1	NM_002020	NM_002020.1	FLT4	2782	ACCAAGAAGCTGA GGACCTG	301	CCTGGAAGCTGT AGCAGACA	1033	AGCCCGCTGACCAT GGAAGATCT	1765
FN1.1	NM_002026	NM_002026.2	FN1	4528	GGAAGTGACACAC GTGAAGGT	302	ACACGCTAGCCG GTCACCT	1034	ACTCTCAGGCGGTG TCCACATGAT	1766
FOUR1.1	NM_016730	NM_016730.1	FOUR1	859	GAACGCCAAGCAC CACAAAG	303	CCAGGTTCGACA CTGCTCAT	1035	AAGCCAGGCCCCGA GGACAAGTT	1767
FOS.1	NM_005252	NM_005252.2	FOS	2418	CGAGCCCTTTGATG ACTTCCT	304	GGAGCGGGCTG TCTCAGA	1036	TCCCAGCATCATCCA GGCCACAG	1768

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
FRAP1.1	NM_004958	NM_004958.2	MTOR	3095	AGCCTAGAGACT GTGGACC	305	ATGATCCGGGAG GCATAGT	1037	CCTGACGAGTCCC TGGATTTCAC	1769
FRP1.3	NM_003012	NM_003012.2	SFRP1	645	TTGTAACCTGTGG GTTAGCA	306	CACATCCAAATG CAAATGG	1038	TCCCAGGGTAGAAT TCAATCAGAGC	1770
FST.1	NM_006350	NM_006350.2	FST	2306	GTAAGTCGGATGA GCCTGTCTGT	307	CAGCTTCCTTCA TGGCACACT	1039	CCAGTGACAATGCCA CTTATGCCAGC	1771
FXD2.2	NM_001466	NM_001466.2	FZD2	3760	TGGATCCTCACTG GTCCG	308	GGCTGCATGTC TACCAA	1040	TGCGCTTCCACCTTC TTCACCTGTC	1772
G-Catenin.1	NM_002230	NM_002230.1	JUP	770	TCAGCAGCAAGG CATCAT	309	GGTGGTTTCTT GAGCGTGTACT	1041	CGCCCGCAGGCCTC ATCCT	1773
GADD45B.1	NM_015675	NM_015675.1	GADD45B	2481	ACCCTCACAAGAC CACACT	310	TGGAGTTTCATG GGTACAGA	1042	AACCTCAGCCCCAGC TCCCAAGTC	1174
GAS2.1	NM_005256	NM_005256.2	GAS2	6451	AACATGTCATGGTC CGTGTG	311	GGGTCGTGTTT CAACAAAT	1043	CCTGC AAAAGTTTCC CAGCCTCCT	1775
GATA3.3	NM_002051	NM_002051.1	GATA3	1	CAAAAGAGCTCACT GTGGTGCT	312	GAGTCAGAATGG CTTATTACAGAT G	1044	TGTTCCAAACCACTGA ATCTGGACC	1776
GATM.1	NM_001482	NM_001482.2	GATM	6296	GATCTCGCCTTGG ACGAAC	313	GTAGTCGCCTGG GTGCTCT	1045	AAAGTTGCTGCACC CATCCTGTC	1777
GBL.1	NM_022372	NM_022372.3	MLST8	6302	GCTGTCAATAGCAC	314	GFTCACTCGTC ACCAATG	1046	CCCCCGTCAGATTCCC AGACATAGC	1778
GBP2.2	NM_004120	NM_004120.2	GBP2	2060	GCATGGGAACCAT CAACCA	315	TGAGGAGTTTGC CTTGATTCG	1047	CCATGGACCAACTTC ACTATGTGACAGAGC	1779
GCLC.3	NM_001498	NM_001498.1	GCLC	330	CTGTTGCAGGAAG GCATTGA	316	GTCAATGGGTCT CTAATAAGAGA TGAG	1048	CATCTCCTGGCCCA GCATGTT	1780
GCLM.2	NM_002061	NM_002061.1	GCLM	704	TGTAGAATCAAACT CTTCATCATCAACT AG	317	CACAGAATCCAG CTGTGCAACT	1049	TGCAGTTGACATGG CCTGTTCACTCC	1781
GFRAL.1	NM_005264	NM_005264.3	GFRAL	6179	TCCGGTTAGAAC AAGCC	318	GTGGCAAAACAT GAGTGGG	1050	TTTCATTCTCAGACC CTGCTGGCC	1782
GJAL.1	NM_000165	NM_000165.2	GJAL	2600	GTTCACTGGGGT GTATGG	319	AAATACAACAT GCACCTCTCTT	1051	ATCCCCCTCCCTCTCC ACCCATCTA	1783

103

104

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
GLYAT.1	NM_201648	NM_201648.2	GLYAT	6180	TACCATTCGCAAGT GCC	320	GGATGCTGGGA GGCTCTT	1052	AGGATTTCTCCAGCA TCTGCAGCA	1784
GMN.1	NM_015895	NM_015895.3	GMN	3880	GTTGCTACGAGG ATTGAGC	321	TGCTACCCACT TCCTGC	1053	CCTCTGCCACTTA CTGGGTGA	1785
GNAS.1	NM_000516	NM_000516.3	GNAS	2665	GAACGTGCCTGAC TTTGACTT	322	ACTCTTTCATCC TCCACAG	1054	CCTCCCAGAAATCTAT GAGCATGCC	1786
GPC3.1	NM_004484	NM_004484.2	GPC3	659	TGATGCCTGGA AACAGT	323	CGAGGTGTGAA AGGTGCTTATC	1055	AGCAGCAACTCCG AAGGACAACG	1787
GPX1.2	NM_000581	NM_000581.2	GPX1	2955	GCTTATGACCGACC CCAA	324	AAAGTTCAGGC AACATCGT	1056	CTCATACCTGGTCT CCGGTGTGT	1788
GPX2.2	NM_002083	NM_002083.1	GPX2	890	CACACAGATCTCCT ACTCCATCCA	325	GGTCCAGCAGTG TCTCTGAA	1057	CATGTCATCCTAA GGCTCCTCAGG	1789
GPX3.1	NM_002084	NM_002084.3	GPX3	6271	GCTCTAGTCCAAT TGTTCTGC	326	TGAGGCAAGTG GGAGATG	1058	ACTGATACCTCAAC TTGGGGCCA	1790
GRB14.1	NM_004490	NM_004490.1	SRG14	2784	TCCACATGAAGCCC TTTCAG	327	AGTGCCAGGCG TAAACATC	1059	CCTCCAGCAGATC CTTCTTCAACCG	1791
GRB7.2	NM_005310	NM_005310.1	GRB7	20	CCATCTGCATCCAT CTTGT	328	GGCCACCAGGT ATTATCTG	1060	CTCCCCACCCTTGAG AAGTGCCT	1792
GR01.2	NM_001511	NM_001511.1	CXCL1	86	CGAAAAGATGCTGA ACAGTGACA	329	TCAGAACAGCC ACCAGTGA	1061	CTTCTCTCTCCCTTC TGGTCAGTTGGAT	1793
GSTM1.1	NM_000561	NM_000561.1	GSTM1	727	AAGCTATGAGAAA AGAAATACAGAT	330	GGCCAGCTTGA ATTTTCA	1062	TCAGCCACTGGCTTC TGTCAATATCAGAG	1794
GSTM3.2	NM_000849	NM_000849.3	GSTM3	731	CAATGCCATCTTGC GCTACAT	331	TCCACTCGAAT CTTTCTTCTTCA	1063	CTCGCAGCACACA TGTGTGTTGAGA	1795
GSTP.3	NM_000852	NM_000852.2	GSTP1	66	GAGACCCTGCTGT CCAGAA	332	GGTTGTAGTCAG CGAAGGAGATC	1064	TCCACAAATGAAGT CTTGCTCTCCCT	1796
GSTT1.3	NM_000853	NM_000853.1	GSTT1	813	CACATCCCAACC TGCT	333	GGCTCAGTGTG CATCATCT	1065	CACAGCCCTGAA AGCCACAT	1797
GZMA.1	NM_6144	NM_6144.2	GZMA	4111	GAAAGATTCCCT ATCCATGC	334	TGCTTTTTCGT CAGCTGTAA	1066	AGCCACACGCGAAG GTGACCTTAA	1798
HADH.1	NM_005327	NM_005327.2	HADH	6181	CCACACACAGA CCGATTC	335	CCACAAGTTCA TGACGGC	1067	CTGGCTCCATTTCT TCAACCCAG	1799

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
HAVCR1.1	NM_012206	NM_012206.2	HAVCR1	6284	CCACCCAAAGGTCA CGACTAC	336	GAACAGTGGTGC TCGTTTCG	1068	TCACAACTGTTCCAA CCGTCAAGA	1800
HDAC1.1	NM_004964	NM_004964.2	HDAC1	2602	CAAGTACCACAGC GATGACTACATTAA	337	GCTTGCTGTACT CCGACATGTT	1069	TTCTTGGCTCCATC CGTCCAGA	1801
Hepsin.1	NM_002151	NM_002151.1	HPN	814	AGGCTGCTGGAGG TCATCTC	338	CTTCTCTCGGCC ACAGTCT	1070	CCAGAGGCCGTTTC TTGGCCG	1802
HER2.3	NM_004448	NM_004448.1	ERBB2	13	CGGTGTGAGAAGT GCAGCAA	339	CTCTCTCGAAGT GCTCCAT	1071	CCAGACCATAGCACA CTCGGGCAC	1803
HGD.1	NM_000187	NM_000187.2	HGD	6303	CTCAGGTCTGCC CTACAAT	340	TTATTGGTGTCTC CGTGGAC	1072	CTGAGCAGCTCTCA GGATCGGCTT	1804
HGF.4	M29145			457	CCGAATCCAGATG ATGATG	341	CCCAAGGAATGA GTGGATT	1073	CTCATGGACCCCTGG TGCTACACG	1805
HGFAC.1	NM_001528	NM_001528.2	HGFAC	2704	CAGACACAAAGTG CCAGATT	342	GCAGGAGCTG GAGTAGC	1074	CGTCAACGTTCTCAT CCAAAGTGG	1806
HIF1A.3	NM_001530	NM_001530.1	HIF1A	399	TGAACATAAAGTCT GCAACATGGA	343	TGAGGTTGGTTA CTGTTGGTATCA TATA	1075	TTGACTGCACAGG CCACATTCAAC	1807
HIFAN.1	NM_17902	NM_17902.2	HIF1AN	7211	TGTTGGCCAGGTC TCACTG	344	GCATCATAGGGC CTGGAG	1076	CTCTAGCCAGTTAGC CTCGGGCAG	1808
HIST1H1D1.1	NM_005320	NM_005320.2	HIST1H1D	4013	AAAAGGCGAAGAA GGCAG	345	GCTCAGATACTG GGGGTCC	1077	AACTGCTGGGAAC GCAAGGATC	1809
HLA-B.1	NM_005514	NM_005514.6	HLA-B	6334	CTTGTGAGGGACT GAGATGC	346	TGCAGAAAGAGA TGCACAGAG	1078	TCITTCACGCCTCCCC TTTGTGA	1810
HLA-DPA1.1	NM_033554	NM_033554.2	HLA-DPA1	6314	CGCCCTGAAGACA GAATGT	347	TCGGAGACTCAG CAGGAAA	1079	TGATCTTGAGAGCCC TCTCCTTGGC	1811
HLA-DPB1.1	NM_002121	NM_002121.4	HLA-DPB1	1740	TCCATGATGGTTCT GCAGGTT	348	TGAGCAGCACCA TCAGTAACG	1080	CCCCGGACAGTGGC TCTGACG	1812
HLA-DQB1.1	NM_002123	NM_002123.3	HLA-DQB1	6304	GGTCTGCTCGGTG ACAGATT	349	CTCTGATCATTT CCGAAACC	1081	TATCCAGGCCAGATC AAAGTCCGG	1813
HLADQA1.2	NM_002122	NM_002122.3	HLA-DQA1	4071	CATCTTCTCTCTG TGGTCA	350	GCTGGTCTCAGA AACACCTTC	1082	TGTGACTGACTGCC CATTGCTCAG	1814
HMB1.1	NM_002128	NM_002128.3	HMB1	2162	TGGCTGTGTCATTG GTGAT	351	GCTTGTATCTGT CAGCAGTGT	1083	TTCCACATCTCTCCC AGTTTCTTCGCAA	1815

107

108

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
HNRPA1.1	NM_004499	NM_004499.3	HNRPA1	6051	AGCAGGACGACC AACTGA	352	GTTGCCAAGTT AAATTTGGTACAT AAT	1084	CTCCATATCCAAACA AAGCATGTGTGCG	1816
HPCAL1.1	NM_002149	NM_002149.2	HPCAL1	6182	CAGCAGATGGAC ACCAA	353	GTCGCTCTTGGC ACCTCT	1085	TCTTCCAGGACAGT TTGCCGTCA	1817
HPD.1	NM_002150	NM_002150.2	HPD	6183	AGCTGAAGACGGC CAAGAT	354	CGTCGTAGTCCA CAGGAT	1086	AGTCCTCCAGGGC ATCAATGTTT	1818
HSD11B2.1	NM_000196	NM_000196.3	HSD11B2	6185	CCAACCTGCCTCAA GAGC	355	GGAAGTGCCTGG GCAAGT	1087	CTGCAGGCTACGG CAAGGACTAC	1819
HSP90AB1.1	NM_007355	NM_007355.2	HSP90AB1	5456	GCATTGTGACCCAG CACCTAC	356	GAAATGCTGGG CTTTCTAT	1088	ATCCGCTCCATATTG GCTGTCCAG	1820
HSPA1A.1	NM_005345	NM_005345.4	HSPA1A	2412	CTGCTGCGACAGT CCACTA	357	CAGTTTCGCTCT GGGAAG	1089	AGAGTACTCCCGTT GTCCCAAG	1821
HSPA8.1	NM_006597	NM_006597.3	HSPA8	2563	CCTCCCTCTGGTG GTGCTT	358	GCTACATCTACA CTTGGTTGGCTT AA	1090	CTCAGGGCCACCA TTGAAGAGTTG	1822
HSPB1.1	NM_001540	NM_001540.2	HSPB1	2416	CCGACTGGAGGAG CATAAA	359	ATGCTGGCTGAC TCTGCTC	1091	CGCACTTTTCTGAGC AGACGTCCA	1823
HSPG2.1	NM_005529	NM_005529.2	HSPG2	1783	CAGTACGTGTGCC GAGTGTT	360	CTCAATGGTGAC CAGACA	1092	CAGCTCCGTGCTC TAGAGGCCT	1824
HTATIP.1	NM_006388	NM_006388.2	KAT5	3893	TCGAATTGTTTGGG CACTG	361	GCGTGTGCTGA CGGTAT	1093	TGAGGACTCCCGAG ACAGCTCTGA	1825
HYAL1.1	NM_153281	NM_153281.1	HYAL1	4524	TGGCTGTGGAGTT CAAAATG	362	CCAATCACCCACA TGCTCTTC	1094	CGATGCTACCCCTGG CTGGCAG	1826
HYAL2.1	NM_033158	NM_033158.2	HYAL2	5192	CAACCATGCACTCC CAGTC	363	ACTAAGCCCCGT GAGCCT	1095	TCTTCACACGACCCA CCTACAGCC	1827
HYAL3.1	NM_003549	NM_003549.2	HYAL3	6298	TATGTCCGCTCAC ACACC	364	CAATGACTGCA CAAGGTCA	1096	TGGACAGGAACCT CCCAGATCTC	1828
ICAM1.1	NM_000201	NM_000201.1	ICAM1	1761	GCAGACAGTGACC ATCTACAGCTT	365	CTTCTGAGACCT CTGGCTTCGT	1097	CCGGCGCCCAACGT GATTCT	1829
ICAM2.1	NM_000873	NM_000873.2	ICAM2	2472	GGTCATCTGACAC TGCAAC	366	TGCACCTCAATGG TGAGGAC	1098	TTGCCACAGCCAC CAAAGTG	1830
ICAM3.1	NM_002162	NM_002162.3	ICAM3	7219	GCCTTCAATCTCAG CAACG	367	GAGAGCAATTGC AGTACAC	1099	CAGAGGATCCGACT GTTGCCAGTC	1831

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ID1.1	NM_002165	NM_002165.1	ID1	354	AGAACCGCAAGGT GAGCAA	368	TCCAACCTGAAGG TCCTCGATG	1100	TGAGATTCTCCAGC ACGTCAATCGAC	1832
ID2.4	NM_002166	NM_002166.1	ID2	37	AACGACTGCTACTC CAAGCTCAA	369	GGATTTCATCT TGCTCACCTT	1101	TGCCAGCATCCCC CAGAACAA	1833
ID3.1	NM_002167	NM_002167.3	ID3	6052	CTTCACCAATFCCC TTCTTG	370	CTCTGGCTCTTC AGCCACA	1102	TCACAGTCCTTCGCT CCTGAGCAC	1834
IFI27.1	NM_005532	NM_005532.2	IFI27	2770	CTCTCCGATTGAC CAAGTT	371	TAGAACCTCGCA ATGACAGC	1103	CAGACCCCAATGGAG CCCAGGAT	1835
IGF1.2	NM_000618	NM_000618.1	IGF1	60	TCCGAGAGCTGTGA TCTAAGGA	372	CGGACAGAGCGA GCTGACTT	1104	TGTATTGGCACCACC TCAAGCCTG	1836
IGF1R.3	NM_000875	NM_000875.2	IGF1R	413	GCATGGTAGCCGA AGATTCA	373	TTTCCGGTAATA GTCTGTCTCATA GATATC	1105	CGCGTCATACCAGAA TCTCCGATTTGA	1837
IGF2.2	NM_000612	NM_000612.2	IGF2	166	CCGTGCTTCCGGA CAACTT	374	TGAGCTGCTTCC AGGTGTCA	1106	TACCCCGTGGCAA GTTCTTCCAA	1838
IGFBP2.1	NM_000597	NM_000597.1	IGFBP2	373	GTGGACAGCACCA TGAACA	375	CCTTCATACCCG ACTTGAGG	1107	CTTCGGCCAGCAC TGCCTC	1839
IGFBP3.1	NM_000598	NM_000598.4	IGFBP3	6657	ACATCCCAACGCAT GCTC	376	CCACGCCCTTGT TTCAGA	1108	ACACCACAGAGGC TGTGAGCTCC	1840
IGFBP5.1	NM_000599	NM_000599.1	IGFBP5	594	TGGACAAAGTACGG GATGAAGCT	377	CGAAGGTGTGGC ACTGAAAGT	1109	CCCGTCAACGTACTC CATGCCTGG	1841
IGFBP6.1	NM_002178	NM_002178.1	IGFBP6	836	TGAACCGCAGAGA CCAACAG	378	GTCTTGACACC CCGAGAA	1110	ATCCAGGCACCTCTA CCACGCCCTC	1842
IL-7.1	NM_000880	NM_000880.2	IL7	2084	GCGGTGATTCCGA AATTCG	379	CTCTCCTGGGCA CTTGCTT	1111	CTCTGGTCTCATCC AGGTGCGC	1843
IL-8.1	NM_000584	NM_000584.2	IL8	2087	AAGGAACCATCTCA CTGTGTGTAAAC	380	ATCAGGAAGGCT GCCAAGAG	1112	TGACTTCCAGCTGG CCGTGGC	1844
IL10.3	NM_000572	NM_000572.1	IL10	909	GGCGCTGTTCATCG ATTTCCT	381	TGGAGCTTATTA AAGGCATCTTC	1113	CTGCTCCACCGCCT TGCTCTTG	1845
IL11.2	NM_000641	NM_000641.2	IL11	2166	TGGAAGGTTCCACA AGTCAC	382	TCTTGACCTTGC AGCTTTGT	1114	CCTGTGATCAACAGT ACCCGTATGGG	1846
IL15.1	NM_000585	NM_000585.2	IL15	6187	GGCTGGGTACCAA TGCTG	383	TGAGACCCAGTA GTCAGTGGTT	1115	CAGTATGCTGGTA GGCTCCTGCC	1847

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
IL1B.1	NM_000576	NM_000576.2	IL1B	2755	AGCTGAGGAGAT GCTGGTT	384	GGAAGAAGGTG CTCAGGTC	1116	TGCCACAGACCTTC CAGGAGAAT	1848
IL6.3	NM_000600	NM_000600.1	IL6	324	CCTGAACCTTCCAA AGATGG	385	ACCAGCAAGTC TCCTCAT	1117	CCAGATTGGAAGCAT CCATCTTTTCA	1849
IL6ST.3	NM_002184	NM_002184.2	IL6ST	2317	GGCCTAATGTTCCA GATCCT	386	AAAATTGTGCCT TGGAGGAG	1118	CATATTGCCAGTGG TCACCTACA	1850
ILT-2.2	NM_006669	NM_006669.1	LILRB1	583	AGCCATCACTCTCA GTGCAG	387	ACTGCAGATCA GGGTCTCC	1119	CAGTCTCTATCGTG GCCCTGA	1851
IMP3.1	NM_0018285	NM_0018285.2	IMP3	4751	GTGGACTGTCTCA AGATCAA	388	GGTTCAGATC GAAGTCAT	1120	CTCATTTGTAATCTAG CACGTGCCGC	1852
INDO.1	NM_002164	NM_002164.3	IDO1	5124	CGCCTTGACGTCT AGTTC	389	ATCTCATGACC TTTGCCCC	1121	ACATATGCCATGGTG ATGCATCCC	1853
INHBA.1	NM_002192	NM_002192.1	INHBA	2635	GTGCCGAGCCAT ATAGCA	390	CGTAGTGTGTG ATGACTGTTGA	1122	ACGTCCGGGTCTC ACTGTCTCTCC	1854
INHBB.1	NM_002193	NM_002193.1	INHBB	2636	AGCTCCAGGATAC CAGCAA	391	TCCTCCAGTGAC AGGCATTG	1123	AGCTAAGTGCCATT TGTCAACG	1855
INSR	NM_0010798	NM_0010798.17	INSR	6455	CAGTCTCCGAGAG CGGATT	392	GTGATGCGAGGT GAAGCC	1124	AGTTCTCTCAATGAGG CCTCGGTCA	1856
IQGAP2.1	NM_006633	NM_006633.2	IQGAP2	6453	AGAGACACCAGCA ACTGCG	393	ATCATTTGCACGG CTCACC	1125	CCGTGGCATGTGCT ACCTCCTGTT	1857
ISG20.1	NM_002201	NM_002201.4	ISG20	6189	GTGCAGACTGAA GCCCCAT	394	GTTGCTGTCCCA AAAAGCC	1126	AAAGCTCTAGTCCC TGCAGACG	1858
ITGA3.2	NM_002204	NM_002204.1	ITGA3	840	CCATGATCCTCACT CTGCTG	395	GAAGCTTTGTAG CCGGTGAT	1127	CACTCCAGACCTCG CTTAGCATGG	1859
ITGA4.2	NM_000885	NM_000885.2	ITGA4	2867	CAACGCTTCAGTGA TCAATCC	396	GTCTGGCCGGG ATTCTTT	1128	CGATCCTGCATCTGT AAATCGCCC	1860
ITGA5.1	NM_002205	NM_002205.1	ITGA5	2668	AGGCAGCCCTAC ATTATCA	397	GTCTTCTCCACA GTCCAGCA	1129	TCTGAGCCTTGTCT CTATCCGGC	1861
ITGA6.2	NM_000210	NM_000210.1	ITGA6	2791	CAGTGACAAACAGC CCTTCC	398	GTTTAGCCTCAT GGCGTC	1130	TCGCCATCTTTTGTG GGATTCTTT	1862
ITGA7.1	NM_002206	NM_002206.1	ITGA7	259	GATATGATTGGTCG CTGCTTTG	399	AGACTTCCATT CCCCACCAT	1131	CAGCCAGACCTGG CCATCCG	1863

113

114

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
ITGAV.1	NM_002210	NM_002210.2	ITGAV	2671	ACTCGACTGCACA AGCTATT	400	TGCCATCACCAT TGAATCT	1132	CCGACAGCCACAGA ATAACCCAAA	1864
ITGB1.1	NM_002211	NM_002211.2	ITGB1	2669	TCAGAAATGGATTT GGCTCA	401	CCTGAGCTTAGC TGGTGTG	1133	TGCTAATGTAAGGCA TCACAGTCTTTTCCA	1865
ITGB3.1	NM_000212	NM_000212.2	ITGB3	6056	ACCGGGAGGCCCT ACATGA	402	CTTAAGCTCTTT CACTGACTCAAT CT	1134	AAATACCTGCAACCG TTACTGCCGTGAC	1866
ITGB4.2	NM_000213	NM_000213.2	ITGB4	2793	CAAGTGCCTCA GTGGA	403	GGCACACCTTC ATCTCAT	1135	CACCAACCTGTACCC GTATTGCA	1867
ITGB5.1	NM_002213	NM_002213.3	ITGB5	2670	TCGTGAAAGATGAC GAGGAG	404	GGTGAACATCAT GACGCAGT	1136	TGCTATGTTTCTACA AAACGCCAAGG	1868
JAG1.1	NM_000214	NM_000214.1	JAG1	4190	TGGCTTACACTGGC AATGG	405	GCATAGCTGTGA GATCGG	1137	ACTCGATTGCCAGC CAACCACAG	1869
K-ras.10	NM_033360	NM_033360.2	KRAS	3090	GTCAAAATGGGGA GGGACTA	406	CAGGACCAACAC AGAGTGAG	1138	TGTATCTTTGTTGAGC TATCCAACTGCCCC	1870
KCNJ15.1	NM_002243	NM_002243.3	KCNJ15	6299	GGACGTTTACCTG CCTTGA	407	AGCTCTGGAAA CACTGGTC	1139	TCACTCCGAGGTCA GGTGCTTC	1871
KDR.6	NM_002253	NM_002253.1	KDR	463	GAGGACGAAGGCC TCTACAC	408	AAAAATGCCTCC ACTTTTC	1140	CAGGCATGCAGTGT TCTTGGCTGT	1872
Ki-67.2	NM_002417	NM_002417.1	MKI67	145	CGGACTTTGGGTG CGACTT	409	TTACAACTCTTCC ACTGGACGAT	1141	CCACTTGTGGAACCA CCGCTCGT	1873
KIAA1303 raptor.1	NM_020761	NM_020761.2	RPTOR	6300	ACTACAGCGGAG CAGGAG	410	GGCATCTGAGCA AGAGGT	1142	TGGAGGTAGCTGCA ATCAACCCAA	1874
KIF1A.1	NM_004321	NM_004321.4	KIF1A	4015	CTCCTACTGCTCGC ACACC	411	TCCCGGTACACC TGCTTC	1143	CCTGAGGACATCAAC TACGGTCG	1875
Kitlng.4	NM_000899	NM_000899.1	KITLG	68	GTCCCGGGATGG ATGTT	412	GATCAGTCAAGC TGTCGTACAATT	1144	CATCTCGTTATCCA ACAATGACTTGGCA	1876
KL.1	NM_004795	NM_004795.2	KL	6191	GAGTCTCTGTCTAA ACCCGTG	413	CTATGTCAAGG CCCTCAA	1145	CCTGAGGATCTGT CTCACTGGCA	1877
KLK3.1	NM_001648	NM_001648.2	KLK3	4172	CCAACTTACCACC TGCAC	414	AGGTGAGGAAG ACAACCG	1146	ACCCACATGGTGACA CAGCTCTCC	1878
KLRK1.2	NM_007360	NM_007360.1	KLRK1	3805	TGAGAGCAGGCT TCTTTGA	415	ATCCTGGTCCTC TTTGCTGT	1147	TGTCTAAAATGCCA GCCTTCTGAA	1879

115

116

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
KRT19.3	NM_002276	NM_002276.1	KRT19	521	TGACGGCAGAAAT CAGGAGTA	416	TGCGGTAGGTGG CAATCTC	1148	CTCATGGACATCAAG TCGCGGCTG	1880
KRT5.3	NM_000424	NM_000424.2	KRT5	58	TCAGTGAAGAAG AGTTGGA	417	TGCCATATCCAG AGGAACA	1149	CCAGTCAACATCTT GTTGTACACAGA	1881
KRT7.1	NM_005556	NM_005556.3	KRT7	4016	TTCAGAGATGAACC GGGC	418	ACTTGGCAGCT GGTTCT	1150	ATGTTGTCGATCTCA GCCTGCAGC	1882
L1CAM.1	NM_000425	NM_000425.2	L1CAM	4096	CTTCTGGCCAATG CCTA	419	TGATTTCCGCA GTCAGG	1151	ATCTACGTTGTCCAG CTGCCAGCC	1882
LAMA3.1	NM_000227	NM_000227.2	LAMA3	2529	CAGATGAGGCACAT GGAGAC	420	TGAAATGGCAG AACGGTAG	1152	CTGATTCCTCAGGTC CTTGCGCTG	1884
LAMA4.1	NM_002290	NM_002290.3	KAMA4	5990	GATGCACTGCGGT TAGCAG	421	CAGAGGATACGC TCAGCAC	1153	CTCTCCATCGAGGAA GGCAAAATCC	1885
LAMB1.1	NM_002291	NM_002291.1	LAMB1	3894	CAAGAGACTGGG AGGTGTC	422	CGCAGAACTGA CAGTGTC	1154	CAAGTGCTGTACCA CACGGAAGG	1886
LAMB3.1	NM_000228	NM_000228.1	LAMB3	2530	ACTGACCAAGCCTG AGACCT	423	GTCACTCTTGA GCATTTCA	1155	CCACTCGCCATACTG GGTGCACT	1887
LAMC2.2	NM_005562	NM_005562.1	LAMC2	997	ACTCAAGCGGAAAT TGAAGCA	424	ACTCCCTGAAGC CGAGACACT	1156	AGTCTTTATCAGCAC AGTCTCCGCTCC	1888
LAPTMS.1	NM_006762	NM_006762.1	LAPTMS	4017	TGCTGGACTTCTGC CTGAG	425	TGAGATAGGTGG GCACTTCC	1157	TCCTGACCCCTCTGCA GCTCCTACA	1889
LDB1.2	NM_003893	NM_003893.4	LDB1	6720	AACACCCAGTTTGA CGCAG	426	CAAGTCACAGG GAGTTGT	1158	AAAGCTGTCCTCGTC GTCAATGCC	1890
LDB2.1	NM_001290	NM_001290.2	LDB2	3871	ATCAGGTGGACT GCGAC	427	TACCTTGGTAAA CATGGGCTTC	1159	AGTGATCATTGGTCA CCCAGCACG	1891
LDHA.2	NM_005566	NM_005566.1	LDHA	3935	AGGATACACATCCT GGGCTA	428	CCGCTAAGAT TCTTCATT	1160	TCTGCCAAATCTGCT ACAGAGAGTCCA	1892
LGALS1.1	NM_002305	NM_002305.3	LGALS1	6305	GGGTGGAGTCTTC TGACAGC	429	AGACCAACAGCC ATGATTGA	1161	CCCGGGAACATCCT CCTGGAC	1893
LGALS3.1	NM_002306	NM_002306.1	LGALS3	2371	AGCGAAAATGGC AGACAAAT	430	CTTGAGGTTTG GGTTTCCA	1162	ACCCAGATAACGAT CATGGAGCGA	1894
LGALS9.1	NM_009587	NM_009587.2	LGALS9	5458	AGTACTCCACCCG GTGC	431	GACAGCTGCACA GAGCCAT	1163	CTTCCACCGTGTG ACACCATCTC	1895

117

118

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
LIMK1.1	NM_016735	NM_016735.1		3888	GCTTCAGGTGTGTGACTGC	432	AAGAGTGCCTTCCTTCTC	1164	TGCCTCCTGTGCGACCACTACTA	1896
LMNB1.1	NM_005573	NM_005573.1	LMNB1	1708	TGCAACGCTGGTGTACA	433	CCCAACGAGTTC TGGTTCTTC	1165	CAGCCCCCAACTGACCTCATC	1897
LMO2.1	NM_005574	NM_005574.2	LMO2	5346	GGCTGCCAGCAGACATC	434	CTCAGCCAGTCC TCGTGC	1166	CGTACTTCTCTGAAGGCCATCGAC	1898
LOX.1	NM_002317	NM_002317.3	LOX	3394	CCAATGGGAGAAC AACGG	435	CGGTAGGCTG GTACTGTG	1167	CAGGCTCAGCAAGCTGAACCTG	1899
LRP2.1	NM_004525	NM_004525.1	LRP2	4112	GGCTGTGACTGG GTTTCCA	436	GAGACAAAGAGG CCAATCCAG	1168	CGGGCATCCAAACCA GTAGAGCTTT	1900
LRRC2.1	NM_024512	NM_024512.2	LRRC2	6315	CCAGTGTCCCAATC TGTGTC	437	GGTCAGGTTATT GCTGCTGA	1169	CCACTGCAATTCGA CATCCG	1901
LTF.1	NM_002343	NM_002343.2	LTF	6269	AACGGAAGCCTGT GACTGA	438	AGACACACGGC ATGATTCT	1170	CTAGAAGCTGCCATC TTGCCATGG	1902
LYZ.1	NM_0002939	NM_0002939.1	LY	6268	TTGCTGCAAGATAA CATCCG	439	ACCATGCTCTA ATGCCTTG	1171	CACGACAAACCTCTT TTGCACAAG	1903
MADH2.1	NM_005901	NM_005901.2	SMAD2	2672	GCTGCCTTTGGTAA GAACATGTC	440	ATCCCAGCAGTC TCTTCACAACT	1172	TCCATCTTTGCCATTCCACGCCG	1904
MADH4.1	NM_005359	NM_005359.3	SMAD4	2565	GGCAATTACTGGC CTGTTCA	441	ACCAATACTCAG GAGCAGGATGA	1173	TGCATTCAGCCTCC CATTTCCA	1905
MAL.1	NM_002371	NM_002371.2	MAL	6194	GTTGGGAGCTTGC TGTGTC	442	CACAAACAGGAG GTGACCCCT	1174	ACCTCCAACTGCTGTGCTGTCTGC	1906
MAL2.1	NM_052886	NM_052886.1	MAL2	5113	CCTTCGTCTGCCCTG GAGAT	443	GGAACATTTGAG GAGGCAA	1175	CAAAATCCAGACAAGACCCCGAA	1907
MAP2K1.1	NM_002755	NM_002755.2	MAP2K1	2674	GCCTTTCTTACCCA GAAGCAGAA	444	CAGCCCCCAGCT CACTGAT	1176	TCTCAAAGTCTGTCATCCTTCAGTTCTCCCA	1908
MAP2K3.1	NM_002756	NM_002756.2	MAP2K3	4372	GCCCTCCAATGTCC TTATCA	445	GTAGCCACTGAT GCCAAAATC	1177	CACATCTTCACATGGCCCTCCTTG	1909
MAP4.1	NM_002375	NM_002375.2	MAP4	2066	GCCGGTCAGGCAC ACAAG	446	GCAAGCATACACA CAACAAAATGG	1178	ACCACACAGTCCACGCTCCAGGG	1910
MARCKS.1	NM_002356	NM_002356.4	MARCKS	4021	CCCTCTTGATCT GTTGAG	447	CGGTCTTGGAGA ACTGGG	1179	CCCATGCTGGCTTCTTCAACAAAG	1911

119

120

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
Maipin.2	NM_002639	NM_002639.1	SERPINE5	362	CAGATGCCACTTT GAGAACATT	448	GGCAGCATTAAC CACAGAGATT	1180	AGCTGACACAGTGT GACGACGAGACC	1912
MCAM.1	NM_006500	NM_006500.2	MCAM	3972	CGAGTTCAGTGG CTGAGA	449	TGCACTGAAGC ACAGGC	1181	CTTTCCAGACCTGG CCTGTCTCT	1913
MCM2.2	NM_004526	NM_004526.1	MCM2	580	GACTTTTGCCCGCT ACCTTTC	450	GCCACTAACTGC TTCAGTATGAAG AG	1182	ACAGCTCATTTGTTGT CACGCCGA	1914
MCM3.3	NM_002388	NM_002388.2	MCM3	524	GGAGAACAAATCCC CTTGAGA	451	ATCTCCTGGATG GTGATGGT	1183	TGGCTTTTCTGTCTA CAAGATCACCA	1915
MCM6.3	NM_005915	NM_005915.2	MCM6	614	TGATGGTCTTATGT GTCACATTCA	452	TGGACACGAAA CACACCAA	1184	CAGGTTTCATACCAA CACAGGCTTCAGCA C	1916
MCP1.1	NM_002982	NM_002982.1	CCL2	700	CGCTCAGCCAGAT GCAATC	453	GCACTGAGATCT TCCTATTGGTGA	1185	TGCCCCAGTCACCT GCTGTTA	1917
MDH2.1	NM_005918	NM_005918.2	MDH2	2849	CCAACACCTTTGTT GCAGAG	454	CAATGACAGGGA CGTTGACT	1186	CGAGCTGATCCAA ACCTTCAG	1918
MDK.1	NM_002391	NM_002391.2	MDK	3231	GGAGCCGACTGCA AGTACA	455	GACTTTTGGTGCC TGTGCC	1187	ATCAGCAGCACCCCA GTTCTCAA	1919
MDM2.1	NM_002392	NM_002392.1	MDM2	359	CTACAGGAGCGCC ATCGAA	456	ATCCAACCAATC ACCTGAATGTT	1188	CTTACACCAGCATCA AGATCCGG	1920
MGMT.1	NM_002412	NM_002412.1	MGMT	689	GTGAATGAAACGC ACCACA	457	GACCCCTGCTCAC AACCAGAC	1189	CAGCCCTTTGGGGA AGCTGG	1921
MGST1.2	NM_020300	NM_020300.2	MGST1	806	ACGGATCTACCACA CCATTGC	458	TCCATATCCAAC AAAAAACTCAAA G	1190	TTTGACACCCCTTCC CCAGCCA	1922
MICA.1	NM_000247	NM_000247.1	MICA	5449	ATGTTGAATGTCAC CCGC	459	AAGCCAGAAGCC CTGCAT	1191	CGAGGCTCAGAGG GCAACATTAC	1923
MIF.2	NM_002415	NM_002415.1	MIF	3907	CCGACAGGGTCT ACATCA	460	GGTGGAGTTGTT CCAGCC	1192	CTATTACAGCATGAA CGCGGCCAA	1924
MMP1.1	NM_002421	NM_002421.2	MMP1	2167	GGGAGATCATCGG GACAACTC	461	GGSCCTGGTTGA AAAGCAT	1193	AGCAGATTTCCTCC AGGTCCATCAAAAGG	1925
MMP10.1	NM_002425	NM_002425.1	MMP10	4920	TGTACCCACTCTAC AACTCATCTCA	462	TGATGCCATTTC ACATCATCTTG	1194	AGCTGCCAGTTTC CGCCTTC	1926

121

122

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
MMP14.1	NM_004995	NM_004995.2	MMP14	4022	GCTGTGGAGCTCT CAGGAA	463	AGCAAGGACAGG GACCAA	1195	CCTGAGGAGGCAC ACTTGCTCCT	1927
MMP2.2	NM_004530	NM_004530.1	MMP2	672	CCATGATGGAGAG GCAGACA	464	GGATCCGTCCT TACCGTCAA	1196	CTGGAGCANGCG ATGGATACCC	1928
MMP7.1	NM_002423	NM_002423.2	MMP7	2647	GGATGGTAGCAGT CTAGGATTAACT	465	GGAATGTCCCAT ACCCAAAGAA	1197	CCTGTATGCTGCAAC TCATGAACCTGGC	1929
MMP9.1	NM_004994	NM_004994.1	MMP9	304	GAGAACCAATCTCA CCGACA	466	CACCCGAGTGTA ACCATAGC	1198	ACAGGTATTCCTCTG CCAGCTGCC	1930
MRP1.1	NM_004996	NM_004996.2	ABCC1	15	TCATGGTGCCCGT CAATG	467	CGATGTCTTTG CTCTTCATGTG	1199	ACCTGATACGTCCTG GTCTTCATCGCCAT	1931
MRP2.3	NM_000392	NM_000392.1	ABCC2	55	AGGGATGACTTG GACACAT	468	AAACTGCAATGG CTTTGTCA	1200	CTGCCATTCGACATG ACTGCAATT	1932
MRP3.1	NM_003786	NM_003786.2	ABCC3	8	TCATCTGGGGGAT TACTTCCT	469	CGTTGAGTGGA ATCAGCAA	1201	TCTGTCTGGCTGG AGTCGCTTTCAT	1933
MRP4.2	NM_005845	NM_005845.3	ABCC4	6057	AGCGCTGGGAATC TACAACT	470	AGAGCCCTGGA GAGAGAT	1202	CGGATCCAGTGTTT TCCCACTTA	1934
MSH2.3	NM_000251	NM_000251.1	MSH2	2127	GATGCAGAAATTGAG GCAGAC	471	TCTTGCAAGTC GGTTAAGA	1203	CAAGAAGATTACTTT CGTCGATTCCCGA	1935
MSH3.2	NM_002439	NM_002439.1	MSH3	2132	TGATTACCATCATG GCTCAGA	472	CTTGTGAAAATG CCATCCAC	1204	TCCCAATTGTCGCTT CTTCTGCAG	1936
MSH6.3	NM_000179	NM_000179.1	MSH6	2136	TCTATTGGGGGATT GGTAGG	473	CAAAATTGCGAGT GGTGAAT	1205	CCGTTACCAGCTGG AAATTCTCTGAGA	1937
MT1B.1	NM_005947	NM_005947.1	MT1B	5355	GTGGGCTGTGCCA AGTGT	474	ACAGCAGCGCA CTTCTC	1206	ATGAGCCTTTCGAGA CACAGCCCT	1938
MT1G.1	NM_005950	NM_005950.1	MT1G	6333	GTGACCCACTGC CTCTT	475	AGCAGTTGGGT CCATTG	1207	CCGAGGCGGAGACT AGAGTTCCC	1939
MT1H.1	NM_005951	NM_005951.2	MT1H	6332	CGTGTTCACCTGCC TCATT	476	AGCAGTTGGGT CCATTG	1208	CCGAGGTGAGACTG GAGTTCCCA	1940
MT1X.1	NM_005952	NM_005952.2	MT1X	3897	CTCTGCAATGCA AAGAGTG	477	ACTTGCACAGC CCACAG	1209	CACCTCCTGCAAGAA GAGCTGCTG	1941
MUC1.2	NM_002456	NM_002456.1	MUX1	335	GGCCAGGATCTGT GGTGGTA	478	CTCCAGTCGTG GACATTGA	1210	CTCTGGCCTTCCGA GAGGTACC	1942

123

124

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
MVP.1	NM_017458	NM_017458.1	MVP	30	ACGAGAACGAGGG CATCTATGT	479	GCATGTAGTGC TTCCAATCAC	1211	CGCACCTTTCCGGT CTTGACATCCT	1943
MX1.1	NM_002462	NM_002462.2	MX1	2706	GAAGGAATGGGAA TCAGTCATGA	480	GTCATTAGAGT CAGATCCGGGAC AT	1212	TCACCCCTGGAGATCA GCTCCCGA	1944
MYBL2.1	NM_002466	NM_002466.1	MYBL2	1137	GCCGAGATCGCCA AGATG	481	CTTTTGATGGTA GAGTTCAGTGA TTC	1213	CAGCATTGTCTGTCC TCCCTGGCA	1945
MYH11.1	NM_002474	NM_002474.1	MYH11	1734	CGGTACTTCTCAGG GCTAATATATACG	482	CCGAGTAGATGG GCAGGTGTT	1214	CTCTTCTGCGTGGT GGTCAACCCCTA	1946
MYRIP.2	NM_015460	NM_015460	MYRIP	1704	CC TTCACCTTCCTC GTCAAC	483	AGCAGCTCTTGC AGCAATTG	1215	ATTTGCAATCTCCAC ACTGGCGCT	1947
NBN.1	NM_002485	NM_002485	NBN	4121	GCATCTACTTGCCA GAACCAA	484	TCCCTTGCAGCT GGAGTT	1216	CTTCCAGTTCTTGGC TGCTTGCAG	1948
NCF1.1	NM_000265	NM_000265	NCF1	4676	GACACCTTCATCCG TCACAT	485	ATAGTCTGGCT GGGTACG	1217	AAGCGCTTCTCAAAG CCCAGCAG	1949
NFAT5.1	NM_006599	NM_006599	NFAT5	3071	CTGAACCCCTCTCC TGATC	486	AGGAACGATGG CGAGGT	1218	CGAGAATCAGTCCC CGTGGAGTTC	1950
NFATC2.1	NM_173091	NM_173091	NFATC2	5123	CAGTCAAGGTCAGA GGCTGAG	487	CTTTGGCTCGTG GCATTC	1219	CGGGTTCTTACCCC ACAGTCAATC	1951
NFKBp50.3	NM_003998	NM_003998	NFKB1	3439	CAGACCAAGGAGA TGGACCT	488	AGCTGCCAGTGC TATCCG	1220	AAGCTGTAACATGA GCCGACCA	1952
NFKBp65.3	NM_021975	NM_021975	RELA	39	CTGCCGGGATGGC TTCTAT	489	CCAGGTTCTTGA AACTGTGGAT	1221	CTGAGCTCTGCCCG GACCGCT	1953
NFX1.1	NM_002504	NM_002504	NFX1	4025	CCCTGCCATACCAG CTCA	490	CGTCCACATTCA CACTGTAGC	1222	CCTGCCCTGTGACT GCTTGTAAGC	1954
NME2.1	NM_002512	NM_002512	NME2	3899	ATGCTTGGGGAGA CCAATC	491	CTGAATGCAGAA GTCCCCAC	1223	AGCAGATTCAAAGCC AGGCACCAT	1955
NNMT.1	NM_006169	NM_006169	NNMT	5101	CCTAGGCAGGGA TGGAG	492	CTAGTCCAGCCA AACATCCC	1224	CCCTCTCTCATGCC CAGACTCTC	1956
NOL3.1	NM_003946	NM_003946	NOL3	6307	CAGCTTTGGGAAG TGAGACT	493	ATGATGTGTGTG GCCTTTGT	1225	CTCAAGGTCCCTTTC TGCTCCCCCT	1957
NOS2A.3	NM_000625	NM_000625	NOS2	6509	GGGTCCATTATGAC TCCCAA	494	GCTCATCTGGAG GGTAGG	1226	TGTCCCTGGGTCTT CTGGTCAAAAC	1958

125

126

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
NOS3.1	NM_000603	NM_000603	NOS3	2624	ATCTCCGCTCGCT CATG	495	TCGAGCCATAC AGGATTGTC	1227	TTCACGCTTCGCC ATCACCG	1959
NOTCH1.1	NM_017617	NM_017617	NOTCH1	2403	CGGTCCACCACT TTGAATG	496	GTTGTATTGGTT CGGCACCAT	1228	CCGCTCTGCAGCG GGACA	1960
NOTCH2.1	NM_024408	NM_024408	NOTCH2	2406	CACCTCCCTGCTGG GATTAT	497	AGTTGTCAAACA GGCAGCTCG	1229	CCGTGTTGCACAGC TCATCACACT	1961
NOTCH3.1	NM_000435	NM_000435	NOTCH3	6464	TGTGGACGAGTGT GCTGG	498	ACTCCCTGCCAG GTGGT	1230	ACCCTGTGGCCCTC ATGGTAICTG	1962
NPD009 (ABAT official)	NM_020686	NM_020686	ABAT	1707	GGCTGTGGCTGAG GCTGTAG	499	GGAGCATTCGAG GTCAAATCA	1231	TTCACAGAGTGTCTC ACCTCCAGCAGAG	1963
NPM1.2	NM_002520	NM_002520	NPM1	2328	AATGTTGTCCAGGT TCTATTGC	500	CAAGCAAAGGT GGAGTTC	1232	AACAGGCATTTTGGGA CAACACATTCTTG	1964
NPPB.1	NM_002521	NM_002521.2	NPPB	6196	GACACCTGCTTCTG ATTCCAC	501	TGAGTCACTTCA AAGCGG	1233	AGGGCTTTTTCCTC AACCCTGTG	1965
NPR1.1	NM_000906	NM_000906.2	NPR1	6197	ACATCTGCAGCTCC CCTG	502	CACACAAAGCAG CTTCCA	1234	CCTTCAGAAACCTCA TGCTCTCG	1966
NPY1R.1	NM_000909	NM_000909.4	NPY1R	4513	GGATCTTCCCACT CTGCT	503	TTGCTTTTTCGC TCCTGC	1235	CCTTCCATTCCCAAC CTTCTTCT	1967
NRG1.3	NM_013957	NM_013957.1	NRG1	410	CGAGACTCTCTCA TAGTGAAGGTAT	504	CTTGGCGTGTGG AAATCTACAG	1236	ATGACCCACCCGGC TCGTATGTCA	1968
NUDT1.1	NM_002452	NM_002452.3	NUDT1	4564	ACTGGTTTCCACTC CTGCTT	505	GTCCAGGATGGT GTCCCTGA	1237	CCACGGGTACTTCAA GTTCCAGGG	1969
OGG1.1	NM_002542	NM_002542.4	OGG1	6196	ACCAAGGTGGCTG ACTGC	506	ATATGACATCC ACGGGC	1238	TCTGCCTGATGGCC CTAGACAAGC	1970
OPN, osteopontin.3	NM_000582	NM_000582.1	SPPI	764	CAACCGAAGTTTTC ACTCCAGTT	507	CCTCAGTCCATA AACCACACTATC A	1239	TCCCCACAGTAGACA CATATGATGGCCG	1971
p21.3	NM_000389	NM_000389.1	CDKN1A	33	TGGAGACTCTCAG GGTCGAAA	508	GGCGTTTGGAGT GGTAGAAATC	1240	CGCGCGCAGACCAG CATGAC	1972
p27.3	NM_004064	NM_004064.1	CDKN1B	38	CGGTGGACCAAGA AGAGTTAA	509	GGCTCGCCTCTT CCATGTC	1241	CCGGGACTTGGAGA AGCACTGCA	1973
p53.2	NM_000546	NM_000546.2	TP53	59	CTTTGAACCCCTTGC TTGCA	510	CCCGGACAAAG CAATG	1242	AAGTCTCGGTGCTT CTGACGAC	1974

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
PAH.1	NM_000277	NM_000277.1	PAH	6199	TGCTGATTCATT AACAGTGA	511	CACATTCTGTCC ATGGCTTTAC	1243	ATCCTTTGCAGTGCC CTCCAGAAA	1975
PAI1.3	NM_000602	NM_000602.1	SERPINE1	36	CCGCAACGTGGTTT TCTCA	512	TGCTGGTTTCT CCTCCGTGT	1244	CTCGTGTGGCCA TGCTCCAG	1976
Pak1.2	NM_002576	NM_002576.3	PAK1	3421	GAGTGTGGGTG TTATGGA	513	CCATGCAAGTTT CTGTCACC	1245	ACATCTGTCAAGGAG CCTCCAGCC	1977
PAR6A.1	NM_016948	NM_016948.2	PAR6A	4514	GATCCTCGAGGTC AATGGC	514	ACCATCATGTCC GTCACTTG	1246	TCCAAGGTCTTCCCG GCTACTTCA	1978
PBOB1.1	NM_021635	NM_021635.1	PBOV1	3936	GCAAGCCTTTCCA GAAAAA	515	GGCTGGGCTTAA ACAGTCAT	1247	TGGTAGCAGAATTGC CTTTTCACCA	1979
PCCA.1	NM_000282	NM_000282.2	PCCA	6250	GGTGAATCTGTGC ACTGTCA	516	ATTCCAGCTCCA CGAGCA	1248	TCCCTTCTCCAACT GTGTCTCCA	1980
PCK1.1	NM_002591	NM_002591.2	PCK1	6251	CTTAGCATGGCCCA GCAC	517	CTTCCGGAACCA GTTGACA	1249	CAGCCAAACTGCCC AAGATCTTCC	1981
PNCA.2	NM_002592	NM_002592.1	PCNA	148	GAAGGTGTTGGAG GCACCTCAAG	518	GGTTACACCGC TGGAGCTAA	1250	ATCCCAGCAGGCCT CGTTGATGAG	1982
PCSK6.1	NM_002570	NM_002570.3	PCSK6	6282	ACCTTGAGTAGCAG AGGCC	519	GTTGCTGGAGCC ATTTCCAC	1251	CACACCTTCTCTCAGA ATGGACCCC	1983
PDCD1.1	NM_005018	NM_005018.2	PDCD1	6286	GACAAAGCCACCTT CACC	520	GGCTCATGCGGT ACCACT	1252	TCTCCAACACATCGG AGAGCTTCG	1984
PDE4DIP.1	NM_014644	NM_014644.4	PDE4DIP	6417	GCCTTCGTTTGCTG TGAGAG	521	AGCTTCATTGGA GGAGAGGA	1253	TCGGCGAGTCTCTCT AAGTCATGATCTC	1985
PDGFA.3	NM_002607	NM_002607.2	PDGFA	56	TTGTTGGTGTGCCC TGGTG	522	TGGGTTCTGTCC AAACACTGG	1254	TGGTGGCGGTCACT CCCTCTGC	1986
PDGFB.3	NM_002608	NM_002608.1	PDGFB	67	ACTGAAGAGAGACC CTTGGAG	523	TAAATAACCCCTG CCACACACA	1255	TCTCTGCCGATGC CCCTAGG	1987
PDGFC.3	NM_016205	NM_016205.1	PDGFC	29	AGTTACTAAAAAT ACCACAGGTCCTT	524	GTCGGTGAAGTA TTTTGTGCAA	1256	CCCTGACACCGGTC TTTGTCTCAACT	1988
PDGFD.2	NM_025208	NM_025208.2	PDGFD	31	TATCAGAGGCAGGT CATACCA	525	TAACGCTTGGCA TCATCAT	1257	TCCAGGTCAACTTTT GACTTCCGGT	1989
PDGFRA.2	NM_006206	NM_006206.2	PDGFRA	24	GGGAGTTTCCAAG AGATGGA	526	CTTCAACCACT TCCCAAAAC	1258	CCCAGACCCGACC AAGCACTAG	1990

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
PDGFRB.3	NM_002609	NM_002609.2	PDGFRB	464	CCAGCTCTCCTTCC AGCTAC	527	GGGTGGCTCTCA CTTAGCTC	1259	ATCAATGTCCCTGTC CGAGTGCTG	1991
PDZK1.1	NM_002614	NM_002614.3	PDZK1	6319	AATGACCTCCACCT TCAACC	528	CGCAGAGAAG CCATAGTT	1260	TGCCCTTCTTGCTTG GACAGTTTACA	1992
PDZK3.1	NM_015022	NM_015022.2		3885	GAGCTGAGAGCCT TGAGCAT	529	CTCGGCCCTGCT GAGTAA	1261	CTCGCTGCAGAGCT TGTCAAGGTC	1993
PF4.1	NM_002619	NM_002619.1	PF4	6326	GCACTGCTCTGTGT GTGAAG	530	GGCCTTGATCAC CTCACG	1262	TCCGTCCAGGCAC ATCACC	1994
PFKP.1	NM_002627	NM_002627.3	PFKP	6252	AGCTGATGCCGCA TACATT	531	GGTGCTCCACGT TGGACT	1263	CAGATCCCTGATGTC GAAGGGCTC	1995
PFN2.1	NM_053024	NM_053024.1	PFN2	3426	TCTATACGTGATG GTGACTGC	532	GCCACAGCCAC ATTGTAT	1264	CTCCCCACCTTGACT CTTTGTCCG	1996
PGF.1	NM_002632	NM_002632.4	PGF	4026	GTGTTTTTCCCTCG GAGC	533	AGCAAGGAACA GCCTCAT	1265	ATCTTCTCAGACGTC CCGAGCCAG	1997
PI3K.2	NM_002646	NM_002646.2	PIK3C2B	368	TGCTACCTGAGCA GCCCG	534	AGCCCTCCTTC AGTAACCA	1266	TCCTCTCGAAACGAG CTGTGTCTGACTT	1998
PI3KC2A.1	NM_002645	NM_002645.2	PI3KC2A	6064	ATACCAATCACC GC AAAACC	535	CACACTAGCATT TTCTCCGCATA	1267	TGTGCTGTGACTGG ACTTAACAAATAGCC	1999
PIK3CA.1	NM_006218	NM_006218.1	PIK3CA	2962	GTGATTGAAGAGCA TGCCAA	536	GTCTCTGCTGGG AATAGC	1268	TCCTGTCTCTCGGA TACAGCCA	2000
PLA2G4C.1	NM_003706	NM_003706.1	PLA2G4C	6249	CCCTTTCCCAAGT AGAAGAG	537	AGGATGTAGCAG CTGGCG	1269	CCTTGGACCACAAAT CCAGCTCAG	2001
PLAT.1	NM_033011	NM_033011.2	PLAT	6459	GATTGTCTGGAA GTGCTGT	538	TAGCTGATGCC TGGTCC	1270	TAGATACCAGGCC ACGTGCTACG	2002
PLAUR.3	NM_002659	NM_002659.1	PLAUR	708	CCCATGGATGCTC CTCTGAA	539	CCGGTGGTACC AGACATTG	1271	CATTGACTGCCGAG GCCCCATG	2003
PLG.1	NM_000301	NM_000301.1	PLG	6310	GGCAAAATTCCAA GACCAT	540	ATGTATCCATGA GCGTGTGG	1272	TGCCAGGCCCTGGGA CTCTCA	2004
PLN.1	NM_002667	NM_002667.2	PLN	3886	TGATGCTTCTCTGA AGTTCTGC	541	CCTGCTGTGATG GGATGAC	1273	AGATCTGCAGCTTC CACATCAGC	2005
PLOD2.1	NM_000935	NM_000935.2	PLOD2	3820	CAGGAGGTGGTT GCAAAAT	542	TCTCCAGGATG CATGAAG	1274	TCCAGCCTTTTCGTG GTGACTCAA	2006

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
PLP1.1	NM_000533	NM_000533.3	PLP1	4027	AGAACAGACTGGC CTGAGGA	543	CAGAGGGCCATC TCAGGTT	1275	CACCATAGCCACCA GCAACTGCT	2007
PMP22.1	NM_000304	NM_000304.2	PMP22	6254	CCATCTACACGGTG AGGCA	544	TGTAGCGAAAC CGTAGGA	1276	AATCCAGTTGAGAT GCCACTCCG	2008
PPAP2B.1	NM_003713	NM_003713.3	PPAP2B	6457	ACAAGACCAATCCC AGTGA	545	CACAGAAGAAAC TATGCAGCAG	1277	ACCAAGGCTCCTTG AGCAAAATCCT	2009
PPARG.3	NM_005037	NM_005037.3	PPARG	1086	TGACTTTATGGAGC CCAAGTT	546	GCCAAGTCGCTG TCATCTAA	1278	TTCAGTGCATTGAA CTTCACAGCA	2010
PPP1R3C.1	NM_005398	NM_005398.4	PPP1R3C	6320	TTCCTTCCCTCTCA ATCCAC	547	CACAGCTTTCCA TCACCATC	1279	CCTTCCTCACTTTT CCTTGCCCA	2011
PPP2CA.1	NM_002715	NM_002715.2	PPP2CA	3879	GCAATCATGGAAC TGACGA	548	AGTGGCTCGCC TCTACG	1280	TTTCTTGCAGTTGA CCCAGCAC	2012
PRCC.1	NM_005973	NM_005973.4	PRCC	6002	GAGGAAGAGGAGG CGGTG	549	CAGGAGAGAAG CGAACAA	1281	CTACATCTGGGCC GCTTTAGGG	2013
PRKCA.1	NM_002737	NM_002737.1	PRKCA	2626	CAAGCAATGCGTCA TCAATGT	550	GTAATTCGGCC CCTCTTCT	1282	CAGCCTCTGCGAA TGGATCACACT	2014
PRLCB1.1	NM_002738	NM_002738.5	PRLCB1	3739	GACCCAGCTCCACT CCTG	551	CCCAATTCAGTA CTCCATCA	1283	CCAGACATGGACC GCCTGTACTT	2015
PRKCD.2	NM_006254	NM_006254.1	PRKCD	626	CTGACACTTGCCG CAGAGAA	552	AGTGGTCTCTTG GTCTGGAA	1284	CCCTTCTCACCCAC CTCATCTGCAC	2016
PRKCH.1	NM_006255	NM_006255.3	PRKCH	4370	CTCCACCATGAGC GTCTGTC	553	CACACTTTCCT CCTTTTGG	1285	TCCTGTTAAGATCCC AAGCCACA	2017
PRO2000.3	NM_014109	NM_014109.2	ATAD2	1666	ATTGGAAAAACCTC GTCACC	554	TGGGTATCTTGG TCTTGCA	1286	CCCAACATATTTAT AGTGGCCGAG	2018
PROM1.1	NM_006017	NM_006017.1	PROM1	4516	CTATGACAGGAG CCACC	555	CTCCAACATGA GGAAGACG	1287	ACCCGAGGCTGTG CTCCAACAC	2019
PROM2.1	NM_144707	NM_144707.1	PROM2	5108	CTTCAGCGCATCCA CTACC	556	CAATGCTGTCT TCACCAC	1288	CTTCTCGTTCAGAT CCAGAGGCC	2020
PRPS2.1	NM_00103901	NM_00103901	PRPS2	4694	CACGACCAAGAT TCAGGT	557	ATTGTGTCTCT TCGGATTG	1289	TTGACATTTCCATGA TCTTGGCCG	2021
PRSS8.1	NM_002773	NM_002773.2	PRSS8	4742	GTACACTCTGGCCT CCAGTCA	558	CCACACAGGCT GGAGTT	1290	TCCTGGATCCAAGC AAGGTGACA	2022

133

134

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
PSMA7.1	NM_002792	NM_002792.2	PSMA7	6255	GCCAACTGCAGG ATGAAAG	559	AGCCATGCAGA CGTTGT	1291	CCAAAGCACAGATCT TCCGCACTG	2023
PSMB8.1	NM_148919	NM_148919.3	PSMB8	6461	CAGTGGCTATCGG CCTAATC	560	TAAGCAATAGCC CTGCGG	1292	CAAGTTCATAGGCCT CTTCAGGGC	2024
PSMB9.1	NM_148954	NM_148954.2	PSMB9	6462	GGGTGTCATCTA CCTGGTC	561	CATGCCCCAAGA TGACTCG	1293	TGGTCCACACCGC AGCTGTAATA	2025
PTEN.2	NM_000314	NM_000314.1	PTEN	54	TGGCTAAGTGAAGA TGACAATCATG	562	TGCACATATCATT ACACCAAGTTCGT	1294	CCTTCCAGCTTTAC AGTGAATTGCTGCA	2026
PTGIS.1	NM_000961	NM_000961.3	PTGIS	5429	CCACACTGGCATCT CCCT	563	GCCATGGGATG AGAAACT	1295	CCTTCTCCAGGGACA GAAGCAGGA	2027
PTHR1.1	NM_000316	NM_000316.1	PTHR1	2375	CGAGGTACAGCT GAGATCAAGAA	564	GCGTGCTTTTCG CTTGAA	1296	CCAGTGCCAGTGTC CAGCGGCT	2028
PTK2.1	NM_005607	NM_005607.3	PTK2	2678	GACCGTTCGAATG ATAAGGT	565	CTGGACATCTCG ATGACAGC	1297	ACGAGGCCGCTCAC ATTCTCGTAC	2029
PTK2B.1	NM_004103	NM_004103.3	PTK2B	2883	CAAGCCAGCCCGA CCTAAG	566	GAACCTTGGAAT GCAGCTTTG	1298	CTCCGCAAAACCAAC TCCTGGCT	2030
PTN.1	NM_002825	NM_002825.5	PTN	3964	CCTTCCAGTCCAAA AATCCC	567	CCCTCTCTCCA CTTTGGAT	1299	TTCCTCTGCTCTGGG GCTCTCTTG	2031
PTPNS1.1	NM_080792	NM_080792.1	SIRPA	2896	CTCAGCTAGCACT AAGCACATC	568	TTTCAAGATTGC ACGTTTCACAT	1300	TCTCAGTAAATTACA GGCGTCCACAG	2032
PTPRB.1	NM_002837	NM_002837.2	PTPRB	3881	GATATGCGGTGAG GAACAGC	569	CTGGCCACACCG TATAGTGA	1301	ATGCACACAGACTCA TCCGCCACT	2033
PTPRC.1	NM_080921	NM_080921.2	PTPRC	6450	TGGCCGTCATGG AAGA	570	GGACATCTTTTG TGCTGGTTG	1302	CAACATCTCCAAAAG CCCGAGTGC	2034
PTPRG.1	NM_002841	NM_002841.2	PTPRG	2682	GGACAGCACAAG GACTTGA	571	GGACTCGGAACA GGTAAAGG	1303	CACCATTAGCCATGT CTCACCCGA	2035
PTTG1.2	NM_004219	NM_004219.2	PTTG1	1724	GGTACTCTGATCT ATGTTGATAAGGAA	572	GCTTAGCCCAT CCTTAGCA	1304	CACAGGGTGCTGT GTTCTCCA	2036
PVALB.1	NM_002854	NM_002854.2	PVALB	4316	AAACCAAGATGCTG ATGGCT	573	CAGCCACCAGAG TGGAGAA	1305	TTTTGGCGTCCCAT CTTTGTCTC	2037
PXDN.1	NM_012293	NM_012293.1	PXDN	6257	GCTGCTCAAGCTG AATCCC	574	ACCCAGATCTT CCTGGTC	1306	ACTGGACGGCGAC ACCATCTACT	2038

135

136

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
RAC1.3	NM_006908	NM_006908.3	RAC1	2698	TGTTGTAAATGTCT CAGCCCC	575	TTGAGCAAGCG TACAAAGG	1307	CGTTCTTGGTCTCTGT CCCTTGGG	2039
RAD51.1	NM_002875	NM_002875.2	RAD51	3976	AGACTACTCGGGT CGAGGTG	576	AGCATCCGAGA AACCTG	1308	CTTTCAGCCAGCA GATGCACTTG	2040
RAF1.3	NM_002880	NM_002880.1	RAF1	2130	CGTCGTATGCGAG AGTCTGT	577	TGAAGCGTGAG GTGTAGAA	1309	TCCAGGATGCCTGTT AGTTCTCAGCA	2041
RALBP1.1	NM_006788	NM_006788.2	RALBP1	2105	GGTGTCAATATAA ATGTGCAATGC	578	TTGATATTTGCC AGCAGCTATAAA	1310	TGCTGTCTCTGTCGG TCTCAGTACGTTCA	2042
RARB.2	NM_016152	NM_016152.2	RARB	687	TGCCTGGACATCCT GATTCT	579	AAGCCGTCTGA GAAAGTCA	1311	TGCACCAGGTATACC CCAGAACAAAG	2043
RASSF1.1	NM_007182	NM_007182.4	RASSF1	6658	AGGCACGTGAAG TCATTG	580	AAAGATGCAAA CTTGCGG	1312	CACCACCAAGAACTT TCGCAGCAG	2044
RB1.1	NM_000321	NM_000321.1	RB1	956	CGAAGCCCTTACAA GTTTTC	581	GGACTCTTCAGG GGTGAAT	1313	CCCTTACGGATTCCT GGAGGGGAC	2045
RBM35A.1	NM_017697	NM_017697.2	ESRP1	5109	TGTTTGAATCAC CAGGG	582	CTCTGTCCGAG ACTTCATCT	1314	CGCCCATCAGGAGA TGCCCTTATC	2046
REG4.1	NM_032044	NM_032044.2	REG4	3226	TGCTAACTCCTGCA CAGCC	583	TGCTAGGTTTCC CCTCTGAA	1315	TCCTCTTCCCTTCTG CTAGCCTGGC	2047
RET.1	NM_020630	NM_020630.3	RET	5001	GCCTGTGCGATTCT TGTGC	584	GGAAGGCAGA CCCTCAC	1316	AACATCAGCGTGGC CTACAGGCTC	2048
RGS1.1	NM_002922	NM_002922.3	RGS1	6258	TGCCCTGTAAAGCA GAAGAGAT	585	CTCGAGTGCAGA AGTCAATA	1317	AGCAGCATCTGAATG CACAAATGCT	2049
RGS5.1	NM_003617	NM_003617.1	RGS5	2004	TTCAAACGGAGGCT CCTAAGAG	586	GAAAGTTCCACC AGGTTCTTCA	1318	AATATTGACCACTTC ACTAAGGACATCACA	2050
RHEB.2	NM_005614	NM_005614.3	RHEB	6609	GATGATTGAGAAC GCCTTGC	587	GCTCCCAAGACT CTGACACA	1319	TGTCATGTCTCTAGA ACACCTGGAGTT	2051
RhoB.1	NM_004040	NM_004040.2	RHOB	2951	AAGCATGAACAGGA CTTGACC	588	CCTCCCAAGTC AGTTGC	1320	CTTTCCAAACCCCTGG GGAAGACAT	2052
rhoC.1	NM_175744	NM_175744.1	RHOC	773	CCCGTTTGGTCTG AGGAA	589	GAGCATCAAGG TAGCCAAAGG	1321	TCCGGTTCGCCATGT CCCG	2053
RIPK1.1	NM_003804	NM_003804.3	RIPK1	6259	AGTACCTTCAAGCC GGTCAA	590	AAGTCCCTGGGA ACTGTGC	1322	CAGCCACAGAACAG CCTGGTTTAC	2054

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
RND3.1	NM_005168	NM_005168.3	RND3	5381	TCGGAATTGGACTT GGGAG	591	CTGGTTACTCCC CTCCAACA	1323	TTTAAAGCCTGACTC CTCACCAGC	2055
ROCK1.1	NM_005406	NM_005406.1	ROCK1	2959	TGTCACATAGGAA TGAGCTTC	592	GTTTACACGCA ATTGCTCA	1324	TCACTCTCTTTGCTG GCCAACTGC	2056
ROCK2.1	NM_004850	NM_004850.3	ROCK2	2992	GATCCGAGACCTT CGCTC	593	AGGACCAAGGAA TTTAAAGCA	1325	CCCATCAACGTGGA GAGCTTGCT	2057
RPLP1.1	NM_213725	NM_213725.1	RPLP1	5478	CAAGGTGCTCGGT CCTTC	594	GTCGCCGATGA AGTGAG	1326	CCTCACCCCAACGC AGCCTTAGCT	2058
RPS23.1	NM_001025	NM_001025.1	RPS23	3320	GTTCTGTTGCTG GATTTGG	595	CCTTAAAGCGGA CTCCAGG	1327	ATCACCAACAGCATG ACCTTTGCG	2059
RPS27A.1	NM_02954	NM_02954.3	RPS27A	6329	CTTACGGGGAAGA CCATCAC	596	TCTGTGATCTTG GCCTTTAC	1328	TCGTATCCGAGGGTT CAACCTCG	2060
RPS6KA1.1	NM_002953	NM_002953.3	RPS6KA1	3865	GCTCATGAGGCTA GTGCCTC	597	CGCTGAAGTCC AGCTTCT	1329	ACCCGAGAAATGGA CAGACCTCAG	2061
RPS6KB1.3	NM_003161	NM_003161.1	RPS6KB1	928	GCTCATTTATGAAA ACATCCCAAAAC	598	AAGAAACAGAAG TTCTCTGGCTTT CT	1330	CACACCAACCAATAA TTTGCATT	2062
RRM1.2	NM_001033	NM_001033.1	RRM1	1000	GGGCTACTGGCAG CTACATT	599	CTCTAGCATCG GTACAAGG	1331	CATTGGAATTGCCAT TAGTCCCAGC	2063
RRM2.1	NM_001034	NM_001034.1	RRM2	2546	CAGCGGATTAAAC AGTCCT	600	ATCTGGCTTGAA GCAGTGAG	1332	CCAGCACAGCCAGT TAAAGATGCA	2064
RUNX1.1	NM_001754	NM_001754.3	RUNX1	6067	AACAGAGACATTGC CAACCA	601	GTGATTTGCCCA GGAAAGTTT	1333	TTGGATCTGTTTGCT GTCCAAACC	2065
S100A1.1	NM_006271	NM_006271.1	S100A1	2851	TGGACAAAGTGAT GAAGGAG	602	AGCACACATAC TCCTGGAA	1334	CCTCCCCGTCTCCAT TCTCGTCTA	2066
S100A10.1	NM_002966	NM_002966.1	S100A10	3579	ACACCAAAATGCCA TCTCAA	603	TTTATCCCCAGC GAATTTGT	1335	CACGCCATGGAAC CATGATGTTT	2067
S100A2.1	NM_005978	NM_005978.2	S100A2	2369	TGCTGTGCTGGT CAGTACCT	604	TCCCCCTTACTC AGCTTGAAC	1336	CACAAGTACTCCTGC CAAGAGGGCGAC	2068
SAA2.2	NM_030754	NM_030754.2	SAA2	6655	CTACAGCACAGATC AGCACCA	605	TGCTGACACTCA GGACCAAG	1337	AGCTTCTCACGGGC CTGGTTTTCT	2069
SCN4B.1	NM_174934	NM_174934.3	SCN4B	7223	GCCTTCCTGGAGTA CCCG	606	GTGGCCCAATTC CCCAAGT	1338	TGCTCCCTATGCCCT TCCAAGCAT	2070

139

140

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
SCNN1A.2	NM_001038	NM_001038.4	SCNN1A	3263	ATCAACATCCTGTC GAGGCT	607	GAAGTTGCCCAG CGTGTC	1339	AGAGACTTGCCATC CCTGGAGGA	2071
SDHA.1	NM_004168	NM_004168.1	SDHA	5443	GCAGAACTGAAGAT GGGAAGAT	608	CCCTTTCCAAAC TTGAGGC	1340	CTGTCCACCAAATGC ACGCTGATA	2072
SDPR.1	NM_004657	NM_004657.4	SDPR	3877	ACCAAGCAAGATG GAGCA	609	GGTCATTCTGGA TGCCCTT	1341	CGGAGCCCTCCAAA CTGATCTGTC	2073
SELE.1	NM_000450	NM_000450.1	SELE	5383	ACACTGGTCTGGC CTGCTAC	610	AAAGTCCAGCTA CCAAGGGA	1342	CCTGTGAAGTCCCA CTGAGTCCA	2074
SELENBP1.1	NM_003944	NM_003944.2	SELENBP1	6200	GGTACAGCCTCG ACACAA	611	CCATCTCGTAAG ACATTGGGA	1343	TGCCCCTCACTGCT GATCATGAC	2075
SELL.1	NM_000655	NM_000655.3	SELL	5483	TGCAACTGTGATGT GGGG	612	CTTCCAAAGGCT CACACTG	1344	CACAACTGACACTG GGGCCATA	2076
SELPF.1	NM_003006	NM_003006.3	SELPF	6316	TGGCACATATCTTC TTCTGTTG	613	CTAATTACGCAC GGGTACA	1345	CACTGTGCTGCTGG CGGTCC	2077
SEMA3B.1	NM_004636	NM_004636.1	SEMA3B	1013	GCTCCAGATGTG TTTCTGTTG	614	ACGTGGAGAAGA CGGCATAGA	1346	TCGCGGGACCAACG GACC	2078
SEMA3C.1	NM_006379	NM_006379.2	SEMA3C	5409	ATGGCCATTCCTGT TCCAG	615	GTCTCACATCTT GTCTTCGGC	1347	CCTCCGTTTCCAGT TGGGTAGAA	2079
SEMA3F.3	NM_004186	NM_004186.1	SEMA3F	1008	CGGAGGCCCTCA TTATACA	616	CAGTCGCCGTTG ACATCCT	1348	CTCCCCACAGCGCA TCGAGGA	2080
SEMA5B.1	NM_001031702	NM_001031702	SEMA5B	5003	CTCAGAGACAGCT CCAACAT	617	TCACATTCCGCA CAGGAC	1349	AGCCTCTGGACCCA GAACATCAC	2081
SERPINA5.1	NM_000624	NM_000624.3	SERPINA5	5201	CAGCATGGTAGTG GCAAGA	618	CTTTGTGGCACT GAGCTGG	1350	AGGTCCAGAGTCCT GGCCCTTGAT	2082
SFN.1	NM_006142	NM_006142.3	SFN	3580	GAGAGAGCCAGTC TGATCCA	619	AGGCTGCCATGT CCTCATA	1351	CTGCTCTGCCAGCTT GGCCTTC	2083
SGK.1	NM_005627	NM_005627.2	SGK	2960	TCCGACAGACACCT CCTG	620	TGAAGTCATCCT TGGCCC	1352	TGTCCTGTCTTCTTG CAGGAGGC	2084
SHANK3.1	XM_037493	XM_037493.5		3887	CTGTGCCCTCTACA ACCAGG	621	GGACATCCCTGT TAGCTCCA	1353	AGCTGTGCTGCTGT CCTGCTCTTC	2085
SHC1.1	NM_003029	NM_003029.3	SHC1	2342	CCAACACCTTCTTG GCTTCT	622	CTGTTATCCCAA CCAAACC	1354	CCTGTGTTCTTGTG AGCACCTTC	2086

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
SILV.1	NM_006928	NM_006928.3	SILV	4113	CCGCATCTTCTGCT CTTGT	623	ACTCAGACCTGC TGCCCA	1355	TTGGTGAGAAATAGCC CCCTCCTCA	2087
SKIL.1	NM_005414	NM_005414.2	SKIL	5272	AGAGGCTGAATATG CAGGACA	624	CTATCGCCTCA GCATGG	1356	CCAATCTCTGCCTCA GTTCTGCCA	2088
SLC13A3.1	NM_022829	NM_022829.4	SLC13A3	6202	CTTGCCCTCCAACA AGGTC	625	AGCCCACTGAGG AAGAGGA	1357	CCCCCACTACTTCTCT CGACACCAA	2089
SLC16A3.1	NM_004207	NM_004207.1	SLC16A3	4569	ATGCGACCCACGT CTACAT	626	AATCAGGGAGGA GGTGAGC	1358	CCCCGCCAGGATGA ACACGTAC	2090
SLC22A3.1	NM_021977	NM_021977.2	SLC22A3	6506	ATCGTCAGCGAGTT TGACCT	627	CAGATGGCTTG GGTGAG	1359	CAGCATCCACGCAAT GACACAGAC	2091
SLC22A6.1	NM_153277	NM_153277.1	SLC22A6	6463	TCCGCCACCTCTTCT CTCT	628	GACCAGCCCAT GTATGCAAG	1360	CTCTCCATGTGTGG TTTGCCACT	2092
SLC2A1.1	NM_006516	NM_006516.1	SLC2A1	2966	GCCTGAGTCTCTCT GTGCC	629	AGTCTCCACCT CAGGCAT	1361	ACATCCCAGGCTTCA CCCTGAATG	2093
SLC34A1.1	NM_003052	NM_003052.3	SLC34A1	6203	GCTGAGACCCACT GACCTG	630	AGCCTCTCTCCG TAGGACAA	1362	TCCTGGGCAACCAC TATGAGGTCT	2094
SLC7A5.2	NM_003486	NM_003486.4	SLC7A5	3268	GCGCAGAGGCCAG TTAAA	631	AGCTGAGCTGTG GGTTGC	1363	AGATCACCTCTCTCGA ACCCACTCC	2095
SLC9A1.1	NM_003047	NM_003047.2	SLC9A1	5385	CTTCGAGATCTCCC TCTGGA	632	AGTGGGATCAC ATGGAAC	1364	CCTTCTGGCCTGCCT CATGAAGAT	2096
SLIT2.2	NM_004787	NM_004787.1	SLIT2	3708	TTTACCGATGCACC TGTC	633	TGCAGGCATGA ATTGGG	1365	CACAGTCTGTGCCCT TGAAACCAT	2097
SNAIL.1	NM_005985	NM_005985.2	SNAIL	2205	CCCAATCGAAGC CTAACTA	634	GTAGGCTGTG GAAGGTAA	1366	TCTGATTAGAGTCC TGCAGCTCGC	2098
SNRK.1	NM_017719	NM_017719.4	SNRK	6710	GAGAAAAGTCAG GGCCG	635	GCCGCTTTCAG AATCATC	1367	CCAGTGCAGTAGTT CGGAGCCA	2099
SOD1.1	NM_000454	NM_000454.3	SOD1	2742	TGAAGAGAGGCAT GTTGGAG	636	AATAGACATC GGCCACAC	1368	TTTGTACAGAGTCAC ATTGCCCAA	2100
SP3.1	NM_00101731	NM_00101731	SP3.1	5430	TCAAGATCTCAGC AGCCAA	637	CCATGGATTGTC TGTGGTGT	1369	CAGTCAAGCCCAAT TGTGCAAGG	2101
SPARC.1	NM_003118	NM_003118.1	SPARC	2378	TCTTCCCTGTACAC TGGCAGTTC	638	AGCTCGTGTGG GAGAGTA	1370	TGGACCAGACCCC ATTGACGG	2102

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
SPARCL1.1	NM_004684	NM_004684.2	SPARCL1	3904	GGCACAGTGAAG TGATGA	639	GATGAGCTCTC TCGGCCT	1371	ACTTCATCCCAAGCC AGGCCTTTC	2103
SPAST.1	NM_014946	NM_014946.3	SPAST	4033	CCTGAGTGTTCAC AGGC	640	ATTCCAGGTGG ACCAAAG	1372	TACAGCCCTCTGGC AGGAGCTCT	2104
SPHK1.1	NM_021972	NM_021972.2	SPHK1	4178	GGCAGTTCCTTGA ACCAT	641	GCAATTGGTCAG GAGGTCTT	1373	TGGTGAOCTGCTCAT AGCCAGCAT	2105
SPRY1.1	AK026960	AK026960.1	SPRY1	1051	CAGACCAAGTCCCT GGTCATAGG	642	CCTTCAAGTCAT CCACAATCAGTT	1374	CTGGTCCGGATTG CCCTTTTCA	2106
SQSTM1.1	NM_003900	NM_003900.3	SQSTM1	4662	GGACCCGCTTACA GGTGAAC	643	GGTCCAGAGAG CTTTGGC	1375	CAGTCCCTACAGATG CCAGAATCCG	2107
STAT1.3	NM_007315	NM_007315.1	STAT1	530	GGGCTCAGCTTTCA GAAGTG	644	ACATGTTCACT GGTCCACA	1376	TGGCAGTTTCTTCTT GTCACCAAAA	2108
STAT3.1	NM_003150	NM_003150.1	STAT3	537	TCACATGCCATTT GGTGTT	645	CTTGCAGGAAGC GGCTATAC	1377	TCCTGGGAGAGATT GACCAGCA	2109
STAT5A.1	NM_003152	NM_003152.1	STAT5A	403	GAGGCGCTCAACA TGAATTC	646	GCAGGAACACG AGGTTCTC	1378	CGGTTGCTCTGCAC TCGGCCT	2110
STAT5B.2	NM_012448	NM_012448.1	STAT5B	857	CCAATGGTGGTGA TCGTTCA	647	GCAAAAGCATTG TCCAGAGAGA	1379	CAGCCAGAGACA ATGCGACGG	2111
STC2.1	NM_003714	NM_003714.2	STC2	6507	AAGAGAGCCATCA CCCAAC	648	AGATGGAGCACA GGCTTCC	1380	TTCTGCTCACACTGA ACCTGCACG	2112
STK11.1	NM_000455	NM_000455.3	STK11	3383	GGACTCGAGACG CTGTG	649	GGATCCTTCGC AACTTCTT	1381	TTCTTGAGGATCTTG ACGGCCCTC	2113
STK15.2	NM_003600	NM_003600.1	AURKA	341	CATCTTCAGGAG GACCACT	650	TCCGACCTCAA TCATTTCA	1382	CTCTGTGGCACCCT GGACTACCTG	2114
STK4.1	NM_006282	NM_006282.2	STK4	5424	GAGCATCTTCCTG CAACTT	651	CTGAGGTGAAC CCAGTCA	1383	ACCTCTTCCCTCAG ATGGGGAGC	2115
STMN3.3	NM_005940	NM_005940.2	MMP11	741	CCTGGAGCTGCA ACATACC	652	TACAATGGCTTT GGAGGATAGCA	1384	ATCCTCTCTGAAGCC TTTTTCGACG	2116
SUCLG1.1	NM_003849	NM_003849.2	SUCLG1	6205	CCAGCCTGTAGT GTCCTTCA	653	CGCATGACCCA TTCTTC	1385	CCGAGGAGGACAG TTAAACCAGC	2117
SULT1C2.1	NM_001056	NM_001056.3	SULT1C2	6206	GGGACCCAAAGCA TGAAT	654	AGCAGTGTTC TCCACCTTC	1386	TTCCCATGAAGTGA TCACCTTCC	2118

145

146

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
SURV.2	NM_001168	NM_001168.1	BIRC5	81	TGTTTGTATTCCCG GGCTTA	655	CAAGCTGTCAG CTCTAGCAAAAG	1387	TGCCTTCTTCTCTCC TCACTTCTCACT	2119
TACSTD2.1	NM_002353	NM_002353.2	TACSTD2	6335	ATCACCAACCGGA GAAAGTC	656	AAGCTCGTTTC TTTCTCAA	1388	CCCCAGTTCCTTGA TCTCCACC	2120
TAGLN.1	NM_003186	NM_003186.3	TAGLN	6073	GATGAGCAGGTG GCTCAGT	657	AGTCTGGAACAT GTCAGTCTTGAT G	1389	CCCATAGTCTCTCAGC CGCCTTCAG	2121
TAP1.1	NM_000593	NM_000593.5	TAP1	3966	GTATGCTGCTGAAA GTGGGAA	658	TCCCACTGCTTA CAGCCC	1390	CACCAGCTGCCAC CAATGTAGAG	2122
TCF4.1	NM_003199	NM_003199.1	TCF4	4097	CACACCCTGGAATG GGAG	659	ATGTGGCAACTT GGACCT	1391	CGCATCGAATCACAT GGGACAGAT	2123
TCOF1.2	NM_0010086	NM_001008656	TCOF1	6719	AGCGAGGATGAGG ACGTG	660	CACCACATTGGT CTGTATGC	1392	TCCCCGCTTACACAGT GCTTGACTC	2124
TEK.1	NM_000459	NM_000459.1	TEK	2345	ACTTCGGTGTCTACT TAACAACCTTACATC	661	CCTGGCCTTGG TGTTGAC	1393	AGCTGGGACCACGT CTGCTCCCTG	2125
TERT.1	NM_003219	NM_003219.1		992	GACATGGAGAACAA GCTCTTTGC	662	AGGTGTTCACCA ACAAGAAATCAT	1394	ACCAAAACGACGAG CCAGCCCG	2126
TFAP2B.1	NM_003221	NM_003221.3	TFAP2B	6207	CGTGTGACGTGCG AGAGA	663	CCACACGCTCTC AGGACC	1395	ATGGACGCGCCTTG CTCTTACTGT	2127
TFAP2C.1	NM_003222	NM_003222.3	TFAP2C	4663	CATGCCTCACACAGA TGGA	664	CTGTCTGATCGT GCAGCAAC	1396	CTGGTCTGTGACATT CTGCACCTC	2128
TFPI.1	NM_006287	NM_006287.4	TFPI	6270	CCGAATGGTTTCCA GGTG	665	TTGCGGAGTCAG GGAGTTA	1397	ATGGAACCCAGCTCA ATGCTGTGA	2129
TGFA.2	NM_003236	NM_003236.1	TGFA	161	GGTGTGCCACAGA CCTTCCT	666	ACGGAGTTCTTG ACAGAGTTTGA	1398	TTGGCCTGTAATCAC CTGTGCAGCCTT	2130
TGFB1.1	NM_000660	NM_000660.3	TGFB1	4041	CTGTATTTAAGGAC ACCCGTGC	667	TGACACAGAGAT CCGCACTC	1399	TCTCTCCATCTTTAA TGGGGCCCC	2131
TGFB2.2	NM_003238	NM_003238.1	TGFB2	2017	ACCATCCCCCAG AAGACTA	668	CCTGGTCTGTTT GTAGATGG	1400	TCCTGAGCCCGGG AAGTCCC	2132
TGFB1.1	NM_000358	NM_000358.1	TGFB1	3768	GCTACGAGTGCTG TCCTGG	669	AGTGGTAGGGCT GCTGGAC	1401	CCTTCTCCCCAGGG ACCTTTTCAT	2133
TGFB1.1	NM_004612	NM_004612.1	TGFB1	3385	GTCATCACTGGC CTTGG	670	GCACACGAAGCA CACTGGT	1402	AGCAATGACAGCTG CCAGTTCCAC	2134

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
TGFB2.3	NM_003242	NM_003242.2	TGFB2	864	AACACCAATGGTT CCATCT	671	CCTTTTCATCAG GCCAAACT	1403	TTCTGGCTCCTGAT TGCTCAAGC	2135
THBD.1	NM_000361	NM_000361.2	THBD	4050	AGATCTGCGACGG ACTGC	672	GGAATGACATC GGCAGC	1404	CACCTAATGACAGTG CGCTCCTCG	2136
THBS1.1	NM_0003246	NM_0003246.1	THBS1	2348	CATCCGAAAGTGA CTGAAGAG	673	GTACTGAATCC GTTGTGATAGCA TAG	1405	CCAAATGAGCTGAGG CGGCCTCC	2137
TIMP1.1	NM_003254	NM_003254.2	TIMP1	6075	TCCTGCGGTCCC AGATAG	674	GTGGGAACAGG GTGGACACT	1406	ATCCTGCCCGAGT GGAAGCTGAAGC	2138
TIMP2.1	NM_003255	NM_003255.2	TIMP2	606	TCACCTCTGTGAC TTCATCGT	675	TGTGGTTCAGGC TCTTCTCTG	1407	CCCTGGGACACCT GAGCACCA	2139
TIMP3.3	NM_000362	NM_000362.2	TIMP3	593	CTACCTGCCCTTGCT TTGTGA	676	ACCGAAATTGGA GAGCATGT	1408	CCAAAGAACGAGTGT CTCTGGACCG	2140
TK11.2	NM_003258	NM_003258.1	TK11	264	GCCGGGAAGACCG TAATGT	677	CAGCGGACCCAG GTTGAG	1409	CAAATGGCTTCTCTCT GGAAGTCCCA	2141
TLR3.1	NM_003265	NM_003265.2	TLR3	6289	GGTTGGGCCACCT AGAAAT	678	CCATTCCTGGCC TGTGAG	1410	CTTGCCCAATTTCAT TAAGSCCA	2142
TMEM27.1	NM_020665	NM_020665.2	TMEM27	3878	CCCTGAAAGAATGT TGTGGC	679	TCTGCACCTGGT TGACAGAG	1411	TCTGTGACTGCCAT TCATGCTGA	2143
TMEM47.1	NM_031442	NM_031442.3	TMEM47	6713	GGATTCCACTGTTA GAGCCCTT	680	GCAATAAACCAA CAGCCAAATG	1412	CCGCTGCTTATCCT ACCCAAATGA	2144
TMSB10.1	NM_021103	NM_021103.1	TMSB10	6076	GAATCGCCAGCTT CGATAA	681	CTCGCAGGGT GTTCTTCT	1413	CGTCTCGTTTCTT CAGCTTGGC	2145
TNF.1	NM_000594	NM_000594.1	TNF	2852	GGAGAAGGTTGAC CGACTCA	682	TGCCAGACTCG GCAAAAG	1414	CGCTGAGATCAATCG GCCCGACTA	2146
TNFAIP3.1	NM_006290	NM_006290.2	TNFAIP3	6290	ATCGTCTTGGCTGA GAAAGG	683	GTGGAATGGTC TGGCTTC	1415	CAACCCACGCGACTT GTGTGTCTT	2147
TNFAIP6.1	NM_007115	NM_007115.2	TNFAIP6	6291	AGGAGTGAAGAT GGGATGC	684	CTGTAAGAGCG CACCACAC	1416	ATTGTACACACCCAC ACGCAAGG	2148
TNFRSF10C.3	NM_003841	NM_003841.2	TNFRSF10C	6652	GGAGTTTGACCAG AGATGCAA	685	CTCTGTCCCAG AGTTCCC	1417	AACGGTAGGAAGCG CTCCTTCCAC	2149
TNFRSF10D.1	NM_003840	NM_003840.3	TNFRSF10D	4406	CCTCTCGCTTCTGG TGGTC	686	GCTCAGGAATCT CTGCCCTCA	1418	AGGCATCCAGGGA CTCAGTTCCAC	2150

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
TNFRSF11B.1	NM_002546	NM_002546.2	TNFRSF11B	4675	TGGCGACCAAGAC ACCTT	687	GGGAAAGTGGTA CGTCTTTGAG	1419	AGGGCCTAATGCAC GCACTAAAGC	2151
TNFRSF1A.1	NM_001065	NM_001065.2	TNFRSF1A	4943	ACTGCCCTGAGCC CAAAAT	688	GTCAGCACGCGT GGAGAG	1420	TGCCAGACAGCTATG GCCTCTCAC	2152
TNFSF12.1	NM_003809	NM_003809.2	TNFSF12	2987	TAGCCGAGAGTT CCCAA	689	CACAGGGAATTC TCAAGGGA	1421	TTGTCTTGTGTTTCTCG CCCCCTACA	2153
TNFSF13B.1	NM_006573	NM_006573.3	TNFSF13B	4944	CCTACGCCATGGG ACATC	690	TGAAAACAAAGT CACCAGACTC	1422	TCCCCAAAGACATGG ACCTTCTTCC	2154
TNFSF7.1	NM_001252	NM_001252.2	CD70	4101	CCAACTCACTGG GACACTT	691	ACCCACTGCACT CCAAAGAA	1423	TGCCTTCCCAGAAACA CTGATGAGA	2155
TNIP2.1	NM_024309	NM_024309.2	TNIP2	3872	CATCTCAGAAAGG GCCCGA	692	GGACACTTTTCC TCCAGTT	1424	AATCTACTTTTGAGC CCGTTCCTG	2156
TOP2A.4	NM_001067	NM_001067.1	TOP2A	74	AATCAAGGGGGA GAGTGAT	693	GTCAGATTTTG CCCGAGGA	1425	CATATGGACTTTTGAC TCAGCTGTGGC	2157
TOP2B.2	NM_001068	NM_001068.1	TOP2B	75	TGTGGACATCTTCC CCTCAGA	694	CTAGCCCGACCG GTTCTGT	1426	TTCCCTACTGAGCCA CCTTCTCTG	2158
TP.3	NM_001953	NM_001953.2	TYMP	91	CTATATGAGCCAG AGATGTGACA	695	CCACGAGTTTCT TACTGAGAATGG	1427	ACAGCCTGCCACTCA TCACAGCC	2159
TRAIL.1	NM_003810	NM_003810.1	TNFSF10	898	CTTCACAGTGCTCC TGCAGTCT	696	CATCTGCTTCAG CTCGTTGGT	1428	AAGTACAGGTAAGTT ACAGCCACACA	2160
TS.1	NM_001071	NM_001071.1	TYMS	76	GCCTCGGTGTGCC TTTCA	697	CGTATGTGCGC AATCATG	1429	CATCGCAGCTACG CCCTGCTC	2161
TSC1.1	NM_000368	NM_000368.3	TSC1	6292	TCACCAAACTCTCAG CCCG	698	GTGTCAGCATAA GGGCTGGT	1430	TTTCTCATCGTTCA GCCGATGTC	2162
TSC2.1	NM_000548	NM_000548	TSC2	5132	CACAGTGGCTCTT TCTCCT	699	CAGGAAACGCTC CTGTGC	1431	TACCAGTCCAGCTGC CAAGGACAG	2163
TSPAN7.2	NM_004615	NM_004615.3	TSPAN7	6721	ATCACTGGGTGAT CCTGC	700	GGAGATATAGG TGCCACAG	1432	AAGTTTGCCCCAGAC TCCAACAGC	2164
TSPAN8.1	NM_004616	NM_004616.2	TSPAN8	6317	CAGAAATCTCTGCA GGCAAGT	701	AATCCAGATGCC GTGAATT	1433	TGCTCCAGAGCATAT TGCAGGACA	2165
TUBB.1	NM_001069	NM_001069.1	TUBB2A	2094	CGAGGACGAGGCT TAAAAAC	702	ACCATGCTTGAG GACAAACAG	1434	TCTCAGATCAATCGT GCATCCTTAGTGAA	2166

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
TUSC2.1	NM_007275	NM_007275.1	TUSC2	6208	CACCAAGAAGGG CAGAA	703	CGATGCCCTGAG GAATCA	1435	TCTTATGCACCTCGCC TCAGCTTGG	2167
tusc4.2	NM_006545	NM_006545.4	TUSC4	3764	GGAGGAGCTAAAT GCCTCAG	704	CCTCAAGTGA TGGTGTG	1436	ACTCATCAATGGGA GAGTGCACC	2168
TXLNA.1	NM_175852	NM_175852.3	TXLNA	6209	GCCAGAAAGGCTC AGTCT	705	ATGCTTTCCAGT TGGCGG	1437	TCCTCAGAGACATCA CGAAGGGCC	2169
UBB.1	NM_018955	NM_018955.1	UBB	3303	GAGTCGACCCCTGC ACCTG	706	GCGAATGCCATG ACTGAA	1438	AATTAAACAGCCACC CTCAGGCG	2170
UBE1C.1	NM_003968	NM_003968.3	UBA3	2575	GAATGCACGCTGG AACTTTA	707	CTGGTAGCCTGG GCATAGA	1439	AATTTCCCATGTGC ACCATTGCA	2171
UBE2C.1	NM_007019	NM_007019.2	UBE2C	2550	TGCTGCGCATAAA GGGAT	708	ATGGTCCCTACC CAATTGAA	1440	TCTGCCTTCCCTGAA TCAGACAACC	2172
UBE2T.1	NM_014176	NM_014176.1	UBE2T	3882	TGTTCTCAAAATGC CACCAA	709	AGAGGTCAACAC AGTTGCGA	1441	AGGTGCTTGGAGAC CATCCTTCAA	2173
UGCG.1	NM_003358	NM_003358.1	UGCG	6210	GGCAACTGACAAAC AGCCT	710	AGAACTACCCC TTTCACTGG	1442	CAAGCTCCAGGTG TCTCTCTTCTGA	2174
UMOD.1	NM_003361	NM_003361.2	UMOD	6211	GCGTGGACCTGGA TGAGT	711	TTACGCAGCTGC TGTGG	1443	CCATTCTGGAGCTC ACAACTGCT	2175
upa.3	NM_002658	NM_002658.1	PLAU	89	GTGGATGTGCCCT GAAGGA	712	CTCGGATCCAG GGTAGAA	1444	AAGCCAGGCGTCTA CACGAGAGTCTCAC	2176
USP34.1	NM_014709	NM_014709.2	USP34	4040	AAGCTGTGATGGC CAAGC	713	GGAATGGCCACA ACTGAGA	1445	TCCAGAGGCCCTGA GGTTGCTTTA	2177
VCAM1.1	NM_001078	NM_001078.2	VCAM1	1220	TGGCTTCAGGAGC TGAATACC	714	TGCTGTCTGTAT GAGAAAATAGTG	1446	CAGGCACACACAGG TGGACACAAAT	2178
VCAN.1	NM_004385	NM_004385.2	VCAN	5979	CCTGCTACACAGCC AACAAAG	715	AGAAAGCGCTG AGGTCC	1447	CCCCTGTGGAAGA CAAAAGGGCC	2179
VDR.2	NM_000376	NM_000376.1	VDR	971	GCCCTGGATTTCAG AAAGAG	716	AGTTACAAGCCA GGGAAGGA	1448	CAAGTCTGGATCTG GGACCCCTTCC	2180
VEGF.1	NM_000376	NM_000376.3	VEGFA	7	CTGCTGTCTTGGGT GCATTG	717	GCAGCTGGGAC CACTTG	1449	TTGCTTGCTGCTCT ACCTCCACCA	2181
VEGFB.1	NM_003377	NM_003377.2	VEGFB	964	TGACGATGGCCTG GAGTGT	718	GGTACCGGATCA TGAGGATCTG	1450	CTGGCAGCACCAA GTCCGGA	2182

TABLE A-continued

Gene	Accession Num	Sequence_ID	Official Symbol	Gene Version ID	F Primer Seq	SEQ ID No.	R Primer Seq	SEQ ID No.	Probe Seq	SEQ ID No.
VHL.1	NM_000551	NM_000551.2	VHL	4102	CGGTGGTGACTT GTCTGC	719	AAGACTGTGCCC TGCTCAG	1451	ATGCTCAGTCTTCC CAAAAGCAGG	2183
VIM.3	NM_0003380	NM_0003380.1	VIM	339	TGCCCTTAAAGAA CCAATGA	720	GCTTCAACGGCA AAGTTCTCTT	1452	ATTTCACGATCTGG CGTTCCA	2184
VTCN1.1	NM_024626	NM_024626.2	VTCN1	4754	ACAGTGGTCTGGG CATCC	721	GCTCAAAGCTGG TATTGGAGAC	1453	CGAGAAGTTGGCTC CCTGGTCAAC	2185
VTN.1	NM_000638	NM_000638.2	VTN	4502	AGTCAATCTTCGCA CACGG	722	GTAAGAGCGAT GGAGCGT	1454	TGGACACTGTGGAC CCTCCTTACC	2186
VWF.1	NM_000552	NM_000552.3	VWF	6212	TGAAGCACAGTGC CCTCTC	723	CCAGTCTCCCAT TCACCGT	1455	CTCCATGTCACTGTG CAGCTCGAC	2187
WIF.1	NM_007191	NM_007191.3	WIF	6077	AACAAGCTGAGTGC CCAGG	724	CAGTCGAGATG CGTCTTT	1456	TACAAAAGCCTCCAT TTCGGCACC	2188
WISP1.1	NM_003882	NM_003882.2	WISP1	603	AGAGGCAATCCATGA ACTTCACA	725	CAAACTCCACAG TACTTGGTTGA	1457	CGGCTGCATCAGC ACACGC	2189
WT1.1	NM_000378	NM_000378.3	WT1	6458	TGTAAGGTCTGGCA TCTGAG	726	TATATGCAGCCT GGTAAAGC	1458	CAGTGAGAAACGCC CCTTCATGTG	2190
WMOX.5	NM_016373	NM_016373.1	WMOX	974	ATCGCAGCTGGTG GGTGAC	727	AGTCCTCTGTGTG CATGGACTT	1459	CTGCTGTTTACCTTG GCGAGGCCTTTC	2191
XDH.1	NM_000379	NM_000379.3	XDH	5089	TGGTGGCAGACAT CCCTT	728	GCCACAACTGTC CCAGTCTT	1460	TGAAGCCAACTTGT ATCTGGCCA	2192
XIAP.1	NM_001167	NM_001167.1	XIAP	80	GCAATTTGAGAGAC ACAGGAAAGT	729	TGCGTGGCAGTA TTTTCAAGA	1461	TCCCAAAATTGCAGA TTTATCAGCGC	2193
XPINPEP2.2	NM_003399	NM_003399.5	XPINPEP2	6503	CACCTCGACTGAA CATACC	730	AAGAGGATGAA TGCAAAGG	1462	CCTGCTGGCCCATTT GCCTAGAA	2194
YB-1.2	NM_004559	NM_004559.1	YBX1	395	AGACTGTGAGTTT GATGTTGTTGA	731	GGAACACACCA GGACCTGTAA	1463	TTGCTGCCTCCGCA CCCTTTTCT	2195
ZHX.1	NM_014943	NM_014943.3	ZHX2	6215	GAGTACACCAAGTT AGCGGC	732	TCTCCTTGAACC AACGCAC	1464	ATCTCAGTTGCGACC AGGCAGTC	2196

TABLE B

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
A-Catenin.2	78	CGTTCGGATCCTCTATACTGTCATCCAGGCATGCTACAGCACCTTGATGTCGAGCCTATAAGGCCAA CAGGACCT	2197
A2M.1	66	CTCTCCGCCCTTCCTAGCTGTCCAGTGGAGAGGAACAAGCGCTCACTGCATCTGTGCAACGG	2198
AMP.1	66	GTGTGGCAGGTGGACACTAGGAGGAGTCTGTCTCTTTGAAGCGGGAGACCTGGAGTGGATGGAG	2199
ABCB1.5	77	AAACACCACTGGAGCAATTGACTACAGGCTCGCAATGATGCTGCTCAAGTTAAAGGGGCTATAGGTTT CAGGCTTG	2200
ACADSB.1	68	TGGCGGAGAACTAGCATCAGCCTCCTGAAGCTGCCATCATTGTTAATTTGAGGACTGGGCTGTCTT	2201
ACE.1	67	CCGCTGTACGAGGATTTCACTGCCCTCAGCAATGAAGCCTACAAGCAGGACGGCTTCACAGACACGG	2202
ACE2.1	66	TACAAATGAGAGGCTCTGGGCTTGGGAAAGCTGGAGATGTGAGGTCGGCAAGCAGCTGAGGCCATTA	2203
ADAM17.1	73	GAGTGGCAGGAGGCGATTAAATGCTACTTGCAAAAGGCGTGTCTACTTGCAAGGTAATAGCAGTGAGTG CCCG	2204
ADAM8.1	67	GTCACTGTGTCCAGCCACCTTCCAGTTCTCTGTCTACACCGGCAGGCACCAAGCAGGTCATCA	2205
ADAMTS1.1	73	GGACAGGTGCAAGTCACTGCCCCAAGCCAAAGGCAATTGGCTACTTCTTCGTTTTCAGCCCCAAGGTTGT AGAT	2206
ADAMTS4.1	66	GAGAAATGTCTGCCCTGGGCCCTACCTCCAGCAGAAGCCAGACACGGGCCACGATGAATACCACGAT	2207
ADAMTS4.1	71	TTTGACAAAGTCAATGGTGTGCGGAGGGGACGGTTCTGGTTGCAGCAAGCAGTCAGGCTCCTTCAGGAA ATT	2208
ADAMTS5.1	79	CACGTGGCTCACGAATCGACATTTACTTGGCCTCTCCCATGACGATTCCAAATTCCTGTGAAGAGACC TTGGTTGC	2209
ADAMTS8.1	72	GCGAGTTCAAAGTGTTCGAGGCCAAGGTGATTGATGGCACCTGTGTGGGCCAGAAACACTGGCCATCT GTG	2210
ADAMTS9.1	66	GCACAGGTTACAAACCAACAGAAATGTCCCTATAACGGGAGCCGCGCGATGACTGCCAATGTCG	2211
ADD1.1	74	GTCTACCCAGCAGCTCCGCAAGGAGGGATGGCTGCCTTAAACATGAGTCTTTGATGGTGACTCCTGTG AACGA	2212
ADPP.1	67	AAGACCATCACTCCGTGGCCCATGACCAGTCTCTGCCATCATCCAGAACTAGAGCCGCAAAATTG	2213
ADH1B.1	84	AAGCCAAACACCTTCCTTTCTTAACCATCTACTGTGTACCTTTGGCATTGAGGAAAAATATTCCTGTGA CTTCTTGCATTTT	2214
ADH6.1	68	TGTTGGGAGTAAACACTTGGACCTCTTTGTATCCACCATCTTGGGCCATGAGGGGCTGGAAATCGTT	2215

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ADM.1	75	TAAGCCACAAGCACACGGGGCTCCAGCCCCCGAGTGGAAAGTCTCCCCACTTCTTTAGGATTTAGGCGCCCA	2216
AGR2.1	70	AGCCACAATGTGACTAATTGGAGAAGAGCAAGGGTGGTACGTGTGTGATGAGGCAGATGGAGATCAGAA	2217
AGT.1	73	GATCCAGCCTCACTATGCTCTGTGACCTGTGACAGGTGGAGGGTCTCACTTTCCAGCAAAAACCTCCCTCAA	2218
AGTR1.1	67	AGCAATTGATCGATACCTGGCTATTGTTACCCAAATGAAGTCCCGCTTCGACGCACAAATGCTTTAG	2219
AHR.1	69	CGGCATAGAGACCGACTTAATACAGAGTTGGACCGTTTGGCTAGCCTGTGCGCTTTCCACACAAGATGT	2220
AIF1.1	71	GACGTTACAGCTACCTTGACTTTCTCAGGATGATGCTGGGCAAGAGATCTGCCATCTCTAAAAATGATCCTGA	2221
AKT1.3	71	CGCTTCTATGGCGCTGAGATTGTGTGAGCCCTGGACTACCTGCTACCTCGAGAGAAACGTTGGTGTACCGGGA	2222
AKT2.3	71	TCCTGCCACCCTTCAAACCTCAGGTCACGTCGAGGTGACACAAAGGTACTTTGGATGATGAATTTACCG	2223
AKT3.2	75	TTGTTCTGTGCTTGGACTATCTACATTCGGGAAAGATTGTGTACCGTATCTCAAGTTGGAGAAATCTAAT	2224
ALDH4.2	68	GGACAGGGTAAGACCGTGATCCAAAGGAGATTGACGCTGCAGCGGAACCTCATCGACTCTTCTCCGGTT	2225
ALDH6A1.1	66	GGCTCTTTCAACACAGAGTCCTTTGTGGGAGAGCCAAAGAGTGGCTGCCAGAGCTGGTGGAGCATGC	2226
ALDOA1	69	GCCTGTACGTGCCAGCTCCCGACTGCCAGAGCCTCAACTGTCTCTGTTTCCAGATCAAGCTCCGATGA	2227
ALDOB.1	80	CCCTCTACAGAAAGACAGCCAGGGAAGCTGTTTCAGAAACATCCTCAAGGAAAAGGGGATCGTGGTG	2228
ALOX12.1	67	AGTTCTCAATGGTGGCAACCCCAATGCTGTTGAGACGCTGACCTCTCTGCCCCTCCAGGCTAGTGCT	2229
ALOX5.1	66	GAGCTGCAGGACTTCGTGACGATGTCTACGTGTACGGCATGCGGGCGCGCAAGTCTCTCAGGCTTC	2230
AMACR1.1	71	GGACAGTCAGTTTGGGTGGCTGTATCCAGTAACCTCGGGGCTGTTTCCCGTGGGTCTCTGGGCTG	2231
ANGPT.1	71	TCTACTTGGGTGACAGTGCTCACGTGGCTCGACTATAGAAAACCTCCACTGACTGTGCGGCTTTAAAAA	2232
ANGPT2.1	69	CCGTGAAAAGCTGCTCTGTAAAGCTGACACAGCCCTCCCAAGTGAGCAGGACTGTTCTTCCCACTGCAA	2233
ANGPTL2.1	66	GCCATCTGCGTCAACTCCAGGAGCCTGAGGTGCTTCTGGAGAACCGAGTGCAATAGCAGGAGCTA	2234
ANGPTL3.3	78	GTTGCAATTACTGGCAATGTCCTCCCAATCCGGAACCAAGATTTGGTGTGTTTTTCTACTTGGGATC	2235

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ANGPTL4.1	66	ATGACCTCAGATGGAGCTGGACAGTAATTGAGAGCGCCACGATGGCTCAGTGGACTTCAACCGG	2236
ANGPTL7.1	67	CTGCACAGACTCCAAACCTCAATGGAGTGTACTACGCTGGGTGAGCACAATAAGCACCTGGATGGC	2237
ANTXR1.1	67	CTCCAGGTGTACCTCCCAACCTAGCCTTCTCCACAGCTGCCTACAAACAGAGTCTCCAGCCTTCTC	2238
ANSA1.2	67	GCCCTATCCTACCTTCAAATCCATCCTCGGATGTGCTGCTGCTTGCATAAGGCCATAATGGTTAAAGG	2239
ANXA2.2	71	CAAGACACTAAGGCGACTACAGAAAAGCGCTGCTGTACCTGTGTGTGGAGATGACTGAAGCCCGAC ACG	2240
ANXA4.1	67	TGGAGGGATGAAGAAAATTATCTGGACGATGCTCTCTGTGAGACAGGATGGCCAGGACCTGTATGAG	2241
ANXA5.1	67	GCTCAAGCGTGGAGATGACGTGTGGGGGACACTTCAGGGTACTACAGCGGATGTTGGTGTCT	2242
AP-1 (JUN official).2	81	GACTGCAAAGATGGAAAACGACCTTCTATACGATGCCCTCAACGCCCTCGTTCTCTCCGTCGGAGAGCGG ACCTTATGGCTA	2243
APLM2.1	67	ACAAACGACCGCACCATCTCCTTTCATCCCGCTGATGGTGTACTTTGAGCTCATGTACACGCCCTCAG	2244
APAF1.2	66	CACAAAGGAAGCTGGTAATGCAATTACAGAGAAGCTCTCCAAATTGAAAGGTGAACCAAGGATG	2245
APC.4	69	GGACAGCAGGAATGTGTTCTCCATACAGGTCACGGGAGCCATGGTTCAGAAACAAATCGAGTGGGT	2246
APOC1.3	70	CCAGCCTGATAAAGGTCTCTCGGGCAGGACAGGACCTCCCAACCAAGCCCTCCAGCAAGGATTCAGAG TG	2247
APOE.1	75	GCCTCAAGAGCTGTTCTGAGCCCCCTGGTGGAAACATGACAGCCAGTGGGCCGGGCTGGTGAGAA GGTGCAGG	2248
APOL1.1	73	CGGACCAAGAACTGTGACCACAGGCGAGGCGAGCCACCAAGAGAGATATGCTGGCAGGGGCCAGGA CAAAAT	2249
APOLD1.1	66	GAGCAGCTGGAGTCTCGGGTTTCACTCTGCACCAAGTCCAGTCTGTGGCCACGACCTCAAGATCTCT	2250
AQP1.1	66	GCTTGCTGTATGACCCCTGGCCACAGCCTTCCCTCTCGATTGACCTGGAGGGGAGAGGTGAGCCTT	2251
AREB.2	82	TGTAGTGAATGCTTCTAGTAGTGAACCGTCTCTCGGAGCCGACTATGACTACTCAGAAGAGTATGA TAACGAACCAAA	2252
ARF1.1	64	CAGTAGAGATCCCGCAACTCGCTTGTCTTGGTTCACCCCTGCAATTCATAGCCATGTGCTTGT	2253
ARG9.1	67	GCATGGGCTACTGCATCCTTTTGTGCACGGACTGAGCAAGCTCTGCACCTTGGCTGAATCGATGTGG	2254
ARGHEF18.1	71	ACTCTGCTTCCCAAGGCAACCGTTGCTGTTCACACGCTCAGCCTGTCTGGGGAGCGGGCCTTAGC TTC	2255

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ARHA.1	73	GGTCTCCGTCGGTTCTCTCATTAGTCCACGGTCTGCTTTCAGCTACCCGCTTCGTCTCCGAGTTTGCGAC	2256
ARHGDUB.1	66	TGGTCCCTAGAACAAAGAGGCTTAAACCGGGCTTTCACCCAACTGCTCCCTCTGATCCTCCATCA	2257
ARRB1.1	69	TGCAGGAACGCTCATCAAGAGCTGGCGAGCAGGTTTACCTTTTACGATCCCTCCAAACC	2258
ASS1.1	85	CCCCAGATAAAGTCAATTGCTCCCTGGAGGATGCTGAATTTCTACAACCGGTTCAAGGGCCGCAATGACCTGATGGAGTACGCA	2259
ATP1A1.1	67	AGAACGCCATTATTGGAGCTGGGGGCCCTCGGAAACGAGTCCTAGGTTTCTGCCACCTCTTTCTGCC	2260
ATP5E.1	66	CCGCTTTGCTACAGCATGGTGGCTACTGGAGACAGGCTGGACTCACTACATCCGATACTCCCA	2261
ATP6V1B1.1	67	AACCATGGGAACGCTGCTCTTCTGAACTTGGCCAAATGACCCACGATCCGAGCGGATCATCACC	2262
AXL.1	66	TTGCAGCCCTGTCTTCTCATCCTATCCACCTCCATCCAGACAGGTCCCTCCCTTCTCTGTGCAG	2263
AZU1.1	74	CCGAGGCCCTGACTTCTTCACCGAGTGGGCTCTTCCGAGACTGGATCGGATGGTGTCTCAACAACCCGGGAC	2264
B-Catenin.3	80	GGCTCTTGTCGCTACTGTCTTCTCGGGCTGGTGACAGGAGAGACATCACTGAGCCTGCCATCTGTGCTCTCGTCACTGGA	2265
B2M.4	67	GGGATCGAGACATGTAAGCAGCATCATGGAGGTTTGAAGATGCCGCAATTGGATTGGATGAATTCCA	2266
BAD.1	73	GGGTACGGGGCTCGAGATGGGCCACAGAGCATGTTCCAGATCCCAGAGTTTGAGCCCGAGTGAGCAG	2267
BAG1.2	81	CGTTGTGAGCACTTGGAAATACAGATGGTTGCCGGGTCAATGTTAATTGGGAAAAGAACAGTCCACAGGAAGAGGTTGAAC	2268
BAG2.1	69	CTAGGGCAAAAAGCATGACTGCTTTTTTCTGTCTGGCATGGAATCAGCAGTCACCTTGGGCATTTAG	2269
Bak2	66	CCATTCCACCAATCTACTCTGAGGCCAGGACGCTGGGGTGTGGGGATTGGTGGGTCTATGTTCCC	2270
Bax.1	70	CCGCCGTGGACACAGACTCCCCCGAGAGGTCTTTTCCGAGTGGGAGCTGACATGTTTTCTGACGGCA	2271
BBC3.2	83	CCTGGAGGGTCTGTACAAATCTCATATGGGACTCTTGCCCTTACCCAGGGGCCACAGAGCCCCCGAGATGGAGCCCAATTAG	2272
Bcl2.2	73	CAGATGGACCTAGTACCCACTGAGATTTCCACGCCGAGAGACAGCATGGGAAAAAATGCCCTTAAATCATAGG	2273
BCL2A1.1	79	CCAGCCTCCATGATCATCATGTGTCTATACTCAGTCAAGCTCAGTGAGCATTTCTCAGCACATTTGCCCTCAACAGCTTCA	2274

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
BCL2L12.1	73	AACCCACCCCTGTCTTGGAGCTCCGGGTAGCTCTCAAACTCGAGGCTGCGACCCCTTTCCCGTCAGC TGAG	2275
Bclx.2	70	CTTTTGTGGAATCTATGGGAACAATGCGAGCGCGAGAGCGAAGGCGCAGGAACGCTTCAACCGCT G	2276
BCRP.1	74	TGTA CTGGCGAAGAATAATTGGTAAAGCAGGGCATCGATCTCTCACCCTGGGCTTGTGGAAGAATCAC GTGGC	2277
BFGF.3	77	CCAGGAAGAATGCTTAAGANGTGATGGATCTCAATGACCTGGCGAAGACTGAAAAATACAACTCC CATCACCA	2278
BGN.1	66	GAGTCCGCAAGGATGACTTCAAGGGTCTCCAGCACCTCTACGCCCTCGTCCCTGGTGAACAACAAG	2279
BHLHB3.1	68	AGGAAGATCCCTCGACGCCAGGAAGAGCTCCCTGAACTCTTGCGTCCGAAAGGACGGAGGTTCAA	2280
BIK.1	70	ATTCTATGGCTCTGCAATTGTCAACCGTTAACTGTGGCCTGTGCCAGGAAGAGCCATTCACTCTCTGC	2281
BIN1.3	76	CCTGCAAAAGGGAACAAGACCCTTTCGCTCCAGATGGCTCCCTGCGGCA CCCCAGATCAGAGT CAACCAAG	2282
BLR1.1	67	GACCAAGCAGAAAGCTCAGACTGGTTGAGTTCAGGTAGTGCCTCCCTGGCTGTGACCGAAACAGCGCT	2283
BNIP3.1	68	CTGGACGGAGTAGCTCCAAGAGCTCTCACTGTGACAGCCCACTCGCTCGCAGACACCAAGATACC	2284
BRCA1.1	65	TCAGGGGCTAGAAATCTGTTGCTATGGGCCCTTCAACCAACATGCCCAACAGATCAACTGGAATGG	2285
BTRC.1	63	GTTGGGACACAGTTGGTCTGCAGTCGGCCAGGACGGTCTACTCAGCACAACTGACTGCTTCA	2286
BUB1.1	68	CCGAGGTTAATCCAGCAGGTATGGGCCCAAGTGTAGGCTCCAGCAGGAACCTGAGAGGCCATGCTTT	2287
BUB3.1	73	CTGAAGCAGATGGTTTCATATTCTCGGGCTGTTAAACAAGAGAGGTTAAGGTTAGACTCTTGGGAATC AGC	2288
c-kit.2	75	GAGCAACTGCTTATGGCTTAATTAAGTCAAGTCGGGCCATGACTGTGCTGTAAAGATGCTCAAGCCG AGTGCC	2289
C13orf15.1	84	TAGAACTGTCTGCCAGAGGGGACAAAGACGTGCACTCAACCTTCTACAGGCCACTCTCAGGCTCACCT TAAAAACAGCCCTTG	2290
C1QA.1	66	CGGTCAACCAACAGGAGAACCGTACCAGAACCACTCCGGCCGATTGCTGTGCACCTGTACCCG	2291
C1QB.1	70	CCAGTGGCCTCACAGGACACCAAGCTTCCAGAGGGGCTGTGACACAGTATGATGATGAAGATCCCATGG G	2292
C20 orf1.1	65	TCAGCTGTGAGCTCGGATACCGCCCGCAATGGGACCTGCTCTTAACCTCAACCTAGGACCGT	2293
C3.1	67	CGTGAGGAGTGCAAGAAAGAGGACATCCCACTGACAGACTCAGTGACCAAGTCCCGGACACCGAGT	2294

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID No.
C3AR1.1	66	AAGCCGATCCAGACTTGCTGAATCGGAATCTCTGGGGTTGGACCCACAGAAAGGCACCTTAACA	2295
C7.1	69	ATGTCTGAGTGTAGGGGGCGCTCTGAGATGCAGAGGGCAGACATCTCTGTCCACGACATAAGGCC T	2296
CAL2.1	66	CTCTCTGAGGTGTCCTGGCCAGCCCTGGAGAGCAGCTGGTCTCTGCAGCACCCCTCAGTTCCTGT	2297
CA2.1	69	CAACGTGGAGTTTGATGACTCTCAGGACAAAGCAGTGTCTAAGGGAGGACCCCTGGATGGCAGCTTACAG	2298
CA9.3	72	ATCCTAGCCCTGGTTTTTGGCCCTCCTTTTTTGTCTGTCAACAGCGTCGCGTTCCCTTGTGCAGATGAGAAG CAG	2299
CACNA2D1.1	68	CAAACTTACGTGGCCCTGTTCCATGGCATAACTAAGGCGCAGACTCCTTAAGGCACCCACTGGCTG	2300
CALD1.2	78	CACCTAAGTTTGAACAGTTCACAGAAAGACCCCAAGCTCAAGACGCAGGACGAGCTCAGTTGTAGAGGG CTAATTCGC	2301
CASP1.1	77	AGAAAGCCCATAGAGAAGATTATTATCGCTTCTGCTCTTCCACACAGATAATGTTTCTTGGAGACA TCCCACA	2302
CASP10.1	66	ACCTTTCTCTTGGCCGGATGTCCTCAGGGCTGGCAGATGCAGTAGACTGCAGTGGACAGTCCCCAC	2303
CASP6.1	67	CCTCACACTGTTGAACAGAAAAGTTTCTCAGCGCCGAGTGGACTTTTTCAAAGACCCCAAGTGCATTT	2304
Capase 3.1	66	TGAGCCTGAGCAGAGACATGACTCAGCCCTGTTCCATGAAGGCAGAGCCATGGACACGCAGAGAAAG	2305
CAT.1	78	ATCCATTGATCTCACCAAGGTTTGGCCTCACAAAGGACTACCCCTCTCATCCAGTTGGTAAACTGGTCTT AAACCGGA	2306
CAV1.1	74	GTGGCTCAACATTTGTGTTCCATTTACGCTGATCAGTGGGCTCCAAAGAGGGGCTGTAAATGGAGGC CATTG	2307
CAV2.1	66	CTTCCCTGGGACGACTTGCCAGCTCTGAGGCATGCACAGTACGGGCCCCACAGAGGGTGACAGGAG	2308
CCL18.1	68	GCTCCTGTGCACAAAGTTGGTACCAACAAAGAGCTCTGTGCTCGTCTATACCTCCTGGCAGATTCCA	2309
CCL19.1	78	GAACGCATCATCCAGAGACTGCAGAGGACCTCAGCCCAAGATGAAGCGCCGACAGTTAACCTATGACC GTGCAGAGG	2310
CCL20.1	69	CCATGTGCTTACCAAGAGTTTGCTCCTGGCTGTTGATGTCTAGTGTGCTACTCCACCTCTCCGGGG	2311
CCL4.2	70	GGGTCCAGGAGTACGTGTATGACCTGGAACTGAACCTGAGCTGCTCAGAGACAGGAAGTCTTTCAGGGAA GG	2312
CCL5.2	65	AGGTTCTGAGCTCTGGCTTTTGCCTTGGCTTTGCCAGGGCTCTGTGACCAGGAAGGAAGTCAGCAT2313	2313
CNNB1.2	84	TTCAAGTTGTTGCAGGAGACCATGTACATGACTGTCTCCATTATTGATCGGTTTCATGCAGAAATAATTGTG TGCCCAAGAGATG	2314

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
CND1.1.3	69	GCATGTTCTGTGGCCTCTAAGATGAAGGAGACCATCCCTCGACGCCGAGAAGCTGTGCATCTACACC G	2315
CONE1.1	71	AAAGAAGATGATACCGGTTTACCCAAACTCAACGTCGAAGCCTCGGATTAITGCACCATCCAGAGGC TC	2316
CONE2 variant 1.1	85	GGTCAACCAAGAAAATCATCAGTATGAAATTAGGAATTGTTGGCCACCTGTATTAATCTGGGGGATCAGTCCT TGCATTATCATTGAA	2317
CONE2.2	82	ATGCTGTGGTCTCTTCCCTAATCTGGGGCTTCTTGACATGTAGTTGCTTGGTAATAACCTTTTGTATATC ACAATTTGGGT	2318
CCR1.1	66	TCCAAGACCCAAATGGGAATTCACCTCACACACCTGCACCTTCACTTTCCCTCACGAAGCCTACGA	2319
CCR2.1	67	CTCGGGAATCCTGAAAACCTGCTTCGGTGTGGAACGAGAGAGGATAGGGCAGTCAGAGTC	2320
CCR4.2	82	AGACCTGGTGGAGCTAGAGTCCCTTCAGGACTGCACCTTTGAAAAGATACTTGGACTATGCCATCCAGG CCACAGAACTCT	2321
CCR5.1	67	CAGACTGAATGGGGTGGGGGGCGCTTAGGTACTTAATCCAGATGCCCTTCTCCAGACAAACCAG	2322
CCR7.1	64	GGATGACATGACATCAGCTCTTGGCTCCACTGGGATGGGAGGAGGACAAGGGAATGTGAGG	2323
CD105.1	75	GCAGGTGTCACAGATGATGATCAGCAATGAGGCGGTGGTCAATATCCTGTGAGCTCATCACCACAGGG GAAAAA	2324
CD14.1	66	GTGTGCTAGCGTACTCCGCGCTCAAGGAACGTGACGCTCGAGGACCTAAAGATAACCGGCACCATGC	2325
CD18.2	81	CGTCAGGACCCACCATGTCTGCCCATACGCGGCCGAGACATGGCTTGGCCACAGCTCTTGAGGATG TCACCAATTAAAC	2326
CD1A.1	78	GGAGTGAAGGAACCTGGAACATTAATTCGGTATACGCACCATTCCGTCATTTGAGGGAATTCGTAGATAC GCCCATGA	2327
CD24.1	77	TCCAACTAATGCCACCAAGCGGCTGGTGTGCCCTGCAGTCAAACAGCCAGTCTCTTGGTGGTCTC ACTCTCTC	2328
CD274.2	69	GCTGCATGATCAGCTATGGTGGTCCGACTACAAGCGAATTAAGTGAAGTCAATGCCCATACAACA	2329
CD31.3	75	TGTATTTCAAGACCTCTGTGCACTTATTTATGAACCTGCCCTGTCTCCACAGAACACAGCAATTCCTCAG GCTAA	2330
CD34.1	67	CCACTGCACACCTCAGAGGCTGTTCTTGGGGCCCTACACCTTGAGAGGGGCGAGTAACCTCTG	2331
CD36.1	67	GTAACCCAGGACGCTGAGGACACACAGTCTCTTTCTTGACGCCCAATGGTGCCATCTTCGAACCTT	2332
CD3z.1	65	AGATGAAGTGAAGCGCTTTTCAACCGGCCATCTCTGCAGGCACAGTTGCCGATTACAGAGGCA	2333
CD4.1	67	GTGCTGGAGTCGGACTAAACCCAGGTCCCTTGTCACAGTTCCACTGCTGCTCTTGAATGCAGGGA	2334

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
CD44 . 1	67	GGCACCAGCTGTTATGAAGGAACTGGAAACCCAGAGACACACCCCTCCCTCATTCACCATGAGCATG	2335
CD44s . 1	78	GACGAAGACAGTCCCTGGATCACCGACAGCACAGACAATCCCTGTACAGAGACCAAGACACATTC CACCCAGT	2336
CD44v6 . 1	78	CTCATACCAAGCATCCAAATGCAAGGAAGGACACACCAAGCCCAGAGACAGTTCTCTGGAGTATTCT TCAACCCAA	2337
CD53 . 1	72	CGACAGCATCCACCGTTACCACTCAGACAAATAGCACCAGGACGCTGGGACTCCATCCAGTCATTCT GCA	2338
CD68 . 2	74	TGTTTCCAGCCCTGTGTCTCCTCCACCTCCAAAGCCAGATTGAGATTCGAGTCAATGTACACAACCCAGGTGG AGGAG	2339
CD82 . 3	84	GTGAGGCTCAGGTGAAGTGTGTCGGCTGGGTCAAGTTCTACAACTGGACAGACAACGCTGAGCTCAT GAATCGCCCTGAGGTC	2340
CD8A . 1	68	AGGTGAGGTGCTTGAGTCTCCAAAGGCAAGGAACAAGTACTTCTTGATACCTGGGATACTGTGCCC	2341
CD99 . 1	77	GTTCTCCGCTAGCTTTTCAGATGCTGACCTTGCAGATGGGCTTTCAGGTGGAGAGGAAAGGAGGCA GTGATGGT	2342
cdc25A . 4	71	TCCTTGCTGGCTACGCCTCTTCTGTCCCTGTTAGAGTCTCTCCGTCCATATCAGAACTGTGCCCACATGCA G	2343
CDC25B . 1	85	AAACGACAGTTTGCATCAGACGCTTCCAGTCTATGCCGTGAGGCTGCTGGGCCACAGCCCCGTGC TTCGGAACATCACCAAC	2344
CDH1 . 3	81	TGAGTGTCCTCCCGTATCTTCCCGCCCTGCCAATCCCGATGAAATTGGAAATTTTATTGATGAAATCT GAAAGCGGCTG	2345
CDH13 . 1	67	GCTACTTCTCCACTGTCCCGTTCAAGTCTGAATGCTGCCACACAGCCAGGACAGTCCACAGAGAGG	2346
CDH16 . 1	67	GACTGTCTGAATGCGCCAGGCAAGCTCTAGCTGGGAGCTTGGCCTCCATCTGAGTCCCTGG	2347
CDH2 . 1	66	TGACCGATAAGGATCAACCCCATACACAGCCTGGAACGCAAGTGTACAGAAATCAGTGGCGGAGATC	2348
CDH5 . 1	67	ACAGGAGACGTGTTTCGCATTGAGAGGCTGGACCGGAGAAATATCTCAGAGTACCACTCACTGCTG	2349
CDH6 . 1	66	ACACAGCGACATACAGGCCACCAAGAGGCTGGAAGGGAAGAAAAACCCGTTTACATCTCTTCGAG	2350
CDH4 . 1	66	CCTTCCCATCAGCACAGTTCGTGAGGTGGCTTTACTAGGCGCACTGGAGGCTTTTGAGCATCCCAA	2351
CDH6 . 1	67	AGTGCCCTGTCTCACCATACCTCCAGGACCTGGAAAGGTGCAAGAAACCTGGATTCCCACTGTC	2352
CDKN2A . 2	79	AGCACTCAGCCCTTACGCGCAATTCATGTGGGCAATTTCTTGGAGCCTCGCAGCCTCCGGAAGCTGTC GACTTCATGA	2353

171

172

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
CEACAM1.1	71	ACTTGCCTGTTTCAGAGCAGCTCAATTCCTCCACCCCGAGTCCTGCTATCACTCTAATTGGATTGGCC A	2354
CDBPA.1	66	TTGGTTTTCGCGATACCTTGC AAAATGAGACTCTCCGTGGCAGCTGGGGGAAGGCTCTGAGAC	2355
CENPF.1	68	CTCCCGTCAACAGCGTCTCTTCCAAACACTGGACAGGAGTGCAATCCAGATGAAGGCCAGACTCACCC	2356
CFLAR.1	66	GGACTTTTGTCCAGTGACAGCTGAGACAACAAGGACCACCGGAGGAGGTGTAGGAGAGAAAGCGCCG	2357
CGA (CHGA official)	76	CTGAAGGAGCTCCAAAGACCTTCGCTCTCCAAAGGCGCCAAAGGAGGGCACATCAGCAGAAGAAACACAG CGGTTTGG	2358
Chk1.2	82	GATAAATTGGTACAAAGGATCAGCTTTTCCAGCCCAATGTCTGTGATCATATGCTTTTGAATAGTCAGT TACTTGGCACCC	2359
Chk2.3	78	ATGTGGAACCCCCACCTACTTTGGCGCCTGAAAGTTCTTGTTCGTGGGACTGCTGGGTATAACCGTGC TGTGGACTG	2360
CIAP1.2	72	TGCCTGTGGTGGGAAGCTCAGTAACTGGGAACCAAAGATGATGCTATGTGAGAACACCGGAGGCAITTT TCC	2361
cIAP2.2	86	GGATAATTCCGTGCTCTTATTCAAACTCTCCATCAAAATCCTGTAAACTCCTGTAAGCAAAATCAAGATTTTTC TGCCTTGATGAGAAG	2362
CLCNKB.1	67	GTGACCCCTGAAGCTGTCCCCAGAGACTTCCCTGCATGAGGCACACAACCTCTTTGAGCTGTTGAACC	2363
CLDN10.1	66	GGTCTGTGGATGAAC TGCGCAGGTAAACGCGTTGGGTCTTTTCCATTGCCGACCGCATTTTACTATC	2364
CLDN7.2	74	GGTCTGCCCTAGTCACTCCTGGGAGGTGCAGTGTCTCTCTGTCTCTCTGGGAATGAGAGCAAGGCT GGGTAC	2365
CLU.3	76	CCCAGGATACCTACCACTACCTGCCCTTCAGCCTGCCCCACCGAGGCGCTCACTTCTTTTCCCAAG TCCCGCA	2366
cMet.2	86	GACATTTCCAGTCTCTGCAGTCAATGCCTCTCTGCCCCACCCCTTTGTTCAGTGTGGCTGGTGCCACGAA AATGTGTGCGATCGGAG	2367
cMYC.3	84	TCCCTCCACTCGGAAGGACTATCCTGTCTGCCAGAGGGTCAAGTTGGACAGTGTCAAGTCTCTGAGACA GATCAGCAACAACCG	2368
COL18A1.1	67	AGCTGCCATCAGCCTACATCGTCTCTGCATTTGAGAACAGCTTTCATGACTGCGCTCCAAAGTAGCCAC	2369
COL1A1.1	68	GTGGCCATCCAGCTGACCTTCTCGCGCTGATGTCCACCCGGGCTCCAGAAACATCACCTACCACCTG	2370
COL1A2.1	80	CAGCAAGAACTGTATAGGAGCTCCAAGGACAAAGAAACAGTCTGGCTAGGAGAACTATCAATGCTG GCAGCCAGTTT	2371
COL4A1.1	66	AGAAAGGCCTCCCGAGATTGGATGGCATCCCTGGTGTCAAAGGAGAAAGCAGGTCTTCTCTGGGACTC	2372

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
COL4A2.1	67	CAACCTGGTGATGTCTGCTACTATGCCAGCCGGACGACAAAGTCCTACTGGCTCTCTACCACTGGC	2373
COL5A2.2	72	GGTCAGGAACCCCAAGTCCGCTGGTGCTACAGGATTTCCTGGTTCTGCGGGCAGAGTTGGACCTCC AGGC	2374
COL7A1.1	66	GGTGACAAAGACCTCGGGGAGACAATGGGGACCCCTGGTGACAAAGGGCAGCAAGGGAGAGCCTGGT	2375
COX2.1	79	TCTGCAGAGTTGGAAGCACCTATGGTGACATCGATGCTGTGGAGCTGTATCTCCCTCTTCTGGTAGAA AAGCCTCGGC	2376
CP.1	73	CGTGAGTACACAGATGCTCTTCCACAAATCGAAAGGAGAGAGGCCCTGAAGAAGAGCATCTTGGCATC CTGG	2377
CPB2.1	67	GGCACATACGGATTCTTGCTGCCGAGCGTTACATCAAAACCCACCTGTAGAGAAGCTTTTGCCGCTG	2378
CRADD.1	69	GATGGTGCTCCAGCAACCCGCTGGGGAGTGTCCCTGAGTCACTGGGGTGAATCCTGACTTTCACTC	2379
cripto (TDGF1 official	65	GGGTCTGTGCCCCATGACACCTGGCTGCCCAAGAGTGTCCCTGTGTAATGCTGGCACGGTCA	2380
CRP.1	66	GACGTGAACACACGGGTGTCTGTTCAGAGGAGCCCATCTCCCATCTCCCCAGCTCCCTATCTGGAG	2381
CSF1.1	74	TGCAGCGGCTGATTGACAGTCAGATGGAGACCTCGTGCACAAATACATTTGAGTTTGTAGACCAGGAAC AGTTG	2382
CSF1R.2	80	GAGCACAAACCAACCTACGAGTGCAGGGCCCAACACACGCTGGGGAGTGGCTCCTGGGCCCTTCATACC CATCTCTGAGG	2383
CSF2.1	76	GAACCTGAAGGACTTCTGTCTTGTATCCCTTTTGACTGCTGGGAGCCAGTCCAGGAGTGAGACCGGCC AGATGAG	2384
CSF2RA.2	67	TACCACACCCAGCATTCCTCTGATCCAGAGAAAATCGGATCTCGAACAGTGGCACCCAGCCTCTAG	2385
CSF3.2	79	CCGAGGCTCTGTCTCTTCCCTGCAATTCGTAGTTTCAATTCCTCGCTGTAGCAGTGAGAAAAAGCTC CTGTCTCTCC	2386
CTGF.1	76	GAGTTCAAAGTCCCTGACGGCGAGGTCAATGAAGAAGACATGATGTTCAACAGACCTGTGCCCTGCCAT TACAACT	2387
CTSB.1	62	GGCCGAGATCTACAAAACGGCCCCGTGGAGGAGCTTTCTCTGTGTATTCGGACTTCCTGC	2388
CTSD.2	80	GTACATGATCCCCGTGAGAAAGGTGTCCACCCTCGCCCGCATCACACTGAAGCTGGGAGGCAAGGCT ACAAGCTGTCCC	2389
CTSH.2	77	GCAAGTTCCAACCTGGAAGGCCATCGGCTTTGTCAAGGATGTAGCCAACATCAATCTATGACGAGG AAGCGATG	2390
CTSL.2	74	GGGAGGCTTATCTCACTGAGTGAGCAGAACTGGTAGACTGCTCTGGGCCCTCAAGGCAATGAAGGCTG CAATGG	2391

175

176

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID No.
CTS12.1	67	TGTCCTCACTGACGACAGAAATCTGGTGGACTGTTCGCTCTCAAGGCAATCAGGGCTGCAATGGT	2392
CTSS.1	76	TGACAACGGCTTTCAGTACATCATTTGATAACAAGGCAATCAGTCAAGCGCTTCCTATCCCTACAAAGC CATGGA	2393
CUBN.1	71	GAGGCCGTTACTTGGCACCGACATGCCCCATCCTATACATCCTTCAGCAGCGCCCTGACGCTGAGAT TC	2394
CUL1.1	71	ATGCCCTGGTAATGTCTGCATTCACATGACGCTGGCTTTGTGGCTGCTCTTGATAAGGCTTGTGTC GC	2395
CUL4A.1	75	AAGCATCTTCCTGTTCTTGGACCGCACCTATGTCTGCAGAACTCCACGCTGCCCTCCATCTGGGATAT GGGATT	2396
CX3CL1.1	66	GACCTTGCCGCTACCTGAGGGGCTCTTATGGCTGGGTTCTACCCAGGTGCTAGGAACACTCC	2397
CX3CR1.1	68	TTCCAGTTGTGACATGAGGAAGGATCTGAGGCTGGCCCTCAGTGTGACTGAGACGGTTGCATTTAGC	2398
CXCL10.1	68	GGAGCAAAATCGATGCACTGCTTCCAAAGATGGAACACACAGAGGCTGCCCTCCTCCATCATCTCCCTA	2399
CXCL12.1	67	GAGCTACAGATGCCATGCCGATTCTTCGAAGCCATGTTGCCAGAGCCACGTCAGCATCTCAAA	2400
CXCL14.1	74	TGCGCCCTTCTCTGTACATATACCCTTAAGAACGGCCCTCCACACACTGCCCCCAGTATATGCCGC ATTG	2401
CXCL9.1	70	ACCAGACCAATTGTCTCAGAGCAGGTGCTGCTCTTCTGGCTACTCCATGTTGGCTAGCCTCTGTAA	2402
CXCR4.3	72	TGACCGCTTCTACCCCAATGACTTGTGGTGGTGTGTGTTCCAGTTTCAGCACATCATGGTTGGCCTTATC CT	2403
CXCR6.1	67	CAGAGCCTGACGGATGTTCCTGGTGAACTACCCCTGGCTGACCTGGTGTGTTGTCTGCACCTGTC	2404
CYP2C8.2	73	CCGTGTTCAAGAGGAGCTCACTGCCCTTGTGGAGAGTTGAGAAAACCAAGGCTTCACCCCTGTGATCC CACT	2405
CYP2C8v2.1	70	GCTGTAGTGACGAGATCCAGAGATACAGTGACCTTGTCCCCACCCGGTGTGCCCATGCACTGACCACT G	2406
CYP3A4.2	79	AGAACAGGACACATAGATCCTTACATATACACACCTTTTGAAGTGGACCCAGAAAACCTGCAATTGGCAT GAGGTTTGC	2407
CYR61.1	76	TGCTCATTTCTTGGAGCATTAGGTATTTTCGAAACTGCCAAGGGTGTCTGGTGGGATGGACACTAATG CAGCCAC	2408
DAG1.1	67	GTGACTGGGCTCATGCCCTCCAAAGTCAGAGTTTCCCTGGTGGCCACAGACAGGAGGCACAAAGTGGGAT	2409
DAPK1.3	77	CGCTGACATCATGAATGTTCTCGACCGGCTGGAGGCGAGTTTGGATATGACAAAAGACACATCGTTGCT GAAAAGA	2410

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
DCBLD2.1	69	TCACGAGGCGAGGAAGTTTATCATGCTATGCTGAACCACTCCCAATTACGGGCGCTGAGTATGCAACC	2411
DCC.3	75	AAATGTCCTCTGACTGCTCCGCGAGTCCGACCGAGGAGTTCAGTGATCAAGTGGAGAGAAAGATG GCATTCA	2412
DCN.1	67	GAAGCCACTATCATCTCTCTTCTGTTGCACAAGTTTCTTGGGCTGGACCGTTTTCAACAGAGAGGC	2413
DCXR.1	66	CCATAGCGTCTACTGCTCCACCAAGGTTGCCGTGACATGCTGACCAAGGTGATGGCCCTAGAGCT	2414
DDC.1	67	CAGAGCCGAGACACCATGAACGCAAGTGAATCCGAAGGAGAGGAAAGGAGATGGTGGATTACGTGG	2415
DEPB1.1	68	GATGGCCTCAGGTGGTAACTTTCTCACAGGCTTTGGCCA CAGATCTGATCATTTACAATTGCGTCAGCA	2416
DET1.1	70	CTTGTGGAGATCACCCAATCAGGTTCTATGCCCGGGACTCGGGCCTGCTCAAAGTTTGAGATCCAGGCG GG	2417
DHPS.3	78	GGGAGACGGGATCAATAGGATCGAAAACCTGTGTTGCCCAATGAGAATTACTGCAAGTTTGAGGACT GGCTGATGC	2418
DIABLQ.1	73	CACAATGGCGGCTCTGAAGAGTTGGCTGTCGCGAGCGTAACTTTCATTCTTCTAGGTACAGACAGTGTTT GTGT	2419
DIAPH1.1	62	CAAGCAGTCAAGGAGAAACAGAGCGCGGAGACAGAGAAGAAAAGATCAGGCGAGCAAAAAC	2420
DICER1.2	68	TCCAAATCCAGCATCACTGTGGAGAAAAGCTGTTGTCTCCCGAGCATACTTTATCGCCTTCACTGCC	2421
DKFZP564O0823.1	66	CAGCTACACTGTCGAGTCCGCTGCTGAGCCTCCACACATCATCTCCCCCTCAAGCTCCAGCCTCAT	2422
DLC1.1	68	GATTACAGACGAGGATGAGCCTTGTGCCATCAGTGGCAAA TGGA CTTTCCAAAGGGA CAGCAAGAGTG	2423
DLL4.1	67	CACGAGGTATAAGCGAGGAGCCTACCTGGACATCCCTGCTCAGCCCCGGGCTGGACCTTCTTCT	2424
DPEP1.1	72	GGACTCCAGATGCCAGGAGCCCTGCTGCCACATGCAAGGACCAAGCATCTCCTGAGAGGACGCCCTGG CTTA	2425
DPYS.1	70	AAAGAA TGGCACCATG CAGCCACCATGT CATGGTCCACCTTTGCGACCAGACCCCTCAACACCCGAC T	2426
DR4.2	83	TGCACAGAGGGTGTGGTTACACCAATGCTTCCACAATTTGTTTGTGCTTGGCCTCCCATGTACAGCTTGTA AATCAGATGAAGA	2427
DR5.2	84	CTCTGAGACAGTGTCTCGATGACTTTGCAGACTTGGTGCCCTTTGACTCTCTGGGAGCGGCTCATGAGGA AGTTGGGCTCATGG	2428
DUSP1.1	76	AGACATCAGCTCCTGGTTTCAACGAGGCCATTGACTTTCATAGACTCCATCAAGAATGCTGGAGGAAGGT GTTTGTG	2429
DUSP9.1	77	CGTCTAATCAACGTGCTATGGCGGACCACTGCTCGAGGCTGCCTCTTCTGCGACTGTTACTTTTTTC TTTGGGG	2430

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
E2F1.3	75	ACTCCCTCTACCCCTTGAGCAAGGCGAGGGTCCTGAGCTGTTCTTCTGCCCCATACTGAAGGAACCTGA GGCCTG	2431
EBAG9.1	66	CGCTCCTGTTTTTCTCATTGTGCACTGGGTTTTGATTCCACCATGGCCATCACCAGTTTCGGT	2432
ECRG4.1	66	GCTCCTGCTGCTGTGCTGGGCCCCAGGTGGCATAAAGTGAATAAATACTCAAGCTGATGCTTCAAAA	2433
EDG2.1	72	ACGAGTCCATTGCTCTTTTATAACCGAAGTGGAAAGCATCTTGCCACAGAAATGGAAACACAGTCAGCAA GC	2434
EDN1 endothelin.1	73	TGCCACCTGGACATCATTTGGGTCAACACTCCCGAGCACGTTGTTCCGTATGGACTTGGAAAGCCCTAGG TCCA	2435
EDN2.1	79	CGACAAGGAGTGCCTACTTCTGCCACTTGGACATCATCTGGGTGAACACTCCTGAAACAGACAGCTCC TTACGGCCTG	2436
EDNR.2	76	TTTCCTCAAAATTTGCCTCAAGATGGAACCCCTTTGCCCTCAGGGCATCCTTTTGGCTGGCACTGGTTGGAT GTGTAA	2437
EDNRB.1	72	ACTGTGAACCTGCCGTGTCAGTGTCCACATGACAAGGGGCGAGGTAGCACCTCTCTCACCCATGCTGT GGT	2438
EEF1A1.1	67	CGAGTGGAGACTGGTGTCTCAAACCCGGTATGGTGTCACTTTTGCTCCAGTCAACGTTACACAGG	2439
EFNB1.2	66	GGAGCCCGTATCCTTGGAGCTCCCTCAACCCCAAGTTCTCTGAGTGGGAAGGCTTGGTGATCTATCC	2440
EFNB2.1	73	TGACATTATCATCCCGCTAAGGACTGCGGACAGCGTCTTCTGCCCCCTCACTACGAGAAGGTCAAGCGGGA CTAC	2441
EGF.3	84	CTTTGCCCTTGCTCTGTACAGTGAAGTCAGCCAGAGCAGGGCTGTTAACTCTCTGTGAAATTTGTCAATAAG GGTGTCAAGTATTT	2442
EGFR.2	62	TGTCGATGGACTTCCAGAACCACTGGGCGCTGCCAAAAGTGTGATCCAGCTGTCCCAAT	2443
EGLN3.1	68	GCTGGTCTCTACTGCGGGAGCGGCTGGGCAATACTACTCAAGGAGAGGTCTAAGGCAATGGTGG	2444
EGR1.1	76	GTCCCCGCTGCAGATCTGTGACCCGTTGCGATCCTTCTCTACTCGCCCCACCATGGACAACTACCTAA GCTGGAG	2445
EIF2C1.1	67	CCCTCACGGACTCTACGCGGCTTCGCTTCAACAGGAGATCAAGGGCTGAAGGTGGAAGTCACCCA	2446
EIF4EBP1.1	66	GGCGTGAAGAGTCAAGTTTGAGATGGACATTTAAAGCACACCATCTGTTGGAGCACTACCA	2447
ELTD1.1	66	AGGTCTTGTGCAAGAGAGCCCTCGCTCTTCTGTCTCTCGGCACCACTGGATCTTTGGGGTT	2448
EMCN.1	73	AGGCACTGAGGGTGGAAAAATGCAAGCACTTCAGCAACCGCCGGTCTTATTCAGTATATTTTGGCG GTG	2449

TABLE B -continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
EMP1.1	75	GCTAGTACTTTGATGCTCCCTTGATGGGTCACAGAGAGCCTCCCTGCAGCCACAGACTTGGCCTCCAGCTGTTC	2450
ENO2.1	67	TCCTTGGCTTACCTGACCTCTTGCTGTCTGTGTCGCCCTCTTTCTGTGCCCTACTCATTTGGGTT	2451
ENPEP.1	67	CACCTACACGGAGAACGACAAGTCAGAGCATAGTGGCCACCCGATCATGAAACCAACAGATGCCAGG	2452
ENPP2.1	67	CTCCTGGCACAATACTTCAGGCCAACCATGCCAGAGGAAGTTACCAGACCCCAATTATCCAGGGA	2453
EPAS1.1	72	AAGCCTTGGAGGGTTTCATNGCCGTGTGACCCCAAGATGGCGACATGATCTTCTGTCAAGAAAACATCA GCA	2454
EPB41L3.1	66	TCAGTGCATACGCTCTCACTCTCTCCTTCCCTGTGCTCTGTGCCCTGTGCTACTGGAGCCCAAG	2455
EPHA2.1	72	CGCCTGTTCCACCAAGATTGACACCATTGCGCCCGATGAGATCACCGTCAGCAGCGACTTCGAGGCACGC CAC	2456
EPHB1.3	67	CCTTGGGAGGAAGATCCCTGTGAGATGGACAGCTCCAGAGGCCATCGCCTACCGCAAGTTCACCTTC	2457
EPHB2.1	66	CAACCAAGCAGCTCCATCGGCAGTGTCCATCATGCATCAGGTGAGCCGCCACCGTGGACAGCATTAAC	2458
EPHB4.1	77	TGAACGGGGTATCTCCTTAGCCACGGGGCCCCGTCCCATTTGAGCCTGTCAATGTCACCACTGACCCGAG AGGTACCT	2459
EPO.1	84	CAGTGCCAGCAATGACATCTCAGGGGCCAGAGAACTGTCCAGAGAGCAACTCTGAGATCTAAGGATGT CACAGGCCCAACTTG	2460
ExDB3.1	81	CGGTTATGTCTATGCCAGATACACACCTCAAGGTACTCCCTCCCTCCCGGGAAGGCACCCCTTTCTTCAGT GGGTCTCAGTTC	2461
ERBB4.3	86	TGGCTCTTAATCAGTTTCGTTACCTGCTCTGGAGAAATTACGCATTATTCTGTGGGACAAACTTTATGAG GATCGATATGCCCTTG	2462
ERCC1.2	67	GTCAGGTGGATGTGAAGATCCCCAGAGCCCTCAAGAGCTGGCTAAGATGTGTATCTCTGGCCCG	2463
ERCC4.1	67	CTGCTGGAGTACGAGCGACAGCTGGTGTGGAAGTCTCGACACTGACGGGCTAGTAGTGTGCGCCC	2464
EREG.1	91	ATAACAAAGTGTAGCTCTGACATGAATGGCTATTGTTTGCATGGACAGTGCATCTATCTGGTGGACATGA GTCAAAACTACTGCAGGTGTG	2465
ERG.1	70	CCAACTACTAGGCTCCCAACAGCCATATGCCTTCTCATCTGGGCACTTACTACTAAAGACCTGGCGGAG G	2466
ERK1.3	67	ACGGATCACAGTGGAGGAAGCGTGGCTCACCTACCTGGAGCAGTACTATGACCCGACGGATGAG	2467
ERK2.3	68	AGTTCTTGACCCCTGGTCTCTCCAGCCCGTCTTGGCTTATCCACTTTGACTCTCTTTGAGCCGTTT	2468
ESPL1.3	70	ACCCCCAGACGGATCAGGCAAGCTGGGCCCTCATGTCCCTTTCACGGTGTGTTGAGGAAGTCTGCCCTAC A	2469

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ESRRG.3	67	CCAGCACCATTTGTTGAAGATCCCCAGACCAAGTGTGAATACATGCTCAACTCGATGCCAAGAGACT	2470
F2.1	77	GCTGCATGTTCTGGAAGGTAACTGTGCTGAGGGTCTGGGTACGAACACTACCGAGGGCATGTGAAACATCAC CCGGTCAGG	2471
F3.1	73	GTGAAGGATGTGAAGCAGACGCTACTTGGCAGGGTCTTCTTACCCGGCAGGGAATGTGGAGAGCAC CGGTT	2472
FABP1.1	66	GGTCCAAAGTGATCCAAAACGAAATCACGGTGGGGAGGAATGTGAGCTGGAGACAAATGACAGGG	2473
FABP7.1	72	GGAGACAAAGTGGTTCATCAGGACTCTCAGCACATTCAAGAACACGGAGATTAGTTTCCAGCTGGAGAA GAG	2474
FAP.1	66	CTGACCAGAACCCAGGCTTATCCGGCTGTCCACGAACCACTTATACACCCACATGACCCACTTCC	2475
fas.1	91	GGATTGCTCAACAAACCACTGCTGGGCATCTGGACCTCTCTACCTCTGGTCTTACGTCGTGTGCTAGATTA TCGTCCAAAAGTGTAAAGCC	2476
fas1.2	80	GCACTTTGGGATTCCTTTCATTATGATTTCTTTGTACAGGCACCGAGAAATGTGTATTCACTGAGGTCT TCTTACATGC	2477
FBXW7.1	73	CCCCAGTTTCAACGAGACTTCATTTTCATTGCTCCCTAAAGAGTTGGCACTCTATGTGCTTTCAATTCCTGG AAC	2478
FCBR1G.2	73	TGCCATCCTGTTTCTGTATGGAATTGTCTCACCTCTCTTACTGTGCACTGAAGATCCAAGTCGAAAG GCA	2479
FCGR3A.1	67	GTCTCCAGTGGAAAGGAAAAGCCCATGATCTTCAAGCAGGGAAGCCCCAGTGAAGTAGCTGCATTCTCT	2480
FDPS.1	77	GGATGATTACCTTGACCTCTTTGGGGACCCCAAGTGTGACCGGCAAAATTTGGCACTGACATCCAGGACAA CAAAATGCA	2481
FEN1.1	66	GTGGAGAAGGTAAGCCAGGGTCGCTGAGAGACTCTGTTCTCCCTGGAGGACTGGTTGCCATGAG	2482
FGF1.1	66	GACACCGACGGGCTTTTATACGGCTCACAGACACCAATGAGGAATGTTTCTCTGGAAAGGCTG	2483
FGF2.2	76	AGATGCAGGAGAGAGAGCCCTTGCAAACTGCAGACTGCTTTTGGCCCAATATAGATTGGGTAAAGCT GCAAAAC	2484
FGF9.1	67	CACAGCTGCCATACTTCGACTTATCAGGAATTCGGCTGGTGGCTGCGCGAGGGTGCACTTTACTT	2485
FGFR1.3	74	CACGGACATTCACACATCGACTACTATAAAAAGACAAACCAACGGCCGACTGCCTGTGAAGTGGATGG CACCC	2486
FGFR2 isoform 1.1	80	GAGGACTGTTGGCATGCGTGGCCCTCCAGAGACCAACGTTCAAGCAGTTGGTAGAAGACTTGGATC GAATTCCTACTC	2487
FH.1	67	ATGGTTGCAGCCCAAGTCATGGGGAACCATGTTGCTGTCTACTGTCTGGAGGCAGCAATGGACATTTTG	2488

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
FHL1	67	CCAGTGAGCGCTTCATGACCTGCGTCTGTGATGAAGTGGCGGATTTGTTTCAGACGACCCAGAGAG	2489
	66	ATCCAGCCTTTTCGGAATACATCCTATCTGCCACACATCCAGCGTGAGGTCCCTCAGCTACAAAG	2490
	72	GGTTCCAGCTTTCTGTAGCTGTAAGCATTTGGTGGCCACACCACTCCTTACAAGCAACTAGAACCTGC GGC	2491
FILIP1	66	ACACCGGTACAAACGTCATCTGCTCGAGGAACCCAGTCAGTGTCAAGACAAAGACGGGTCAATCCAG	2492
FKBP1A	76	CTGCCCCGACTGAATGTGTTCTGTCACTCAGCTTTGCTTCCGACACCTCTGTGTTTCTTCCCTTTTC CTCGTA	2493
FLJ22655	82	CTCCTTCACACGAACCTTTTCATTTATTGTACAAATCTACAGCACCAAGACGCGTGTGGTCTTACGGAGTATTGCTG CCCACTTTGCCT	2494
FLT1	75	GGCTCTGGAATCTATCTTTGACAAAATCTACAGCACCAAGACGCGTGTGGTCTTACGGAGTATTGCTG TGGGA	2495
FLT3LG	71	TGGGTCCAAGATGCAAGGCTTTGCTGGAGCGGTGAACACGGAGATACACTTTGTCACCAAATGTGCCTT TC	2496
FLT4	69	ACCAAGAGCTGAGGACCTGTGGCTGAGCCCGCTGACCATGGAAGATCTTTGTCTGCTACAGCTTCCAGG	2497
FN1	69	GGAAGTGACAGACGTGAAGCTCACCATCATGTGGACACCCGCTGAGAGTGCACTGACCCGGCTACCGTG T	2498
FOLR1	67	GAACGCCAAGCACCAAGGAAAAAGCCAGGCCCCGAGGACAAGTTGCATGAGCAGTGTGACCCCTGG	2499
FOS	67	CGAGCCCTTTTGATCTTCCTGTTCCAGCATCATCCAGGCCCAAGTGGCTCTGAGACAGCCCGCTCC	2500
FRAP1	66	AGCGCTAGACACTGTGGACCGCTGACGGAGTCCCTGGATTTCACCTGACTATGCCCTCCCGGATCAT	2501
PRP1	75	TTGGTACCTCTGTGGTTAGCATCAAGTTCTCCCCAGGGTAGAATTCAATCAGAGCTCCAGTTTGCATTTGG ATGTG	2502
FST	72	GTAAGTCGGATGAGCCTGTCTGTGCCAGTGACAAATGCCACTTATGCCAGCGAGTGTGCCATGAAGGAAG CTG	2503
FZD2	78	TGGATCCTCACCTGGTGGTGCTGTGCTGCGCTTCCACCTTCTTCACCTGTCAACCGTACTTTGGTATGAC ATGCAGCGC	2504
G-Catenin	68	TCAGCAAGAAGGCATCATGGAGGAGGATGAGGCCTGCGGGCGCCAGTACACGCTCAAGAAAAACCCACC	2505
GADD45B	70	ACCCCTGCACAAGACCACTTTTGGGACTTTGGGAGCTGGGGCTGAAGTTGCTCTGTATCCCATGAACCTCCC A	2506
GAS2	68	AACATGTCAATGGTCCGTGTGGGAGGAGGCTGGAAAACCTTTTGCAAGGGTATTTGTTGAAAACACGACCCC	2507

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
GATA3.3	75	CAAGGAGCTCACTGTGTGGTGTCTGTGTTCAAACCACTGAATCTGGACCCCATCTGTGAATAAGCCATTCTG GACTC	2508
GATM.1	67	GATCTCGGCTTTGGACAACCTTGACAGGATGGGTGACGCAACTTTCCAGAGACCCAGGACAGCTAC	2509
GBL.1	66	GCTGTCAATAGCACCGGAACTGCTATGTCTGGAATCTGACGGGGGGCATTTGGTGACGAGGTGACC	2510
GBP2.2	83	GCATGGGAACCATCAACAGCAGGCCCATGGACCAACTTCACTATGTGACAGAGCTGACAGATCGAATCA AGGCAACTCCTCA	2511
GLC3.3	71	CTGTTGAGGAAGGCATTGATCATCTCCTGGCCAGCATGTTGCTCATCTCTTTATTAGAGACCCACTGA C	2512
GLM.2	85	TGTAAATCAAACTCTTTCATCATCAACTAGAAAGTCAGTTGACATGGCCTGTTCAAGTCTTGGAGTTGCA CAGCTGGATTCTGTG	2513
GFRA1.1	69	TCCGGTTAAGAACAAAGCCCTGGGGCCAGCAGGGTCTGAGAATGAAAATCCCACATCATGTTTGGCCAC	2514
GJAL.1	68	GTTCACTGGGGGTGTATGGGGTAGATGGGTGGAGAGGGGGATAGAGAGGTGCATGTTGGTATTT	2515
GLYAT.1	68	TACCATTGCRAAGGTGCCAGATGCTGCAGATGCTGGAGAAATCCTTGAGGAAGAGCCTCCAGCATCC	2516
GMN.1	67	GTTCCGTACGAGGATGAGCGTCTCCACCAGTAAGTGGCAAGAGGCGCAGGAAGTGGGTACGCA	2517
GNAS.1	72	GAACGTGCTGACTTTGACTTCCCTCCCGAATCTATAGCATGCCAAGGCTCTGTGGGAGGATGAAAGG AGT	2518
GPC3.1	68	TGATGCGCTGGAAACAGTCAGCAGGCAACTCCGAAGGACAAACAGAGATAAGCACCCTTTCAACCTCG	2519
GPX1.2	67	GCTTATGACCGACCCCAAGCTCATCACTGGTCTCCGGTGTGTCGCAACGATGTTGCTGGAACTTT	2520
GPX2.2	75	CACACAGATCTCTACTCCATCCAGTCCCTGAGAGCCCTTAGGATGCAGCATGCCTTCAGGAGACACTGC TGGACC	2521
GPX3.1	69	GCTCTAGGTCCAAATTGTTCTGCTCTAACTGATACCTCAACCTTGGGGCCAGCATCTCCCACTGCCCTCA	2522
GRB14.1	76	TCCCACGTGAAGCCCTTTCAAGTTGCGGTTGAAGAAGGACTCGCTTGGAGGAAAAAAGGATGTTTACGCCT GGGCACT	2523
GRB7.2	67	CCATCTGCATCCATCTGTTTGGGCTCCCCACCTTGAGAAGTGCCTCAGATAATACCTGGTGGCC	2524
GR01.2	73	CGAAAAGATGCTGAACAGTGACAAATCCAACTGACACAGAGGAGGAGGAAGCTCACTGGTGGCTGTTT CTGA	2525
GSTM1.1	86	AAGCTATGAGGAAAAGAGTACAGATGGGGACGCTCTGATTATGACAGAAGCCAGTGGCTGAATGA AAAATTCAAGCTGGGCC	2526
GSTM3.2	76	CAATGCCATCTTGGCTACATCGCTCGCAAGCACACACATGTGTGGTGAGACTGAAGAAGAAAAAGATTCTG AGTGGAC	2527

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
GSTp . 3	76	GAGACCTGTGTCCAGAACAGGAGGCAAGACCTTCATTGTGGGAGACAGATCTCCTTCGCTGA CTACAACC	2528
GSTT1 . 3	66	CACCATCCCCACCCTGCTTCCACAGCGCCTGAAAGCACAAATAGAATGATGCACACTGAGGCC	2529
GZMA . 1	79	GAAGAAGTTTCCCTATCCANGCTATGACCCAGCCACACGCGAAGGTGACCTTAAACTTTTACAGCTGACG GAAAAAGCA	2530
HADH . 1	66	CCACGAGACAAGACCGAATCGCTGGCCTCCATTCTTCAACCCAGTCGCTGTCAATGAACACTTGTGG	2531
HAVCR1 . 1	76	CCACCCAAAGTTCACGACTACTCCAAATTGTCACAACTGTTCCAAACCGTCACGACTGTTTCGACGAGCAGG ACTGTTT	2532
HDAC1 . 1	74	CAAGTACCAACAGCGATGACTACATTAATAATCTTGGCTCCATCCGTCACAGATAACATGTCGGAGTACAGC AAGC	2533
Hepsin . 1	84	AGGCTGCTGGAGGTCAATCTCCGTGTGATTGCCCCAGAGGCCGCTTCTTGGCCGCCATCTGCCAAGA CTGTGGCCGCAGGAAG	2534
HER2 . 3	70	CGGTGTGAGAAGTGCAGCAGCCCTGTGCCCGAGTGTCTATGTTCTGGGCATGGAGCACTTGGCAGA GG	2535
HGD . 1	76	CTCAGTCTGCCCTACAATCTCTATGCTGAGCAGCTCTCAGGATCGGCTTTCACCTTGTCCACGAGCA CCAATAA	2536
HGF . 4	65	CCGAAATCCAGATGATGCTCATGGACCCTGGTGTCTACACGGGAAATCCACTCAATTCCTTTGGG	2537
HGFAC . 1	72	CAGGACACAAGTGCAGATTGCGGGCTGGGCCACTTGGATGAGAACGTGAGCGGCTACTCCAGCTCC CTGC	2538
HIF1A . 3	82	TGAACATAAAGTCTGCACATGGAAGGTATTGCACTGCA CAGGCCACATTACGTAATATGATACCAACAG TAACCAACCTCA	2539
HIF1AN . 1	66	TGTTGGCCAGTCTCAGTCAGCCTGCCGAGGCTAACTGGCTAGAGCTCCAGGCCCTATGATGC	2540
HIST1H1D . 1	67	AAAAAGCGAAGAAGGACGCGCAACTGCTGGGAAACGCAAAAGCATCCGGACCCCCAGTATCTGAGC	2541
HLA-B . 1	78	CTTGTGAGGGA CTGAGATGCAGGATTTCTTCAGCCTCCCCCTTTGTGACTTCAAGAGCCTCTGGCATCT CTTTCGTCA	2542
HLA-DPA1 . 1	78	CGCCTGAAGACAGAAATGTTCCATATCAGAGCTGTGATCTTGAGAGCCCTCTCCTTGGCTTTCCTGCTGA GTCTCCGA	2543
HLA-DPB1 . 1	73	TCCATGATGTTCTCGAGTTTCTCGGCCCCCCCGACAGTGGCTCTGACGGCGTTACTGATGGTGGT GCTCA	2544
HLA-DQB1 . 1	67	GGTCTGCTCGGTGACAGATTTCTATCCAGGCCAGATCAAAGTCCGGTGGTTTCGGAATGATCAGGAG	2545

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
HLADQA1.2	76	CATCTTTCTCTCTGTGTTCAACATCACATGGCTGAGCAATGGCAGTCAGTCACAGAAGGTGTTTCTGA GACCAGC	2546
HMB1.1	71	TGGCTGTTCATTGGTGATGTTGCGAAGAACTGGGAGAGATGTGGAATAACACTGTGCAGATGACAA GC	2547
HNRPA8.1	84	AGCAGGACGCGCCAACTGATCGCACATGCTTTGTTTGGATATGAGTGAACAATAATTATGTACCAAA TTAACTTGGCAAC	2548
HPCAL1.1	70	CAGGCAGATGGACACCAACAATGACGGCAAACTGCTCCTTGGAGAATTTCATCAGAGGTGCCAAGAGCGA C	2549
HPD.1	78	AGCTGAAGACGCCAAGATCAAGGTGAAGGAGAAACATTGATGCCCTGGAGAGCTGAAAAATCCTGGTG GACTACGACG	2550
HSD11B2.1	69	CCAACTGCTCAAGAGCTGCTCAGGCTACGGCAAGGACTACATCAGCACTTGATGGGCGAGTTC C	2551
HSP90AB1.1	66	GCATTGTACAGCACCTACGGCTGGACAGCCAAATATGGAGCGGATCATGAAGCCACAGGCACTTC	2552
HSPAL1.1	70	CTGCTGGACAGTCCACTACCTTTTTCGAGAGTGACTCCCGTTGTCCTCCAGGCTTCCAGAGCGAACCT G	2553
HSPA8.1	73	CCTCCTCTGTTGTTGCTTCTCAGGGCCACCATTTGAAGAGTTGATTAGCCAAACCAAGTGTAGATG TAGC	2554
HSPB1.1	84	CCGACTGGAGAGCATAAAAGCGCAGCCGAGCCAGCGCCCCGCACTTTCTGAGCAGACGTCCAGAG CAGAGTCAGCCAGCAT	2555
HSPG2.1	66	GAGTACGTGTGCCAGTGTGGGCAGCTCCGTGCTCTAGAGGCCCTCTGTCTGTGTCACCATTTGAG	2556
HTATIP.1	68	TCGAATTGTTTGGCACTGATAGGACTCCAGGACAGCTCTGATGGAATACCGTCAGCACCAAGC	2557
HYAL1.1	78	TGGCTGTGGAGTTCAAATGTCGATGCTACCTGCTGGCAGGACCGTGTGTGAGCGGAGAGCATG TGGTGATTGG	2558
HYAL2.1	67	CAACCATGCACTCCAGTCTACGTCTTCACAGACCCACCTACAGCCGAGGCTCAGGGGCTTAGT	2559
HYAL3.1	67	TATGTCCGCTTCACACCCGAGATCTGGGAGTTCTGTCCAGGATGACCTTGTGCAGTCCATTG	2560
ICAM1.1	68	GCAGACAGTGACCATCTACAGCTTTCCGGCGCCCAACAGTGTCTGACGAAGCCAGAGGTCTCAGAA G	2561
ICAM2.1	62	GGTCATCCTGACACTGCAACCCACTTTGGTGGCTGTGGGCAAGTCTTCACCATTTAGTGCA	2562
ICAM3.1	67	GCCTTCAATCTCAGCAACGTGACTGGCAACAGTCCGGATCCTCTGCTCAGTGTACTGCAATGGCTCTC	2563
ID1.1	70	AGAACCGCAAGGTGAGCAAGGTGGAGATTCTCCAGCAGCTCATCGACTACATCAGGGACCTTCAGTTGG A	2564

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ID2.4	76	AACGACTGCTACTCCAAGCTCAGGAGCTGGTCCCGACATCCCCCAGAAACAGAGAGGTGAGCAAGATG GAAATCC	2565
ID3.1	80	CTTCACCAATFCCCTTCCCTGGAGACTAAACCTGGTGTCTCAGGACGGAAGGACTGTGAACCTTGTGGCCTG AAGAGCCAGAG	2566
IFI27.1	71	CTCTCCGGATTGACCAAGTTTCATCTGGGCTCCATTGGGTCTGCCATTGGCGGTGTCAATTGCGAGGTTT TA	2567
IGF1.2	76	TCCGAGCTGTGATCTTAAGGAGGCTGGAGATGTATTGGCACCCCTCAAGCCTGCCAAGTCAGCTCGCT CTGTCCG	2568
IGF1R.3	83	GCATGGTAGCCGAAAGATTTCAAGTCAAAATCGGAGATTTTGGTATGACGCGAGATATCTATGAGACAGA CTATTACCGGAAA	2569
IGF2.2	72	CCGTGCTTCCGGAACAATTCGCCAGATACCCCGTGGGCAGTTCTTCCAAATGACACACCTGGAAGCAGT CCA	2570
IGFBP2.1	73	GTGGACAGCACCATGAACAATGTTGGCGGGGGAGGCAGTGCTGGCCGGAAAGCCCTCAAGTCGGGTA TGAAGG	2571
IGFBP3.1	66	ACATCCCAACCATGTCTCTGGAGCTCACAGCCTTCTGTGGTGTCAATTTCTGAACAAGGGCGTGG	2572
IGFBP5.1	69	TGGACAAGTACGGGATGAAGCTGCCAGGCTGGAGTAGTACGTTTGACGGGACTTTTCAGTGGCCACACCTTC	2573
IGFBP6.1	77	TGAACCCGACAGACCAACACAGAGGAATCCAGGCACCTCTACCAAGCCCTCCAGGCCCAATTCGTGGGGT GTCCAAGC	2574
IL-7.1	71	GCGGTGATTCGGAATTCGCGAATTCCTCTGGTCTCATCCAGGTGCGGGGAAGCAGGTGCCCAGGA GAG	2575
IL-8.1	70	AAGGAACCATCTCACTGTGTGTAAACATGACTTCCAAGCTGGCCGTGGCTCTCTTGGCAGCCTTCTCTGA	2576
IL10.3	79	GGCGTGTATCGATTTCTTCCCTGTGAAAACAAGACAGAGCCGTGGAGCAGGTGAAGAATGCCTTTA ATAAGCTCCA	2577
IL11.2	66	TGGAAGTTCCACAAGTCAACCCTGTGATCAACAGTACCCGTATGGGACAAAGCTGCAAGGTCAAGA	2578
IL15.1	79	GGCTGGGTACCAATGTCTGACGTTCAACAGCTATGCTGTGTAGGCTCCTGCCAGTGTGGAAACCCACTGACTA CTGGCTCTCA	2579
IL1B.1	67	AGCTGAGGAAGATGCTGGTTCCCTGCCACAGACCTTCCAGGAGAATGACCTGAGCACCTTCTTTTCC	2580
IL6.3	72	CCTGAACCTTCCAAAAGATGGCTGAAAAAAGATGGATGCTTCCAATCTGGATTCAATGAGGAGACTTGCTG GT	2581
IL6ST.3	74	GGCCTAATGTTCCAGATCCTTCAAAAGAGTCATATTGCCAGTGGTCACTCCTCACACTCCTCCAGGCAAA TTTT	2582

TABLE B -continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
ILT-2.2	63	AGCCATCACCTCTAGTGCAGCCAGGTCCTATCTGCGCCCTGAGGAGACCCCTGACTCTGCAGT	2583
IMP3.1	72	GTGACTCGTCCAAGATCAAGCGCAGTGTCTAGATACATGAGGAGCGGATGACTTCGATCTGGA AGCC	2584
INDO.1	66	CGCCTTGACGCTCTAGTTCTGGGATGCATCACCATGGCATATGTGTGGGCAAGGTCATGGAGAT	2585
INHBA.1	72	GTGCCCAGGCATATAGCAGGCACGTCCGGGTCCCTCACTGTCTTCCACTCACAGTCATCAACCACTA CCG	2586
INHBB.1	72	AGCCTCAGGATACCAAGCAATGGATCGGTGCACAAATGGCAGCTTAGCTACAAATGCCTGTCACTCGG AGA	2587
INSR.1	67	CAGTCTCCGAGCGGATTGAGTTCTCTAATGAGGCCTCGGTCAATGAAGGCTTCACTGCGCATCAC	2588
IQAP2.1	68	AGAGACACCACTGCGCAACAGGAGGTAGACCATGCCCACGGACATGGTGAAGCCGTGCAATGAT	2589
ISG20.1	70	GTGTCACTGAAGCCCCATCCAGCCCCGTTCCGAGGGACTAGAGGCTTTCGGGCTTTTGGGACAGCAA C	2590
ITGA3.2	77	CCATGATCCTCACCTCTGCTGGTGACTATACATCCAGACCTCCGCTTAGCATGGTAATCACCGGCTACA AAGCTTC	2591
ITGA4.2	66	CAACGCTTCAGTGATCAATCCCGGGCGATTTACAGATGACAGGATCGGAAGAATCCCGGCCAGAC	2592
ITGA5.1	75	AGGCAGCCCTTACATTATTCAGAGCAAGAGCCGGATAGAGGACAAGGCTCAGATCTTGCTGACTGTGGA GAAGAC	2593
ITGA6.2	69	CAGTGACAAAACAGCCCTTCCAAACCAAGGAATCCACAAAAGATGGCGATGACGCCCATGAGGCTAAAC	2594
ITGA7.1	79	GATATGATGGTCTGCTTTGTGCTCAGCCAGGACCTGGCCATCCGGGATGAGTTGGATGGTGGGA ATGGAAGTTCT	2595
ITGAV.1	79	ACTCGACTGCACAAGCTATTTTGTGATGACAGCTATTTGGGTTATTTCTGTGGCTGTGGAGATTTCAATG GTGATGGCA	2596
ITGB1.1	74	TCAGAAATTGGATTTGGCTCATTTGTGGAAAAGACTGTGATGCCCTTACATTAGCACAAACACAGCTAAGCT CAGG	2597
ITGB3.1	78	ACCGGGAGCCCTACATGACGAAAATACCTGCACCGTTACTGCCGTGACGAGATTGAGTCAGTGAAGAAG AGCTTAAGG	2598
ITGB4.2	68	CAAGTGCCCTCAGTGAGCTCACCAACCTGTACCCGTAATTCGACTATGAGATGAAGGTGCGC	2599
ITGB5.1	71	TCGTGAAAAGATGACAGGAGGCTGTGCTATGTTTCTACAAAACCCCAAGGACTGCCGTCAATGATTTCA CC	2600
JAG1.1	69	TGGCTTACTGCGAATGTTTCTGTGTTGGCTGGGAAATCGAGTGCCGCATCTCACAGCTATGC	2601

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
K-ras.10	71	GTCAAAATGGGGAGGAGCTAGGGCAGTTTGGATAGCTCAACAAGATACAAATCTCACTCTGTGTGGTCC TG	2602
KCNJ15.1	67	GGACGTTCTACCTCCCTTGAAGAAGACACCTGACCTCGGAGTGAGTGACCAAGTGTTCAGAGCCT	2603
KDR.6	68	GAGACAGAGCCCTCTACACCTGCAGGCATGCAGTGTTCCTGGCTGTGCAAAAGTGGAGGCATTTT	2604
Ki-67.2	80	CGACTTTGGGTGCGACTTGACGAGCGGTGTTCCACAAGTGGCTTCGGGGCCGGATCGTCCCAGTG GAAGATTGTAA	2605
KIAA1303 raptor.1	66	ACTACAGCGGAGCAGGAGCTGGAGTAGCTGCAATCAACCCAAATCACCTCTTGCTCAGATGCC	2606
KIF1A.1	66	CTCCTACTGTGTCGACACCTCACTGAGGACATCAACTACGCGTCGCAGAAAGCAGGTGTACCGGGA	2607
Kiting.4	79	GTCCCCGGATGGATGTTTCCCAAGTCATTGTTGGATAGCGAGATGGTAGTACAAATTGTCACACAGC TTGACTGATC	2608
KL.1	72	GAGGTCTGTCTAAACCTGTGTCCCTGAGGATCTGTCTCACTGGCATCTTGTGAGGGCCTTGACA TAG	2609
KLK3.1	66	CCAAGCTTACCACCTGCACCCGGAGAGCTGTCTACCATGTGGTCCCGTGTGCTTCCTCCCTCACCTT	2610
KLRK1.2	70	TGAGAGCCAGGCTTCTTGATGTCTCAAAATGCCAGCCTTCTGAAAGTATACAGCAAAAGAGGACCAAGAT	2611
KRT19.3	77	TGAGCGGCAGAAATCAGGAGTACCAGCGGCTCATGGACATCAAGTCGCGGCTGGAGCAGGAGATTGCCA CCTACCGCA	2612
KRT5.3	69	TCAGTGGAGAAAGGAGTTGGACCAAGTCAACATCTCTGTTGTCAAGCAGTGTTCCTCTGGATATGGCA	2613
KRT7.1	71	TTCAAGATGAACCGGGCCATCCAGAGGCTGCAGGCTGAGATCGACAAACATCAAGAACCAGCGTGCCA AGT	2614
L1CAM.1	66	CTTGCTGGCCAAATGCCTACATCTAGTTTCCAGCTGCCAGCCAAAGATCCCTGACTGCGGACAATCA	2615
LAMA3.1	73	CAGATGAGGCACATGGAGACCCAGGCCAAGGACCTGAGGAATCAGTTGCTCAACTACCGTTCTGCCATT TCAA	2616
LAMA4.1	67	GATGCACTGCGGTTAGCAGCGCTCTCCATCGAGGAAGGCAAAATCCGGGGTGTGAGCGTATCCTCTG	2617
LAMB1.1	66	CAAGAGACTGGGAGGTGTCTCAAGTGGCTGTACACACGGAAGGGGAACACTGTAGTTCGTGCCG	2618
LAMB3.1	67	ACTGACCAAGCCTGAGACCTACTGCACCCAGTATGGCGAGTGGCAGATGAAATGCTGCAAGTGTGAC	2619
LAMC2.2	80	ACTCAAGCGGAAATTGAAGCAGATAGGTCTTTATCAGCACAGTCTCCGCTCTCTGGATTCAAGTGTCTCGG CTTTCAGGGAGT	2620
LAPTMS.1	66	TGCTGGACTTCTGTGCTGAGCATCTGTGACCTCTGCAGCTCCTACATGGAAGTGCCCACTATCTCA	2621
LDB1.2	67	AACACCCAGTTTGACGAGCCACGGCATTCACGACGAGGACAGCTTTAACAACCTCCCTGCACTGG	2622

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
LDB2.1	68	ATCACGGTGGACTCGGACCACTGTACCATGGTCACCCAGCACGGGAAAGCCCATGTTTACCAAGGTA	2623
LDHA.2	74	AGGCTACACATCCTGGGCTAATGGACTCTCTGTAGCAGATTTCGCAGAGAGTATAATGAAGAATCTTAGGCGGG	2624
LGALS1.1	72	GGGTGGAGTCTTCTGACACAGCTGGTGCGCCTGCCCGGGAACATCCTCTGGACTCAATCATGGCTTTGTGCTCT	2625
LGALS3.1	69	AGCGAAAAATGGCAGACAAATTTTCGTCCATGATGCGTTATCTGGGTCGTGAAACCCAAACCCCTCAAG	2626
LGALS9.1	67	AGTACTTCACCGCGTGCCCTTCCACCGTGTGGACACCATCTCCGTCAATGGCTCTGTGCAGCTGTC	2627
LIMK1.1	67	GCTTCAGGTGTTGTGACTGTGCAGTGCCTCCTGTCCGACCACTACTATGAGAAGGATGGGCAGCTCTT	2628
LMNB1.1	66	TGCAAAAGCTGGTGTCAAGCCAGCCCCCAACTGACCTCATCTGGAAGAACCAGAACTCGTGGGG	2629
LMO2.1	74	GGCTGCCAGCAGAACATCGGGACCGCTACTTCTCTGAAGGCCATCGACCAGTACTGGCACGAGGACTGCCTGAG	2630
LOX.1	66	CCAATGGAGAAACAACGGGCAGGTGTTTCAGCTTGCTGAGCCTGGGCTCACAGTACCAGCCTCAGCG	2631
LRP2.1	66	GGCTGTAGACTGGGTTTCCAGAAAAGCTCTACTGGTTGGATGCCCGCCTGGATGGCCTCTTTGTCCCTC	2632
LRRC2.1	71	CCAGTGTCCTCAATCTGTCTCCTCGGATGTCGAATTTGCAAGTGGTTGGATATCAGCAGCAATAACCTGA	2633
LTF.1	68	AACGGAAGCCTGTGACTGAGGCTAGAAGCTGCCATCTTTGCCATGGCCCCGGAATCATGCCGTGGTGTCTT	2634
LYZ.1	80	TTGCTGCAAGATAACATCGCTGATGCTGTAGCTTGTGCAAGAGGGTTGTCCGTGATCCACAGGCATTAGAGCATGGGT	2635
MADH2.1	70	GCTGCCTTTGGTAAGAACATGTCGTCCATCTTGCCATTCACGCCGCCAGTTGTGGAAGAGACTGCTGGGA	2636
MADH4.1	76	GGACATTACTGGCCTGTTCACAATGAGCTTGCAATCCAGCCTCCCAATTCOAATCATCTGCTCCTGAGTATTGGT	2637
MAL.1	66	GTTGGGAGCTTGTGTGTCTAACCTCCAACCTGCTGTGCTGTCTGTAGGGTCACTCCTGTTTGTG	2638
MAL2.1	67	CCTTCGTCGTGCTGGAGATTCCTGTTCGGGGTCTTGTCTGGATTTTGGTTGCTCCTCCCAATGTTCC	2639
MAP2K1.1	76	GCCTTTCTTACCCAGAAAGCAGAGAGGTGGGAGAACTGAAGGATGACGACTTTTGAGAAGATCAGTGAGCTGGGGCTG	2640
MAP2K3.1	67	GCCCTCCAATGCTCTATCAACAAGGAGGGCCATGTGAAGATGTGACTTTTGGCATCAGTGGGTAC	2641
MAP4.1	72	GCCGGTCAGGCACACAAGGGGCCCTTGGAGCGTGGACTGGTTGTTTGGCAATTTTGTGTGTATGCTGC	2642

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
MAPK3 . 1	67	CCCCTCTTGGATCTGTTGAGTTCTTTTGGAGAGCCAGCATGGGTGCCAGTTCTCCAAGACCG	2643
Maipin . 2	77	CAGATGGCCACTTTGAGAACATTTTAGCTGACACAGTGTGAACACGACCAAAATCCTTGTGGTTAA TGCTGCC	2644
MCAM . 1	66	CGAGTTCAGTGGCTGAGAGAAGACAGGCCAGGTGCTGGAAGGGGGCTGTGCTTCAGTTGCA	2645
MCN2 . 2	75	GACTTTTGGCCGCTACCTTTTCATTCCGGCGTGACACAAATGAGCTGTGCTCTTCATCTGAGCAGTTA GTGGC	2646
MCN3 . 3	75	GGAGAACAAATCCCTTGAGACAGAATATGGCCTTCTGTCTACAAAGGATCACAGACCATCACCATCCAG GAGAT	2647
MCN6 . 3	82	TGATGGTCCATATGTGTCACATTCATCAGGTTTCATACCAACACAGGCTTCAGCACTTCCTTTGGTGTG TTTCCTGTCCCA	2648
MCP1 . 1	71	CGCTCAGCCAGATGCAATCAATGCCCCAGTCACCTGCTGTTATTAATTCACCAATAGGAAGATCTCAGTG C	2649
MDH2 . 1	63	CCAAACACCTTTGTGCAGAGCTGAAGGTTTGGATCCAGCTCGAGTCAACGTCCCTGTCAATTG	2650
MDK . 1	66	GGAGCCGACTCAAGTACAAGTTTGAGAACTGGGGTGGTGTGATGGGGGCACAGGCACCAAAATC	2651
MDM2 . 1	68	CTACAGGGAGCCCATCGAATCCGGATCTTGATGCTGGTGAAGTGAACATTCAGGTGATGGTTGGAT	2652
MGMT . 1	69	GTGAAATGAACGACCACTGACAGCCCTTTTGGGGAAGCTGGAGCTGCTGGTTGTGAGCAGGCT	2653
mGST1 . 2	79	ACGGATCTACACACCATTTGCATATTTGACACCCCTTCCCCAGCCAAATAGAGCTTTTGAGTTTTTTGTG GATATGGA	2654
MICA . 1	68	ATGGTGAATGTCACCGACGAGGCCTCAGAGGGCAACATTACCGTGACATGCAGGGCTTCTGGCTT	2655
MIF . 2	66	CCGGACAGGGTCTACATCACTATTACGACATGAACGGGCCAATGTGGGCTGGAAACAACCTCCACC	2656
MMP1 . 1	72	GGGAGATCATCGGACAACTCTCCTTTTGATGACCTGGAGAAATCTTGCTCATGCTTTTCAACCAGG CCC	2657
MMP10 . 1	66	TGGAGGTGACAGGAAGCTAGACACTGACACTCTGGAGGTGATCGCAAGCCAGGTGTGGAGTTC	2658
MMP14 . 1	66	GCTGTGAGCTCTCAGGAAGGGCCCTGAGGAAGCACACTTGCTCTGTGGTCCCTGTCTCTTGCT	2659
MMP2 . 2	86	CCATGATGGAGGACAGACATCATGATCAACTTTTGGCGCTGGGAGCATGGCGATGGATACCCCTTTGA CGGTAAAGACGGACTCC	2660
MMP7 . 1	79	GGATGGTAGCAGTCTAGGGATTAACTTCTGTATGCTGCAACTCATGAACCTTGCCCATTTCTTTGGGTATG GGACATTCC	2661
MMP9 . 1	67	GAGAACCAATCTCACGACAGGCAGCTGGCAGAGGAATACCTGTACCGCTATGGTTTACACTCGGGTG	2662

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
MRP1.1	79	TCATGGTCCCGTCAATGCTGTGTATGGCGATGAAGACCAAGACGCTATCAGGTGGCCCCACATGAAGACA AAGACAATTCG	2663
MRP2.3	65	AGGGATGACTTGGACACANCTGCCATTCGACATGACTGCAATTTTGACAAAGCCATGCAGTTTT	2664
MRP3.1	91	TCATCCTGGCGATCTACTTCTCTGGCAGAACCTAGGTCCCTCTGTCTGTGGCTGGAGTCGCTTTCATGG TCCTTGCTGATTCCTCAACGG	2665
MRP4.2	66	AGCGCTGGAATCTACAACTCGGAGTCCAGTGTTCCTCCACTTGTCACTTCTCTCCAGGGGCTCT	2666
MSH2.3	73	GATGCAGAAATTGAGGCAGACTTTACAAGAAGATTTACTTCGTGATTCCTCCAGATCTTTAACCGACTTGCCA AGA	2667
MSH3.2	82	TGATTACCATCATGGCTCAGATTGGCTCCTATGTTCTCTGAGAAAGCGACAATTGGGATTTGTGATG GCATTTTCAGAAG	2668
MSH6.3	68	TCTATTGGGGATTGGTAGGAACCGTTACCAGCTGGAAATTCCTGAGAATTTCAACCACTCGCAATTTG	2669
MT1B.1	66	GTGGGCTGTGCAAGATGTGCCAGGGCTGTGTCGCAAAAGGCTCATCAGAGAAGTGCCGCTGTGTGT	2670
MT1G.1	74	GTGCACCCACTGCCTCTTCCCTTCTCGCTTGGGAACCTAGTCTCGCCTCGGTTTCAATGGACCCCAA CTGCT	2671
MT1H.1	74	CGTGTTCACCTGCCTCTTCTCTCTCGCTTGGGAACCTCAGTCTCACCTCGGCTTGCAATGGACCCCAA CTGCT	2672
MT1X.1	80	CTCTGCAAAATGCAAGAGTGCAAATGCACCTCCTGCAAGAGAGCTGCTGCTCTCTGCTGCCCTGTGGG CTGTGCCAAGT	2673
MUC1.2	71	GGCCAGGATCTGTGGTGTACATTTGACTCTGGCCTTCCGAGAGGTACCATCAATGTCACACGACGTGG AG	2674
MVP.1	75	ACGAGAACGAGGGCATCTATGTGCAGGATGTCAAGACCGGAAGGTTGGCGCTGTGATTGGAAGCACC TACATGC	2675
MX1.1	78	GAAGGAATGGGAATCAGTCAATGAGCTAATCACCTGGAGATCAGCTCCCGAGATGTCCCGGATCTGACT CTAATAGAC	2676
MYBL2.1	74	GCCGAGATCGCAAGATGTTGCCAGGGAGGACAGACAATGCTGTGAAGAATCACTGGAACCTCTTACCATC AAAAG	2677
MYH11.1	85	CGGTACTTCTCAGGGCTAAATATACGTACTCTGGCCTTCTCTCGTGGTGGTCAACCCCTTATAACACC TGCCATCTACTCG	2678
MHRIP.2	69	CCTTCACCTTCTCGTCAACACCAAGCCAGTGTGGAGATTGCAAAATTCAAATGCTCTGCAAGAGCTGCT	2679
NBN.1	76	GCATCTACTTCCAGAACCAAAATTAATTAATCTTCCAAAGTTCTGGCTGCTTGCAGGTGGAACTCCAGTGC AAGGGA	2680

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
NCF1.1	66	GACACCTTCATCCGCTCACAATCGCCCTGCTGGGCTTTGAGAAGCGCTTCGTACCCAGCCAGCACTAT	2681
NFAT5.1	70	CTGAACCCCTCTCTGTGTACCCGAGAAATCAGTCCCGTGGAGTTCCCCCTCCACCTCGCCATCGTTTCC T	2682
NFATC2.1	72	CAGTCAAGGTCAGAGGCTGAGCCCGGGTTCTACCCACAGTCATTCAGCAGCAGAATGCCACAGCC AAAG	2683
NFKBp50.3	73	CAGACCAAGGAGATGGACCTCAGCGTGGTGGGCTCATGTTTACAGCTTTTCTTCCGATAGCACTGGC AGCT	2684
NFKBp65.3	68	CTGCCGGATGGCTTCTATGAGGCTGAGCTCTGCCGGACCGCTGCATCCACAGTTTCCAGAACTGG	2685
NFX1.1	74	CCCTGCCATACCACTCAGCTCACCCTGCCCTGTGACTGCTTGTAAAGCTAAAGTAGAGCTACAGTGTGATGT GGAG	2686
NME2.1	66	ATGCTTGGGGAGACCAATCCAGCAGATTCAAAGCCAGGCACCATTCGTGGGACTTCTGCATTGAG	2687
NNMT.1	67	CCTAGGGAGGGATGGAGAGAGAGTCTGGGCAATGAGAGAGGGTCTCGGGATGTTTGGCTGGACTAG	2688
NOL3.1	72	CAGCCTTGGGAAGTGAGACTAGAAGAGGGGAGCAGAAGGACCTTGAGTAGACAAGGCCACACACA TCAT	2689
NOS2A.3	67	GGGTCCATTATGACTCCCAAAGTTTGACCAGAGGAGCCAGGGACAAGCCTACCCCTCCAGATGAGC	2690
NOS3.1	68	ATCTCCGCTCGCTCATGGGCACGGTGATGGCGAAGCGAGTGAAGGCGACAATCCTGTATGGCTCCGA	2691
NOTCH1.1	76	CGGGTCACCAAGTTTGAATGGTCAATGGGAGTGGCTGTCCCGCTGCAGAGCGGCAATGGTCCGAACC AATACAC	2692
NOTCH2.1	75	CACCTCCCTGCTGGGATTATATCAACAACAGTGTGATGAGCTGTGCAACACGGTCGAGTGCCTGTTTG ACAACT	2693
NOTCH3.1	67	TGTGACGAGTGTCTGGCCCCGCCACCCCTGTGGCCCTCATGGTATCTGCACCAACCTGGCAGGGAGT	2694
NPD009 (ABAT_offici	73	GGCTGTGGCTGAGGCTGTAGCATCTCTGCTGGAGGTGAGACACTCGGAACATGATTTGACCTCGAAT GCTCC	2695
NPM1.2	84	AATGTTGTCCAGGTTCTATTGCCAAAGATGTGTTGTCCAAAATGCCTGTTTAGTTTTTAAAGATGGAATC CACCTTTGCTTG	2696
NPBB.1	66	GACACCTGCTTCTGATTCCACAGGGGCTTTTCTCTCAACCCCTGTGGCCGCTTTTGAAGTGACTCA	2697
NPR1.1	66	ACATCTGCAGTCCCTCATGACCTTCAGAACCTCATGCTCTCGGCCCTGGAAGCTGGCTTGTGTG	2698
NPY1R.1	70	GGATCTTCCCCACTCTGCTCCCTTCCATTCACACCTTCTCTTCTTTAATAGCAGGAGCGAAGACAA	2699
NRG1.3	83	CGAGACTCTCTCATAGTGAAGGTATGTGTACGCCATGACACCCCGGCTCGTATGTCACTGTAGAT TTCACACGCCAAG	2700

207

208

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
NUDT1.1	77	ACTGGTTTCCACTCCTGCTTCAGAAGAAGAAATTCACGGGTACTTCAAGTTCAGGGTCAGGACACCAT CCTGGAC	2701
OGG1.1	71	ACCAAGGTGGCTGACTGCATCTGCTGATGCCCTAGACAAGCCCCAGGCTGTCCCGTGGATGTCCA TAT	2702
OPN, osteopontin.3	80	CAACCGAAGTTTTCACCTCCAGTTGTCCCCACAGTAGACACATATGATGGCGAGGTGATAGTGTGGTTTA TGGA CTGAGG	2703
p21.3	65	TGGAGACTCTCAGGGTCGAAAAACGGCGGCAGACCAGCATGACAGATTTCTACCACTCCAAAACGCC	2704
p27.3	66	CGGTGGACACGAAGAGTTAAACCCGGGACTTTGGAGAAGCACTGCAGAGACATGGAAGGCGAGCC	2705
P53.2	68	CTTTGAACCCCTTGCTTGCAATAGGTGTGCGTCAGAAGCACCAGGACTTCCATTTGCTTTGTCCTCCGGG	2706
PAH.1	80	TGGCTGATTCCATTAAACAGTGAAATTTGGAATCTTTGCAGTGCCCTCCAGAAAAATAAAGTAAAGCCATGG ACAGAA TG TG	2707
PAI1.3	81	CCGCAACGTGTGTTTCTCACCCATATGGGGTGGCCCTCGGTGTTGGCCATGTCTCAGCTGACAACAGGAG GAGAAA CCCAGCA	2708
Pak1.2	70	GAGCTGTGGGTGTTATGGAATACTTGGCTGGAGGCTCCTTGACAGATGTGGTGACAGAAAATTGTCAT	2709
PARD6A.1	66	GATCCTCGAGTCAATGGCATTTGAAGTAGCCGGGAAGACCTTGGACCAAGTGCAGGACATGATGGT	2710
PBOV1.1	72	GCAAGCCCTTTCCAGAAAAATAAAAATGGTTGAAAAGGCAATTCTGCTACCAATGACTGTTTAAGCCAG CC	2711
PCCA.1	68	GGTGAAATCTGTGCACCTGTCAAGCTGGAGACACAGTTGGAGAAGGGGATCTGCTCGTGGAGCTGGAAT	2712
PCK1.1	66	CTTAGCATGGCCCCAGCACCCAGCAGCCAAACTGCCCAAGATCTTCCATGTCAACTGGTTCCGGAAG	2713
PCNA.2	71	GAAGGTGTGGAGGCACTCAAGGACCTCATCACAGGCGCTGTGGGATATTAGCTCCAGCGGTGTAAA CC	2714
PCSK6.1	67	ACCTTGAGTAGCAGAGGCCCTCACACCTTCCTCAGAAATGHAACCCCAAGGTGAATGGCTCCAGCAAC	2715
PDCD1.1	73	GACAACGCCACCTTCACCTGCAGCTTCTCCAAACACATCGGAGAGCTTCGTGCTAAACTGGTACCGCATG AGCC	2716
PDB4DIP.1	73	GCTTCGCTTCTGTGTGAGAGAGCGAGATCATGACTTAGAGAGACTGCGCGATGTCTCTCTCCCAATGA AGCT	2717
PDGFA.3	67	TTGTTGGTGTGCCCTGTGCGGTGGTGGCGGTCACTCCCTCTGCTGCCAGTTTGGACAGAACCCA	2718
PDGFB.3	62	ACTGAAGGAGACCTTGGAGCCTAGGGGCAATCGGACGAGAGTGTGTGGCAGGGTTATTTTA	2719

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
PDGFC.3	79	AGTTACTAATAAATATACACGAGGTCCTTCAGTTGAGACCAAGACCGGTGTGAGGGATTGCACAAATCA CTCACCGAC	2720
PDGFD.2	74	TATCGAGCAGGTATACCATGACCGGAAGTCAAAAGTTGACCTGGATAGGCTCAATGATGATGCCAAG CGTTA	2721
PDGFra.2	72	GGGAGTTTCCAAGAGATGGACTAGTGTCTGGTCCGGTCTTGGGTCTGGAGCGTTTGGGAAGGTGGTT GAAG	2722
PDGFRb.3	66	CCAGCTCTCCTTCAGCTACAGATCAATGTCCCTGTCCGAGTGTGGAGCTAAGTGAGAGCCACCC	2723
PDZK1.1	75	AATGACCTCCACCTTCAACCCCCGAGAAATGTAAGTGTCCAGCAAGAGGGCAAACTATGGCTTCTTC CTGCG	2724
PDZK3.1	68	GAGCTGAGAGCCTTGAGCATGCCTGACCTTCACAAGCTCTCAGCGAGGATTACTCAGAGGGCCGAG	2725
PF4.1	73	GCAGTGCCTGTGTGAAGACCACTCCAGGTCCGTCCAGGCACATCACAGCCTGGAGGTGATCA AGGCC	2726
PFKP.1	68	AGCTGATCCGCATACATATTTTCGAAGAGCCCTTCGACATCAGGGATCTGCAGTCCCAACGTGGAGCACC	2727
PFN2.1	82	TCTATAGTGCATGGTACTGCACAATGGACATCCGGACAAAGAGTCAAGTGGGGAGCCCAACATACAA TGTGGCTGTCGGC	2728
PGF.1	71	GTGGTTTTCCTCGAGCCCCCTGGCTCGGACGCTCTGAGAAGATGCCGGTCATGAGGCTGTTCCCTT GCT	2729
PI3K.2	98	TGCTACCTGGACAGCCCGTTGGTGGCTTCTCCTGAAACGAGCTGTGTGACTTGAGAGTGACTCAC TACTTCTTCTGGTTACTGAGGACGGCT	2730
PI3KC2A.1	83	ATACCAATCACCGCACAAAACCCAGGCTATTTGTTAAGTCCAGTCACAGCAAAAGAAACATATCGGAGA AAATGCTAGTGTG	2731
PIK3CA.1	67	GTGATTGAAGAGCATGCCAATGGTCTGTATCCCGAGAAAGAGGATTTAGTATTTCCACCGCAGAC	2732
PLA2G4C.1	68	CCCTTTCCCAAGTAGAAGAGGCTGAGCTGGATTTGTGTCCAAGGCCCGCCGCTGCTACATCCT	2733
PLAT.1	67	GATTGTGGGAAGTGTGTGAATAGATACAGGGCCACGTGTACGAGACCCAGGGCATCAGCTA	2734
PLAUR.3	76	CCCATGGATGCTCCTCTGAGAGAGACTTTCCTCATTTGACTGCCGAGGCCCATGAATCAATGCTCTGGTAG CCACCGG	2735
PLG.1	77	GGCAAAATTTCCAGACCATGCTCTGGACTGGAATGCCAGGCCTGGACTCTCAGAGCCCAACGCTCAT GGATACAT	2736
PLN.1	84	TGATGCTTCTCTGAAGTTCCTGCTACAACCTCTAGATCTGACGCTTGCCACATCAGCTTAAATCTGTCTATC CCATGCAGACAG	2737

TABLE B - continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
PLD2.1	84	CAGGAGGTGGTTGCCAAATTTCTAAGGTACAAATGCTCTATTGAGTCACCACGAAAAAGGCTGGAGCTTCATGCATCCTGGGAGA	2738
PLP1.1	66	AGAAACAGACTGGCCTGAGGACAGCAGTTGCTGGTGCTAATGCTGAACCTGAGATGGCCCTCTG	2739
PPM22.1	66	CCATCTACCGGTGAGGCACCCGAGTGGCATCTCAACTCGGATTACTCTACGGTTTCGCCTTACA	2740
PPAP2B.1	77	ACAAACCATCCCTCAGTGATGTTCTGTCAGGATTTGCTCAAGGAGCCCTGGTGGCCTGCTGCATAGTATTCTTTCGTG	2741
PPARG.3	72	TGACTTTATGGAGCCCAAGTTTGAGTTTGCTGTGAAGTTCAAGCACTGGAATTAGATGACAGCGACTTGGC	2742
PPP1R3C.1	82	TTCCCTTCCCTCTCAATCCACTAGCTTTTCATGTTGGGCAAGGAAAAAGTTGAGGAAGGATGGCTGATGGTGA TGGAAGCTGTG	2743
PPP2CA.1	78	GCAATCATGGAACTTGACGATACTCTAAAAATCTCTTTCTTGCACTTTGACCCAGCACCTCGTAGAGCGGAGCCACAT	2744
PRCC.1	67	GAGGAAGAGGAGCGGTGGCTCTTACATCTGGGCCCGCTTTAGGGGGCTTGTTCGCTTCTCTCCCTG	2745
PRKCA.1	70	CAAGCAATGCGTCATCAATGTCCCCAGCCTCTGCGGAATGGATCACACTGAGAAAGGGGGCGGATTTAC	2746
PRKCB1.1	67	GACCCAGCTCACCTCCTGCTTCCAGACCAATGGACCGCCTGTACTTTGTGATGAGGTACGTGAATGGG	2747
PRKCD.2	68	CTGACACTTGGCCGAGAGATCCCTTTCTCACCCACCTCATCTGCACCTTCCAGACCACAGGACCACCT	2748
PRKCH.1	68	CTCCACCTATGAGCGTCTGTCTCTGTGGGCTTGGGATGTTAACAGAGGCCAAAAAGGAGGAAAGTGTG	2749
PRO2000.3	79	ATTGGAAAAACCTCGTCACACAGAGAAAGCCCAACATAATTTTATAGTGGCCAGCTTCTCTGTCAAGACCAAGATACCGA	2750
PROM1.1	74	CTATGACAGGGCATGCCACCCGACCAACCGAGGCTGTGTCTCCAACACCGAGGCGTCTTCCTCATGGTTGGAG	2751
PROM2.1	67	CTTCAGCGCATCCACTACCCCGACTTCCTCGTTCAGATCCAGAGGCCCGTGGTGAAGACCAGCATGG	2752
PRPS2.1	69	CACCTGCAACCAAGATTACGGTCAATTGACATTTCCATGATCTTGGCCGAAGCAATCCGAGGACACACAAT	2753
PRSS8.1	68	GTACACTTGGCCTCCAGCTATGCTCTCTGGATCCAAAAGCAAGGTGACAGAACTCCAGCCTCGTGTGG	2754
PSMA7.1	67	GCCAACTGACAGGATGAAGAACAGTGGGAGAGATCTGTGCTTTGGATGACAACTCTGCATGSCCT	2755
PSMB8.1	66	CAGTGGCTATCGGCCATAATCTTAGCCCTGAAGAGGCCTATGACCTTGGCCGACGGGCTATTGCTTATG	2756
PSMB9.1	66	GGGTGTCAATCACTGGTCACTATTACAGCTGCCGGTGTGACCATCGAGTCAATCTTGGGCAATG	2757

213

214

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
PTEN.2	81	TGGCTAAGTGAAGATGACAATCATGTTGACAGCAATCATTGTAAGCTGGAAAGGACGAACACTGGTGTA ATGATATGTGCA	2758
PTGIS.1	66	CCACACTGGCATCTCCCTGACCTTCTCCAGGGACAGAAGCAGAGAGTAAGTTTCTCATCCCATGGGC	2759
PTHR1.1	73	CGAGGTACAAGCTGAGATCAAGAAATCTTGAGCCGCTGGACACTGGCACTTCAAGCCGAAGG CACGC	2760
PTK2.1	68	GACCGTTCGAATGATAAGGTGTACGAAATGTGACGGCCTGGTGAAGCTGTCTCATCGAGATGTCCAG	2761
PTK2B.1	74	CAAGCCAGCCGACCTTAAGTACAGACCCCTCCGAAAACCAACCTCCTGGCTCCAAAGCTGCAGTTCCA GGTTC	2762
PTN.1	67	CCTTCCAGTCCAAAAATCCCGCCAAGAGAGCCCCAGAGCAGAGGAAAAATCCAAAGTGGAGAGAGGGG	2763
PTPNS1.1	77	CTCCAGCTAGACATAAGCAACATCTCGCTGTGACGCCCTGTAAATTAATCTGAGAAATGTGAAACGTGCAAT CTTGAAA	2764
PTPRB.1	68	GATATGGCGTGAGAAACAGCTTGTATGCACACAGACTCATCCGCCACTTTCACTATACGGTGTGCCAG	2765
PTPRC.1	74	TGSCGTCAATGGAAGAGGGCACTCGGGCTTTTGGAGATGTTGTTGTAAGATCAACCAGCACAAAAGA TGTC	2766
PTPRG.1	71	GGACAGCACAAGACTTGAAAGCCCAACATTAGCCATGTCTCACCCGATAGCCTTTACCCCTGTTCCGAGT CC	2767
PTTG1.2	74	GGCTACTCTGATCTATGTTGATAAGGAAATGGAGAACCCAGGCACCCGTGTGTTGCTAAGGATGGGT GAAGC	2768
PVALB.1	81	AAACCAAGATGCTGATGGCTGCTGGAGACAAAGATGGGACGGCAAAATTTGGGTTTGACGAATTCCTCA CTCTGGTGGCTG	2769
PXDN.1	67	GCTGCTCAAGCTGAACCGCACTGGACGGCGACACCATCTACTATGAGACCAGGAAGATCGTGGT	2770
RAC1.3	66	TGTTGTAATGCTCTAGCCCTCGTTCTTGCTCCTGTCCCTTGGAACTTTGTACGCTTTGCTCAA	2771
RAD51.1	66	AGACTACTCGGGTCGAGGTGAGCTTTCAGCCAGCAGATGCACCTGGCCAGGTTTCTGCGGATGCT	2772
RAF1.3	73	CGTCGTATGCGAGAGTCTGTTTTCCAGGATGCCGTAGTTCTCAGCACAGATAATTCACCTCACGCCT TCA	2773
RALBP1.1	84	GGTGTCAATATAAATGTGCAATGCCTTCTTGCTGCTCTGCTCGGTCTCAGTACGTTTCACTTTATAGCTG CTGCAATATCGAA	2774
RARB.2	78	TGCCTGGACATCCTGATTCTTTAGAATTTGCACAGGTATACCCAGAACAAGACACCATGACTTTCTCAG ACGGCCTT	2775
RASSF1.1	75	AGGCACGTGAAGTCAATGAGGCCCTGCTGCGAAAAGTTCTTGTTGGTGGATGATACCCCGCAAGTTTGC ACTCTTT	2776

215

216

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
RB1.1	77	CGAAGCCCTTACAAGTTTCCTAGTTTACCCCTTACGGATTCTCTGGAGGGAACATCTATATTTCACCCCTGA AGAGTCC	2777
RBM35A.1	66	TGGTTTGAATCACAGGGCGCCATCAGGAGATGCTTTATCCAGATGAAGTCTCGGACAGAG	2778
REG4.1	83	TGCTAACTCTCTGCACAGCCCGTCTCTTCTCTTCTGCTAGCCTGGCTAAATCTGCTCATTTATTTCAAG GGGAAACCTAGCA	2779
RET.1	71	GCCTGTGCAGTTCTTGTGCCCCAAATCAGCGTGGCCTACAGGCTCCTGGAGGGTGAAGGTCTGCCCT TCC	2780
RGS1.1	84	TGCCCTGTAAAGCAGAAAGAGATATATAAGCATTTGTGCATTCAGATGCTGTCTAAACAAATCAAATATTGAC TTCGCACCTCGAG	2781
RGS5.1	79	TTCAAACGGAGGCTCTAAAGAGGTGAATATTGACCCTTCACCTAAGGACATCAATGAAGAACCTGGT GGAACTTC	2782
RHEB.2	78	GATGATTGAGAACAGCCTTGCTGTCTACTGTCTAGAACACCCCTGGAGTTTAGTGTCTGTGTCAAGATC TTGGGAGC	2783
RhoB.1	67	AAGCATGAACAGGACTTGACCATCTTTCCAAACCCCTGGGGAAGACATTTGCAACTGACTTGGGGAGG	2784
rhoC.1	68	CCGTTCCGTCTGAGGAAGGCCGGGACATGGCGAACCGGATCAGTGCCTTTGGCTACCTTGAGTGCTC	2785
RIPK1.1	67	AGTACCTTCAAGCCGGTCAAAATTCAGCCACAGAACAGCCTGGTTCACTGTCACAGTTCCTCCAGGGACTT	2786
RND3.1	66	TCGGAAATTGGACTTTGGGAGGCGCGTGAGGAGTCAGGCTTAAAACTTTGTTGGAGGGGAGTAACCAAG	2787
ROCK1.1	73	TGTGCATAGGAATGAGCTTCAGATGCAGTTGGCCACGAAAGAGAGTGATATTGAGCAATTGCGTGT AAAC	2788
ROCK2.1	66	GATCCGAGACCTTCGCTCCCCATCAAGTGGAGAGCTTGCTGGATGGCTTAAATTCCTTGGTCCCT	2789
RPLP1.1	68	CAAGGTGCTCGGTCTTCCGAGGAAGCTAAGGCTGCGTTGGGGTGAGGCCCTCACTTCATCCGGCGAC	2790
RPS23.1	67	GTTCTGTTGTGTGATTGTGTCGCAAGGTCACTGCTGTTGGTGATATTCCTGGAGTCCGCTTTAAGG	2791
RPS27A.1	74	CTTACGGGGAAGACCATCACCCCTCGAGTTGACCCCTCGGATACGATAGAAAATGTAAAGGCCAAGATC CAGGA	2792
RPS6KAI.1	70	GCTCATGGAGCTAGTGCTCTGGACCCGGAGATGGACAGACCTCAGGGGAAGAGCTGGACTTCAGC CG	2793
RPS6KB1.3	81	GCTCATTATGAAAACATCCCAAACCTTTAAAAATGCGAAATTTATTGGTTGGTGTGAAGAAAGCCAGACAAC TCTGTTTCTT	2794
RRM1.2	66	GGGCTACTGGCAGCTACATGCTGGGACTAATGGCAATTCCAATGGCCTTGTACCGATGCTGAGAG	2795

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
RRM2.1	71	CAGCGGGATTAAACAGTCTTTTAAACGACACAGCCAGTCTAAAGATGCGCCTCAGCTTCAACGCAG AT	2796
RUNX1.1	70	AACAGAGCAFTGCCAACCAATATTGGATCTGCTTGTCTCCAACAGCAAACTTCTCGGCAAAATCAC	2797
S100A1.1	70	TGGACAAGGTGATGAAGAGCTAGACGAGAATGGAGACGGGAGGTGACTTCCAGGAGTATGTGGTG CT	2798
S100A10.1	77	ACACCAAAATGCCATCTCAATGGAAACAGCCATGGAAACCATGATGTTTACATTTTCAAAATTCGCTGG GGATAAA	2799
S100A2.1	73	TGGCTGTGCTGGTCACTACTTCCACAAGTACTCTGCGAAGGGCGACAAGTTCAGCTGAGTAAGG GGGA	2800
SAA2.2	72	CTACAGCACAGATCAGCACCATGAAGCTTCTACGGGGCTGGTTTTCTGCTCCTTGGTCTGAGTGTC GCA	2801
SCN4B.1	67	GCCTTCCTGGAGTACCGAGTGTCTCCCTATGCCCTTCCAGCATTTCTACTTGGGGAATTGGGCCAC	2802
SCNN1A.2	66	ATCAACATCTGTGAGGCTGCCAGAGACTCTGCCATCCCTGGAGAGGACACGCTGGGCAACTTC	2803
SDHA.1	67	GCAGAACTGAAGATGGGAAGATTTATCAGCGTGCAATTTGGTGGACAGAGCCTCAAGTTTGGAAAGG	2804
SDPR.1	66	ACCAGCACAAATGGAGCAGCGACAGATCAGTTTGGAGGGCTCCGTGAAGGGCATCCAGATGACC	2805
SELE.1	71	ACACTGGTCTGGCCTGCTACCTACCTGTGAAGTCCCACTGAGTCCAAACATTCCTTGGTAGCTGGACT TT	2806
SELENBP1.1	67	GGTACCAGCCTCGACACAAATGTCATGATCAGCACTGAGTGGGCAGCTCCCAATGCTTACGAGATGG	2807
SELL.1	67	TGCAACTGTGATGTGGGTACTATGGGCCCCAGTGTGAGTTTGTGATTCAGTGAGCCTTTGGAGG	2808
SELP1G.1	83	TGGCCACTATCTTCTCGTGTGCACGTGTGGTGCTGGCGTCCGCCCTCTCCCGCAAGGGCCACATGTAC CCCGTGGTAAATTAC	2809
SEMA3B.1	71	GCTCCAGGATGTGTTTCTGTGTTGTCCTCGGGGACCACGGGACCCCGCTGCTCTATGCCGTCTTCTCCAC GT	2810
SEMA3C.1	66	ATGGCCATTCCTGTTCCAGATTTACCCAACTGGGAAACGGAGGAGCCGAAGACAGATGTGAGAC	2811
SEMA3F.3	86	CGGAGCCCCCTCATATACACTGGGCGAGCCTCCCCACAGCGCATCGAGGAATGCGTCTCAGGCAA GGATGTCAACGGCGAGTG	2812
SEMA5B.1	67	CTCGAGGACAGCTCCAACATGAGCCTCTGGACCCAGAACATCACCCGCTGTCTGTGCGGAATGTGA	2813
SERPINA5.1	66	CAGCATGGTAGTGGCAAGAGAGGTCCAGAGTCTTGGCCCTTGATGCCCCAGCTCAGTGCCACAAAG	2814
SFN.1	70	GAGAGAGCCAGTCTGATCCAGAGGCCAAGCTGGCAGAGGCGGCGAAGCGCTATGAGACATGGCAGC CT	2815

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID No.
ISGK.1	73	TCGCAAGACACCTCTGGAGGCGCTCCTGCAGAAGCAGAGCAAAAGCGGCTCGGGGCCAAGGATGA CTTCA	2816
SHANK3.1	68	CTGTGCCCTCTACAACCAAGAGAGCTGTGCTCGTGCTCTCTTCGTGGAGCTAACAGGGATGTCC	2817
SHC1.1	71	CCAAACACCTTCTGGGCTTCTGGGACCTGTGTTCTTGTGTAGCAACCTCTCCGGTTTGGGTGGGATAAC AG	2818
STILV.1	66	CCGCAATCTTCTGCTTTGTCCCATTTGGTGAGAAATAGCCCCCTCTCTCAGTGGGCAGCAGGTCCTGAGT	2819
SKIL.1	66	AGAGGCTGAATATGCAGGACAGTTGGCAGAACTGAGGCAGAGATTGGACCATCTGAGGCCGATAG	2820
ISLC13A3.1	66	CTTGCCCTCCACAAGGCTCTGCCCCAGTACTTCTCTCGACACCAACTTCTCTCTTCCCTCAGTGGGCT	2821
ISLC16A3.1	68	ATGCGACCCACGCTACATGACTAGTGTTCATCTCTGGGGGGGGCGAGGTGCTCACCTCTCTCCCTGATTT	2822
ISLC22A3.1	66	ATCGTCAGCGAGTTTGACCTTTGCTGTGTCAATGCGTGGATGCTGGACCTCACCCAGCCATCCTG	2823
ISLC22A6.1	68	TCGGCCACCTCTTCTCTGCTCTCCATGCTGTGTGTTTGGCACTAGCTTTGCACTATATGGGCTGGTC	2824
ISLC2A1.1	67	GCCTGAGTCTCCTGTGCCCCACATCCAGGCTTCACCCCTGAATGGTTCCATGCTGAGGGTGGAGACT	2825
ISLC34A1.1	66	GCTCAGACCCACTGACTGCAGACCTCATAGTGGGTGCCCAGGATGTTGTCTCTACGGAGAGAGGCT	2826
ISLC7A5.2	70	GCGCAGAGGCCAGTTAAAGTAGATCACCTCTCGAAACCCACTCCGGTTCCCCGCAACCCACACAGCTCAGC T	2827
ISLC9A1.1	67	CTTCGAGATCTCCCTCTGGATCCTTCTGGCTGCCTCATGAGATAGGTTTCCATGTGATCCTCCCACT	2828
SLIT2.2	67	TTTACCGATCGACCTGTCCATATGGTTTCAAGGGGAGGACTGTGATGTCCCAATTCAATGCTGTGAT	2829
SNAIL.1	69	CCCAATCGGAAGCTTAATACAGGAGCTCGAGGACTTAATCCAGAGTTTACCTTCCAGCAGCCCTAC	2830
SNRK.1	71	GAGGAAAAGTCAGGGCCGGGGCTCCAGCTGCACTAGTTTCGGAGACCAGTGATGATGATTCTGAAAGCC GGC	2831
SOD1.1	70	TGAAGAGAGGCATGTTGGAGACTTGGCAATGTGACTGTGACAAAGATGGTGTGGCCGATGTGTCTAT	2832
SP3.1	69	TCAAGAGTCTCAGCAGCAACCACTCAAGCCCCAAATTGTGTAAGGTAATTACACCAAGACAATCCATGG	2833
SPARC.1	73	TCCTCCCTGTACTGGCAGTTGCGCCAGCTGGACAGACCCCAATTGACGGGTACCTCTCCACACCG AGCT	2834
SPARCL1.1	67	GGCACAGTCAAGTGATGACTACTTCAATCCAAAGCAGGCCCTTCTTGAGGCCGAGAGAGCTCAATC	2835
SPAST.1	66	CCTGAGTTGTTCAAGGGCTTAGAGCTCTGCCAGAGGGCTGTTACTCTTTTGGTCCACCTGGGAAT	2836
SPHK1.1	67	GGCAGCTTCCTTGAAACCAATTATGCTGGCTATGACAGGTCACCAATGAAGACCTCTCTGACCAACTGC	2837

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
SPRY1.1	77	CAGACCACTCCCTGGTCATAGGTCGAAAGGGCAATCCGGACCCAGCCCAAGCAACTGATTGTGGATGA CTTGAAGG	2838
SQSTM1.1	69	GGACCCGTCCTACAGTGAACTCCAGTCCCTACAGATGCCAGAAATCCGAAGGGCCAAAGCTCTCTGGACC C	2839
STAT1.3	81	GGGCTCAGCTTTTCAGAAAGTGTGAGTTGGCAGTTTCTTCTGTCAACAAAAGAGGTCTCAATGTGGACC AGCTGAACATGT	2840
STAT3.1	70	TCACATGCCACTTTTGGTGTTCATAATCTCCTGGGAGAGATTGACCAGCAGTATAGCCGCTTCCTCGCAAG	2841
STAT5A.1	77	GAGCGCTCAACATGAAATTCAGGCCCGAAGTGCAGAGCAACCCGGGGCCTGACACAGGAGAACCTCGT CGTTCTTGGC	2842
STAT5B.2	74	CCAGTGTGTGTGATCGTTTCATGGCAGCCAGGACAACAATGCGACGGCCACTGTCTCTGGACAATGCT TTTGC	2843
STC2.1	67	AAGAGGCAATCAACCCACAGCGTCAGGTTCAAGTGTAGCAGAACTGGGGAAGCCTGTGTCTCATCT	2844
STK11.1	66	GGACTCGAGACGCTGTGCAGGAGGGCCGTCAAGATCCTCAAGAAAGAAAGTTGCCGAAGGATCCC	2845
STK15.2	69	CATCTTCAGGAGGACCACCTCTGTGTGGCAACCCTGTGACTACTGTGCCCCCTGAAATGATTGAAGGTGCGA	2846
STK4.1	66	GAGCCATCTCCTGCAACTTTACCTCTTTCCCTCAGATGGGGAGCCATGACTGGGTTGCACCTCAG	2847
STMY3.3	90	CCTGGAGCTGCAACATACCTCAATCTGTCTCCAGGCCGGATCCTCCTGAAAGCCCTTTTCGCACACTG CTATCTCCAAAGCCATTGTA	2848
SUCLG1.1	66	CCAAAGCCTGTAGTGTCTTCAATTGCTGTTTAACTGCTCCTCTGGGAGAAGAATGGGTCAATGCCG	2849
SULT1C2.1	67	GGGACCCCAAGCATGAAATTCGAAAGGTGATGCAAGTTCAATGGGAAAGAAGGTGGATGAAACAGTGTCT	2850
SURV.2	80	TGTTTGTATTCCTCCGGGCTTACCAAGTGAGAAGTGAGGAGGAAGAAGGCAGTGTCCCTTTTGTAGAGC TGACAGCTTTG	2851
TACSTD2.1	80	ATCACCAACCGGAGAAAGTCGGGGAAGTACAAGAAGGTGGAGATCAAGGAACCTGGGGAGTTGAGAAA GGAAACCGAGCTT	2852
TAGLN.1	73	GATGAGCAGGTGGCTCAGTTCTCTGAAGGGGGCTGAGGACTATGGGGTCAACAAGACTGACATGTTCC AGACT	2853
TAP1.1	72	GTATGCTGCTGAAGTGGGAATCCTCTACATTGGTGGCAGCTGGTACCAGTGGGCTGTAAACAGT GGGA	2854
TCF4.1	67	CACACCTGGAAATGGGAGACGCATCGAATCACATGGGACAGATGTAAAGGGTCCAAGTTGCCACAT	2855
TCOF1.2	66	AGCGAGGATGAGGACGTGATCCCCGCTACACAGTGTCTTACTCTCTGGCATCAGAACCACAATGTGTGTG	2856

TABLE B -continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
TEK.1	76	ACTTCGGTGTACTTAACAACCTTACATCCAGGAGCAGTACGTGGTCCGAGCTAGAGTCAACACCAAG GCCCAGG	2857
TERT.1	85	GACATGGAGAAACAAGCTGTTTCGGGGATTTCGGCGGACGGGCTGCTCCTGCGTTTGGTGGATGATTT CTTGTGGTGACCTC	2858
TFAP2B.1	67	CGTGTGACGTCCGAGACGCGATGGACGCCCTTGCTCTTTACTGTGCAGGTCTGTGAGAGCGTGTGG	2859
TFAP2C.1	68	CATGCTCACCAGATGGACGAGGTGCAGATGTCGACGACCAGCACCTGTTGTGTCGACGATCAGACAG	2860
TFPI.1	69	CCGAATGGTTTCCAGGTGGATAATTATGGAACCCAGCTCAATGCTGTGAATACTCCCTGACTCCGCAA	2861
TGFA.2	83	GGTGTGCCACAGACCTTCTACTTTGGCCTGTAATCACTGTGCAGCCTTTTGTGGGCCCTTCAAACTCT GTCAAGAACTCCGT	2862
TGFB1.1	80	CTGTATTTAAGGACACCGTGCCTCCCAAGCCCACTGGGGCCCCATTAAAGATGGAGAGAGACTGCGG ATCTCTGTGCA	2863
TGFB2.2	75	ACCAGTCCCCAGAAAGACTATCTGAGCCCGAGGAAGTCCCCCGGAGGTGATTTCCATCTACACAGC ACCAGG	2864
TGFB1.1	67	GCTACGAGTGTCTGCGATATGAAAAGTCCCTGGGGAGAAAGGCTGTCCAGCAGCCCTACCACT	2865
TGFB1.1	67	GTCAATCACCCTGGCCTTGGTCTGTGGAACTGGCAGCTGTCAATTCGTGACCAAGTGTCTTGTCTGC	2866
TGFB2.3	66	AACACCAATGGTTTCCATCTTTCTGGGCTCCTGATTTGCTCAAGCACAGTTTGGCTGATGAAGAGG	2867
THBD.1	68	AGATCTGGACGAGACTGGGGGCCACCTAATGACAGTGGCTCCTCGGTGGCTGCCGATGTCAATTC	2868
THBS1.1	85	CATCGCAAGTGACTGAAGAGAAACAAGGTTGCCCAATGAGCTGAGGGGCCCTCCCTATGCTATCA CAACGAGTTCAGTAC	2869
TIMP1.1	76	TCCCTGCGTCCCAGATAGCCTGHAATCCTGCGGAGTGGAGCTGAAGCCTGCACAGTGTCCACCCT GTTCCAC	2870
TIMP2.1	69	TCACCTCTGTGACTTCATCGTGCCCTGGGACACCTGAGCACCAACCAGAGAAGAGCCTGAACCACA	2871
TIMP3.3	67	CTACCTGCTTGTGTTGTGACTTCCAGAACGAGTGTCTCTGGACCGACATGCTCTCCAAATTCGGT	2872
TK1.2	84	GCCGGGAAGACCGTAATTTGGCTGCTGACTGGAGCTTCCAGAGGAAGCCATTTGGGGCCATCCT GAACCTGGTGCCGTG	2873
TLR3.1	71	GGTTGGCCCACTAGAAGTACTTGACCTGGGCCCTTAATGAAATTGGGCAGAACTCACAGGCCAGGAAT GG	2874
TMEM27.1	75	CCCTGAAGAATGTTGGCTGCTCTTTTCTGTGACTGCCATTCATGCTGAACCTGTGTCAACACAGGT GCAGA	2875
TMEM47.1	71	GGATCCACTGTTAGAGCCCTTACCGCTGCTTATCTTACCCTATGACTACATTTGGCTGTGGTTAATTG	2876

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID NO.
TMSB10.1	68	GAAATCGCCAGCTTCGATAGGCCAAGCTGAAGAAAACGGAGACGACAGGAAGAAACACCCCTGCCGAC	2877
TNF.1	69	GGAGAAGGTTGACCGACTCAGCGCTGAGATCAATCGGCCCGACTATCTCGACTTTTGCCGAGTCTGGGC A	2878
TNFAIP3.1	68	ATCGTCTTTGGCTGAGAAAAGGAAAAAGACACAAAGTCGCGTTGGTTGGAGAAGCCAGAGCCATTCCAC	2879
TNFAIP6.1	67	AGGAGTGAAGATGGGATGCCATTATTGCTACAAACCACACGCAAGAGAGTGTGGTGGCGTCTTTACAG	2880
TNFRSF10C.3	67	GGAGTTTGACACAGAGATGCAAGGGGTGAAGGAGCGCTTCCTACCGTTAGGGAAGTCTGCGGACAGAG	2881
TNFRSF10D.1	66	CCTCTCGCTTCTGTGTGTCGTGAACTGAGTCCCTGGGATGCCCTTTTAGGGCAGAGATTCTGTGAGC	2882
TNFRSF11B.1	67	TGGCGACCAAGACACCTTGAAGGGCCTAATGCACGCACTAAAGCACTCAAAGACGTACCACCTTTCCC	2883
TNFRSF1A.1	71	ACTGCCCTGAGCCCCAAATGGGGGAGTGAGAGGCCATAGCTGCTGGCATGGGCCCTCTCCACCGTGCCCT GAC	2884
TNPSF12.1	68	TAGGCCAGGAGTTCCCAAATGTGAGGGCGAGAAAACAAGACAAGCTCCTCCTTTGAGAAATTCCTGTG	2885
TNPSF13B.1	80	CCTACGCTGGGACATCTAATATCAGAGAAAGAGTCCATGCTCTTTGGGGATGAATTGAGTCTGGTGA CTTTGTTTCGA	2886
TNPSF7.1	67	CCAACTCCTACTGGGACACTTTTTCCTTCCCGAAAACACTGATGAGACCTTCTTTGGAGTGAGTGGGT	2887
TNIP2.1	66	CATGTCAGAAAGGCCGATCGGGAACGGGCTCAAAGTAGGATTCAGAAGCTGGAGGAAAAGGTGCG	2888
TOP2A.4	72	AATCCAAAGGGGGAGTGATGACTTCCATATGGACTTTTGACTCAGCTGTGGCTCCTCGGGCAAAATCTG TAC	2889
TOP2B.2	66	TGTGGACATCTTCCCTCAGACTTCCCTACTGAGCCACCTTCTCTGCCACGAACCGGTGCGGCTAG	2890
TP.3	82	CTATATGCAGCCAGAGATGTGACAGCCACCGTGGACAGCTGCCACTCATCAGCCTCCATTCTCAGT AAGAACTCGTGG	2891
TRAIL.1	73	CTTCACAGTGTCTCTGAGTCTCTCTGTGTGGCTGTAACTTACGTGTACTTTACCAACGAGCTGAAGCAG ATG	2892
TS.1	65	GCCTCGGTGTGCCCTTTCAAATCGCCAGCTAGCCCTGCTCAGCTACATGATTTGCGCACATCAG	2893
TSC1.1	66	TCACCAATCTCAGCCCGCTTTCTCTCATCGTTCAGCCGATGTCACCACCCCTTATGCTGACAC	2894
TSC2.1	69	CACAGTGGCCTCTTTCTCTCCCTGTACCAGTCCAGCTGCCAAGGACAGCTGCAAGGAGCGTTTCTCTG	2895
TSPAN7.2	67	ATCACTGGGTGATCTCTGCTGGCTGTTGGAGTCTGGGGCAAACTTACTCTGGGACCATATATCTCCC	2896
TSPAM8.1	83	CAGAAATCTCTGACGGCAAGTTGCTCCAGAGCATATTGACGACAGCCTGTAAACGAATAGTTAAATTCA CGCATCTGATT	2897

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID No.
TUBB.1	73	CGAGGACGAGGCTTAAAAAATTCTCAGATCAATCGTGCAATCCTTAGTGAACCTTCTGTTGTCTCAAGCAT GGT	2898
TUSC2.1	68	CACCAAGAACGGCAGAGAGCGGCCAAGCTGAGGCGAGTGCCATAAGAAATCTGAATTCCTCAGGGCATCG	2899
tusc4.2	68	GGAGGAGCTAAATGCCCTCAGGCCGGTGCACTCGCCCATTGATGAGTCCAAACACCATCCACTTGAAGG	2900
TXLNA.1	68	GCCAGAACGGCTCAGTCTGGGGCCCTTCGTGATGTCTCTGAGGAGCTGAGCCGCCAACTGGAAGACAT	2901
UBB.1	522	GAGTCGACCCCTGCACCTGGTCTCGCGTCTGAGAGGTGGTATGCAGATCTTGTGAAGACCCCTGACCGG CAAGCATCACCCTGGAGTGGAGCCCACTGACACCATCGAAAAATGTGAAGCCCAAGATCCAGGATAA AGAAGCATCCCTCCGACCAGCAGAGGCTCATCTTTGAGGCAAGCAGCTGGAATGGCCGCACTC TTTCTGACTACAACTCCAGAGGAGTCGACCTGCACCTGGTCTCGGCTCTGAGAAGGTGGTATCAGAA TCTTTCGTAAGACCTCCAGCCGACACCATCACTCTGGAAGTGGAGCCCACTGACACCATCGAAAAATG TGAAGCCAGATCCAAAGATAAAGAGGCATCCCTCCGACACAGCAGAGGCTCATCTTTGCAGGCAAGC AGCTGGAAGATGGCGCACCTTTCTGACTACAAATCCAGAGGAGTCGACCTGCACCTGGTCTCTGC GCCTGAGGGTGGCTGTTAATTTCTCAGTCATGGCATTCGC	2902
UBE1C.1	76	GAATGCACGCTGGAACTTTATCCACCACAGGTTAAATTTCCCATGTGCACCATTGCAATCTATGCCCCAGGC TACCAG	2903
UBE2C.1	67	TGTCGTGGCGATAAAGGATTTCTGCCTTCCTGAATCAGACAAACCTTTTCAAATGGGTAGGGACCCAG	2904
UBE2T.1	67	TGTTCTCAAATGCCACCAAAAGGTGCTTGGAGACCATCCCTCAAATCGCAACTGTGTTGACCTCT	2905
UGCG.1	73	GGCAACTGACAAAACAGCCTTATAGCAAGCTCCCAAGGTGTCTCTCTTCTGAAACCACTGAAAGGGGTAGA TCCT	2906
UMOD.1	66	GCCTGGACCTGGATGAGTGCGCCATTCTTGGAGCTCACAACTGCTCCGCCAAACAGCAGCTGGGTAA	2907
up a.3	70	GTGGATGGCCCTGAAGGACAAGCCAGGCGTCTACACGAGTCTCACACTTCTTACCCCTGGATCCGCA G	2908
UBP34.1	70	AAGCTGTGATGGCCAAAGCTTTGGCCCTCCAGGACCTCTGAGGTGGTATTATCTCTCAGTTGTGGCCATTC	2909
VCAM1.1	89	TGGCTTCAGGAGCTGAATACCTCCAGGCACACAGGTGGGACACAAATAGGGTTTTTGGAACCACT ATTTCTCATCACGACAGA	2910
VCAN.1	69	CCTGCTACACAGCCAACAGACCACCAATGTGGAAGACAAAGAGGCCTTTGGACCTCAGGCGCTTCT	2911
VDR.2	67	GCCTTGAATTTCAAGAGACCAAGTCTGGATCTGGGACCCCTTTCCTTCCTTCCTGGCTTGTAACT	2912
VEGF.1	71	CTGCTGTCTTGGGTGCATTTGGAGCCTTGCTGCTTACCTCCACCATGCCAAAGTGGTCCAGGCT GC	2913
VEGFB.1	71	TGACGATGGCCTGAGTGTGTGCCCACTGGGCAGCACCAAGTCCGGATGCCAGATCCTCATGATCCCGT ACC	2914

TABLE B-continued

Gene	Target Sequence Length	Amplicon Sequence	SEQ ID No.
VHL.1	67	CGTTGGTGACTTGTCTGCTCCTCTGCTTTGGGAAGACTGAGGCATCCGTGAGCGAGGACAAGTCCTT	2915
VIM.3	72	TGCCCTTAAGGAACCAATGATGTCCTGGAAAGCCAGATGCGTGAATGGAAGAACTTTGCCGTTGA AGC	2916
VTCN1.1	70	ACAGTGGTCTGGGCATCCCAAGTTGACAGGGAGCCAACCTTCGGAAGTCTCCAATACCAGCTTTTGAG C	2917
VTN.1	67	AGTCAATCTTCGCACACGGGAGTGGACACTGTGGAGCCCTCCCTACCCACGCTCCATCGCTCAGTAC	2918
VWF.1	66	TGAAGCACAGTGCCCTCTCCTCGAGCTGCACAGTGACATGGAGGTGACGGTGAATGGGAGACTGG	2919
WIF.1	67	AACAAGCTGAGTGCCCGAGGGGTGCCGAAATGGAGGCTTTTGTAAATGAAAGACGCATCTGCGAGTG	2920
WISP1.1	75	AGAGGCATCCATGAACTTCACACTTGGGGCTGCATCAGCACACGCTCCTATCAACCCAGTACTGTGG AGTTTG	2921
WT1.1	66	TGTACGGTCGGCATCTGAGACCACTGAGAACGCCCTTCATGTGTGCTTACCCAGGCTGCAATAA	2922
WWOX.5	74	ATCGCAGCTGGTGGGTGTACACTGTGTTTACCTTGGCGAGGCTTTCACCAAGTCCATGCAACAGG GAGCT	2923
XDH.1	66	TGTTGGCAGACATCCCTTCCTGGCCAGATACAGGTTGGCTTCATGAAGACTGGGACAGTTGTGGC	2924
XIAP.1	77	GCAGTTGGAAGACACAGGAAAGTATCCCCAAATTGCAGATTTATCAACGGCTTTTATCTTGAAAAATAGTG CCACGCA	2925
XPNPEP2.2	72	CACCTGCACTGAACATACCCCAAGAGCCCTGCTGGCCCATTCGCTAGAAAACTTTTGCAATTCATCCTCC TT	2926
YB-1.2	76	AGACTGTGGAGTTTGATGTTGTTGAAGAGAAAAGGGTGGGAGGCGACGAAATGTTACAGGTCCTGGTG GTGTTCC	2927
ZHX2.1	67	GAGTACGACCAAGTTAGCGGCCAAGACTGGCCCTGGTCCGAAGTGTGATGTCGCTGGTTCAGGAGA	2928

SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (<https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12412642B2>). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

What is claimed is:

1. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising:
 - measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of:
 - a set of genes consisting of one or more angiogenesis genes selected from: ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; and
 - a set of genes consisting of one or more immune response genes selected from: CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1; and
 - a set of genes consisting of one or more cell cycle genes selected from: BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; and optionally EIF4EBP1.
2. The method of claim 1, wherein the renal cancer is renal cell carcinoma (RCC).
3. The method of claim 2, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
4. The method of claim 1, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
5. The method of claim 1, wherein the renal tumor sample is paraffin-embedded and fixed.
6. The method of claim 1, wherein the panel consists of: at least two angiogenesis genes, at least two immune response genes, and at least two cell cycle genes, and optionally EIF4EBP1.
7. The method of claim 1, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.
8. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising:
 - measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of:
 - a set of genes consisting of one or more angiogenesis genes selected from: ADD1, ANGPTL3, APOLD1, CEACAM1, EDNRB, EMCN, ENG, EPAS1, FLT1, JAG1, KDR, KL, LDB2, NOS3, NUDT6, PPAP2B, PRKCH, PTPRB, RGS5, SHANK3, SNRK, TEK, ICAM2, and VCAM1; and
 - a set of genes consisting of one or more immune response genes selected from: CCL5, CCR7, CD8A, CX3CL1, CXCL10, CXCL9, HLA-DPB1, IL6, IL8, and SPP1; and
 - a set of genes consisting of one or more cell cycle genes selected from: BUB1, C13orf15, CCNB1, PTTG1, TPX2, LMNB1, and TUBB2A; and
 - a set of genes consisting of one or more transport genes selected from: AQP1, SGK1, and both AQP1 and SGK1; and
 - a set of genes consisting of one or more cell adhesion/extracellular matrix genes selected from: ITGB1, A2M, ITGB5, LAMB1, LOX, MMP14, TGFBR2, TIMP3, and TSPAN7; and optionally EIF4EBP1.
9. The method of claim 8, wherein the renal cancer is renal cell carcinoma (RCC).
10. The method of claim 9, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
11. The method of claim 8, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
12. The method of claim 8, wherein the renal tumor sample is obtained from a biopsy.
13. The method of claim 8, wherein the renal tumor sample is paraffin-embedded and fixed.
14. The method of claim 8, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.
15. A method of analyzing the expression of RNA transcripts of genes in a human renal cancer patient, the method comprising:
 - measuring levels of RNA transcripts, in a renal tumor sample from the patient, of a panel of genes consisting of:
 - at least five genes selected from the group consisting of: EMCN, PPAP2B, CCL5, CXCL9, NOS3, NUDT6, AQP1, APOLD1, ITGB5, CCR7, CX3CL1, CEACAM1, PRKCH, CASP10, SNRK, and PTPRB.
16. The method of claim 15, wherein the renal cancer is renal cell carcinoma (RCC).
17. The method of claim 16, wherein the RCC is clear cell renal cell carcinoma (ccRCC).
18. The method of claim 15, wherein the level of the RNA transcripts is measured by quantitative RT-PCR.
19. The method of claim 15, wherein the renal tumor sample is paraffin-embedded and fixed.
20. The method of claim 15, wherein the method further comprises performing global normalization of the levels of the RNA transcripts of the panel of genes.

* * * * *